

MAINTENANCE MANUAL - CONTENTS

Altivar 61-71

1. General

1.1 Objective	Page 5
1.2 Concerned Products	Page 5
1.3 Product design and fabrication	Page 8
1.4 Exchange, analysis and repair	Page 8
1.5 Assistance	Page 8

2. Required skills - training

2.1 Training	Page 9
--------------	--------

3. Documentation

3.1 Catalogue	Page 10
3.2 User's guide	Page 10
3.3 Intranet	Page 10
3.4 Vantive	Page 11

4. Repair service equipment

4.1 Material	Page 12
4.2 Dialogue and diagnostic tool	Page 17
4.3 Test bench	Page 17

5. Spare Parts List

5.1 General presentation	Page 18
5.2 Using of Platform	Page 18

6. Software

6.1 ATV61-71 Drive Software	Page 26
6.2 Graphic Terminal (VW3A1101)	Page 28
6.3 Logic I/O Boards (VW3A3201)	Page 28
6.4 Software of the extended input/output boards (VW3A3202)	Page 29
6.5 Fipio board software (VW3A3311 and VW3A3301)	Page 29
6.6 Modbus Plus board software (VW3A3302)	Page 29
6.7 Interbus-S board software (VW3A3304)	Page 30
6.8 Ethernet board software (VW3A3310)	Page 30
6.9 Unitelway / Modbus board software (VW3A3303)	Page 30
6.10 Profibus-DP board software (VW3A3307)	Page 31
6.11 Encoder interface board (VW3A3401 to VW3A3407)	Page 31
6.12 Devicenet board software (VW3A3309)	Page 31
6.13 Controller inside board software (VW3A3501)	Page 32

7. Troubleshooting the installation

7.1 Problem with the installation	Page 33
7.1.1 Unstable Speed	Page 33
7.1.2 The motor turns the wrong way	Page 34
7.1.3 Excessive Motor Temperature	Page 34
7.1.4 The motor accelerates too slowly	Page 35
7.1.5 Bad control of the motor stop or of deceleration	Page 36
7.1.6 The motor does not start	Page 36

8. Troubleshooting and repair guide

8.1 Schematics	
8.1.1 Power schematic for the Altivar < 18.5 kW	Page 38
8.1.2 Power schematic for Altivar > 18.5kW	Page 38
8.2 Diagnostics	
8.2.1 Introduction	Page 39
8.2.2 Preliminary static verification of an ATV	Page 39
8.2.3 Checking the input diode bridge and the output bridge	Page 40
8.2.4 Checking the thyristors of the input bridge	Page 40
8.2.5 Checking the braking transistor and the free-wheeling diode	Page 41
8.2.6 Checking the precharge fuses for drives > 110KW	Page 41
8.2.7 Checking the DC bus capacitors	Page 42
8.3 Electrical verification	
8.3.1 Product identification	Page 46
8.3.2 Input-output and communication word monitoring	Page 47
8.3.3 Display of past faults and the state of the drive	Page 48
8.3.4 Drive Auto-test	Page 48
8.3.5 Tests available only for SCHNEIDER services	Page 50
8.3.5.1 I / O Test	Page 51
8.3.5.2 Current calibration	Page 51
8.3.5.3 LIC test	Page 52
8.3.5.4 Voltage calibration	Page 53
8.3.5.5 Braking transistor test	Page 53
8.3.5.6 ETR	Page 53
8.3.5.7 Displaying a maintenance message (customisable)	Page 53
8.4 Description of LEDs status	
8.4.1 Power board and measuring board	Page 54
8.4.2 LED location on the power board or measuring board	Page 57
8.4.2.1 Measuring Board	Page 57
8.4.2.2 Power board VX5A1HC1316	Page 57
8.4.2.3 Power board VX5A1HC1622 & VX5A1HC2025	Page 57
8.4.2.4 Power board VX5A1HC3140 & VX5A1HC4050	Page 58
8.4.3 Fan Control Board (VX5A1400)	Page 58
8.5 Keypad Contrast Calibration	Page 59
8.6 Drives State	Page 59
8.7 Displayed Faults	Page 59

9. Changing the faulty component - drawings - parts lists

9.1	Safety instructions and precautions	
9.2	<u>ATV61/71 Size 2 (size, refer to 1.2)</u>	
9.2.1	Dismantling and reassembling	Page 110
9.2.2	Product Assembling Drawing	Page 116
9.2.3	Product Cabling Drawing	Page 116
9.3	<u>ATV61/71 Size 3 (size, refer to 1.2)</u>	
9.3.1	Dismantling and reassembling	Page 117
9.3.2	Product Assembling Drawing	Page 123
9.3.3	Product Cabling Drawing	Page 123
9.4	<u>ATV61/71 Size 4 (size, refer to 1.2)</u>	
9.4.1	Dismantling and reassembling	Page 124
9.4.2	Product Assembling Drawing	Page 129
9.4.3	Product Cabling Drawing	Page 129
9.5	<u>ATV61/71 Size 5A (size, refer to 1.2)</u>	
9.5.1	Dismantling and reassembling	Page 130
9.5.2	Product Assembling Drawing	Page 135
9.5.3	Product Cabling Drawing	Page 135
9.6	<u>ATV61/71 Size 5B (size, refer to 1.2)</u>	
9.6.1	Dismantling and reassembling	Page 136
9.6.2	Product Assembling Drawing	Page 142
9.6.3	Product Cabling Drawing	Page 142
9.7	<u>ATV61/71 Size 6 (size, refer to 1.2)</u>	
9.7.1	Dismantling and reassembling	Page 143
9.7.2	Product Assembling Drawing	Page 154
9.7.3	Product Cabling Drawing	Page 154
9.8	<u>ATV61/71 Size 7A (size, refer to 1.2)</u>	
9.8.1	Dismantling and reassembling	Page 155
9.8.2	Product Assembling Drawing	Page 166
9.8.3	Product Cabling Drawing	Page 166
9.9	<u>ATV61/71 Size 7B (size, refer to 1.2)</u>	
9.9.1	Dismantling and reassembling	Page 167
9.9.2	Product Assembling Drawing	Page 178
9.9.3	Product Cabling Drawing	Page 178
9.10	<u>ATV61/71 Size 8 (size, refer to 1.2)</u>	
9.10.1	Dismantling and reassembling	Page 179
9.10.2	Product Assembling Drawing	Page 190
9.10.3	Product Cabling Drawing	Page 190
9.11	<u>ATV61/71 Size A2 (size, refer to 1.2)</u>	
9.11.1	Dismantling and reassembling	Page 191
9.11.2	Product Assembling Drawing	Page 198
9.11.3	Product Cabling Drawing	Page 198
9.12	<u>ATV61/71 Size A3 (size, refer to 1.2)</u>	
9.12.1	Dismantling and reassembling	Page 199
9.12.2	Product Assembling Drawing	Page 205
9.12.3	Product Cabling Drawing	Page 205
9.13	<u>ATV61/71 Size B (size, refer to 1.2)</u>	
9.13.1	Dismantling and reassembling	Page 206
9.13.2	Product Assembling Drawing	Page 212
9.13.3	Product Cabling Drawing	Page 212

9.14	<u>ATV61/71 Size C (size, refer to 1.2)</u>	
9.14.1	Dismantling and reassembling	Page 213
9.14.2	Product Assembling Drawing	Page 219
9.14.3	Product Cabling Drawing	Page 219
9.15	<u>ATV61/71 Size D (size, refer to 1.2)</u>	
9.15.1	Dismantling and reassembling	Page 220
9.15.2	Product Assembling Drawing	Page 229
9.15.3	Product Cabling Drawing	Page 229
9.16	<u>ATV61/71 Size E (size, refer to 1.2)</u>	
9.16.1	Dismantling and reassembling	Page 230
9.16.2	Product Assembling Drawing	Page 244
9.16.3	Product Cabling Drawing	Page 244
9.17	<u>ATV61/71 Size F (size, refer to 1.2)</u>	
9.17.1	Dismantling and reassembling	Page 245
9.17.2	Product Assembling Drawing	Page 259
9.17.3	Product Cabling Drawing	Page 259
9.18	<u>ATV61/71 Size G (size, refer to 1.2)</u>	
9.18.1	Dismantling and reassembling	Page 260
9.18.2	Product Assembling Drawing	Page 277
9.18.3	Product Cabling Drawing	Page 277
9.19	<u>ATV61/71 Size S6Y (size, refer to 1.2)</u>	
9.19.1	Dismantling and reassembling	Page 278
9.19.2	Product Assembling Drawing	Page 292
9.19.3	Product Cabling Drawing	Page 292
9.20	<u>ATV61/71 Size S8Y (size, refer to 1.2)</u>	
9.20.1	Dismantling and reassembling	Page 293
9.20.2	Product Assembling Drawing	Page 305
9.20.3	Product Cabling Drawing	Page 305
9.21	<u>ATV61/71 Size 9 (size, refer to 1.2)</u>	
9.21.1	Dismantling and reassembling	Page 306
9.21.2	Product Assembling Drawing	Page 335
9.21.3	Product Cabling Drawing	Page 335
9.22	<u>ATV61/71 Size 10 (size, refer to 1.2)</u>	
9.22.1	Dismantling and reassembling	Page 336
9.22.2	Product Assembling Drawing	Page 365
9.22.3	Product Cabling Drawing	Page 365
9.23	<u>ATV61/71 Size 11 (size, refer to 1.2)</u>	
9.23.1	Dismantling and reassembling	Page 366
9.23.2	Product Assembling Drawing	Page 393
9.23.3	Product Cabling Drawing	Page 393
9.24	<u>ATV61/71 Size 12 (size, refer to 1.2)</u>	
9.24.1	Dismantling and reassembling	Page 394
9.24.2	Product Assembling Drawing	Page 421
9.24.3	Product Cabling Drawing	Page 421
9.25	<u>ATV61/71 Size 13 (size, refer to 1.2)</u>	
9.25.1	Dismantling and reassembling	Page 422
9.25.2	Product Assembling Drawing	Page 449
9.25.3	Product Cabling Drawing	Page 449
9.26	<u>ATV61/71 Size 14 (size, refer to 1.2)</u>	
9.26.1	Dismantling and reassembling	Page 450
9.26.2	Product Assembling Drawing	Page 478
9.26.3	Product Cabling Drawing	Page 478

9.27 <u>ATV61/71 Size 15 (size, refer to 1.2)</u>	
9.27.1 Dismantling and reassembling	Page 479
9.27.2 Product Assembling Drawing	Page 506
9.27.3 Product Cabling Drawing	Page 506
9.28 <u>ATV61/71 Size 11V (size, refer to 1.2)</u>	
9.28.1 Dismantling and reassembling	Page 507
9.28.2 Product Assembling Drawing	Page 533
9.28.3 Product Cabling Drawing	Page 533
9.29 <u>ATV61/71 Size 13V (size, refer to 1.2)</u>	
9.29.1 Dismantling and reassembling	Page 534
9.29.2 Product Assembling Drawing	Page 563
9.29.3 Product Cabling Drawing	Page 563
9.30 <u>ATV61/71 Size 15V (size, refer to 1.2)</u>	
9.30.1 Dismantling and reassembling	Page 567
9.30.2 Product Assembling Drawing	Page 600
9.30.3 Product Cabling Drawing	Page 600

10. Procedure for validation after repair

10.1 ATV61/71 Sizes (S2 to S15)	Page 601
10.2 Quality reporting to the VSD division	Page 601

11. Structure Drives

11.1 Common part: Control Bloc	Page 602
11.2 Electronic & mechanical structure of all drives	Page 602

12. Inspection and Maintenance

12.1 Regular Inspection	Page 610
12.2 Periodical Inspection	Page 611
12.3 Keeping the drive in storage	Page 614
12.4 Disposal of the drive	Page 615

1. In General

1.1 Objective

The present document defines the policies and the methods of implementation of local repair for the Altivar 61/71.

The repair will be done by the replacement of defective parts by new components furnished by SCHNEIDER. It lies exclusively within the scope of the GRAP (Guarantee Return Analysis Process) project.

1.2 Concerned Products

Catalogue number structure IP20:

ATV71 H D18 N4 Z
 1 2 3 4 5



1 – Product family

ATV71: variable speed drive for asynchronous three phase motors.
 ATV61: variable speed drive for asynchronous three phase motors.

2 – Variations of the technology

H: standard product on a heatsink.
 E5: ≤ 75 kW (with switch).
 E5: ≥ 90 kW (with switch + fast acting fuses).
 P: product on base plate (only for ATV71).

3- Rating

The rating is expressed in motor kW.
 The letter of the prefix gives the position of the decimal point.
 0: 0.01
 U: 0.1
 D: 1
 C: 10

4 – Distribution network

N4: 380- 480V three phase.
 M3: 200-240V three phase.
 Y: 690V three phase.

5 – Terminal

Nothing: Removable graphic terminal included.
 Z: Only integrated 7 segment terminal (only up to 75KW).

6 – EMC filter

Nothing: Class A EMC filter integrated.
 X: without integrated EMC filter.

7 – Specification for aggressive ambient

Nothing: standard protection.
 337: protected by conformable coating.

Catalogue number structure IP54:

ATV61 **W** **U** **N4** **Z**
 1 2 3 4 5



1 – Product family

ATV71: variable speed drive for asynchronous three phase motors.
 ATV61: variable speed drive for asynchronous three phase motors.

2 – Variations of the technology

W: IP54
 E5: ≤ 75 kW (with switch)
 E5: ≥ 90 kW (with switch + fast acting fuses)

3- Rating

The rating is expressed in motor kW. The letter of the prefix gives the position of the decimal point.
 0: 0.01
 U: 0.1
 D: 1

4 – Distribution network

N4: 380- 480V three phase

5 – EMC filter

Nothing: Class A EMC filter integrated
 Z: Only integrated 7 segment terminal (only up to 75KW)
 C: Drive class B
 A24: Drive with power supply allowing an additional consumption of 250mA.
 C24: Drive class B with power supply allowing an additional consumption of 250mA.
 337: Protected by conformable coating.

IP20 Frame Size	Catalogue Number
T 2	ATV71H037M3, ATV71H075M3, ATV71HU15M3, ATV71H075N4, ATV71HU15N4, ATV71HU22N4 ATV61H075M3, ATV61HU15M3, ATV61H075N4, ATV61HU15N4, ATV61HU22N4
T 3	ATV71HU22M3, ATV71HU30M3, ATV71HU40M3, ATV71HU30N4, ATV71HU40N4 ATV61HU22M3, ATV61HU30M3, ATV61HU40M3, ATV61HU30N4, ATV61HU40N4
T 4	ATV71HU55M3, ATV71HU55N4, ATV71HU75N4, ATV61HU55M3, ATV61HU55N4, ATV61HU75N4
T 5A	ATV71HU75M3, ATV71HD11N4
T 5B	ATV71HD11M3X, ATV71HD15M3X, ATV71HD15N4, ATV71HD18N4 ATV61HD11M3X, ATV61HD15M3X, ATV61HD15N4, ATV61HD18N4
T 6	ATV71HD18M3X, ATV71HD22M3X, ATV71HD22N4 ATV61HD18M3, ATV61HD22M3, ATV61HD22N4
T 7A	ATV71HD30N4, ATV71HD37N4, ATV61HD30N4, ATV61HD37N4 ATV71HD30M3X, ATV71HD37M3X, ATV71HD45M3X, ATV61HD30M3X, ATV61HD37M3X, ATV61HD45M3X
T 7B	ATV71HD30M3X, ATV71HD37M3X, ATV71HD45M3X
T 8	ATV71HD45N4, ATV71HD55N4, ATV71HD75N4 ATV61HD45N4, ATV61HD55N4, ATV61HD75N4
T 9	ATV71HD55M3X, ATV71HD90N4 ATV61HD55M3X, ATV61HD75M3X, ATV61HD90N4, ATV61HC11N4
T 10	ATV71HD75M3X, ATV71HC11N4, ATV61HD90M3, ATV61HC13N4
T 11	ATV71HC13N4, ATV61HC16N4
T 12	ATV71HC16N4, ATV61HC22N4
T 13	ATV71HC20N4, ATV71HC25N4, ATV71HC28N4, ATV61HC25N4, ATV61HC31N4
T 14	ATV71HC31N4, ATV71HC40N4, ATV61HC40N4, ATV61HC50N4
T 15	ATV71HC50N4, ATV61HC63N4

IP54 Frame Size	Catalogue Number
A2	ATV71W0.37N4, ATV71WU15N4, ATV71WU22N4, ATV61W0.75N4, ATV61W0.75N4C, ATV61WU15N4, ATV61WU15N4C, ATV61WU22N4, ATV61WU22N4C, ATV61WU30N4, ATV61WU30N4C
A3	ATV71WU30N4, ATV71WU40N4, ATV61WU40N4, ATV61WU40N4C, ATV61WU55N4, ATV61WU55N4C
B	ATV71WU40N4, ATV71WU55N4, ATV61WU75N4, ATV61WU75N4, ATV61WU75N4C, ATV61WD11N4, ATV61WD11N4C
C	ATV71WD11N4, ATV61WD15N4, ATV61WD15N4C
D	ATV71WD15N4, ATV71WD18N4, ATV61WD18N4, ATV61WD18N4C, ATV61WD22N4, ATV61WD22N4C
SE	ATV61WD30N4, ATV61WD30N4C, ATV71WD22N4
SF	ATV61WD37N4, ATV61WD45N4, ATV61WD37N4C, ATV61WD45N4C, ATV71WD30N4, ATV71WD37N4
SG	ATV61WD55N4, ATV61WD75N4, ATV61WD90N4, ATV61WD55N4C, ATV61WD75N4C, ATV61WD90N4C ATV71WD45N4, ATV71WD55N4, ATV71WD75N4

690V Frame Size	Catalogue Number
S6	ATV71HU22Y, ATV71HU30Y, ATV71HU40Y, ATV71HU55Y, ATV71HU75Y, ATV71HD11Y, ATV71HD15Y, ATV71HD18Y, ATV71HD22Y, ATV71HD30Y, ATV61HU30Y, ATV61HU40Y, ATV61HU55Y, ATV61HU75Y, ATV61HD11Y, ATV61HD15Y, ATV61HD18Y, ATV61HD22Y, ATV61HD30Y
S8	ATV71HD37Y, ATV71HD45Y, ATV71HD55Y, ATV71HD75Y, ATV71HD90Y, ATV61HD37Y, ATV61HD45Y, ATV61HD55Y, ATV61HD75Y, ATV61HD90Y
S11	ATV71HC11Y, ATV71HC13Y, ATV71HC16Y, ATV61HC11Y, ATV61HC13Y, ATV61HC16Y, ATV61HC20Y
S13	ATV71HC20Y, ATV71HC25Y, ATV71HC31Y, ATV61HC25Y, ATV61HC31Y, ATV61HC40Y
S15	ATV71HC40Y, ATV71HC50Y, ATV71HC63Y, ATV61HC50Y, ATV61HC63Y, ATV61HC80Y

1.3 Product design and fabrication

All the modules as well as the accessories and software are conceived and developed by Schneider Electric or in collaboration with certain Schneider partners.

At every step of design and fabrication, the product is treated with the greatest care. The products are continually monitored to insure a high level of quality.

The ATV61/71 has been developed according to the principals of Eco-design. It meets the norms ISO-14001, ROHS, WEEE. It does not include any heavy metals, its components are recyclable.

1.4 Exchange, analysis and repair

The Altivar 61/71 is a repairable product by:

- Local repair for countries that have been trained.
- Factory repair for the other countries.

For factory repair, analysis or expert assistance, consult the Intranet site: (<http://planet.schneider-electric.com>) menu quality

1.5 Assistance

1.5.1 Intervention on site

On site, assistance will be provided by the technicians and engineers of the SCHNEIDER «services" group of the customer's country.

1.5.2 Technical support

The first level of technical support is done by the level 1 and 2 forums upon which the customer depends.

The third level of technical support is done by the Schneider Vantive intranet site: Vantive self service (<http://planet.schneider-electric.com>) then technical support & logistic or in case of emergency by the Global Help Desk direct line:

Tel: +33 (0)1 41 39 39 00

Fax: +33 (0)2 32 78 17 12

1.5.3 Customer Returns

The return of failed materials (complete products or sub-assemblies) is done following the guarantee policy of SCHNEIDER.

For any product return, fill-out and attach the following card. This form will soon be available on the Intranet.

1.5.4 Update of the installed base

In case of a major fault or the addition of new functionality, the update of products may become necessary.

In this case, information will be distributed throughout the SCHNEIDER network.

2. Required skills - training

2.1 Training

2.2.1. Customer and Schneider technical support level 1

The I.S.F. (Schneider Training Institute) organizes training courses on ATV for people to become familiar with these products and learn how to install them.

The catalogue for this course is available on the Internet site:

<http://www.schneiderformation.com>

ATV71 ISF inclass 3 days

Offer, programming and commissioning

ATV71 controller inside ISF inclass 3 days

Programming of controller inside board

2.2.2. Technical support level 2 and after sales services

The Expert training and After Sales Training are provided by STIE and reserved exclusively to Schneider people

The schedule is available on the Intranet site

<http://planet.schneider-electric.com/Intranet/planet-home.nsf>

ATV71 Expert STIE inclass 5 days

Advanced programming and commissioning, application functions

ATV71 After sales STIE inclass 3.5 days

Diagnostic, repairing, test

ATV71 After sales (690V) SEPD inclass 2 days

Diagnostic, repairing, test

ATV71 Communication GTS inclass 3 days

Communication bus applied to ATV

For more information contact STIE training department

Tel 00 33 (0) 2 32 78 11 00

Fax 00 33 (0) 2 32 78 11 64

2.2.4. Training material

Training materials are available from the following sources:

<http://planet.schneider-electric.com/Intranet/planet-home.nsf>

For more information, contact STIE training department

Tel 00 33 (0) 2 32 78 11 00

Fax 00 33 (0) 2 32 78 11 64

3. Documentation

3.1 Catalogue

English version : réf. DIA2ED2051209EN article number : 815133

French version: ref. DIA2ED2051209FR article number: 815141

3.2 User's Guide:

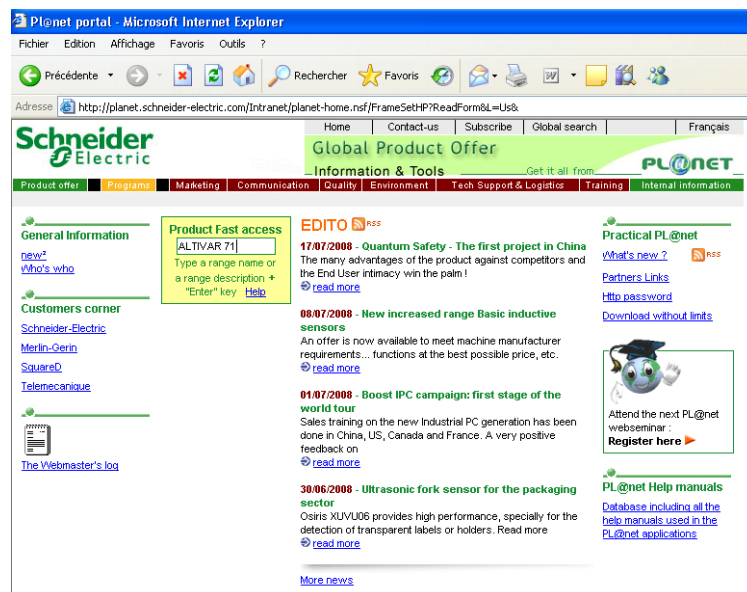
The user's guides exist on the CD-ROM delivered with the drive:

- Simplified manual
- Installation manual
- Programming manual
- Profibus manual
- Interbus manual
- Integrated modbus manual
- Integrated Canopen manual
- Controller inside manual
- Communication parameters

3.3 Intranet

There are two possibilities to access the data through the SCHNEIDER intranet:

- On the web site <http://planet.schneider-electric.com>
Select "product offer", "products by organization", "Automation Business Unit", "Motion & drives".
- On the web site <http://planet.schneider-electric.com>
Directly type the product name



This site manages some technical and marketing information files concerning VVD products (software versions, application files ...).

3.4 Vantive

Access to this database is through the SCHNEIDER intranet.

- <http://planet.schneider-electric.com> , "technical support & logistic", "Access to Schneider Technical support".

Access to this database allows:

- Ask a precise question to have a personalised response.
- For a country, consult its file of questions in course of response or completed.
- Interrogate the database for all the questions already posed to find an appropriate response or a path for further research.

4. Repair service methods

4.1 Material

List of tools necessary to repair the ATV61-71:

➤ **General materials:**

Multimeter

Laptop

➤ **Tools**

Picture	Equipments	Reference RadioSpares	2008 Catalog Page
	Tools Box	611-4183	2-4496
	S/steel shackle brass padlock,40mm	339-9505	2-3864
	Bent nose insulated plier	539-621	2-4717
	Big pliers with beak isolated	162-1563	2-4717
	Combination insulated plier,185mm L	621-584	2-4717
	Big isolated sharp pliers	343-8953	2-4716
	Waterpump insulated plier,250mm L	621-635	2-4717
	Screwdriver insulate AY2,5X75VE	348-2843	2-4747
	Screwdriver insulate AY3,5X75VE	348-2859	2-4747
	Screwdriver insulate AY4X100VE	348-2871	2-4747
	Screwdriver insulate AY5,5X125VE	348-2887	2-4747
	Screwdriver insulate APY.0X75VE	348-2786	2-4747
	Screwdriver insulate APY.1X100VE	348-2792	2-4747
	Screwdriver insulate APY.2X125VE	348-2809	2-4747
	Screwdriver insulate ADY.0X75VE	348-2815	2-4747
	Screwdriver insulate ADY.1X100VE	348-2821	2-4747
	Screwdriver insulate ADY.2X125VE	348-2837	2-4747
	8 L-SHAPED Torx Drivers	250-4267	2-4786
	9pcs L-shape metric hex ball driver set	297-1544	2-4763
	12 flat keys mix	454-4201	2-4795
	Cartridges and tips	162-4360	2-4780
	Small presser	194-2652	2-4711
	Small sharp pliers	194-2618	2-4696

Picture	Equipments	Reference RadioSpares	2008 Catalog Page
	box spanners	454-4217	2-4795
	Bent Point Tweezers	192-2024	2-4720
	Standard Tweezers	420-5805	2-4720
	1 plier straight	269-4297	2-4720
	Toukan 2 lames	454-4239	2-4729
	Automatic wire stripper, 1-3.2mm strip	663-617	2-4735
	Magnetic inspection torch, 450mm long arm	493-7439	2-4664

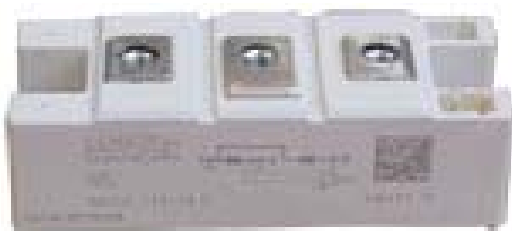
➤ **Mounting of power modules**

If semiconductor power modules (diodes, thyristors, IGBTs) are exchanged, it is very important to mount the new modules in the correct way. This refers to the use of thermal grease as well as the correct way to tighten the modules on the heat sink.

➤ **Thermal grease**

To ensure a good thermal connection of the modules to the heat sink, the use of thermal grease is mandatory.

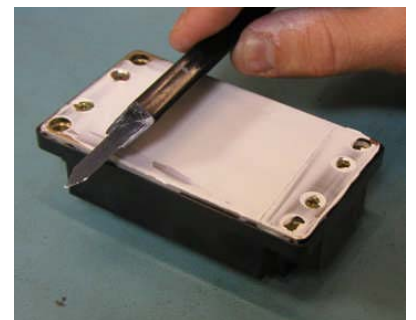
The following pictures show, how the grease has to be smeared on the modules:



Thyristor



Transistor

Width of the module 5cm**Width of the module 6,2cm**

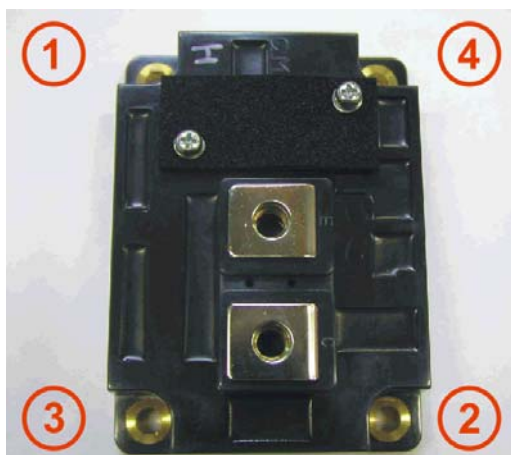
Put the grease in stripes on the module and dispense it evenly with a spatula to approx. 0.3 mm thickness.

Tightening of modules

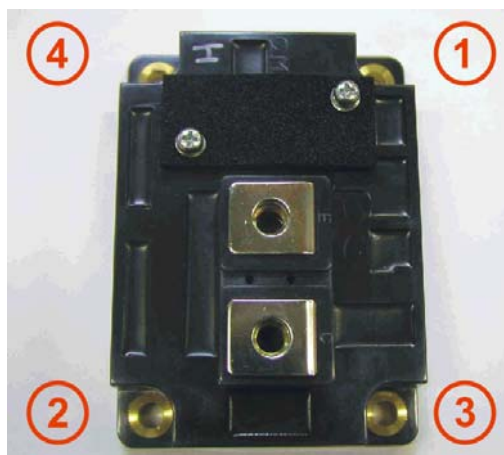
After putting thermal grease on the module, it is important to fasten the modules in the correct way and with the right torque.

The following table shows the necessary mounting torques and has to be followed exactly:

Size	Thyr/Diod. modul (M5)		Thyristor (M5)		Diode (M5)		Brake (M6)		IGBT (M6)	
	Nm (pre)	Nm (fix)	Nm (pre)	Nm (fix)	Nm (pre)	Nm (fix)	Nm (pre)	Nm (fix)	Nm (pre)	Nm (fix)
6	?	?	?	?	?	?	?	?	?	?
7	?	?	?	?	?	?	?	?	?	?
8	?	?	?	?	?	?	?	?	?	?
9	1.0	2.6	x	x	x	x	1.5	3.0	1.5	4.4
10	1.0	2.6	x	x	x	x	1.5	3.0	1.5	4.4
11	2.0	5.0	x	x	x	x	1.5	3.0	1.5	3.0
12	2.0	5.0	x	x	x	x	1.5	3.0	1.5	3.0
13	x	x	2.0	5.0	2.0	5.0	x	x	1.5	3.0
14	x	x	2.0	5.0	2.0	5.0	x	x	1.5	3.0
15	x	x	2.0	5.0	2.0	5.0	x	x	1.5	3.0
BU 13	x	x	x	x	x	x	1.5	3.0	1.5	3.0
BU 14/15	x	x	x	x	x	x	1.5	3.0	1.5	3.0



Pre-fix all screws in the correct order with the torque stated in the table: Nm (pre)



Fix the screws in the correct order with the torque stated in the table: Nm (fix)

➤ ***Specific tools***

To update the software version of controls blocks and graphic terminal, there is a specific tool: **Flash KIT**

You will find in the catalog, the both commercial reference.

Flash kit: VW3A9703 (flash the main μ P, flash the motor control μ P)

Flash kit: VW3A9702 (flash the graphic terminal)

4.2 *Dialog and diagnostic tools*

4.2.1 Dialog tools

A first level of diagnostic is possible by using the graphic terminal or the commissioning software Powersuite.

These two tools allow to:

- Check the state of the inputs/outputs.
- Check the image of the communication words.
- Read the fault history and see the operating state when a fault appears.
- Save and transfer the drive configurations.
- Perform some product diagnostic checks.

4.3 *Test bench*

To test the ATV71 under power after repairing, you will find at the end of this chapter the diagrams of a test bench. The same was used to test the other ATV. Only the cable linking the board connector to the 15 point Sub-D connector is different.

5. Spare Parts List

5.1 General presentation

Introduction

This manual describes how to use the platform (Interface VBA) under Microsoft Office Excel for the ATV61-71.

The Spare Part Package

The spare part package is used by the platform. You will find the following elements:

- Pictures used by the platform.
- Drives Pictures.
- Pictures used for the spare parts (screws, boards, mechanic parts, wires, etc...).

Recommendation

We recommend for that the platform correctly run:

- A laptop with Microsoft Windows XP and Microsoft Office 2003.

5.2 Using of Platform

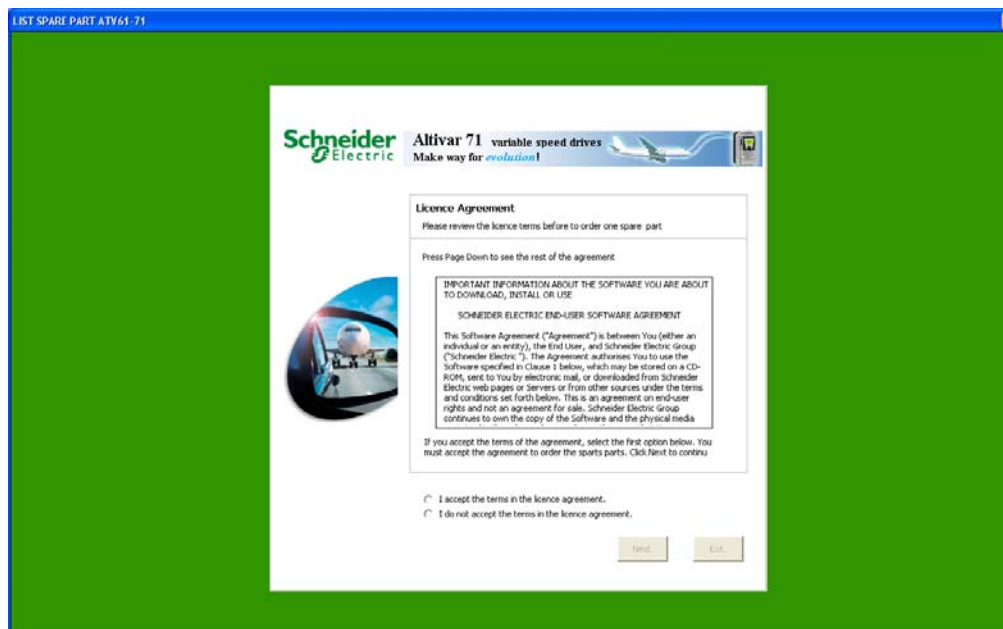
Introduction

This following section explains how to use the platform (Interface VBA) to find the reference of spares part on the ATV61-71 to order to repair them correctly.

Using Interface

Warning, to use correctly the platform, you must configure your computer (display resolution: 1280 x 800).

- Launch the following file: Spare part list ATV61-71 Release V1.5.xls
- The following window appears.



☐ I accept the terms in the licence agreement.

☐ I do not accept the terms in the licence agreement.

Next Exit

- ☐ I accept the terms in the licence agreement.
- ☒ I do not accept the terms in the licence agreement.
- Next Exit

☒ I accept the terms in the licence agreement.

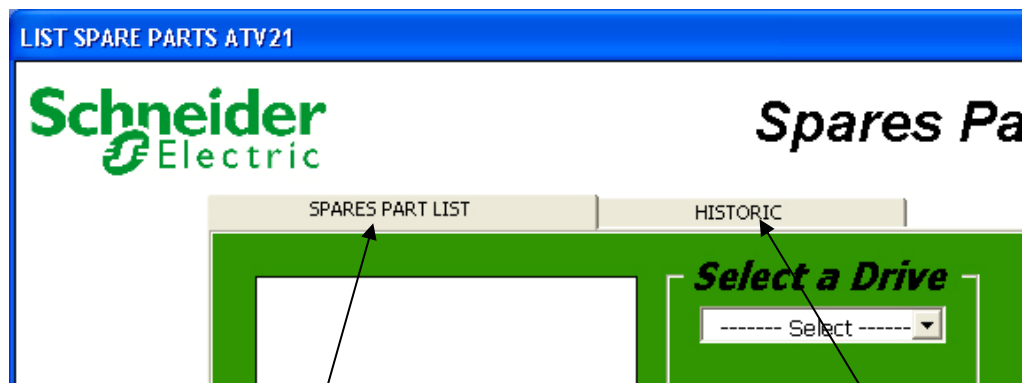
☐ I do not accept the terms in the licence agreement.

Next Exit

[illegible]

Now, you can use the interface...

Descriptive Tabs



Select the tab « SPARES PART LIST » to get the commercial references of spares part.

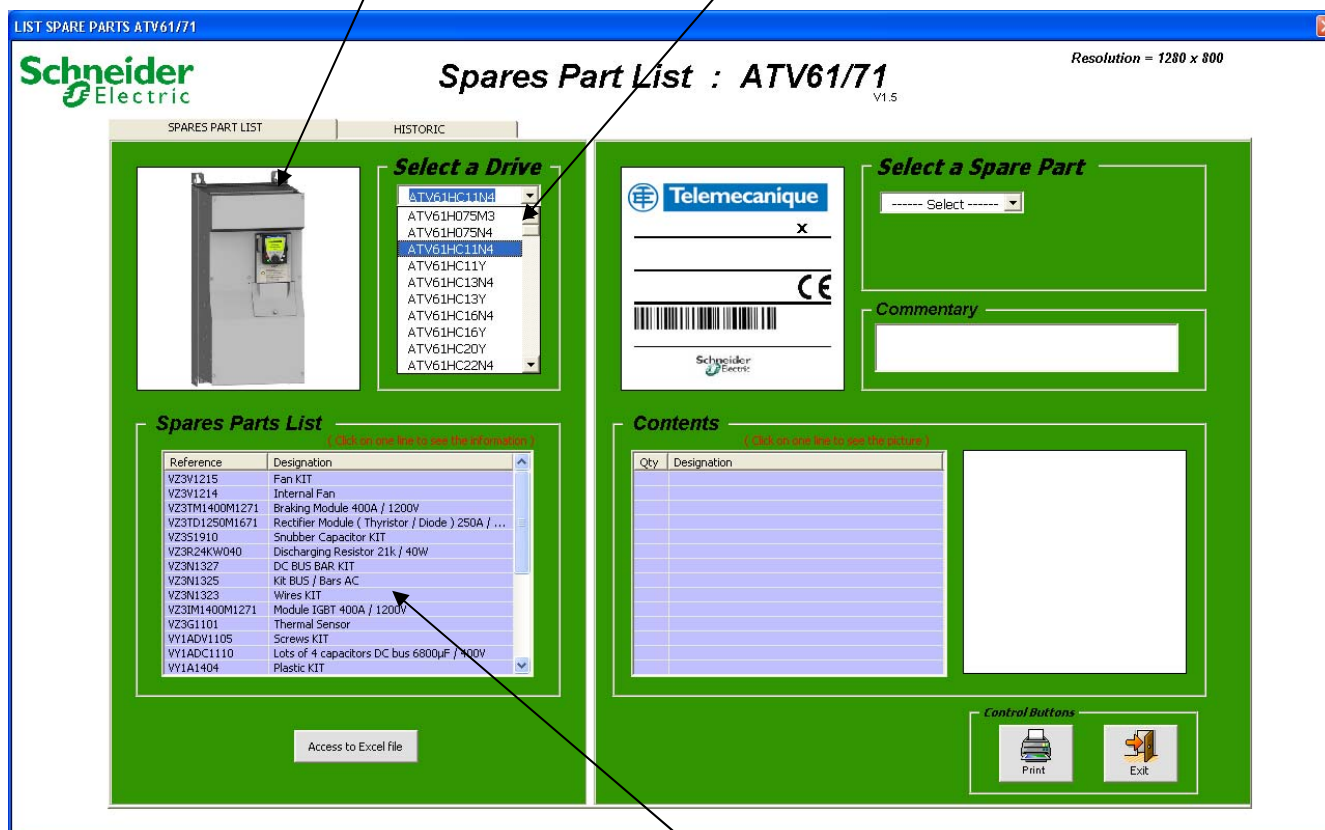
Select the tab « HISTORIC » to see updates.

Descriptive buttons

- One click on the “**Select one drive**” button allows to show the list of all drives.

The drive picture appears.

Choose the drive reference.



The references spares part appears with them designations.

LIST SPARE PARTS ATV61/71

Schneider Electric

Spares Part List : ATV61/71 V1.5

Resolution = 1280 x 800

SPARES PART LIST HISTORIC

Select a Drive
ATV61HC11M4

Select a Spare Part
----- Select -----

Spares Parts List (Click on one line to see the picture)

Reference	Designation
VZ3N1215	Fan KIT
VZ3N1214	Internal Fan
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3TD1250M1671	Rectifier Module 250A / 1600V
VZ3N1310	Snubber Capacitor KIT
VZ3R24KWO40	Discharging Resistor 21k / 40W
VZ3N1307	DC BUS BAR KIT
VZ3N1305	VA BUS / Bars AC
VZ3N1303	Wires KIT
VZ3BM1400M1271	Module 10BT 400A / 1200V
VZ3G1101	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1110	Lots of 4 capacitors DC bus 6000uF / 400V
VY1A1404	Posidr KIT

Access to Excel file

Contents (Click on one line to see the picture)

Qty	Designation
1	Module EUPEC TD250N16KOF
1	Heat Conductive Paste

Control Buttons
Print Exit

Unity of KIT.
Warning, the number can also indicate the quantity of peace inside of the box.

Click on one line.

Commercial Reference

- One click on one of lines of menu “**contents**” allows to show the picture of the part.

Telemechanique

920861 x 3
Rectifier Module (Thyristor / Diode) 250A / 1600V
VZ3TD1250M1671
03389119208611

Select a Spare Part
----- Select -----

Commentary
Nothing Commentary

Contents (Click on one line to see the picture)

Qty	Designation
1	Module EUPEC TD250N16KOF
1	Heat Conductive Paste

Picture of selected part.

Quantity for the selected part.

Designation of the selected part.

- One click on the “**Select one spare part**” button allows to show the list of all spare parts.

Unity of KIT.
Warning, the number can also indicate the quantity of peace inside of the box.

Select a Spare Part

Choose the spare part reference.

Commentary on the spare part.

Commercial Reference

Designation of the selected spare part.

Select a Spare Part

Selected a product to see the quantity used in the drive.

The commercial reference (spare part) chooses is used by these products.

- One click on one of lines of menu “**contents**” allows to show the picture of the part.

Contents
(Click on one line to see the picture)

Qty	Designation
1	R145 Trap
3	Screw M3X8
1	Display Support
1	Control Trap
1	Display PCBA
1	Keyboard

Quantity for the selected part.

Designation of the selected part.

Picture of selected part.

- 
- Control Buttons**
- The diagram shows two buttons: a 'Print' button with a printer icon and an 'Exit' button with a door icon. An arrow points from the 'Print' button to the 'Exit' button.

The following window appears.

[illegible]

- The 'Control Buttons' section contains two buttons: 'Print' (represented by a printer icon) and 'Exit' (represented by a door icon with an arrow pointing out). A black arrow points from the 'Exit' button to the 'Exit' text in the adjacent 'Exit Confirmation' section.

Click on the button.

-
- The screenshot displays two side-by-side application windows. The left window, titled "Spares Parts List", contains a table with two columns: "Reference" and "Designation". The right window, titled "Contents", contains a table with two columns: "Qty" and "Designation". Both tables have multiple empty rows. A "Microsoft Excel" password dialog box is overlaid in the center, asking the user to "Enter the password" and providing "OK" and "Annuler" buttons. Below the dialog box, a button labeled "Access to Excel file" has an arrow pointing to the password input field in the Excel dialog.

The following window appears.

Warning, the password is used only by the developer.

6. Software

6.1 ATV61-71 Drive Software

Any anomalies corrected in a version IE x are also corrected in version IE x + 1.

➤ Version of control block:

ATV71 product release	Promotion Level	ATV71 Application SwVersion	ATV71 MotorControl SwVersion	ATV71 HHP MotorControl SwVersion	Product type	Short description	Known anomalies
V1.1IE01#7	PRODUCTION	A1.1IE01#9	A1.1IE01#9		Standard	First commercial release	Trip in 'INFE' fault if encoder board is installed with switching frequency 16 KHz and input phase loos to stop controlled. This problem solved in the version V1.1IE02. In torque control mode if the reference is negative after switch-on, the drive does not run with the first start command. This problem solved in the version V1.1IE02.
V1.1IE02#2	PRODUCTION	A1.1IE02#5	A1.1IE02#2		Standard		
V1.1IE04#2	PRODUCTION	A1.1IE03#21	A1.1IE03#9	P1.1IE02#1	Standard	Second commercial release for HHP	
V1.1IE05#6	PRODUCTION	A1.1IE04#25	A1.1IE04#9	P1.1IE03#44	Standard	HHP MotorControl with support of Sizes14/15, HHP200V, Standard Indus and Variable Torque for ATV61. Switching frequency fixed to 2,5kHz for HHP drives.	
V1.1IE06#2	PRODUCTION	A1.1IE05#9	A1.1IE06#3	P1.1IE04#48	Standard	New release for all sizes	
V1.1IE07#4	PRODUCTION	A1.1IE06#18	A1.1IE07#14	P1.1IE05#53	Standard	Commercial release with MotorControl Masked	
V1.1IE08#2	PRODUCTION	A1.1IE06#18	A1.1IE07#14	P1.1IE06#62	Standard	Correction HHP : - Use of Voltage Calibration - HDF fault if PowerRemoval asked while the motor is running - Suppress INFB fault when there is no brake module and BRA = No	

ATV71 product release	Promotion Level	ATV71 Application SwVersion	ATV71 MotorControl SwVersion	ATV71 HHP MotorControl SwVersion	Product type	Short description	Known anomalies
V1.1IE09#3	PRODUCTION	A1.1IE06#18	A1.1IE07#14	P1.1IE07#77	Standard	Correction HHP : - Unexpected interruption of fluxing - Unexpected BLF faults	
V1.1IE11#1	PRODUCTION	A1.1IE06#18	A1.1IE08#2	P1.1IE07#77	Standard	Standard release with correction for 75 kW 400V broken modules	
V1.2IE12#12	PRODUCTION	A1.2IE09#51	A1.2IE09#1	P1.2IE08#85	Standard	V1.2 Commercial release : iteration 12	
V1.3IE13#6	PRODUCTION	D1.3IE10#120	C1.3IE10#22	-	S383	RSX-C step 1 release (Resolver) : iteration 6 Based on ATV71 V1.2IE12#10 product	
V1.2IE16#2	PRODUCTION	A1.2IE13#1	A1.2IE09#1	P1.2IE09#87	Standard	V1.2 Commercial release : maintenance T15#2 Ré-integred in the standard version	
MC_FOR TSIJ	PRODUCTION	-	A1.2IE12#9	-	Standard	Based on MC LHP A1.2IE09#1 Specific corrections for TSIJ	
V1.2IE18#4	PRODUCTION	A1.2IE16#14	A1.2IE13#2	P1.2IE10#93	Standard	V1.2 Commercial release : maintenance mars	
V1.7IE20#2	PRODUCTION	D1.7IE18#2	C1.4IE14#2		S383	RSX-C release : step 2 (Univ Encoder)	
V1.9IE22#6	PRODUCTION	D1.9IE20#6	C1.6IE16#3	P1.3IE11#100	S383	RSX-C release : step 3	
V1.6IE25#2	PRODUCTION	A1.6IE23#1	A1.5IE17#2	P1.3IE11#100	Standard	V1.6 Commercial release : maintenance juin	
V2.3IE26#1	PRODUCTION	E2.3IE30#1	-	P1.2IE09#90	S381	Concast official release, iteration 2	
V2.4IE27#4	PRODUCTION	D2.4IE25#4	C1.7IE18#3	-	S383	RSX-C release : step 4, iteration 4	

ATV61 product release	Promotion Level	ATV61 Application SwVersion	ATV61 MotorControl SwVersion	ATV71 HHP MotorControl SwVersion	Product type	Short description	Known anomalies
V1.1IE02#2	PRODUCTION	B1.1IE02#2	A1.1IE05#1	P1.1IE03#45	Standard	Application based on ATV71 V1.1ie04#25 Target : first commercial release for ATV61 from size 2 up to size 12	
V1.1IE03#1	PRODUCTION	B1.1IE03#1	A1.1IE06#3	P1.1IE04#48	Standard	Application based on ATV71 V1.1ie05#9. Target : first commercial release for ATV61 HHP drives	
V1.1IE04#1	PRODUCTION	B1.1IE04#1	A1.1IE06#3	P1.1IE04#48	Standard		
V1.2IE05#3	PRODUCTION	B1.2IE05#12	A1.1IE07#14	P1.1IE06#62	Standard	Application based on ATV71 V1.1ie06#18. Target : Commercial release - supporting BACnet option board - using masked MotorControl software - External fault active at 0 - HHP : TR fixing : DSAT, INFB and use of calibration	
V1.2IE08#1	PRODUCTION	B1.2IE05#12	A1.1IE08#2	P1.1IE06#62	Standard	Standard release with correction for 75 kW 400V broken modules	
V1.4IE09#4	PRODUCTION	B1.4IE08#6	A1.2IE09#1	P1.2IE08#85	Standard	V1.4 Commercial release : iteration 4	
V1.4IE11#2	PRODUCTION	B1.4IE10#1	A1.2IE09#1	P1.2IE09#87	Standard	V1.4 Commercial release :Maintenance T15 Ré-integred in the standard version	
V1.4IE12#4	PRODUCTION	B1.4IE11#5	A1.2IE13#2	P1.2IE10#93	Standard	V1.4 Commercial release : version pour mars	
V1.5IE14#2	PRODUCTION	B1.5IE13#1	A1.5IE17#2	P1.3IE11#100	Standard	V1.5 Commercial release : version pour juin	

6.2 Graphic Terminal (VW3A1101):

The graphic terminal allows the viewing of the various parameters.

- It is the drive that authorizes the access to a particular parameter.
- It is the graphic terminal that writes the values if necessary.
- It is the graphic terminal that contains the messages in 5 languages.
- The graphic terminal is Modbus master in the exchanges with the drive

Graphic Terminal product release	Promotion Level	Graphic Terminal SwVersion	Product type	Short description	Known anomalies
V1.1IE03					None till today

6.3 Logic I/O Boards (VW3A3201):

The extended input output cards can only send information appearing on their terminals to the ATV71, except for the functions, which are only accessible if the option card is present.

Logic I/O option release	Promotion Level	Logic I/O option SwVersion	Product type	Short description	Known anomalies
					None till today

6.4 Software of the extended input/output boards (VW3A3202):

The extended input output cards can only send information appearing on their terminals to the ATV71, except for the functions, which are only accessible if the option card is present.

Extended Input / Output release	Promotion Level	Extended Input / Output SwVersion	Product type	Short description	Known anomalies
V1.1IE03					None till today

6.5 Fipio board software (VW3A3311 and VW3A3301):

Fipio Board release	Promotion Level	Fipio Board SwVersion	Product type	Short description	Known anomalies
V1.1IE01					None till today

6.6 Modbus Plus board software (VW3A3302):

Modbus Plus release	Promotion Level	Modbus Plus SwVersion	Product type	Short description	Known anomalies
V1.1IE02					None till today

6.7 Interbus-S board software (VW3A3304):

Interbus-S release	Promotion Level	Interbus-S SwVersion	Product type	Short description	Known anomalies
V1.1IE01					None till today

6.8 Ethernet board software (VW3A3310):

Ethernet release	Promotion Level	Ethernet SwVersion	Product type	Short description	Known anomalies
V1.1IE04					None till today

6.9 Unitelway / Modbus board software (VW3A3303):

Unitelway / Modbus release	Promotion Level	Unitelway / Modbus SwVersion	Product type	Short description	Known anomalies
V1.1IE01					None till today

6.11 Encoder interface board (VW3A3401 to VW3A3407):

Encoder release	Promotion Level	Product type	Short description	Known anomalies

Modbus Plus release	Promotion Level	Modbus Plus SwVersion	Product type	Short description	Known anomalies

6.13 Controller inside board software (VW3A3501):

Controller Inside release	Promotion Level	Controller Inside SwVersion	Product type	Short description	Known anomalies
V1.1IE03	Production				

7. Troubleshooting the Installation

7.1 Problem with the installation

7.1.1 Problem with the installation “UNSTABLE SPEED”

Possible causes	* Frequency reference signals unstable.	* Motor control algorithm parameters incorrect.	* Incorrect frequency feedback settings.	* Incorrect setting of the power supply, for example : 400V/87Hz	* Drive sized incorrectly for the motor. * Load changing.
Verification	* Verify the frequency reference signals.				* The torque fluctuates as a function of speed or whenever the machine is loaded. * The drive is in current limit when The speed is unstable.
Result	* Incorrect frequency reference. * Wired incorrectly.				* Torque point greater than the capability of the drive. * Current point greater than the capability of the drive.
Solution	* Correct the incorrect wiring. * Repair or replace the reference generator.	* In the motor control menu, enter the motor nameplate data and do an auto-tune.	* Review the setting of SPG (proportional gain) and of SIT (Integral time).	* Motor connected in star (230V), motor nameplate information in delta (460V).	* Re-size the drive. * Re-set the torque limit settings.

Problem with the installation “UNSTABLE SPEED” (continued)

Possible causes	* Oscillations due to mechanical interactions.	* Bad electrical coupling of the motor.	* Bad electrical coupling due to incorrect encoder feedback. * Number of pulses per turn too small.
Verification	* The mechanical couplings. * Resonant frequencies.	* Motor connected in star or delta.	* The mechanical attachment of the encoder to the motor. * The number of pulses per turn and the parameter PGI.
Result	* Torque, current, and motor speed oscillations.		
Solution	* Correct mechanical alignment. * Skip over the resonant frequency.	* Correct the motor connections.	* Align the encoder correctly. * Change the encoder to obtain more pulses per turn.

7.1.2 Problem with the installation “THE MOTOR TURNS THE WRONG WAY”

Possible causes	* Due to loading effects.	* Incorrect motor connections.	* Bad direction of rotation selected.	* Negative signal on the reference inputs.
Verification	* Is this a desired condition of use ? * Safety equipment.	* Output wiring. * Motor wiring.	* Check the direction of rotation on the terminal. * Verify the assignment of the logic inputs.	* Verify the sign and the assignment in the monitoring menu.
Result	* Incorrect frequency reference. * Wired incorrectly.		* Incorrect rotation direction or incorrect command.	
Solution	* If this use is normal, nothing to do. * Replace the damaged safety equipment.	* Respect the wiring of the U, V, and W phases.	* Correct the selection or the command.	* Correct the assignments. * Correct the reference signal * Inhibit reverse operation (Rrs) * Delinearize the ramp.

7.1.3 Problem with the installation “EXCESSIVE MOTOR TEMPERATURE”

Possible causes	* Motor temperature.	* Motor ambient temperature.	* The motor ventilation is blocked or damaged.	* Drive parameters incorrect.	* Bad sizing of the motor for the application.	* Ith protection incorrect.
Verification	* The motor insulation class. * The motor temperature using a thermocouple.	* The ambient air around the motor. * The motor class. * The motor insulation.			* Sizing.	* The value of Ith and the level of DC injection and its duration.
Result		* If the ambient temperature is within the specified allowable motor limits, verify the other causes of motor overtemperature.				
Solution	* If the motor temperature is incorrect, no action is necessary, replace the failed element.	* Use a motor rated for higher temperatures. * Use a forced ventilated motor. * Use a higher power motor.	* Clean out the air passages. * Verify that the fan motor operates correctly.	* In the Motor Control menu, enter the motor nameplate data and do an auto-tuning. * Switching frequency too high.	* Choose a higher power motor.	* Set Ith to the motor nameplate current. * Set Idc permanent < 0.5 In or set S d C.

7.1.4 Problem with the installation “THE MOTOR ACCELERATES TOO SLOWLY”

Possible causes	* Incorrect ramp output.	* Incorrect motor control algorithm parameters.	* Acceleration ramp set incorrectly.	* Acceleration of the motor while in current limit.
Verification	* The frequency reference. * Type of ramp used (parameter RPt). * If the ramp switching function is active.			* The current limit setting (CLI). * If the current limit adaptation function is active (FdB).
Result	* Type or ramp time incorrect.			* Current limit too low.
Solution	* Change the speed reference. * Change the ramp type. * Switch the time of the desired ramp.	* In the motor control algorithm menu, enter the motor nameplate data and do an auto-tuning.	* Change the acceleration setting Acc.	* Increase the current limit setting (CLI). * Current limit is set to its maximum level. * Choose a larger motor within the limits of the drive.

Problem with the installation “THE MOTOR ACCELERATES TOO SLOWLY” (continued)

Possible causes	* Bas setting of the frequency feedback.	* Bad sizing of the motor / bad for the bad.
Verification	* Verify the proportional gain setting (SPG) and the integral time setting (SIT).	* Current point. * Torque point.
Result	* The speed increase is not linear: rounded at the beginning and the end of acceleration.	* Current point greater than the capacity of the drive. * Torque point greater than the capacity of the drive.
Solution	* Review the proportional gain settings (SPG) and the integral time settings (SIT).	* Re-size the drive.

7.1.5 Problem with the installation “BAD CONTROL OF THE MOTOR STOP OF DECELERATION”

Possible causes	* Stop type inadequate.	* Run command or reference always present.	* Ramp not followed.	* Deceleration of the motor in current or torque limit.	* Drive / motor poorly sized the load.
Verification	* The time and the amplitude of DC current injection. * The selected stop type.	* Verify the logic and analog inputs in the monitoring menu.	* Braking torque.	* The current limit (CLI) or torque limit (TLA) settings. * If the Ilim Adaptation function is active (FdB).	* DC Bus Voltage. * Braking torque.
Result	* Stop the motor in free wheel or braking torque insufficient.			* Current or torque limit too low.	* Too much energy returned the drive. * Drive braking torque insufficient.
Solution	* Adjust the time and amplitude of DC injection. * Select another stop type.	* Set the run command or the reference to 0.	* Adjust the decel time. * De-select the autoadaptation of the deceleration ramp function (BrA).	* Increase the current (CLI) or torque limit.	* Use a larger power drive.

7.1.6 Problem with the installation “THE MOTOR DOES NOT START”

Possible causes	* Drive not powered.	* Motor not connected to the drive.	* Drive in free wheel stop. * Drive in DC stop. * Drive in fast stop.	* Push the keypad Stop button	* In 3-wire command (Tct), the Stop logic input is at 0.
Verification	* The drive power supply and its voltage.	* The wires between the motor and the drive. * The command of the contactor between the drive and motor.	* The logic input assignments in the monitoring menu.		* The state of the LI1 input.
Result		* Motor not connected. * Wire damaged.			
Solution	* Apply power to the drive. * Correct the sequence.	* Connect the motor. * Verify its connections. * Correct the command of the output contactor.	* Set the inputs assigned to free wheel and rapid stop to 1. * Set the input assigned to DC stop to 0.	* Put the direction command inputs to 0 then reset them to 1.	* Set LI1 to 1.

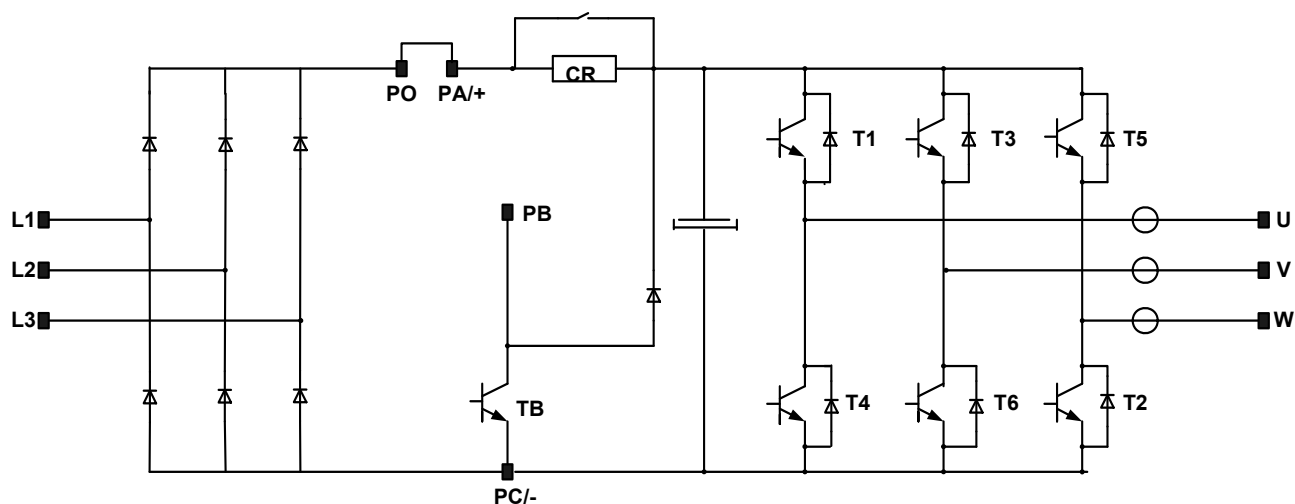
Problem with the installation “THE MOTOR DOES NOT START” (continued)

Possible causes	* The reference frequency is equal to 0.	* Terminal command selected.	* - In keypad command, the frequency reference is 0.	* Current or torque limit too low.	* Bad sizing of the motor / drive for the load.
Verification	* The signals on the analog inputs assigned to summed frequency references.	* If terminal command is active (LCC=ON).	* The parameter LFR in the surveillance menu.	* Limitation of current and torque.	* Current point * Torque point.
Result					* Current point greater than the drive capacity. * Torque point greater than the drive capacity.
Solution	* Send a non-zero frequency reference.	* De-select terminal command.	* Set LFR to a non-zero value and push the terminal run command.	* Increase the current or torque limit value.	* Re-size the drive or the motor.

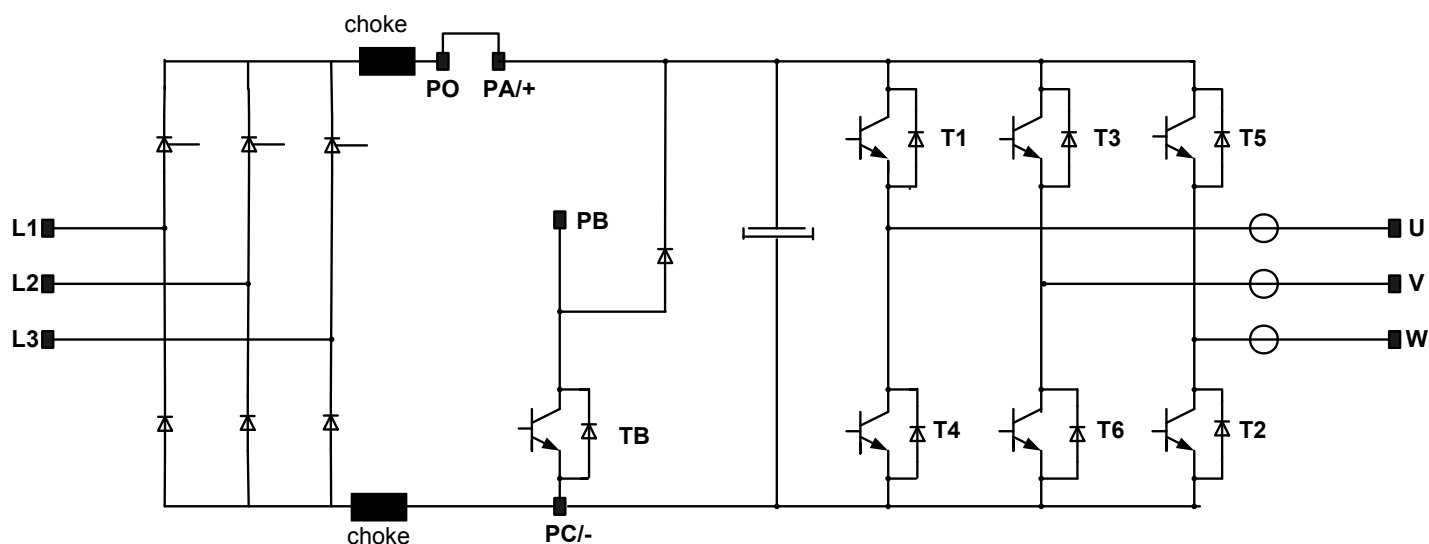
8. Troubleshooting and repair guide

8.1 Schematics

8.1.1 Power schematic for the Altivar < 18.5 kW (230V & 480V)



8.1.2 Power schematic for the Altivar >18.5 kW (230V, 480V & 690V (range up to 90kW))



8.2 Diagnostics

8.2.1 Introduction

Upon reception of the product, carefully note the following:

- The reference of option board and the software version if they exist.
- The software version of drive and of the graphic terminal.
Generally, the software version is indicated on the nameplate label.

Then, replace the components (the option board and the graphic terminal)

If the product is broken and has been returned with a description of the failure observed by the customer, use this information to determine the broken component.

In any other case, follow these instructions:

- Visually examine the product (before applying voltage).
 - Check the external state of the drive:
 - Look for traces of burning or overheating, parts that have been submitted to shocks.
 - Check the connectors.
 - Check the state of the internal screws to determine if a part of drive has been dismantled.

8.2.2 Preliminary static verification of an ATV

Before checking the power electronics, please make sure, that there exists no dangerous voltage in the DC-Link. So, please pay attention to the following security hints:

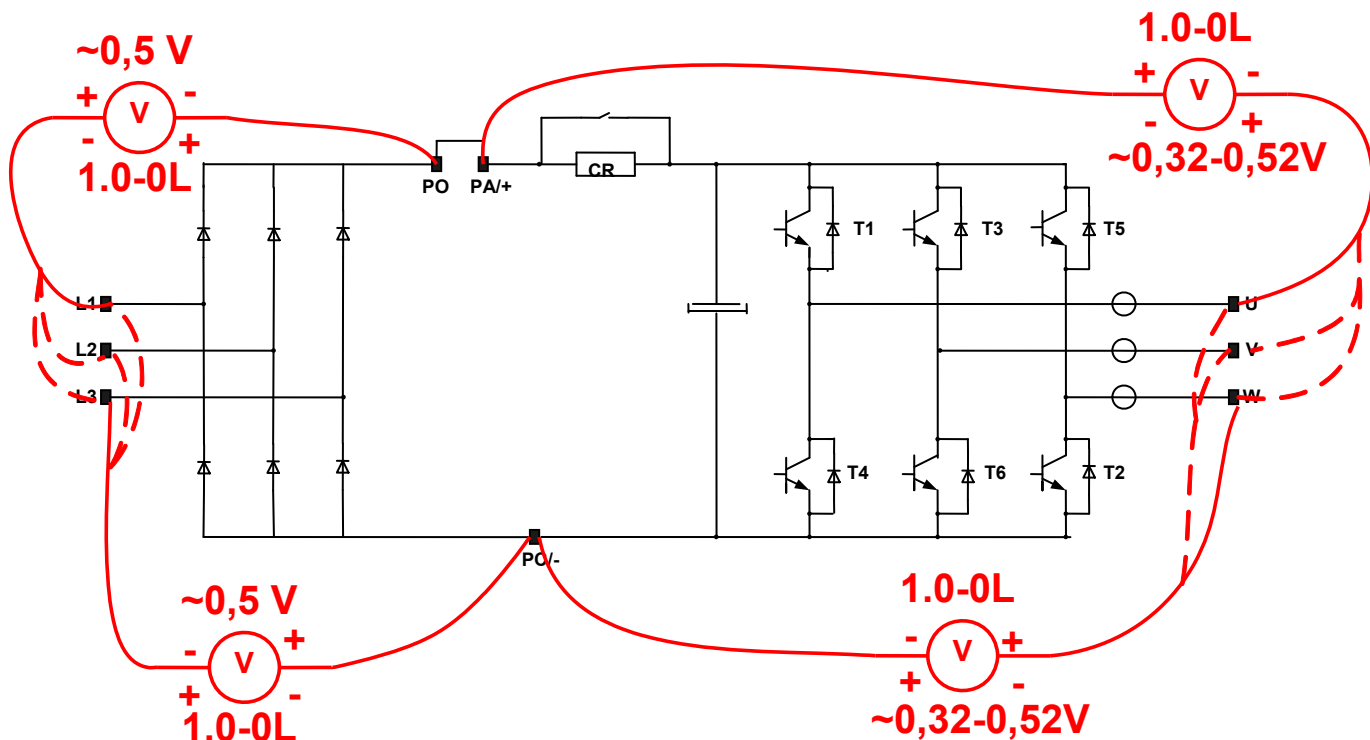
- Clear voltage.
- Wait for capacitor discharge.
- Check capacitors for neutrality.

When checking the power part, the following procedure is recommended:

- Disconnect mains and motor lines.
 - Test the diodes, thyristors and IGBTs using a universal tester with a diode test function and load-independent current.
-

8.2.3 Checking the input diode bridge and the output bridge:

This test is done using a digital multi-meter in diode test mode to check that the transistors and diodes are not short-circuited. The absolute values are not so important because they depend strongly on the test device. However, in function of each module, it is possible to find values include in like indicate the following schematic. The uniformity of the measured values is more important. Please further note, that the drives requires some time to load the intermediate circuit capacity. There are 24 measurements to be made:



8.2.4 Checking the thyristors of the input bridge:

(Only for drives > 15KW 230V, P > 18.5kW 480V, and 690V range up to 90kW)

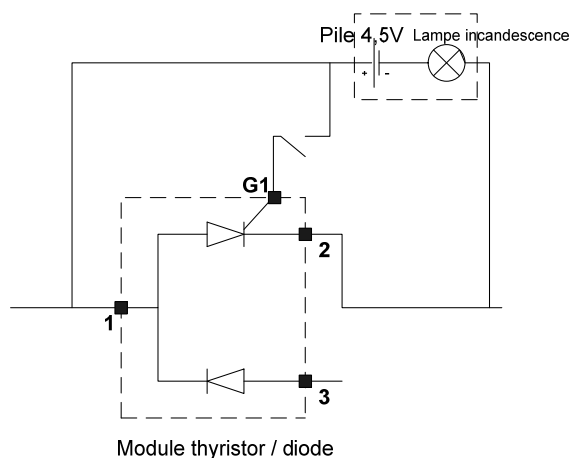
This test is done using a multi-meter in the ohmmeter position to check that the thyristors are not short-circuited. The test is done like the preceding test between PO and L1, L2, L3.

Checking of the thyristor module only:

This test is done using an incandescent lamp and a 4.5 V battery connected. Using the schematic below.

- Switch open $\Rightarrow$ the thyristor is in open circuit (lamp off).
- Close the switch $\Rightarrow$ the thyristor conducts (lamp on).
- Open the switch $\Rightarrow$ the thyristor remains conductive (lamp on).
- Lower the voltage to turn off the thyristor.

If the operation is different than described, the thyristor is probably out of service.

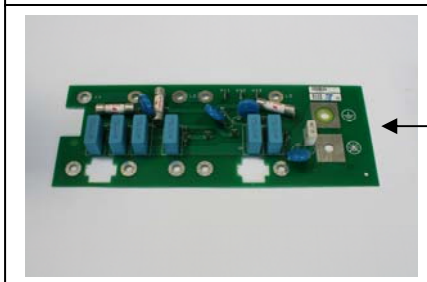
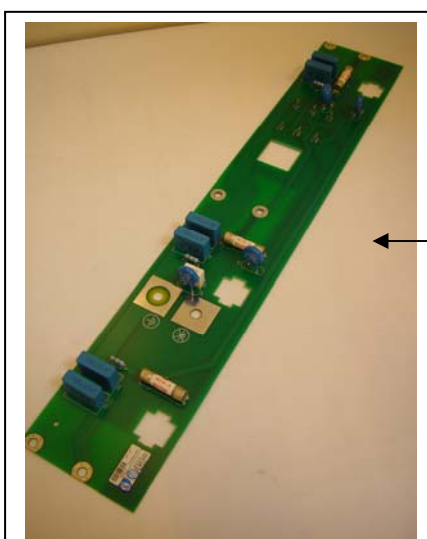
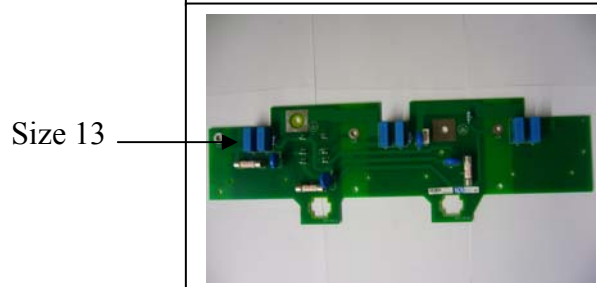
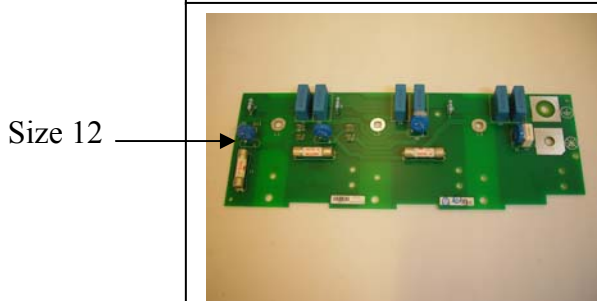
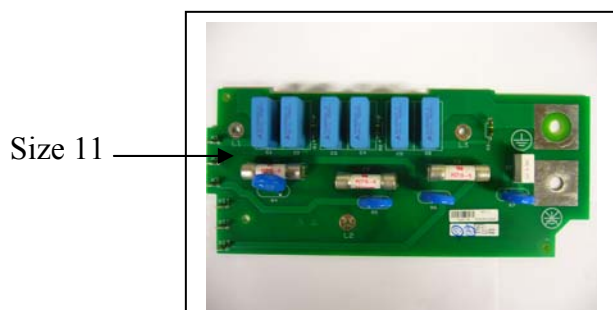


8.2.5 Checking the braking transistor and the free-wheeling diode.

To perform this test, disconnect the braking resistor. The test is done using a digital multi-meter in diode test mode. If the result is zero the transistor and/or diodes are shorted.

8.2.6 Checking the precharge fuses for drives > 110KW

The Fuses F1, F2 and F3 (precharge- accordingly varistor fuses) are located on the RFI filter board. The exact location where the fuses are placed depends on the size of inverter. The measurement of the fuses has to be done with a multi-meter in the ohmmeter position to have a value of zero Ohm:



8.2.7 Checking the DC bus capacitors:

Perform this test only if the converter bridge and inverter bridge tests are good.

Test the capacitors using a multi-meter in ohmmeter mode between the terminals PC and PA. The resistance should begin at a low level and should increase progressively (as the capacitors charge). If this does not happen, the charging resistor or the capacitors are defective.

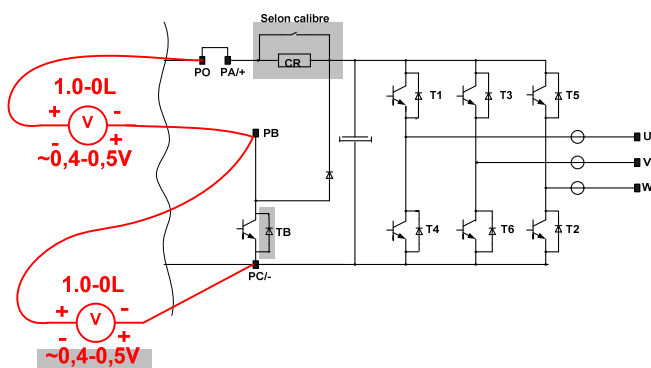


Table of capacitor values and discharge times.

ATV61/71 – Range 240V – IP00 / IP20

STANDARD PRODUCT ON A HEAT SINK					
Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.	Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.
ATV71H037M3	1*390 / 390	< 60 sec	ATV61H037M3	1*390 / 390	< 60 sec
ATV71H075M3	1*560 / 560	< 90 sec	ATV61H075M3	1*560 / 560	< 90 sec
ATV71HU15M3	2*390 P / 780	< 120 sec	ATV61HU15M3	2*390 P / 780	< 120 sec
ATV71HU22M3	2*780 / 1560	< 250 sec	ATV61HU22M3	2*780 / 1560	< 250 sec
ATV71HU30M3	2*1200 P / 2400	< 360 sec	ATV61HU30M3	2*1200 P / 2400	< 360 sec
ATV71HU40M3	2*1200 P / 2400	< 360 sec	ATV61HU40M3	2*1200 P / 2400	< 360 sec
ATV71HU55M3	6*550 P / 3300	< 560 sec	ATV61HU55M3	6*550 P / 3300	< 560 sec
ATV71HU75M3	4*950 P / 3800	< 650 sec	ATV61HU75M3	4*950 P / 3800	< 650 sec
ATV71HD11M3X	5*950 P / 4750	< 650 sec	ATV61HD11M3X	5*950 P / 4750	< 650 sec
ATV71HD15M3X	6*950 P / 5700	< 760 sec	ATV61HD15M3X	6*950 P / 5700	< 760 sec
ATV71HD18M3X	2*2700 P / 5400	< 294 sec	ATV61HD18M3X	2*2700 P / 5400	< 294 sec
ATV71HD22M3X	2*2700 P / 5400	< 294 sec	ATV61HD22M3X	2*2700 P / 5400	< 294 sec
ATV71HD30M3X	2*3900 P / 7800	< 212 sec	ATV61HD30M3X	2*3900 P / 7800	< 212 sec
ATV71HD37M3X	2*4800 P / 9600	< 261 sec	ATV61HD37M3X	2*4800 P / 9600	< 261 sec
ATV71HD45M3X	2*5700 P / 11400	< 310 sec	ATV61HD45M3X	2*5700 P / 11400	< 310 sec

ATV61/71 – Range 480V – IP00 / IP20

STANDARD PRODUCT ON A HEAT SINK					
Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.	Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.
ATV71H075N4	2*390 S / 195	≤ 80 sec	ATV61H075N4	2*390 S / 195	≤ 80 sec
ATV71HU15N4	2*390 S / 195	≤ 80 sec	ATV61HU15N4	2*390 S / 195	≤ 80 sec
ATV71HU22N4	2*560 S / 280	≤ 120 sec	ATV61HU22N4	2*560 S / 280	≤ 120 sec
ATV71HU30N4	2*780 S / 390	≤ 160 sec	ATV61HU30N4	2*780 S / 390	≤ 160 sec
ATV71HU40N4	2*1200 S / 600	≤ 240 sec	ATV61HU40N4	2*1200 S / 600	≤ 240 sec
ATV71HU55N4	3*(2*550 S) P / 825	≤ 360 sec	ATV61HU55N4	3*(2*550 S) P / 825	≤ 360 sec
ATV71HU75N4	4*(2*550 S) P / 1100	≤ 440 sec	ATV61HU75N4	4*(2*550 S) P / 1100	≤ 440 sec
ATV71HD11N4	3*(2*950 S) P / 1425	≤ 570 sec	ATV61HD11N4	3*(2*950 S) P / 1425	≤ 570 sec
ATV71HD15N4	4*(2*950 S) P / 1900	≤ 760 sec	ATV61HD15N4	4*(2*950 S) P / 1900	≤ 760 sec
ATV71HD18N4	4*(2*950 S) P / 1900	≤ 760 sec	ATV61HD18N4	4*(2*950 S) P / 1900	≤ 760 sec
ATV71HD22N4	2*2700 S / 1350	≤ 197 sec	ATV61HD22N4	2*2700 S / 1350	≤ 197 sec
ATV71HD30N4	2*3900 S / 1950	≤ 285 sec	ATV61HD30N4	2*3900 S / 1950	≤ 285 sec
ATV71HD37N4	2*4800 S / 2400	≤ 350 sec	ATV61HD37N4	2*4800 S / 2400	≤ 350 sec
ATV71HD45N4	2*(2*2700 S) P / 2700	≤ 197 sec	ATV61HD45N4	2*(2*2700 S) P / 2700	≤ 197 sec
ATV71HD55N4	2*(2*3900 S) P / 3900	≤ 285 sec	ATV61HD55N4	2*(2*3900 S) P / 3900	≤ 285 sec
ATV71HD75N4	2*(2*4800 S) P / 4800	≤ 350 sec	ATV61HD75N4	2*(2*4800 S) P / 4800	≤ 350 sec
ATV71HD90N4	3*(2*4700 S) P / 7050	≤ 900 sec	ATV61HD90N4	3*(2*4700 S) P / 7050	≤ 900 sec
ATV71HC11N4	3*(2*5200 S) P / 10200	≤ 900 sec	ATV61HC11N4	3*(2*5200 S) P / 10200	≤ 900 sec
ATV71HC13N4	3*(2*5200 S) P / 7800	≤ 900 sec	ATV61HC13N4	3*(2*5200 S) P / 7800	≤ 900 sec
ATV71HC16N4	4*(2*5200 S) P / 10400	≤ 900 sec	ATV61HC16N4	4*(2*5200 S) P / 10400	≤ 900 sec
ATV71HC22N4	6*(2*3600 S) P / 10800	≤ 900 sec	ATV61HC22N4	6*(2*3600 S) P / 10800	≤ 900 sec
ATV71HC25N4	6*(2*5200 S) P / 15600	≤ 900 sec	ATV61HC25N4	6*(2*5200 S) P / 15600	≤ 900 sec
ATV71HC28N4	6*(2*5200 S) P / 15600	≤ 900 sec	ATV61HC31N4	6*(2*5200 S) P / 15600	≤ 900 sec
ATV71HC31N4	9*(2*3600 S) P / 16200	≤ 900 sec	ATV61HC40N4	9*(2*3600 S) P / 16200	≤ 900 sec
ATV71HC40N4	9*(2*5200 S) P / 23400	≤ 900 sec	ATV61HC50N4	9*(2*5200 S) P / 23400	≤ 900 sec
ATV71HC50N4	12*(2*5200 S) P / 31200	≤ 900 sec	ATV61HC63N4	12*(2*5200 S) P / 31200	≤ 900 sec

ATV61/71 – Range 480V – IP54

STANDARD PRODUCT ON A HEAT SINK					
Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.	Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.
ATV71W075N4	2*390 S / 195	≤ 80 sec	ATV61W075N4	2*390 S / 195	≤ 80 sec
ATV71WU15N4	2*390 S / 195	≤ 80 sec	ATV61WU15N4	2*390 S / 195	≤ 80 sec
ATV71WU22N4	2*560 S / 280	≤ 120 sec	ATV61WU22N4	2*560 S / 280	≤ 120 sec
ATV71WU30N4	2*780 S / 390	≤ 160 sec	ATV61WU30N4	2*780 S / 390	≤ 160 sec
ATV71WU40N4	2*1200 S / 600	≤ 240 sec	ATV61WU40N4	2*1200 S / 600	≤ 240 sec
ATV71WU55N4	3*(2*550 S) P / 825	≤ 360 sec	ATV61WU55N4	3*(2*550 S) P / 825	≤ 360 sec
ATV71WU75N4	4*(2*550 S) P / 1100	≤ 440 sec	ATV61WU75N4	4*(2*550 S) P / 1100	≤ 440 sec
ATV71WD11N4	3*(2*950 S) P / 1425	≤ 570 sec	ATV61WD11N4	3*(2*950 S) P / 1425	≤ 570 sec
ATV71WD15N4	4*(2*950 S) P / 1900	≤ 760 sec	ATV61WD15N4	4*(2*950 S) P / 1900	≤ 760 sec

ATV61/71 – Range 480V – IP54 (continued)

STANDARD PRODUCT ON A HEAT SINK					
Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.	Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.
ATV71WD18N4	4*(2*950 S) P / 1900	≤ 760 sec	ATV61WD18N4	4*(2*950 S) P / 1900	≤ 760 sec
ATV71WD22N4	2*2700 S / 1350	≤ 197 sec	ATV61WD22N4	2*2700 S / 1350	≤ 197 sec
ATV71WD30N4	2*3900 S / 1950	≤ 285 sec	ATV61WD30N4	2*3900 S / 1950	≤ 285 sec
ATV71WD37N4	2*4800 S / 2400	≤ 350 sec	ATV61WD37N4	2*4800 S / 2400	≤ 350 sec
ATV71WD45N4	2*(2*2700 S) P / 2700	≤ 197 sec	ATV61WD45N4	2*(2*2700 S) P / 2700	≤ 197 sec
ATV71WD55N4	2*(2*3900 S) P / 3900	≤ 285 sec	ATV61WD55N4	2*(2*3900 S) P / 3900	≤ 285 sec
ATV71WD75N4	2*(2*4800 S) P / 4800	≤ 350 sec	ATV61WD75N4	2*(2*4800 S) P / 4800	≤ 350 sec
			ATV61WD90N4	2*(2*4800 S) P / 4800	≤ 350 sec

ATV61/71 – Range 690V

STANDARD PRODUCT ON A HEAT SINK					
Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.	Catalog Number	Number of capacitors and their value, wired in series (S) or in parallel (P) and the total value (in μF)	Discharge time from 412V (M3), 825V (N4) or 1127V (Y) to 50V.
ATV71HU22Y	2*1100 S / 550	≤ 130 sec	ATV61HU22Y	2*1100 S / 550	≤ 130 sec
ATV71HU30Y	2*1100 S / 550	≤ 130 sec	ATV61HU30Y	2*1100 S / 550	≤ 130 sec
ATV71HU40Y	2*1100 S / 550	≤ 130 sec	ATV61HU40Y	2*1100 S / 550	≤ 130 sec
ATV71HU55Y	2*1100 S / 550	≤ 130 sec	ATV61HU55Y	2*1100 S / 550	≤ 130 sec
ATV71HU75Y	2*1100 S / 550	≤ 130 sec	ATV61HU75Y	2*1100 S / 550	≤ 130 sec
ATV71HD11Y	2*1100 S / 550	≤ 130 sec	ATV61HD11Y	2*1100 S / 550	≤ 130 sec
ATV71HD15Y	2*1100 S / 550	≤ 130 sec	ATV61HD15Y	2*1100 S / 550	≤ 130 sec
ATV71HD18Y	2*1100 S / 550	≤ 130 sec	ATV61HD18Y	2*1100 S / 550	≤ 130 sec
ATV71HD22Y	2*1100 S / 550	≤ 130 sec	ATV61HD22Y	2*1100 S / 550	≤ 130 sec
ATV71HD30Y	2*1100 S / 550	≤ 130 sec	ATV61HD30Y	2*1100 S / 550	≤ 130 sec
ATV71HD37Y	2*(2*1800 S) P / 1800	≤ 180 sec	ATV61HD37Y	2*(2*1800 S) P / 1800	≤ 180 sec
ATV71HD45Y	2*(2*1800 S) P / 1800	≤ 180 sec	ATV61HD45Y	2*(2*1800 S) P / 1800	≤ 180 sec
ATV71HD55Y	2*(2*1800 S) P / 1800	≤ 180 sec	ATV61HD55Y	2*(2*1800 S) P / 1800	≤ 180 sec
ATV71HD75Y	2*(2*1800 S) P / 1800	≤ 180 sec	ATV61HD75Y	2*(2*1800 S) P / 1800	≤ 180 sec
ATV71HD90Y	2*(2*1800 S) P / 1800	≤ 180 sec	ATV61HD90Y	2*(2*1800 S) P / 1800	≤ 180 sec
ATV71HC11Y	3*(2*2100 S) P / 3150	> 900 sec	ATV61HC11Y	3*(2*2100 S) P / 3150	> 900 sec
ATV71HC13Y	3*(2*2100 S) P / 3150	> 900 sec	ATV61HC13Y	3*(2*2100 S) P / 3150	> 900 sec
ATV71HC16Y	3*(2*2100 S) P / 3150	> 900 sec	ATV61HC16Y	3*(2*2100 S) P / 3150	> 900 sec
ATV71HC20Y	6*(2*2100 S) P / 6300	> 900 sec	ATV61HC20Y	3*(2*2100 S) P / 3150	> 900 sec
ATV71HC25Y	6*(2*2100 S) P / 6300	> 900 sec	ATV61HC25Y	6*(2*2100 S) P / 6300	> 900 sec
ATV71HC31Y	6*(2*2100 S) P / 6300	> 900 sec	ATV61HC31Y	6*(2*2100 S) P / 6300	> 900 sec
ATV71HC40Y	12*(2*2100 S) P / 12600	> 900 sec	ATV61HC40Y	6*(2*2100 S) P / 6300	> 900 sec
ATV71HC50Y	12*(2*2100 S) P / 12600	> 900 sec	ATV61HC50Y	12*(2*2100 S) P / 12600	> 900 sec
ATV71HC63Y	12*(2*2100 S) P / 12600	> 900 sec	ATV61HC63Y	12*(2*2100 S) P / 12600	> 900 sec
			ATV61HC80Y	12*(2*2100 S) P / 12600	> 900 sec

Warning, concerning the ATV61/71 (range up to 90kW) there are two suppliers for the capacitors. If you change the capacitors, you should put the same supplier (don't mixed the capacitors). In the table, the values for the capacitors are the **end of lifetime values(C (END))**.

Example: You can see the picture above the nominal value as well as the end of lifetime value.

Nominal value.



End of lifetime value.

To order to gain time during diagnosis of a product, it is possible to discharge the capacitors after application of voltage by connecting a resistance of around 100 ohms 40 watts between terminals PA+ and PC-. A direct short circuit will pre-maturely age the capacitors.

Procedure for applying voltage to an ATV after long duration storage.

As a function of the length of storage, you must re-apply voltage progressively to the product to re-form the capacitors. Calculate the storage time from the date code of the product and not from the date of delivery.

Storage time	Procedure to re-apply voltage
≤ 1 year	Apply voltage in the normal way
Between 1 and 2	Apply voltage to the product for 1 hour
≥ 2 years	Use a variable AC power supply to - 25% of nominal voltage for 30 - 50% of nominal voltage for 30 - 75% of nominal voltage for 30 - 100% of nominal voltage for 210

8.3 Electrical verification of an ATV

If no short circuits have been discovered during the preliminary and static tests it is possible to apply voltage to the controller.

The ATV71 has functions, LEDs (>90KW) and menus that facilitate diagnosis of the product to determine breakdowns in the product or application.

The configuration of drive can be blocked with passwords in the code pin1 or the code pin2 (expert mode) .To disable this function without known the password:

- Enter 6969 for the code pin1.
- Enter 2663 for the code pin2.

These features allow:

- The product identification and its options.
- Monitoring of inputs-outputs and the communication words.
- The display of past faults as well as the state of the drive at the moment the fault appeared.
- Automatic testing of the drive (power and control circuits).
- The display of a maintenance message (customisable).

8.3.1 Product Identification.

This menu is only readable with the graphic terminal

Menu 1 – Drive menu

1.11 – Identification

It allows the determination of the identity of the drive boards and shows:

- The drive catalog number and its size.
- The software version drive.
- The application software version.
- The motor control software version.
- The drive serial number.
- The installed option boards.
- The software versions of the option boards (if present)

All the options are identified by the ATV71 but certain information is Only available for those options using a µP (software version....)

RUN +50.00Hz 1250A +50.00Hz
1.11 IDENTIFICATION
ATV71HU15N4
1,5 kW / 2 HP
380 /480 V
Application V0.4 IE 05
Motor Control V0.4 IE 05
<< >> Quick
6W0410xxxxxxxxxx
PDT V1.1 IE 04
OPTION 1
I/O EXTENSION CARD
Vx.x IE xx
OPTION 2
FIPIO CARD
Vx.x IE xx
CONSOLE
GRAPHIC S
Vx.x IE xx
ENCODER
RS 422

The board identities can be recovered by PowerSuite software and then exported (ASCII format) to be used by other software (maintenance management ...).

8.3.2 Input-output and communication word monitoring.

With the graphic terminal:

Menu 1 – Drive menu

1.2 - Monitoring (SUP)

This menu allows the monitoring of logic and analog inputs/outputs, the state of internal drive values, and data and communication values in order to help in diagnosing a wiring problem.

RUN Term	+ 50.00Hz	1250A
I/O IMAGE		
LOGIC INPUTS IMAGE		
ANALOGUE INPUTS IMAGE		
LOGIC OUTPUTS IMAGE		
ANALOGUE OUTPUTS IMAGE		
FREQ. SIGNALS IMAGE		
CODE	<<	>> Quick

RUN Term	+ 50.00Hz	1250A
Logic Inputs MAP		
PR	LI1	LI2
LI3	LI4	LI5
LI6	LI7	LI8
LI9	LI10	LI11
LI12	LI13	LI14
<< >> Quick		

RUN Term	+ 50.00Hz	1250A
LI1 ASSIGNMENT		
FORWARD		
SWITCHING RAMP 2		
Delay time : 10 ms		
<< >> Quick		

RUN Term	+ 50.00Hz	1250A
LOGIC OUTPUTS IMAGE		
R1	R2	LO1
<< >> Quick		

RUN Term	+ 50.00Hz	1250A
LO1 CONFIGURATION		
FAULT		
Delay time : 10 ms		
Active at : 1		
Hold time : 20 ms		
<< >> Quick		

RUN Term.	+50.00Hz	1250A
COM. IMAGE		
CMD Channel : Modbus		
CMD Value : 8001 Hex		
REF. Channel : CANopen		
REF. Value : 12,5 Hz		
Status word : 2153 Hex		
CODE	<<	>> Quick

RUN Term	+50.00Hz	1250A
COM. IMAGE		
W3142 : 6230 Hex		
W1427 : F230 Hex		
W1123 : +2343		
W1487 : F230 Hex		
COM.SCAN IN		
CODE	<<	>> Quick

RUN Term.	+50.00Hz	1250A
COM. SCAN IN		
Val Com Scan In1 : 64		
Val Com Scan In2 : 8201 Hexa		
Val Com Scan In3 : 0		
Val Com Scan In4 : 0		
Val Com Scan In5 : 0		
CODE << >> Quick		
Val Com Scan In6 :		
Val Com Scan In7 :		
Val Com Scan In8 :		

8.3.3 Display of past faults and the state of the drive at the moment of the appearance of the fault.

Menu 1 – Drive Menu

1.10 - Diagnostics

Fault history

RUN Term. +50.00Hz 1250A
1.10- DIAGNOSTIC
FAULT HISTORY
PRESENT FAULT LIST
ADDITIONAL FAULT INFO
TEST PROCEDURE
SERVICES MESSAGE
HOME << >> Quick

Save 8 faults with 12 parameters saved at the time that the fault appeared:

- Drive state.
- Status word.
- Command word.
- Motor current.
- Output frequency.
- Elapsed time.
- Distribution network voltage.
- Motor thermal state.
- Command channel (terminal, modbus, ...).
- Reference channel.

RUN Term. +50.00Hz 1250A
FAULT HISTORY
SHORT CIRCUIT:40000 h
OVERCURRENT : 3500 h
EXTERNAL FLT : 3102 h
OVERVOLTAGE : 1312 h
UNDERVOLTAGE : 20 h
HELP << >> Quick

If the controller inside board (option) is installed it is possible to have a real-time clock and save the year, month, day, hour, and minute that a fault occurs.

For each saved fault, there is an indication of the probable cause via the Help button.

RUN Term. +50.00Hz 1250A
SHORT CIRCUIT
FREQUENCY REF: 20.0Hz
CURRENT : 200.0 A
MOTOR SPEED : 600 RPM
OUTPUT FREQ : 600 RPM
MACHINE SPEED: 1.2 ms
HELP << >> Quick

RUN Term. +50.00Hz 1250A
SHORT CIRCUIT FAULT
Check the connection cables with the speed controller disconnected, and the motor insulation. Check the speed controller transistor bridge
Quick

8.3.4 Drive Auto-test (power and control circuits).

Warning: Before performing theses tests remove the terminal board to avoid having run commands present during the tests.

Tests available for the customer.

Menu 1 – Drive menu

1.10 - Diagnostics

Test Procedure

RUN Term. +50.00Hz 1250A
1.10- DIAGNOSTIC
FAULT HISTORY
PRESENT FAULT LIST
ADDITIONAL FAULT INFO
TEST PROCEDURE
SERVICES MESSAGE
HOME << >> Quick

Allows the diagnosis of a failed element through various tests such as:

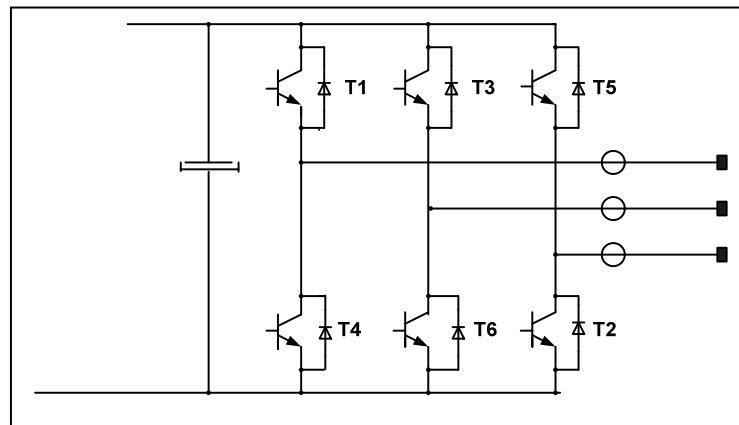
- Thyristor test.
- Transistor test.

RDY	Term	0.00Hz	0A
TEST PROCEDURES			
TRANSISTOR TEST			
THYRISTOR TEST			
Quick			

Transistor test

For this test a motor must be connected and the motor nameplate data entered into the “motor control” menu. The result depends upon the proper matching between the motor and the current information entered. It uses injected DC (**warning**, to the current level, which must be representative of the full load motor current).

Principle:



To detect a short circuit each transistor are command during 6 μ S, if the opposite transistor is in short circuit, there are a short circuit in the DC bus.

To detect an open transistor 2 transistors are command (ex: T1 and T6), the motor is used, and the current is measured via the current sensor. If one is open it's checked by combination (T1/T6 then T1/T2 then T3 / T2 then T5 / T4 and finally T5 /T6).

The results only apply to the IGBTs: possible results are OK, short circuit and open.

If an output contactor is present, be sure it is closed for the test.

To start the test you must push the ENT button for ~2 sec.

RDY	Term	00.0Hz	0A
TRANSISTOR RESULT			
IGBT 1	:	OK	
IGBT 1	:	Open	
IGBT 2	:	OK	
IGBT 2	:	OK	
IGBT 3	:	OK	
Quick			

If the test result is short circuit or open transistor the fault can be caused by:

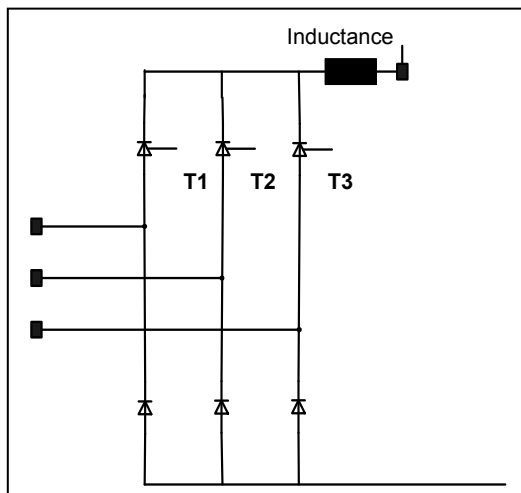
- Bad motor connection = check the wiring.
- Problem with the transistor command = change the power board or the gate drive board.
- Problem with the transistor = change the transistor.

Thyristor test

This menu is only present for sizes larger than 18.5 kW (230V, 480V and 690V range).

The test does not require any specific wiring by the customer: it is only necessary that the product be powered by three-phase power.

Principle:



Each thyristors are blocked one by one then open, the discharging time is measured. If the thyristor is bad the level of the DC bus is not the same.

This test can work for a long time because it depends of the time discharge.

The result generated is a number of pulses that corresponds to an OK / NOK result depending on whether the power supply is 50 / 60 Hz.

RDY	Term	00.0Hz	0A
THYRISTORS RESULT			
Thyristor 1	:		OK
Thyristor 2	:		Fault
Thyristor 3	:		OK
Quick			

Warning, the diodes are never tested.

8.3.5 Tests available only for SCHNEIDER services.

These tests are only available through the hidden menu and for drive < ATV71HC90N4

Procedure to access the hidden menu:

Menu 1 – Drive menu

1.11 - Identification

Graphic terminal

V#.#IE## Bottom

RDY	Term	0.00Hz	0A
1.11 IDENTIFICATION			
GRAPHIC TERMINAL			
GRAPHIC S			
V#.#IE## Bottom			
12WOP4200			
ENCODER			
<< >> Quick			

Place the cursor on the version (V#.#IE##) then push the ENT button for 3 seconds. The display automatically returns to the Identification menu.

Go into the Test procedures menu, some new tests appear:

- Input / Output test.
- Current calibration.
- LIC test.
- Voltage calibration.
- Braking transistor test.

RDY	Term	0.00Hz	0A
TEST PROCEDURES			
I/O TEST			
TRANSISTOR TEST			
I CALIBRATION RESULT			
LIC TEST			
U CALIBRATION RESULT			
Quick			

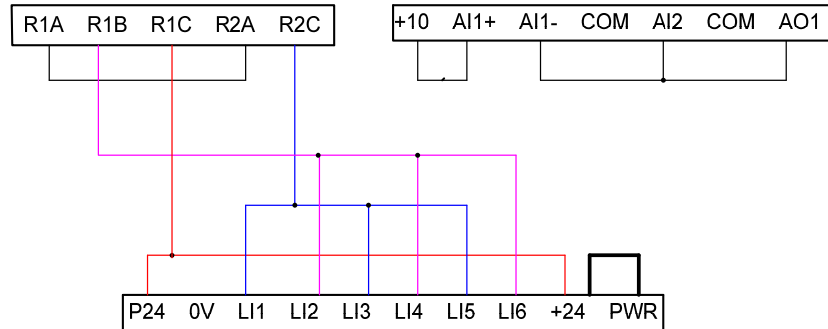
Warning, after switching-ON/OFF the hidden

BRAKING TRANS.TEST
ETR

8.3.5.1 I / O Test

The target of this test is to check the logic input, the output relay, and the visualisation of the LIs directly on the microprocessor and the calibration of the analog input and output.

To do this test, insert the following jumpers on the terminal block.



Warning: The logic input switch SW1 must be in source mode and SW2 to LI6 to avoid having a "Failed result" for the logic input test.

During this test procedure the motor can't start (IGBTs are locked)

To launch the test press ENT for a long time.

If a fault is discovered, change the control block or the terminal board.

RDY	Term	50.0Hz	0A
I/O RESULT			
LI2			OK
LI3			OK
LI4			OK
LI5			Failed
LI6			OK
Quick			

8.3.5.2 Current calibration

This test must be done with precaution.

The target of this test is to adjust the gain and the offset of each of the output phases.

For this test a motor must be connected and the motor nameplate data entered into the "motor control" menu.

For each output phases, there are two parameters, gain and offset.

The offsets are set automatically when the motor run. When the motor is running, the offset can be saved with the calibration test and will available for the next switch-on of the drive.

RDY	Term	00.0Hz	0A
I CALIBRATION RESULT			
Current phase U/T1:		-	10
Current phase V/T2:			0
Current phase W/T3:			0
IPUW	:		0
IPVW	:		0
Quick			

The gain allows to calibrate the output current in each phase. This test must be done if there is an unbalance of the output current.

To make this test configure the DC current injection on a logic input.

During the current injection measure with a current probe calibrated the output current phase by phase (IU, IV, and IW).

This value must be multiplied by Q12 ($2^{12} = 4096$) and divided by the I_{max} (See on the sheet below). Be careful to notice the sign of the values. The result must be entered on the menu (IPUW, IPVW, IPWW).

$$IP \bullet W = I \bullet \times 4096 / I_{max}$$

To finish the test run the “current calibration” with DC injection together. The sign between “current phase U, V, W must be the same as $IP \bullet W$.”

I_{max} values

Rating	240V	480V	690V
ATV71H037	9,1		
ATV71H075	14,8	7	
ATV71HU15	25	12,5	
ATV71HU22	33,3	16,8	12,3
ATV71HU30	42,2	23,8	13,8
ATV71HU40	53,6	32,7	17,8
ATV71HU55	84,4	44,3	23,1
ATV71HU75	100,5	53,6	30,7
ATV71HD11	165,6	84,7	41,5
ATV71HD15	200,7	100,5	56,9

Rating	240V	480V	690V
ATV71HD18	226,1	127	73,8
ATV71HD22	268,3	146,6	89,2
ATV71HD30	336	201,2	107,6
ATV71HD37	442,7	243	144,5
ATV71HD45	538,2	287,3	181,4
ATV71HD55		357,8	209,1
ATV71HD75		489,2	261,3
ATV71HD90			319,7

8.3.5.3 LIC test

Allows the testing of the max current value ($2 \times \sqrt{2} \times I_n$) to be sure that the drive trips on OCF.

The transistors T3/T4/T2 are commanded on. The time of the test is 50 LIC point with the nominal voltage. The LIC current on the result is the average of the current during the duration of the test.

If the LIC test is failed that's mean the LIC level has been not reach. The causes are a problem with motor or with the current feedback.

Connect a motor to the drive. If the motor is used on an application, uncouple the motor because it can turn. The motor size must match the drive size.

To start the test you must push the ENT button for ~2sec.

RDY	Term	00.0Hz	0A
LIC RESULT			
LIC test		OK	
LIC current		##,##A	
Quick			

8.3.5.4 Voltage calibration

This test is necessary if there are inconsistency between ULN on the display and the measurement with multimeter.

Before making the calibration measure the input voltage with multimeter.

Indicate in the second line (mains voltage) the value read on the multimeter. If the value entered is bad, do the calibration again until to have the correct value.

Start the voltage calibration.

RDY	Term	00.0Hz	0A
U CALIBRATION RESULT			
Mains voltage		395.5 v	
Mains voltage		395.5 v	
VOLTAGE CALIBRATION			
Quick			

8.3.5.5 Braking transistor test

The target of this test is to test the braking transistor.

Principle: The braking transistor is command during 5 seconds. There are no results for the test. The user must check with a multimeter the voltage to the terminal PA/PB.

When there is no command of the transistor the voltage is 0V

When the transistor is command the voltage is the DC bus voltage.

If there are nothing (voltage = 0V) it is a problem with the braking transistor, power board or control board.

8.3.5.6 ETR

This menu allows to reset the run time and the KWh of the drive.

RDY	Term	00.0Hz	0A
Operating t.reset			
No			
reset KWh			
rst. Runtime			
Quick			

8.3.5.7 Displaying a maintenance message (customisable).

Menu 1 – Drive Menu

1.10 - Diagnostics

Maintenance Message

Allows the display of a customised message in case of fault (5 lines of 23 characters max).

To write a maintenance message, follow the procedure in chapter 7 of the programming guide – display configuration.

RUN	Term.	+50.00Hz	1250A
1.10- DIAGNOSTIC			
FAULT HISTORY			
PRESENT FAULT LIST			
ADDITIONAL FAULT INFO			
TEST PROCEDURE			
SERVICES MESSAGE			
HOME	<<	>>	Quick

8.4 Description of LEDs status (size > 10KW)

8.4.1 Power board and measuring board

The LED's, placed for size 12 and 13 on the power board (VX5A1HC1622 and VX5A1HC2025) or for size 11 on the measurement board and power board (VX5A1500 and VX5A1HC1316) are indicating the function of power supply of the related part. If a LED is not lighting, the power supply for the corresponding part is defective or the relative supplied part is defective and burdens the supply too much.

Warning, for security and personal safety, do not depend on the displayed LED signals. A no lit LED does not guarantee that no power is supplied to the unit.

Name of board		LED number	Functionality	Indicated supply voltage	Name of connector	Name of target connector
Size 11	Size 12 & 13					
Measuring board VX5A1500	Power board VX5A1HC1622 & VX5A1HC2025	LED 1	Indicates a supplied DC-Link on the auxiliary circuit. LED 1 will be lighted, if the voltage in the Auxiliary-DC-Link is over a range of 60V and will be dim out under 10V.	560 VDC to 740 VDC	-----	-----
		LED 2	Indicates a functioning power supply for the FCB-Card (Fan Control Board).	+ 15 VDC	X14	A7 - FCB X2
		LED 3	Indicates a functioning power supply for the application circuit.	+ 24 VDC	X4	A10 - HHP_I1 X4
		LED 4	Indicates a functioning power supply for the internal fan, to cool the control electronic parts of the frequency inverter.	- 24 VDC	X21 + X22 (size 12 & 13)	Cooling Fan X31
		LED 5 & LED 6	Indicates a functioning power supply for the current transformers.	+ 15 VDC (size 11) +24VDC (size 12 & 13)	X11	T1, T2, T3
				- 15 VDC (size 11) -24VDC (size 12 & 13)		
		LED 7	Indicates a functioning power supply for powering the Motor control MC1.	+ 5 VDC	X1	A2 - HHP_MC1 X1
		LED 8	Indicates a functioning power supply for the SCB-Card (Soft Charge Board).	+ 15 VDC	X30	A6 - SCB CN2A
		LED 9 LED 10	Indicates a functioning power supply for the BUD (Breaking Unit Device).	+ 15 VDC	X91 & X92	A20 BUD12
				- 15 VDC		
Power board VX5A1HC1316						
						

If you detect a non lighting LED on your supply boards due the troubleshooting, then you will get information for each indicator as follows:

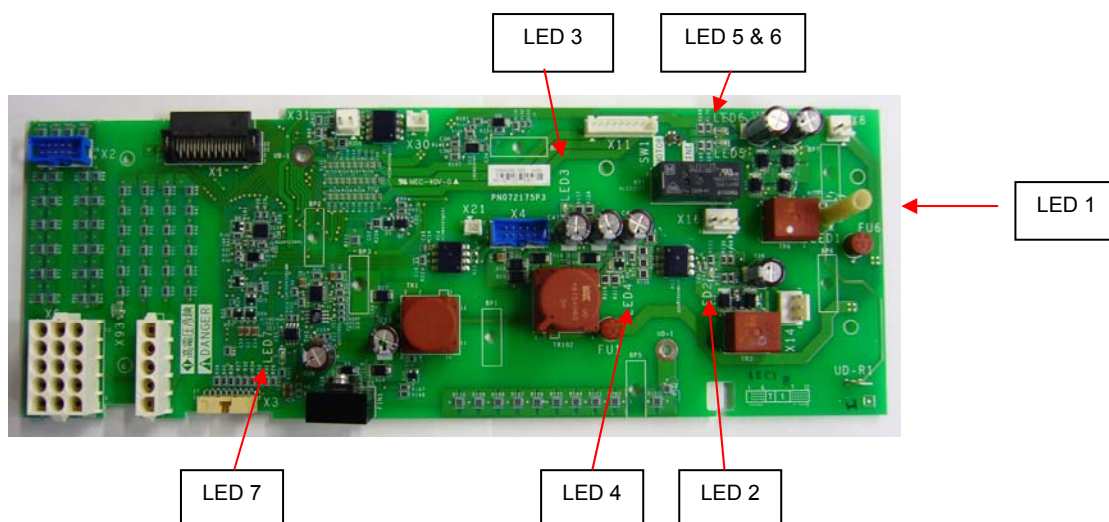
Name of board		LED number	Possible cause	Remedy
Size 11	Size 12 & 13			
Measuring board VX5A1500	Power board VX5A1HC1622 & VX5A1HC2025	LED 1	Power Supply for Auxiliary-DC-Link is defective.	Change the corresponding board.
		LED 2	The fan control board (VX5A1400) is defective and burdens the supply too much.	Disconnect the terminal X14 and supply the inverter again. If the LED2 is now lighting, change only the fan control board (VX5A1400).
			The supply on the corresponding board is defective.	If the LED2 is not lighting for all that (disconnected X14), change the corresponding board and the fan control board (VX5A1400) for sure.
		LED 3	The fault is located in the control block and burdens the supply too much.	Disconnect the terminal X4 and supply the inverter again. If the LED3 is now lighting, change only control block (VX4A71101Y) if option board present remove it step by step.
			The supply on the corresponding board is defective.	If the LED3 is not lighting for all that (disconnected X4), change the corresponding board
		LED 4	The internal cooling Fan (cooling of the control electronic parts) is mechanical blocked or has an electrical defect and burdens the supply too much.	Check the smooth running of the fan. Further, disconnect the terminal X21 or X22 on the corresponding board and supply the inverter again. If the LED4 is now lighting, change only the internal fan 24 VDC.
		LED 5 & LED 6	The current transformers are defective and burden the supply too much.	Disconnect the terminal X11 on the corresponding board and supply the inverter again. If both LED5 and LED 6 are lighting now, change only the current transformers.
			The supply on MB11 is defective.	If both LED5 and LED 6 are not lighting for all that (disconnected X11), change the corresponding board. Make sure, that the current transformers are not electrically damaged. Use an Ohmmeter to have comparison measurements on each transformer as follow:  The comparison measurements have to be same on all 3 installed current transformers. If you can find irregular values, please change the transformers, too.
		LED 7	The motor control is defective and burdens the supply too much.	Disconnect the terminal X1 on the corresponding board and supply the inverter again. If the LED7 is now lighting, change only the motor control board.
		LED 8	The soft charge board (VX5A1300) is defective and burdens the supply too much.	Disconnect the terminal X30 and supply the inverter again. If the LED8 is now lighting, change only the soft charge board (VX5A1300).
			The supply on power board is defective.	If the LED8 is not lighting for all that (disconnected X30), change the power board and the soft charge board (VX5A1300) for sure.
		LED 9 LED 10	The Breaking Unit is defective and burdens the supply too much or the supply on the power board is defective.	If one or both LED's are not lighting, it will be advisable to change the power board and the braking unit kit (VZ3F1110) together, for security reasons.

Name of board		LED number	Functionality	Indicated supply voltage	Name of connector	Name of target connector
Size 14	Size 15					
Power board VX5A1HC3140 & VX5A1HC4050	Power board VX5A1HC5063	LED 1	Indicates a supplied DC-Link on the auxiliary circuit. LED 1 will be lighted, if the voltage in the Auxiliary-DC-Link is over a range of 60V and will be dim out under 10V.	560 VDC to 740 VDC	-----	-----
		LED 2	Indicates a functioning power supply for the FCB-Card (Fan Control Board).	+ 15 VDC	X14 & X15	A70 & A71 – FCB X2
		LED 3	Indicates a functioning power supply for the application circuit.	+ 24 VDC	X4	A10 - HHP_I1 X4
		LED 4	Indicates a functioning power supply for the internal fan, to cool the control electronic parts of the frequency inverter.	- 24 VDC	X21 & X22	Cooling Fan X30,X31,X32
		LED 5 & LED 6	Indicates a functioning power supply for the current transformers.	+ 24VDC - 24VDC	X11	T1, T2, T3
		LED 7	Indicates a functioning power supply for powering the Motor control MC1.	+ 5 VDC	X1	A2 - HHP_MC1 X1
		LED 8	Indicates a functioning power supply for the SCB-Card (Soft Charge Board).	+ 15 VDC	X40	A61 – SCB CN2A
		LED 9 LED 10	Indicates a functioning power supply for the BUD (Breaking Unit Device).	+ 15 VDC - 15 VDC	X30	A60 – SCB CN2A

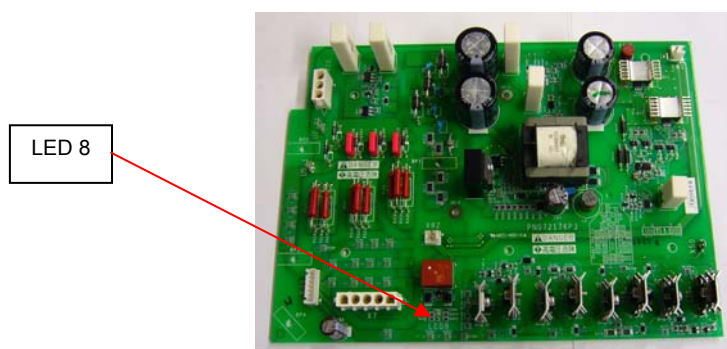
Name of board	Fuse Number	Functionality	Fused supply voltage
Power board VX5A1HC3140 VX5A1HC4050 VX5A1HC5063	FU2	Fuse for the supply Applicative circuit and internal cooling fan.	+/- 24 V=
	FU3	Fuse for the supply of FCB, Motor control and SCB - Charging DS2.	+ 15 V= + 7V=
	FU6	Fuse for the supply of current transformers.	+/- 24 V=
	FU7	Fuse for the supply of SCB - Charging D1.	+ 15 V=

8.4.2 LED location on the power board or measuring board

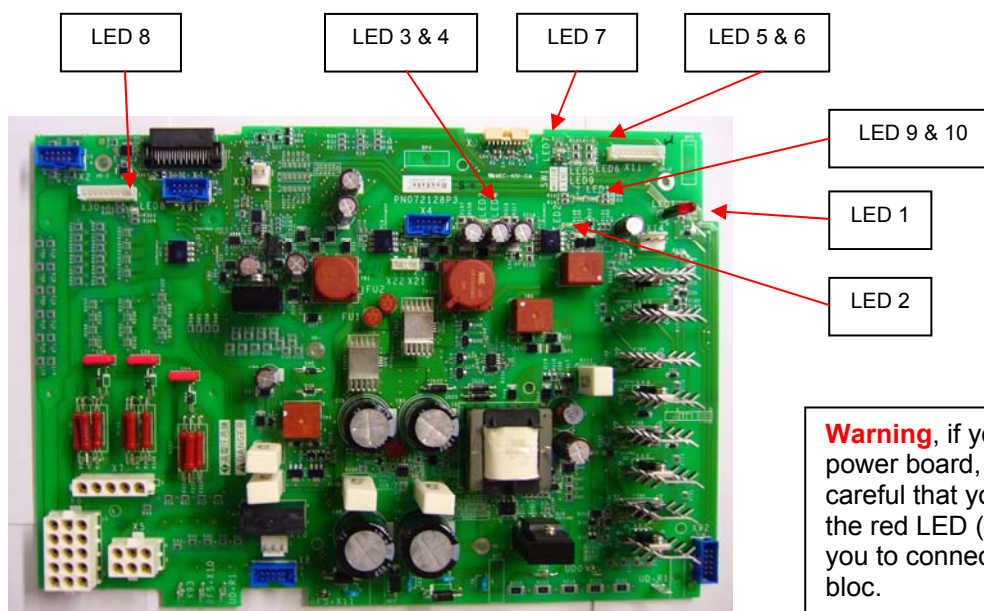
8.4.2.1 Measuring Board (VX5A71HC13N4, only size 11)



8.4.2.2 Power board VX5A1HC1316 (size 11)

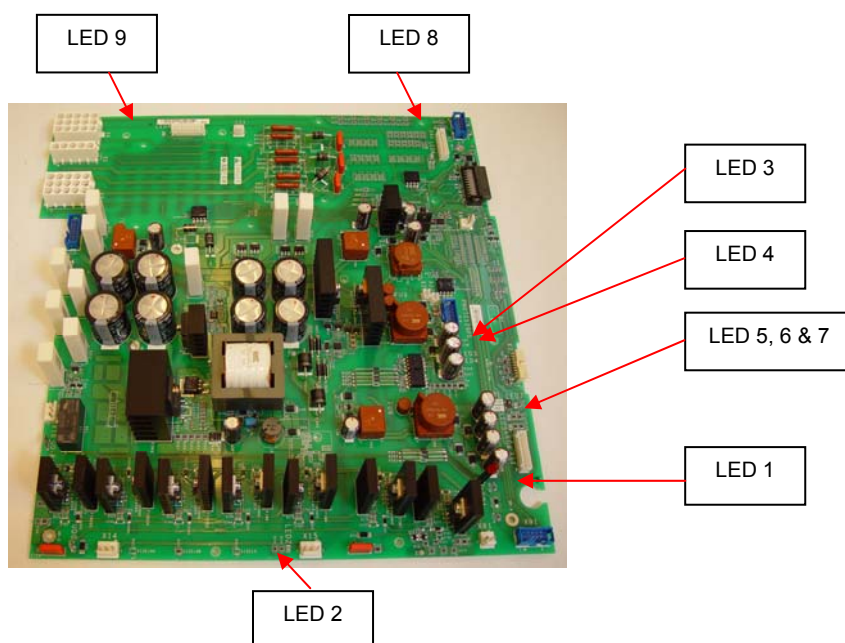


8.4.2.3 Power board VX5A1HC1622 & VX5A1HC2025 (size 12 & 13)



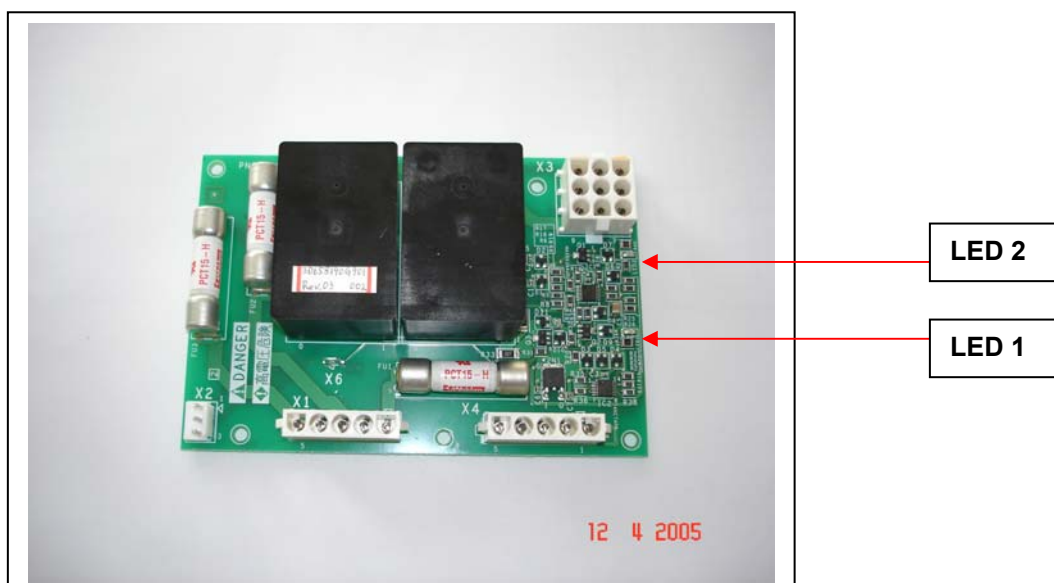
Warning, if you change the power board, you must pay careful that you don't break the red LED (LED1) when you to connect the control bloc.

8.4.2.4 Power board VX5A1HC3140 & VX5A1HC4050 (size 14 & 15)



8.4.3 Fan Control Board (VX5A1400)

Size	Status LED 1	Status LED 2	Functionality	Possible cause	Remedy
11 & 12	1	X	Temperature of the fan is OK.	Mechanically blocked due to defective bearings or dirt.	Exchange the fan or clean.
13 to 15	0	X	The fan motor is overheated;		
	1	1	The fan M1 and M2 are OK;		
	0	0	The fan M1 is overheated		
	0	1	The fan M2 is overheated		
	1	0	LED 2 on the fan control board is defect		



8.5 Keypad Contrast Calibration

These tests are only available through a hidden menu. Procedure to access to this hidden menu.

On the graphic terminal:

- Press the both key ESC and ENT (Wheel) during 5 seconds.
- Select 'Other' and press wheel button.
- Select 'Contrast' and press wheel button.
- Select 'Contrast 128' and press wheel button.
- Adjust value with the wheel to reach the contrast value needed then press on wheel button to valid. Push on "Enter" button in order to valid. The displayed value is in reverse video and is written in external "Flash".
- Finally disconnect keypad.

8.6 Drives State

The following codes are drive states and not fault codes.

<i>Code</i>	<i>Designation</i>
ACC	Acceleration.
CLI	Current limit.
CTL	Controlled stop upon input phase loss.
DCB	Braking by DC injection.
DCFE	Transfer Configuration Impossible.
DEC	Deceleration.
FLU	Motor fluxing in progress.
FST	Fast stop.
NLP	Power not supplied (no input power).
NST	Free wheel stop.
OBR	Automatically adapted deceleration.
PRA	Power Removal Function Active (drive locked out).
RDY	Drive ready.
SOC	Controlled output cut-off in progress.
TUN	Auto-tuning in progress.
USA	Under-voltage alarm.

8.7 Displayed Faults

List of faults that can be read on the display or graphic terminal.

<i>Code</i>	<i>Designation</i>
AI2F	Analogue Input Fault
AnF	Load slipping
APF	Application fault
BUF	Braking unit failure
bLF	Brake command
brF	Mechanical brake
CFF	Incorrect configuration
CF1	Invalid configuration
CnF	Communication network fault
COF	CANopen fault
CrF	Precharge fault
Ecf	Mechanical encoder link
EEFx	Control EEPROM fault

Code	Designation
ENF	Encoder fault
EPFx	External fault
FCF1	Output contactor welded
FCF2	Output contactor open
HCF	Card pairing
HDF	IGBT desaturation.
ILF	Internal link fault
InFx	Internal fault
LCF	Line contactor fault
LFF	4-20mA loss
ObF	Excessive braking
OCF	Overcurrent fault
OHF	Drive overheating
OLF	Motor overheating
OPFx	Motor phase loss
OSF	Input overvoltage
OtFx	PTC _x overheating
PHF	Input phase loss
PrF	Power removal
PrtF	
PtFx	PTC _x fault
SCFx	Short circuit
SLFx	Serial link fault
SOF	Overspeed
SPF	Loss of speed feedback
SSF	Torque / current limit
TJF	IGBT overheating
tnF	Auto-tuning fault
USF	Under voltage

Finding the failure beginning with the displayed fault

The display terminal does not illuminate

Experiences Knew:

Change the display terminal board.

The motor noise is abnormal

Experiences Knew:

The motor nameplate data has not been entered into the motor state and an auto tuning has not been done

AI2F : Analog Input Fault		Automatic Restart : YES
Principe	This fault appears when the current is over at 30mA on the input AI2.	
Probable causes		
Remedies	* Correct mechanical alignment. * Skip over the resonant frequency.	
Experience Knew	Nothing	

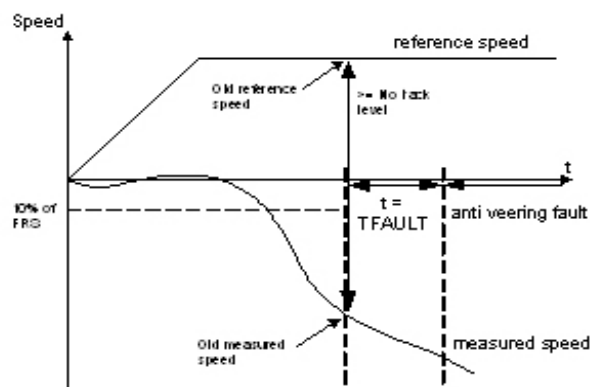
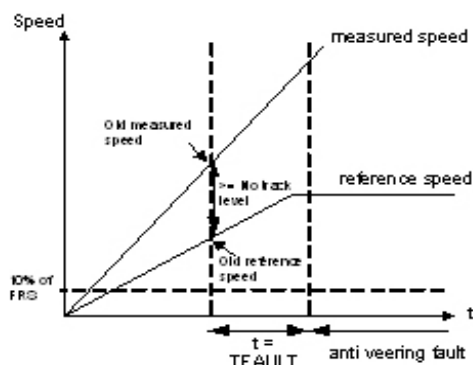
AnF : Load Slipping

Automatic Restart : YES

Principe

The fault appears when the drive has detected a load slipping.
In case of a fault, the drive goes to free wheel stop, if the brake function is configured the brake is closed.
This fault appears when:

- The no following of the ramp is over 10% of the nominal motor frequency(FRS).
- The sign of the speed error is constant.
- The sign of the speed error is opposite of the sign of the feedback variation.
- The measured speed is over 10% of the nominal motor frequency (FRS).



TFAULT = 500mSec

Probable causes

- * Not following the ramp. The speed feedback doesn't match the speed reference.
- * The sign of the speed feedback is different from the sign of the reference.
- * The speed is greater than 10% of nominal speed (FRS).

Remedies

- * Check the motor parameters, gain and stability.
- * Add a braking resistor if the fault appeared during generator mode.
- * Check the compatibility motor / drive / load.
- * Check the mechanical coupling of the encoder and its wiring.

Experience Knew

Nothing

APF : Application Fault		Automatic Restart : YES
Principe	This fault appears when the controller inside board is present. It is generated by the controller inside program (define by the programmer).	
Probable causes	* Controller inside board fault. * The program of the controller inside board generates an application fault.	
Remedy	See the board programming and documentation.	

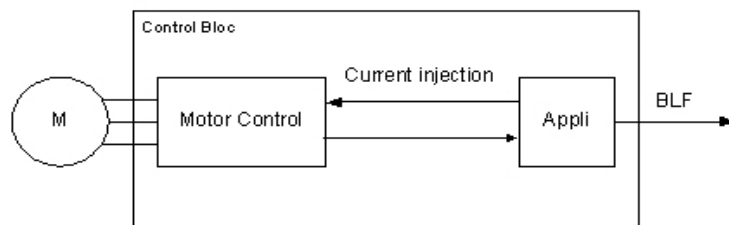
BUF : Braking Unit Fault		Automatic Restart : YES
Principe	* This fault appears only for the drives > 90kW. * This fault appears when there is a short-circuit on the output of the braking transistor (desaturation of the braking IGBT).	
Probable causes	* Short circuit inside the braking module. * Braking resistor in short circuit. * Short circuit to the ground. * BRA=NO (ramp autoadaptation) and no braking resistor.	
Remedies	* Check the wiring of the braking unit and the resistor. * Check the configuration of the drive.	
Experience Knew	Use macro configuration in hosting, in this case the BRA is set automatically to NO. If the braking resistor is not connected the next switch-on the drive trip on a BUF fault. With the version V1.1 IE02: separate control supply 24V then switch-on the drive after less than 1sec with DC bus completely discharged the drive trip on a BUF.	

bIF : Brake Logic Fault

Automatic Restart : YES

Principe

This fault appears when the brake logic is used and the conditions to open the brake are not present.
If the brake release current (parameter IBR) is not attained after a delay (parameter BRR+100ms) or it is impossible to flux the motor during $5 \cdot tr$ (tr = Rotor time constant) then the fault appears.



Probable causes

- * Check the parameter IBR.
- * Bad fluxing of motor ($5 \cdot tr$).(tr = Rotor time constant).
- * Incorrect values of autotuning (check in expert mode TRM).

Remedies

- * Check the drive/motor connections.
- * Check the motor winding.
- * Check the IBR (current for brake opening while ascending) and IRD (current for brake opening while descending) settings.
- * Check the setting of bEn (brake closing frequency).
- * Check the parameter BRR □ Current ramp time (only in expert mode).

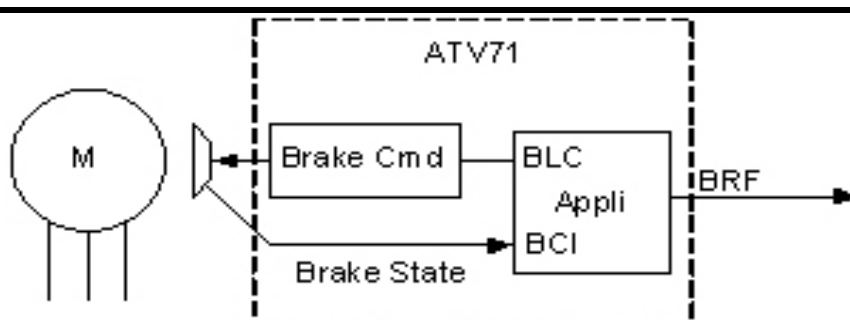
Experience Knew

- * Old motor was rewound (name plate wrong compared to the autotune result).

brF : Brake Fault

Automatic Restart : YES

Principe



Probable causes

* The brake feedback (brake contact assign to a LI) does not match the brake command.

Remedies

- * Check the brake feedback and command circuits.
- * Check the mechanical state of the brake.
- * Check the parameter BCI -> Brake contact.
- * Check the parameter BLC -> Brake contactor control.
- * Check the parameters BRT -> Brake release time.
- * Check the parameters BET -> Brake engage time.

Experience Knew

Nothing

CFF : Incorrect Configuration Fault		Automatic Restart : YES
Principe	At the power ON, the drive checks its configuration. If it detects an anomaly, it triggers a CFF fault.	
Probable causes	* Option Card changed or removed (I/O ; Encoder ; COM ; Application ;). * Control card replaced by a control card configured on a drive with a different rating. * The current configuration is inconsistent.	
Remedies	* Check that there are not card errors. * In the event of the option card being changed/removed deliberately. * Return to factory settings or retrieve the backup configuration.	
Experience Knew	With the software version product V1.1IE02, trip on a CFF fault after switch-off the drive. To resolve the problem changed one of these parameters NCR, ITH, FRS, THT or change to a previous version.	

CFI : Invalid Configuration		Automatic Restart : YES
Principe	If a configuration which has been downloaded is inconsistent (invalid parameter value, the functions configured are incompatible, etc.), the drive will trip on CFI fault. This fault will be reset automatically once a consistent configuration has been downloaded or the fault disappears as soon as one of the parameters of the configuration is written at the correct value.	
Probable causes	* The configuration loaded into the drive via serial link is incoherent (parameters outside limits for example).	
Remedies	* Check the previously loaded configuration. * Load a coherent configuration.	
Experience Knew	Nothing.	

CnF : Communication Network Fault		Automatic Restart : YES
Principe	This fault appears when the Modbus communication between the option card and the master (PLC) is interrupted.	
Probable causes	* Communication fault between option card and the master (protocol interrupted).	
Remedies	* Check the environment (electromagnetic compatibility). * Check the wiring. * Replace the option board. * Check the menu “diagnostic” (more information on the fault).	
Experience Knew	Nothing.	

COF : CANOpen Fault		Automatic Restart : YES
Principe	This fault appears when the CANopen communication between the option card and the master (PLC) is interrupted.	
Probable causes	* Interruption of communication on the CANopen. * Communication driver on the control bloc defect. * Time out in the program too small.	
Remedies	* Check the communication bus. * Check the code error in the menu monitoring. * Check the PLC program.	
Experience Knew	With the first version of terminal board if it is disconnected when the drive is ON, the communication driver are damaged.If this happens, change the control bloc.	

CRF1 : Pre-charge relay Fault		Automatic Restart : YES
Principe	The purpose of this function is to check the closing of the pre-charge relay/contactator. This fault appears only for the size 2 to 5B. If the bus voltage is unstable (5Vdc) after the command to close the relay, with a time of 5s then the fault CRF1 appears.	
Probable causes	* Failure of the command of the pre-charge relay or deterioration of the charge function (resistor). * Unstable DC bus voltage.	
Remedies	* Change the power section or the power board. * Change the control bloc. * Check input voltage.	
Experience Knew	Nothing.	

CRF2 : Thyristor Soft Charge Fault		Automatic Restart : YES
Principe	* This fault appears when DC bus charging is bad. * This fault appears for the size 6 to 15. * On the size 9 to 15, this is the soft charge board which realizes this function.	
Probable causes	* Thyristor, control bloc, soft charge board.	
Remedies	* Change the power section or the power board. * Change the control bloc. * Made a thyristor test.	
Experience Knew	On the size 6 (22kW). Verify where is positionned the cable supplying the control block. If the cable is just situated above the cristal then the product trips in CRF2. To resolve the problem, change the place of the cable.	

Ecf : Encoder Mechanical Link Fault		Automatic Restart : YES
Principe	This fault appears when there is mechanical loss of the coupling between the motor and the encoder. The detection is programmable with the ECC parameter. The product trips if the speed feedback is 0 and the drive is in torque or current limitation during ECT delay. The threshold of feedback speed is: min = 5hz ; maxi = 10% FRS. The detection of this fault is incompatible with an application operating in current or torque limit.	
Probable causes	* Braking of the mechanical coupling of the encoder.	
Remedies	* Check the mechanical encoder coupling. * Check the parameter -> ECC (Encoder coupling check configuration). * Check the parameter -> ECT (Encoder coupling detection time).	
Experience Knew	Nothing.	

EEF1 : Control EEprom Fault		Automatic Restart : YES
Principe	At starting, the microprocessor on the application board tries to read and write to the EEPROM. If it is impossible, the drive trips in EEF1 fault.	
Probable causes	* Internal memory failure.	
Remedies	* Check the environment (electromagnetic compatibility). * Switch off / on the drive, return to factory settings. * Change the control bloc.	
Experience Knew	Nothing	

EEF2 : Power EEprom Fault		Automatic Restart : YES
Principe	At starting, the microprocessor on the control motor board tries to read and write to the EEPROM on the power board. If it is impossible, the drive trips in EEF2 fault.	
Probable causes	* Internal memory failure.	
Remedies	* Check the environment (electromagnetic compatibility). * Switch off / on the drive, return to factory settings. * Change the Power board and check the ribbon cables (40 points).	
Experience Knew	Nothing.	

ENF : Encoder Fault		Automatic Restart : YES
Principe	This fault appears when the encoder test is faulty. The motor is running in open loop (motor control SVCU or SVCI).	
Probable causes	* The difference between the measured and the estimated speed is greater than 4% FRS. * The sign of the encoder feedback (measured speed) in case of AABB and AB hardware signals must be the same as the estimated speed. * The reference must be 10% FRS and stable during 3s.	
Remedies	* Check the encoder board. * Check the encoder wiring. * Check the encoder type and supply.	
Experience Knew	To replace ATV58F the signal A and B or A/ and B/ wiring must be reverse.	

EPF1 : External Fault - LI / Bit		Automatic Restart : YES
Principe	The fault can only appear if a logic input has been configured as ETF (external fault). The stop type when the fault appears is configurable. The acquisition time of LI is 2ms.	
Probable causes	* Fault caused can come from external circuit.	
Remedies	* Check the external circuit in the installation which has caused the fault. * Check the parameter ETF => External fault. * Made I/O diagnostic.	
Experience Knew	* Change the control bloc * Change the I/O terminal.	

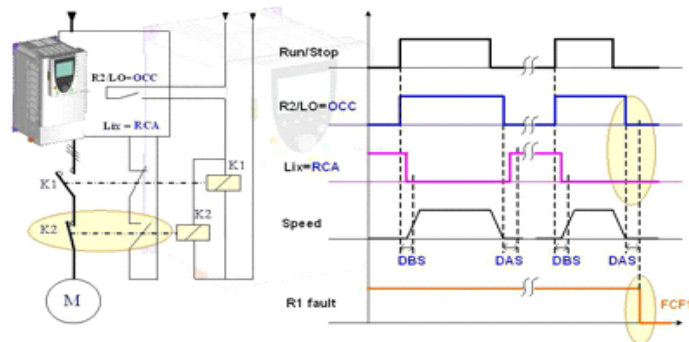
EPF2 : External Communication Fault		Automatic Restart : YES
Principe	This fault is generated by the communication or application program (defined by the programmer) The stop type when the fault appears is configurable. Warning, the Controller inside, Ethernet, Devicenet boards can generate the EPF2 fault.	
Probable causes	* Fault caused by the communication or application program.	
Remedies	* Check the external circuit in the installation which has caused the fault. * Check the communication parameters.	
Experience Knew	Nothing.	

FCF1 : Output Contactor Welded

Automatic Restart : YES

Principe

The auxiliary contact (mirror) of the contactor must be connected to a logic input configured for this effect (RCA=Lix, inverted logic). Following a run or stop command, the function tests that the contactor is closed (or open) after a settable time (TRC).



Probable causes

* The output contactor remains closed when the conditions required for opening are fulfilled.

Remedies

- * Check the contactor and its wiring.
- * Check the feedback circuit.
- * Check the parameter RCA -> Output Contact Fdbk.
- * Check the parameter DBS -> Time to motor run.
- * Check the parameter DAS -> Time to open contactor.

Experience Knew

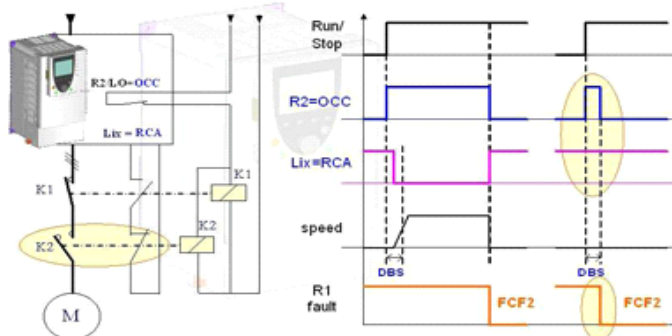
Nothing.

FCF2 : Output Contactor Blocked Open

Automatic Restart : YES

Principe

The auxiliary contact (mirror) of the contactor must be connected to a logic input configured for this effect (RCA=Lix, inverted logic). Following a run or stop command, the function tests that the contactor is closed (or open) after a settable time (TRC). FCF2 corresponds to a contactor-blocked open, this fault is resettable by a simple re-cycling of the run command.



Probable causes

* The output contactor remains open although the conditions for closure have been fulfilled.

Remedies

- * Check the contactor and its wiring.
- * Check the feedback circuit.
- * Check the parameter RCA -> Output Contact Fdbk.
- * Check the parameter DBS -> Time to motor run.
- * Check the parameter DAS -> Time to open contactor.

Experience Knew

Nothing.

HCF : Card Pairing Fault

Automatic Restart : YES

Principe

This fault appears when the parameter PPI is active (see steps following).

```

RUN Term. 0.00Hz 0.0A
MAIN MENU
ACCESS LEVEL
OPEN / SAVE AS
PASSWORD
LANGUAGE
MONITORING CONFIG
HELP << >> Quick
    
```

```

RUN Term. 0.00Hz 0.0A
FAULT MANAGEMENT
DB RES PROTECTION
AUTO TUNING FAULT
CARDS PAIRING
FALLBACK SPEED
RAMP DMDER
HELP << >> Quick
    
```

```

RUN Term. 0.00Hz 0.0A
MAIN MENU
BASIC
STANDART
ADVANCED
EXPERT
HELP << >> Quick
    
```

```

RUN Term. 0.00Hz 0.0A
CARDS PAIRING
Pairing Password : ON
HELP << >> Quick
    
```

```

RUN Term. 0.00Hz 0.0A
DRIVE MENU
MOTOR CONTROL
INPUTS / OUTPUTS CFG
COMMAND
APPLICATION FUNCT
FAULT MANAGEMENT
HELP << >> Quick
    
```

This function is used to detect whenever a card has been replaced or the software has been modified in any way. When a pairing password is entered, the parameters of the cards currently inserted are stored. On the power up, these parameters are verified and if there is a difference then the fault HCF appears.

Probable causes

- * A card has been changed.
- * The parameter PPI is active.

Remedies

- * Reinsert the original card.
- * Validate the configuration in entering the pairing password.
- * To disable this function without knowing the password, enter **7171** in parameter PPI in expert mode.

Experience Knew

Nothing.

HDF : IGBT Desaturation		Automatic Restart : YES
Principe	This fault appears only on the drive >90KW when a short-circuit is present on the IGBT module by measurement of the transistor desaturation.	
Probable causes	* Short circuit or grounding on the output phases. * Transistor IGBT in short circuit.	
Remedies	* Check the insulation motor and the wiring. * Made transistor diagnostic. * Change power board, gate drive board or transistors.	
Experience Knew	Nothing.	

ILF : Internal Link Fault		Automatic Restart : YES
Principe	This fault appears when the communication between the option board and drive is interrupted.	
Probable causes	* Communication fault between the option board and drive.	
Remedies	* Check the environment (electromagnetic compatibility). * Check the connections. * Replace the option board. * Change the control bloc.	
Experience Knew	No 24V supply on the option board (like controller inside) when switch-on the drive.	

INF1 : Rating Error		Automatic Restart : YES
Probable causes	The value saved in the EEPROM power is incorrect regarding the part number of the power board.	
Remedies	* Check the part number of the power board. * Check with the graphic terminal the size and the voltage of product.	
Experience Knew	Nothing.	

INF2 : Incompatible Power Board		Automatic Restart : YES
Probable causes	The drive trips in InF2 fault when the software version is incompatible with the application and control motor boards.	
Remedies	* Check the reference of the power board and its compatibility. * Re-flash the control bloc or change it.	
Experience Knew	Nothing.	

INF3 : Internal Serial Link		Automatic Restart : YES
Principe		
Probable causes	Communication fault between the internal boards (Power, motor control, application).	
Remedies	* Check the internal connections (check the small ribbon cable). * Change the control bloc or version software or terminal board (with the capacitor).	
Experience Knew	With the first version of terminal board if the internal 24V have some disturbances there are a reset of microprocessor (application) and trip in fault. The solution is to modify the wiring (shielded cable) or use a new terminal board (add capacitor C112). With the first version V1.1IE01of ATV71HO37M3DEMO for demo case, INF3 when download a configuration from an-other drive with an other software version. Connection to Power Suite ATV71.	

INF4 : Internal Zone Fabrication Error		Automatic Restart : YES
Principe	This fault appears when the microprocessor doesn't read the reserved industrialisation zone EEPROM.	
Probable causes	Incoherent internal data (EEPROM □ HS).	
Remedies	* Recalibrate the drive using the diagnostic menu. * Change the control bloc or power board.	
Experience Knew	Nothing.	

INF6 : Incompatible Power Board		Automatic Restart : YES
Probable causes	* The option installed in the drive is unknown or incompatible * The terminal board is not connected properly.	
Remedies	* Check the part number and the compatibility of the option. * Change the control bloc or power board.	
Experience Knew	The I/O terminal board is not connected properly.	

INF7 : Internal Hardware Initialisation		Automatic Restart : YES
Probable causes	The initialisation of the drive is incomplete.	
Remedies	* Switch ON/OFF the drive. * Check the ribbon cable (40 points). * Change control bloc or power board.	
Experience Knew	Nothing.	

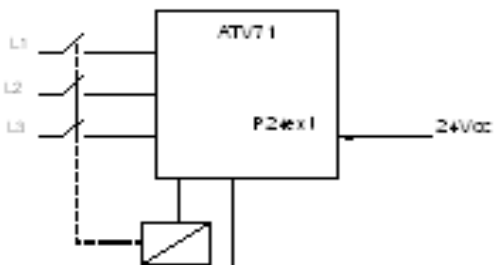
INF8 : Internal Power Supply Fault		Automatic Restart : YES
Principe	This fault appears when the voltage input P24ext of the terminal board or the supply 24Vcc of the power board is over to 32Vcc.	
Probable causes	The control power supply is not correct.	
Remedies	* Check the control power supply. * Check the supply 24Vcc internal or external.	
Experience Knew	Nothing.	

INF9 : Current Transformer Fault		Automatic Restart : YES
Principe	This fault detect only a bad supply of the current sensor.	
Probable causes	* Current measurements are incorrect.	
Remedies	* Check the current sensors. * Check the power board.	
Experience Knew	The I/O terminal board is not connected properly.	

INFB : Internal Thermal Sensor Fault		Automatic Restart : YES
Principe	The IGBT sensor for the size 2 to 6 and probe PTC for the size 7 to 15 gives this fault.	
Probable causes	The thermal probe is not operating correctly or IGBT bad.	
Remedies	* Replace or change the power bloc sub-assembly for the size 2 to 5. * Replace or change the bus board for the size 6. * Change the control bloc, power board or others board as below. * Simulate thermal sensor at the Temperature of 25°C: * For the size 7 & 8, connected on S700 of the power board a resistor of 2,2kOhms. * For the size 9 & 10, connected on CN22 of the power board a resistor of 5kOhms. * For the size 11 & 12, connected on X4 of the braking unit kit a resistor of 5kOhms. * For the size 13, connected on X4 of the gate drive board a resistor of 5kOhms. * For the size 14, connected on X4 of the gate drive board a resistor of 5kOhms. * For the size 15, connected on X4 of the gate drive board a resistor of 5kOhms.	
Experience Knew	Nothing.	

INFC : Moto Thermal State Estimation Fault		Automatic Restart : YES
Principe	The thermal state of the motor is estimated even when the drive is turned off. The function is realised by a capacitor and a counter on the application board. The fault trips if the counter is faulty.	
Probable causes	* Fault on the electronic counter for motor thermal state estimation during turn off.	
Remedy	* Replace the control bloc.	
Experience Knew	Nothing.	

INFE : Internal CPU		Automatic Restart : YES
Principe	The fault appears when the microprocessor can't check his program and data.	
Probable causes	Bad initialisation between motor control board and application board and asic COMPRIV of the power board.	
Remedy	Replace the control bloc.	
Experience Knew	Nothing.	

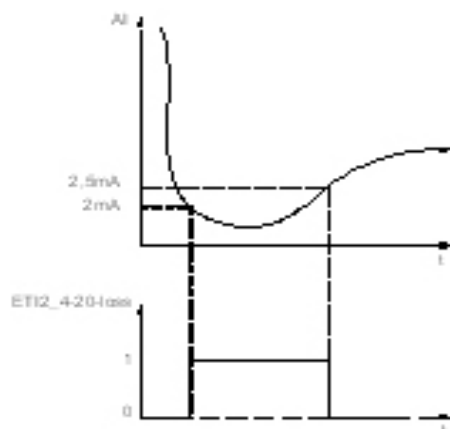
LCF : Line Contactor Fault		Automatic Restart : YES
Principe	A function allows to close the line contactor and open it when the motor is no longer on. The drive's control must be supplied by external 24V. This fault trips if the line contactor is not closed after a settable time.	
		
Probable causes	* Voltage is not applied to the drive although the time set by LCt (line voltage time out) has passed.	
Remedies	* Check the contactor and wiring. * Check the time-out period □ parameter LCT.	
Experience Knew	Nothing.	

LFF2 : Loss of 4-20mA Signal
LFF3 : Loss of 4-20mA Signal
LFF4 : Loss of 4-20mA Signal

Automatic Restart : YES

Principe

This fault appears when one of input (AI2, AI3 and AI4) has loss of the 4-20mA signal.
 If the inputs are configured in 0-20 mA or 0-10 V the fault doesn't appear.
 The drive trips in fault if the inputs current is below 2 mA.



Probable causes

* Loss of the 4-20 mA signal on an analog input AI2 on the terminal board and AI3, AI4 on the option board.

Remedies

- * Check the connection on the analog input.
- * Check the parameter AI2T -> AI2T Type
- * Check the parameter AI3T -> AI3T Type
- * Check the parameter AI4T -> AI4T Type
- * Check the parameters LFL2, LFL3, LFL4 -> Validate the AI fault.

Experience Knew

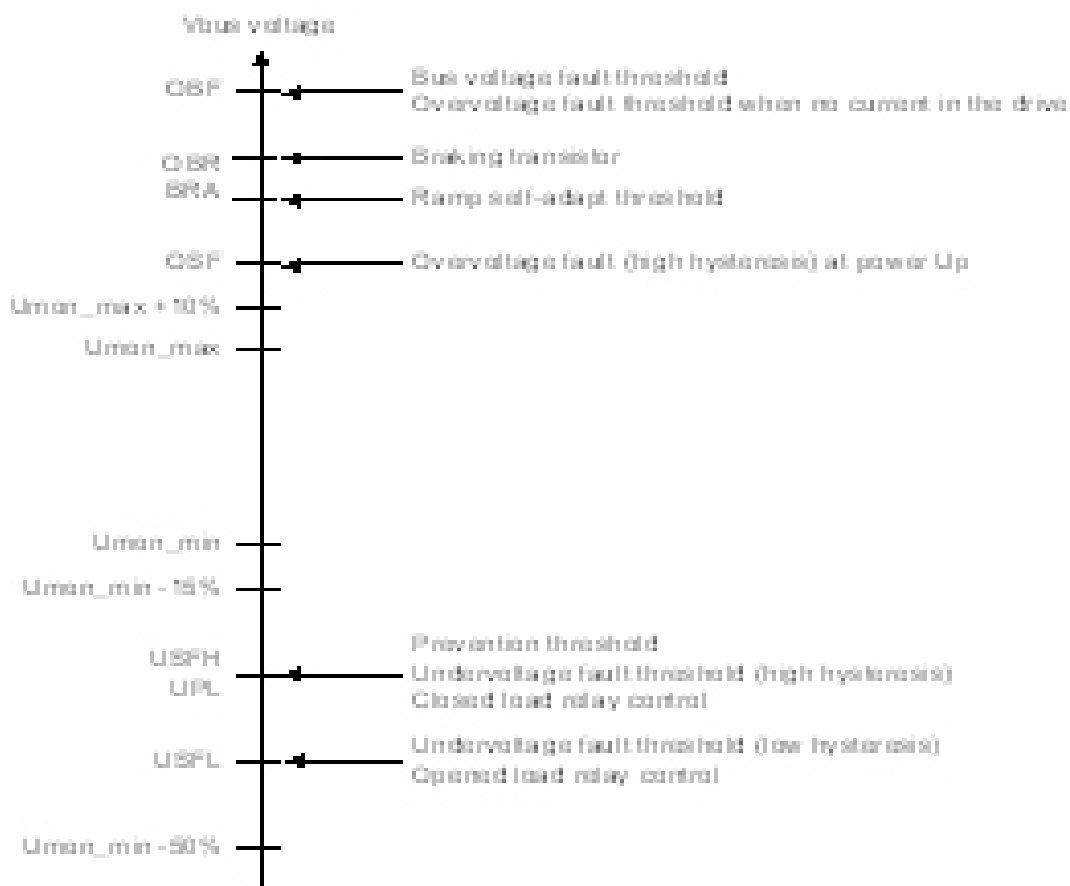
Nothing.

Obf : Over Braking Fault

Automatic Restart : YES

Principe

The fault appears when ramp adaptation is not assigned (BRA=NO), the drive is running and the DC bus voltage pass the OBF level.



		230V range DC Volts	480V range DC Volts	690V range DC Volts
OBF	Bus voltage fault threshold Over-voltage fault threshold when no current in drive	412	825	1174
OBR	Braking transistor external control.	403	805	1127
BRA	Ramp self-adapt threshold.	393	786	1105

Probable causes

* Too much braking energy or overhauling load.

Remedies

- * Increase the deceleration time.
- * Add or adapt a braking resistor.
- * Activate the BRA function (ramp adaptation during braking) if the application permits it.
- * Made diagnostic of braking transistor.
- * Change the braking transistor or IGBT or power board.
- * For P > HC20N4 check the connection with the braking unit.

Experience Knew

Nothing.

OCF : Over Current Failure		Automatic Restart : YES
Principe	OCF fault trips when the drive hardware current limitation is reached (LIC level during 100ms continuously). OCF detection is present only on range up to 75kW.	
Current		

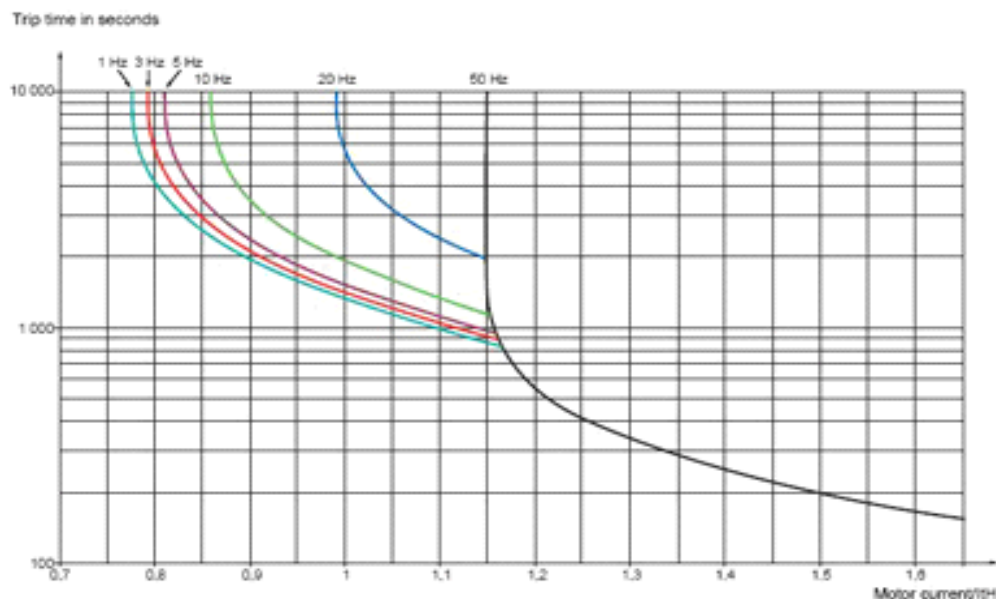
OHF : Drive Over Heating Fault		Automatic Restart : YES
Principe	The information drive thermal state comes from a sensor located on the heatsink. The drive thermal state has given by the parameter THD. If THD=118% then the fault appears and 90% the fault disappears. When THD is at 118% it means 100°C on the heatsink. Fan management: Size 2 to 5b the fan starts on a temperature threshold THD=70%. Size 6 to 15 and 690V range the fan starts as soon as the run command.	
Probable causes	* Ambient temperature too high * Bad cooling. * Filter blocked. * Fan failure.	
Remedies	Check the motor load, the drive ventilation, and the ambient temperature. Wait for the drive to cool down before re-starting.	
Experience Knew	Nothing.	

OLF : Motor OverLoad

Automatic Restart : YES

Principe

The motor overload fault protects the motor following the standard IEC947-4 and NEMA ICS 2-222.
The motor thermal stage is expressed as I^2t .
The thermal motor protection takes into account the copper and iron constants as well as the type of motor ventilation.
The OLF fault is set when the motor thermal state (tHr) is higher than 118%.
The OLF fault is reset when the motor thermal state (tHr) is lower than 100%.
The thermal state is memorized when the drive is switched off and actualised according to the off time when the drive is switched on.



Probable causes

- * Motor overloaded for a long time.
- * Trip due to too high motor current.
- * The type of motor protection programmed does not correspond to the motor.

Remedies

- * Check the motor thermal protection settings.
- * Check the motor load.
- * Wait for the motor to cool down before re-starting.
- * Check the parameter ITH ☐ Motor thermal current.
- * Check the parameter THT ☐ Thermal protection type.
- * Check the parameter THR ☐ Motor thermal state.
- * Check the parameter TSA ☐ Motor thermal alarm.
- * Check the parameter TTD ☐ Motor thermal state threshold parameter in %.
- * Check the parameter OLL ☐ Overload fault behaviour configuration parameter.

Experience Knew

Nothing.

OPF1 : Motor Phase Loss		Automatic Restart : YES
Principe	This fault appears when one motor phase is loss on the output of the drive The minimum time to detect one motor phase loss is 500ms. The threshold by phase is: $I(U,V,W) < \text{Drive nominal current} / 4$ during ODT (Output phase loss detection time)..	
Probable causes	Loss of a phase on the output of the drive.	
Remedy	Check the connections between the drive and motor.	
Experience Knew	Nothing.	

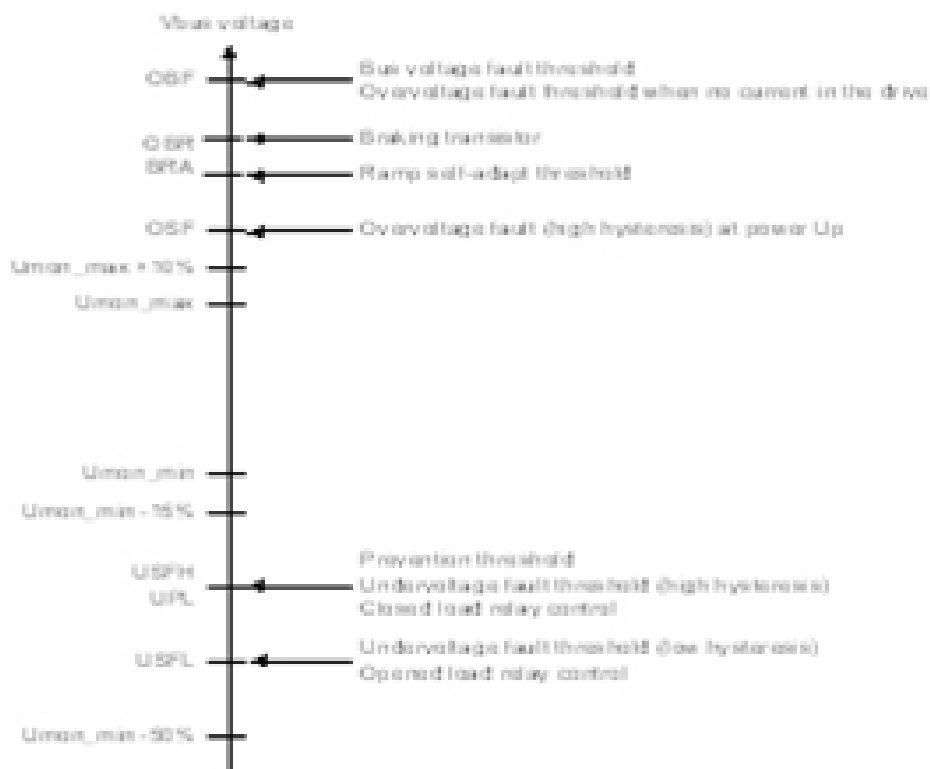
OPF2 : Motor Phase Loss		Automatic Restart : YES
Principe	This fault appears when three motor phases are loss on the output of the drive. The minimum time to detect one motor phase loss is 500ms. The threshold is: $I_{\text{mot measured}} < \text{motor current (NCR)} / 16$ during ODT (Output phase loss detection time).	
Probable causes	* Motor not wired or too small for the drive rating. * Downstream contactor open. * Instantaneous instabilities of the motor current.	
Remedies	* Check the connections between the drive and motor. * If using an output contactor set parameter OPL (motor phase loss) to OAC (downstream contactor). * During a small motor test or testing without a motor set motor phase loss detection OPL (motor phase loss) to no. * Check and optimize the motor parameters than do a tun (auto-tune). * Check the parameter ODT -> Output phase loss detection time.	
Experience Knew	Nothing.	

OSF : Main Over Voltage Fault

Automatic Restart : YES

Principe

This fault appears when the main supply is upper at the maxi supply of product (see table below). There are 2 types of OSF faults.
The drive doesn't measure the main voltage directly the line overvoltage is estimated with the Dc bus voltage.
Warning: OSF detection doesn't protect the drive, when the fault trips the line voltage must be switched off with an external sequence.



		230V range DC Volts	480V range DC Volts	690V range DC Volts
OSFH	Over-voltage fault at power up.	380	756	1087
OSFL	Threshold for reset OBF and OSF fault	375	750	1078
OBF	Bus voltage fault threshold. Over-voltage fault threshold.	412	825	1174

Probable causes

* Input voltage too high.
* Input voltage perturbations.
DC bus charging mode: DC Bus > OSFH during 20ms OSF fault = 1
DC Bus < OSFL OSF fault = 0
Drive not running mode: DC Bus > OBF OSF fault = 1
DC Bus < OSFL OSF fault = 0
When the drive is running OSF fault is not active (only OBF can trip).

Remedy

Check the input voltage.

Experience Knew

Nothing.

OtFx : PTC Over Temperature Fault

Automatic Restart : YES

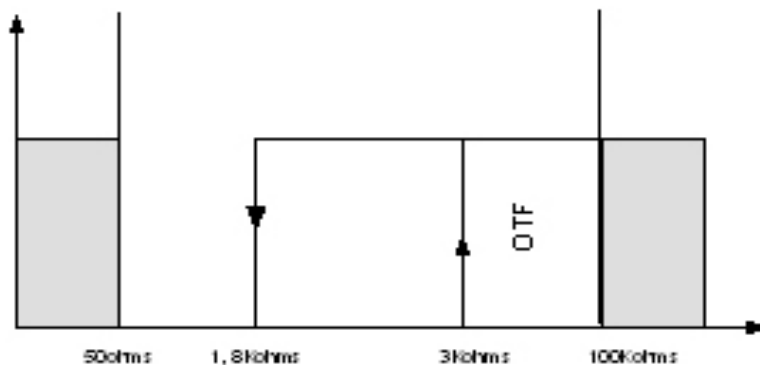
Principe

This fault appears when the PTC probe value is contained between 3KOhms and 100KOhms.

You can connect 3 PTC probes:

- * One on the input LI6 of the terminal board.
- * One on the I/O board.
- * One on the I/O extended.

Probable causes



OtF1 = Overheating detected by motor PTC sensor 1, on LI6 of the terminal board.

OtF2 = Overheating detected by motor PTC sensor 2, on the I/O board.

OtFL = Overheating detected by motor PTC sensor 3, on the extended I/O board.

Remedy

- * Check the load and the motor sizing.
- * Check the motor ventilation.
- * Wait until the motor cools down before restarting.
- * Check the type and state of the PTC sensors.
- * Terminal board or control bloc.

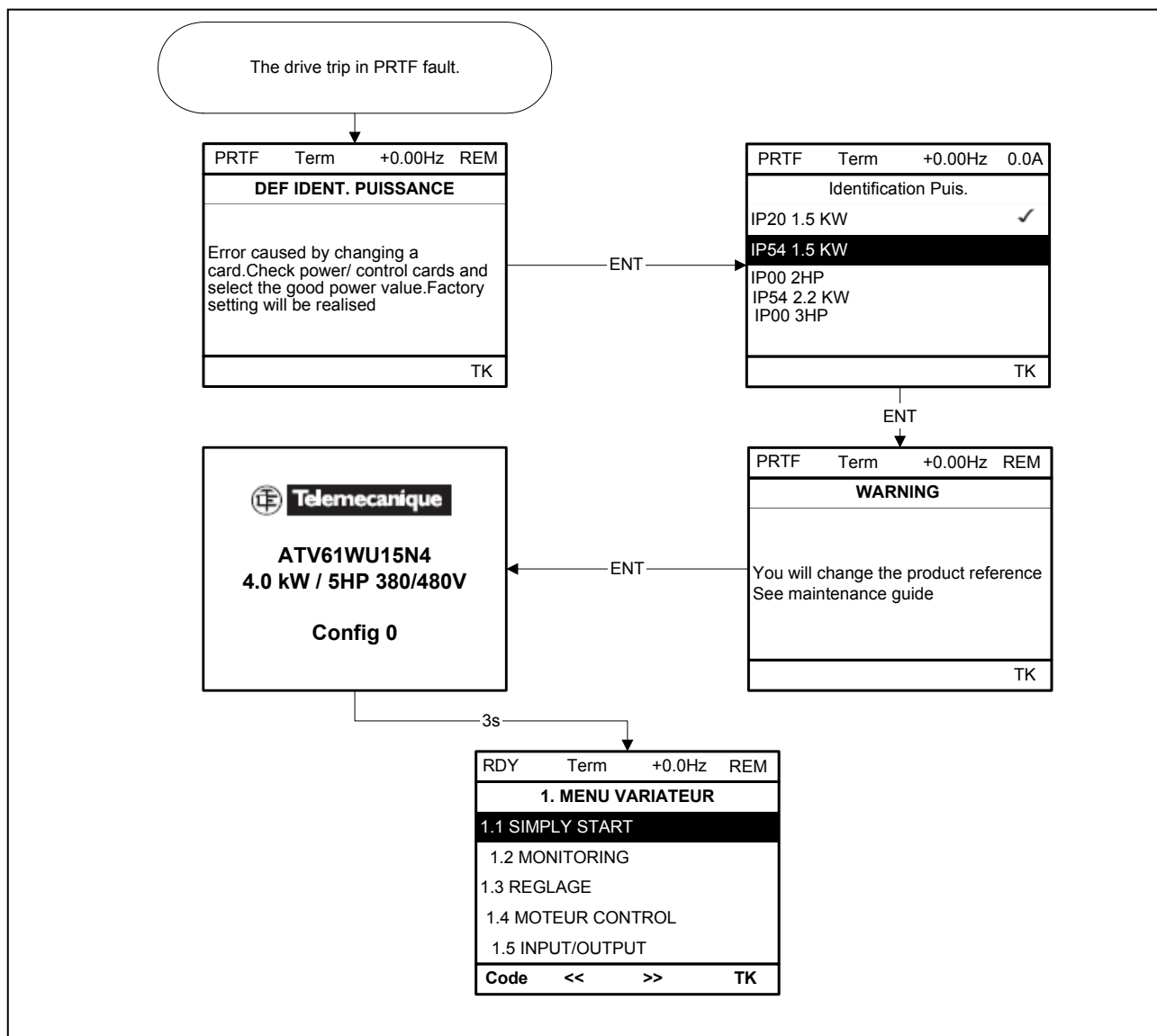
PHF : Input Phase Loss		Automatic Restart : YES
Principe	This fault appears when an input phase is lost. If 2 or 3 phase disappears while the drive is running, the drive will continue to run until the drive trips in another fault (USF for example). This function could be disable by software with the parameter IPL=NO to supply the drive via the DC BUS voltage (range 230V and 480V). Some drives (230V up to 7,5kW) have IPL=NO in factory setting because they can be supply in single phase. Warning, In the case where the motor is in generator mode then the fault doesn't appear. Warning, the fault is stored in the EEPROM during a run.	
Probable causes	* Drive incorrectly powered or fuse opened. * Loss of one input phase. * Use of a single-phase input on a three-phase ATV71.	
Remedies	* Check the power connections and the fuses. * Use a three-phase distribution network. * Inhibit the IPL fault (input phase loss).	
Experience Knew	Nothing.	

PRF : Power Removal		Automatic Restart : YES
Principe	Power removal function allows to prohibit unintended equipment operation this is to protect the person from mechanical movement. It's not for protection from touching the wiring. The fault trips if something is wrong in the internal management of this function (input PR).	
Probable causes	* Fault of the drive "Power Removal" safety function.	
Remedies	* Change the bloc control. * Change the sub-assembly power-bloc for the size 2 to 5B.	
Experience Knew	Nothing.	

PRTF : Power Rating Fault

Automatic Restart : YES

Warning, the following procedure shows how resolve the PRTF fault when the part power or the control bloc has been changed.



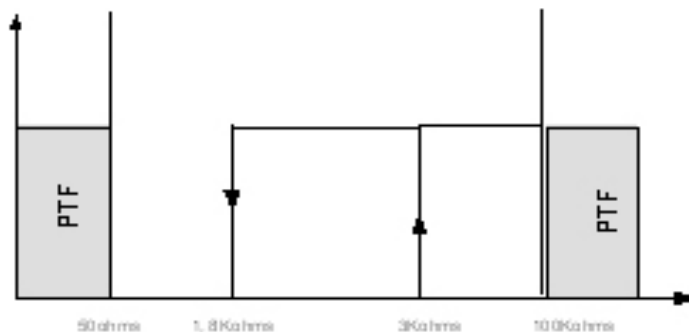
PtFx : PTCx Fault

Automatic Restart : YES

Principe

This fault appears when the PTC probe is in short circuit or open.

Probable causes



- * PtF1 = PTC1 probe .
- * PtF2 = PTC2 probe open or short-circuited.
- * PtFL = PTC probe on input LI6 open or short-circuited.

Remedy

* Check the PTC probe and the drive/motor wiring.

SCF1 : Output Short-Circuit		Automatic Restart : YES
Principe	The drive is protected against short circuit between output phases and phase to the ground. When there is a quick short circuit the drive trips in SCF1 and the fault relay opens. It's a very quick (a few μs) hardware detection, the threshold is about 3 to 4 times the drive nominal current depending on the range.	
Probable causes	* Short circuit or path to ground directly on the output of the drive. * Motor short circuit with short cables. * Short circuit to the earth	
Remedies	* Perform the tests in the drive diagnostic menu. * Check the motor-drive wiring and the motor insulation.	
Experience Knew	Transistor in short circuit.	

SCF2 : Impedant Short-Circuit		Automatic Restart : YES
Principe	The drive is protected against impedant (slow) short circuit between output phase and phase to the ground. When there is a impedant short circuit the drive trips in SCF2 and the fault relay opens. It's a software detection (a few ms): Output current sum > 25% InDrive.	
Probable causes	* (Slow) Short circuit or path to ground on the output of the drive. * Short circuit on a motor with long cables. * Isolation fault in the motor or in the cables.	
Remedy	* Check the motor-drive wiring and the motor insulation.	
Experience Knew	Nothing.	

SCF3 : Short-Circuit to Ground		Automatic Restart : YES
Principe	The purpose of this function is to detect a short circuit to the ground at the start-up or during the running. Quick hardware detection with a current transformer around the three incoming power leads for the smaller ranges and the 3 separate current transformers for the big ranges.	
Probable causes	* High leakage current to ground on the output of the drive when connecting motors in parallel. * Long shielded cables * Bad motor isolation	
Remedies	* Reduce the chopping frequency. * Add inductors in series with the motor.	
Experience Knew	Transistor in short circuit.	

SCF4 : Short-Circuit IGBT		Automatic Restart : YES
Principe	Result of the function "IGON" which detect during the power up if there is a short circuit on the IGBT. This function checks if each transistor are not in short-circuit. The time to check each transistor is between 1µS and 10µs following the size.	
Probable causes	* Power component fault.	
Remedy	* Perform a diagnostic via the menu " Diagnostic".	
Experience Knew	Transistor is short circuited.	

SCF5 : Load Short-Circuit		Automatic Restart : YES
Principe	Result of the function "IGBT test" witch detect during at each run command if there is a short circuit on the IGBT or the output of the drive before starting the motor. In factory setting this detection is disabled (can be enabled in the fault management menu TIT). Warning, this function reduces the response time of the run command (3 ms more)	
Probable causes	* Short-circuit at drive output.	
Remedies	* Check the cables connecting the drive to the motor, and the motor is insulation. * Check the parameter STRT. * Perform a diagnostic via the menu "Diagnostic".	
Experience Knew		

SLF1 : Modbus Communication		Automatic Restart : YES
Principe	This fault can be tripped when the command or the reference come from Modbus protocol.	
Probable causes	* Interruption of communication on the modbus bus.	
Remedies	* Check the parameter ADD -> Modbus Address. * Check the parameter TBR -> Modbus Baud Rate. * Check the parameter TFO -> Modbus format. * Check the parameter TTO -> Modbus Time Out * Check the parameter NTO (bit of parameter CMI).	
Experience Knew		

SLF2 : PowerSuite Communication		Automatic Restart : YES
Principe	This fault can be tripped when the command or the reference come from Powersuite.	
Probable causes	* Communication fault with powersuite.	
Remedy	* Check the Powersuite connection cable.	
Experience Knew	Nothing.	

SLF3 : Graphic Terminal Communication		Automatic Restart : YES
Principle	This fault can be tripped when the command or the reference comes from the graphic terminal (HMI). If the communication is interrupted during 2s (graphic terminal disconnect), the fault SLF3 appears.	
Probable causes	* Communication fault with the graphic terminal.	
Remedy	* Check the terminal connections.	
Experience Knew	Nothing.	

SOF : Over Speed		Reset Fault Remotely : YES
Principe	This fault appears when an overspeed is present on the drive. The feedback speed is used and when the parameter ENU=NO it is the estimated speed. The threshold is 110% of TFR parameter and during 250ms in open or close loop. When the fault occurs, the brake is engaged immediately and the drive is locked (trip) after <BRT> time (if brake sequence is active).	
Probable causes	* Unstable or too large an overhauling load.	
Remedies	* Check the motor parameters and the speed feedback loop. * Add a braking resistor. * Check the sizing of the motor, drive and load.	
Experience Knew	Nothing.	

SPF : Loss of Speed Feedback

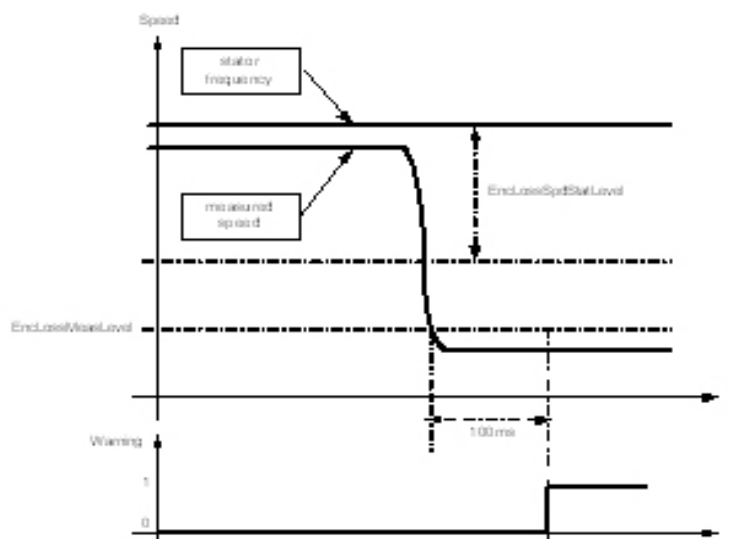
Reset Fault Remotely : YES

Principe

This fault appears when the feedback encoder is lost. There are 2 types of detection, hardware and software.

Detection software:

We use the signal A and AB. The motor is in rotation.



Software detectic

The fault is detected when the measured speed is lower at $n/2$ or $2/n$ Hz during 100ms and when the difference between the stator speed and the measured speed is greater than 20%FRS during 100 ms.

Detection Hardware:

The signal A $\bar{A}$, BB are checked. The motor can be run or stopped.

It trips if a signal is missing during 100 ms in a 500 ms window.

For the encoder board RS422, the threshold is 1.1Vdc, for the pushpull 24 V and 15 Vdc board, the threshold is 5.2 Vdc.

Probable causes

- * Absence of encoder feedback signal.
- * Missing of at least one wire on the encoder connection.

Remedies

- * Check the wiring between the encoder and the drive.
- * Check the encoder.

Experience Knew

Nothing.

SSF : Torque / Current Limit Fault		Automatic Restart : YES
Principe	This fault is active if the function SSB=YES (fault management). The fault trips if the drive is in torque limitation (TLI) or current limitation (CLI) longer than the time set in the parameter STO.	
Probable causes	* Operation in torque limit or current limit. * Overload, too quick acceleration.	
Remedies	* Check there is no mechanical problem in the application. * Check the parameters tLA (torque limit) and tLD (torque/current limit detection). * Check the parameter STO -> Trq /I limit time out. * Change control bloc.	
Experience Knew	Nothing.	

TJF : IGBT Over Heating		Automatic Restart : YES
Principe	This fault appears when the estimation of IGBT temperature is too high. It is an I ² t calculation made regarding the switching frequency, drive frequency and motor current.	
Probable causes	* Drive overload.	
Remedies	* Check if there is a mechanical problem in the application. * Check the sizing of the motor / drive / load. * Reduce the chopping frequency. * Wait for the drive to cool down before re-starting. * Change IGBT.	
Experience Knew	Nothing.	

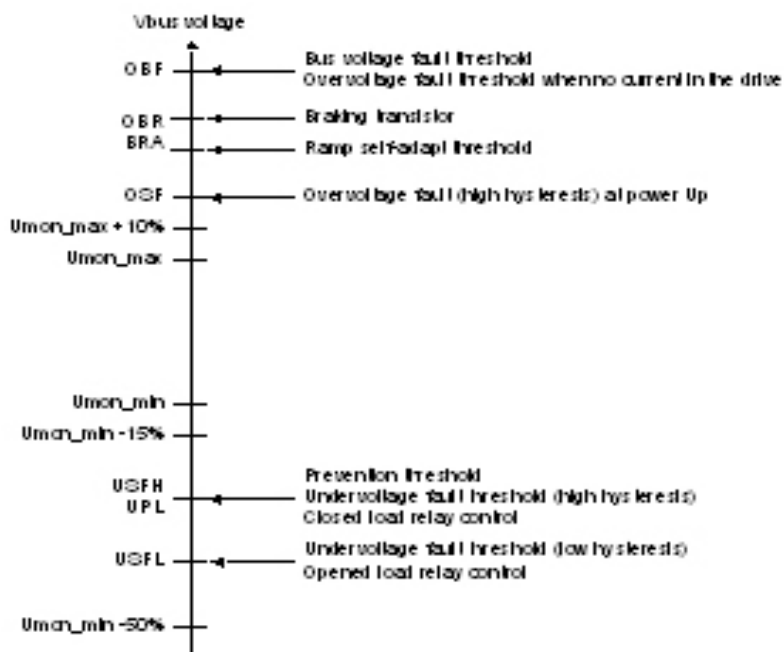
tnF : Auto-Tuning Fault		Automatic Restart : YES
Principe	This fault appears only if a TUNE has been requested. The motor must be de-fluxed before beginning a tune. On a synchronous motor only the stator resistance is measured. For asynchronous motor both the stator resistance and leakage inductance are measured. The tnF fault can be reset by logic input. The output phase loss fault has priority over the tnF fault.	
Probable causes	* Special motor or a motor power that is not within the range of the drive. * Motor not connected to the drive. * Tune with long motor cable and output filter.	
Remedies	* Check the size of the motor compared to the drive * Make sure the motor is connected during the auto-tuning process. * If the output contactor is present be sure to close it during auto tuning.	
Experience Knew	Nothing.	

USF : Under Voltage

Automatic Restart : YES

Principe

This fault appears when the voltage measured on the DC bus is lower than a level depending on the main voltage parameter Ures:



USFH

UPL

Closed load relay control.
Prevention threshold for control stop function.

USFL

Opened load relay control

USF_min - 50%

Opened min load relay control (configuration of parameter USL)

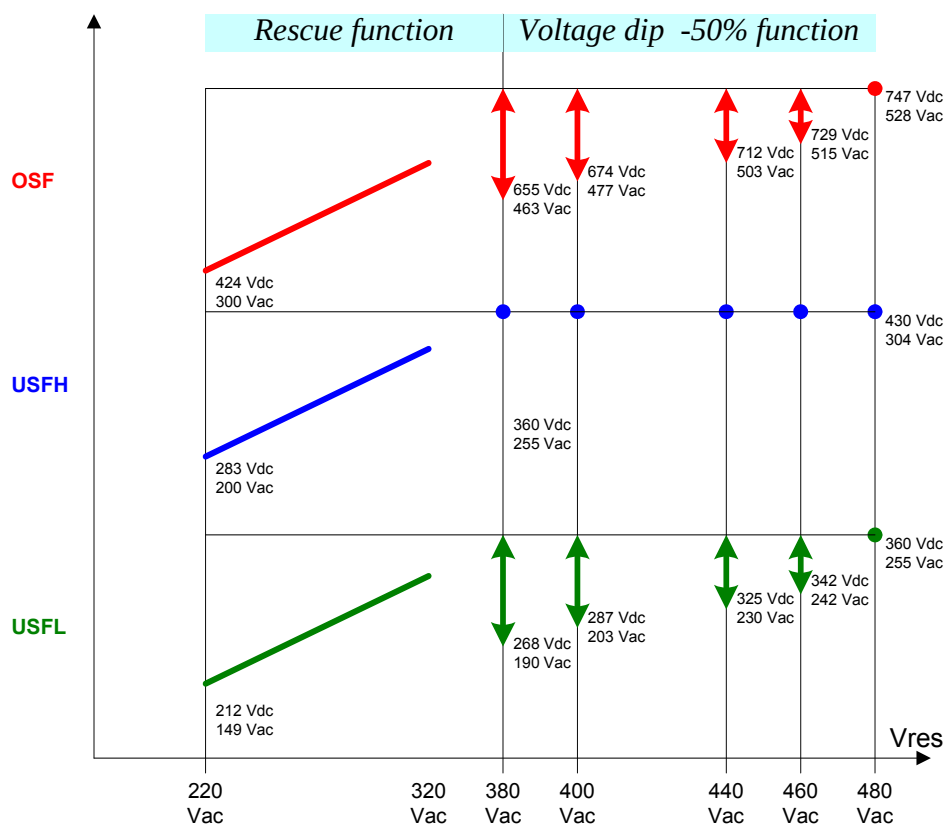
Principe

Range	Ures	USFL min	USFL max	USFH
N4	380Vac	268Vdc - 190Vac	360Vdc - 255Vac	430Vdc- 304Vac
	400Vac	287Vdc - 203Vac	360Vdc - 255Vac	430Vdc- 304Vac
	440Vac	325Vdc - 230Vac	360Vdc - 255Vac	430Vdc- 304Vac
	460Vac	342Vdc - 242Vac	360Vdc - 255Vac	430Vdc- 304Vac
	480Vac *	360Vdc - 255Vac	360Vdc - 255Vac	430Vdc- 304Vac
M3	200Vac	140Vdc - 100Vac	200Vdc - 141Vac	230Vdc- 163Vac
	220Vac	160Vdc - 113Vac	200Vdc - 141Vac	230Vdc- 163Vac
	240Vac	180Vdc - 127Vac	200Vdc - 141Vac	230Vdc- 163Vac
	260Vac *	200Vdc - 141Vac	200Vdc - 141Vac	230Vdc- 163Vac
Y	500Vac	354Vdc- 250Vac	488Vdc- 345Vac	571Vdc- 404Vac
	600Vac	424Vdc- 300Vac	488Vdc- 345Vac	571Vdc- 404Vac
	690Vac *	488Vdc- 345Vac	488Vdc- 345Vac	571Vdc- 404Vac

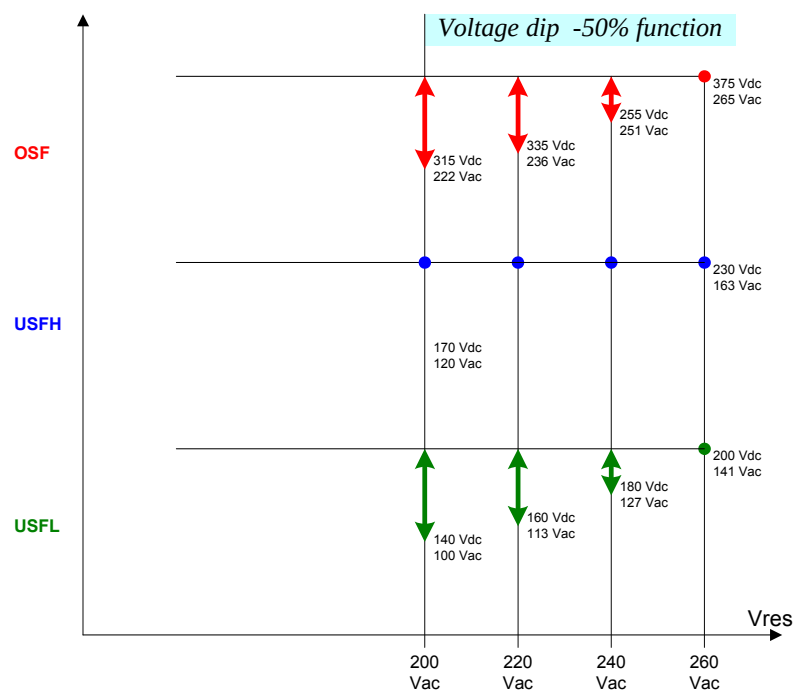
USF : Under Voltage (continued)		Automatic Restart : YES
Probable causes	* Distribution network too small. * Temporary lowering of input voltage. * Deterioration of the charging resistor.	
Remedies	* Check the input voltage and the voltage parameter in the drive.	
Experience Knew	* Shunt between PO/PA+ not tighten (bad contact).	

These diagrams summarise the different threshold OSF, USF:

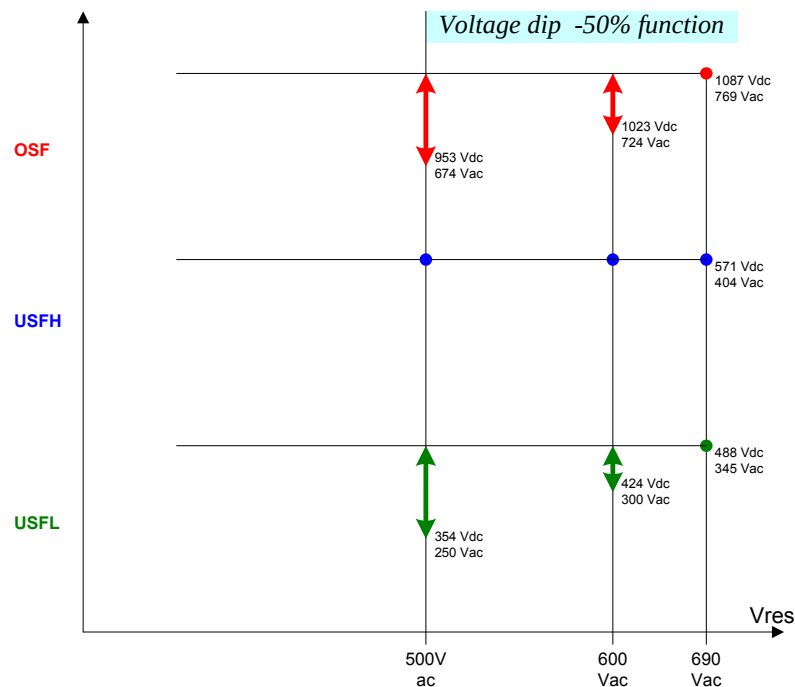
N4 range :



M3 range :



Y range :



9. Changing the faulty component - drawings - parts lists

9.1 Safety instructions and precautions

Safety instructions

Before carrying out any work on an ATV61/71:

- Disconnect the drive power supply.
- Wait for 10 minutes before opening the front cover and check that the Power and Charge LED is off.

On ATV61/71 measure the DC main voltage between the PA/+ and PC/– terminals before starting work.

Never touch the connector on the terminal board/option card when the drive is switched on.

When the product is switched on, the terminal board and option card must always be protected.

Precautions

If you are not wearing anti-static wristbands:

- Handle the electronic boards by taking hold of them by the edge and using the palm of your hand do not touch components with your fingers, as this risks damaging the components due to an electrostatic discharge.
-

9.2 ATV61/71 Size 2 (size, refer to 1.2)

9.2.1 Dismantling and reassembling

Size 2: ATV71H037M3, ATV71H075M3, ATV71HU15M3, ATV71H075N4, ATV71HU15N4, ATV71HU22N4, ATV61H075M3, ATV61HU15M3, ATV61H075N4, ATV61HU15N4, ATV61HU22N4

ATV71H037M3	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1H037M3	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71H075M3	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1H075M3	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU15M3	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU15M3	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV71H075N4	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1H075N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV71HU15N4	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU15N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV71HU15N4	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU22N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV61H075M3	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1H075M3	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV61HU15M3	
Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU15M3	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV61H075N4

Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1H075N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

ATV61HU15N4

Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU15N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

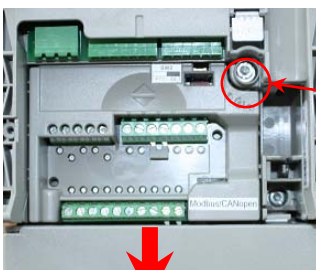
ATV61HU22N4

Reference	Designation
VZ3V1203	Fan KIT
VY1A1202	Plastic Parts
VX5A1HU22N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board

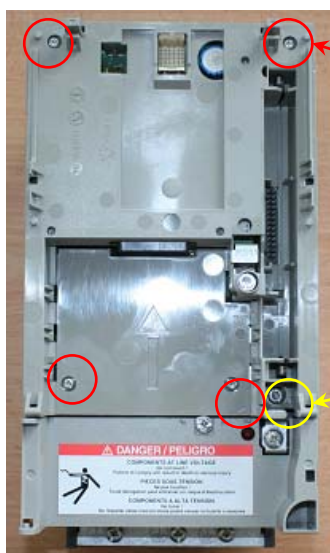
Control Bloc: VX4A71100Y



Press the two clips and pulls forward to take out the display board.



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

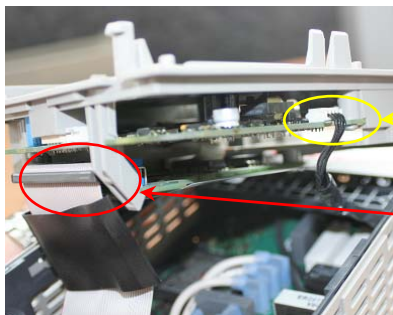


Remove 4 screws (S5 to S8).

Remove this screw (S38).

	Mark	Size	Torque
	S5-S8	M3x12	0,78Nm
	S38	M4x6	1,5Nm

Control Bloc: VX4A71100Y



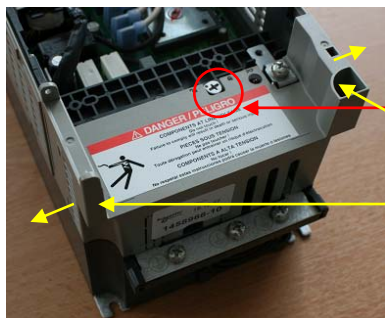
Disconnect the wire.
Application board (S103) → Power board (S103).

Disconnect the ribbon cable.

Plastic Parts: VY1A1202



Remove the power trap.



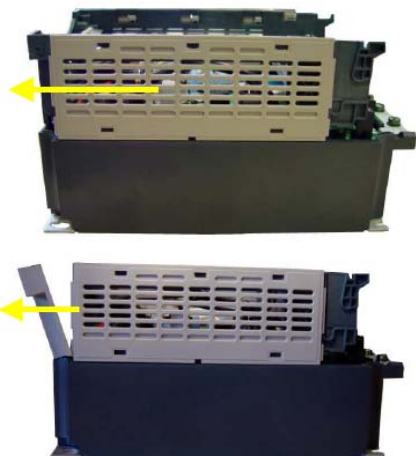
Turn the screw in the unbolt position.

Push on the left and right to remove the wiring trap.



Push the 2 clips to remove the top cap.

Plastic Parts: VY1A1202

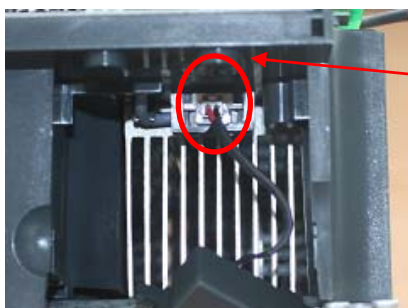


Slide the side cap to the left or to the right to remove it.

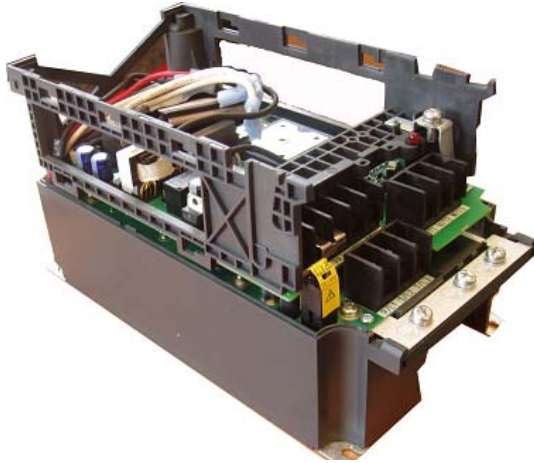
Fan KIT: VZ3V1203



Press on the 2 clips for remove the fan KIT.



Disconnect the wire.

Power Block Sub Assembly: VX5A1HU22N4

The power block sub assembly.

9.2.2 Product Assembling Drawing

Refer to following files: [Assembling_175409400A53_IED06.pdf](#)
[Assembling_175409500A53_IED03.pdf](#)
[Assembling_175409600A53_IED05.pdf](#)

9.2.3 Product Cabling Drawing

Refer to following file: [Cabling_175554400A53_IED04.pdf](#)

9.3 ATV61/71 Size 3 (size, refer to 1.2)

9.3.1 Dismantling and reassembling

Size 3: ATV71HU22M3, ATV71HU30M3, ATV71HU40M3, ATV71HU30N4, ATV71HU40N4, ATV61HU22M3, ATV61HU30M3, ATV61HU40M3, ATV61HU30N4, ATV61HU40N4

ATV71HU22M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU22M3	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU30M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU30M3	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU40M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU40M3	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU30N4

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU30N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU40M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU40M3	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU22M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU22M3	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU30M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU30M3	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU40M3

Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU22M3	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU30N4	
Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU30M3	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU40N4	
Reference	Designation
VZ3V1209	Fan KIT
VY1A1203	Plastic Parts KIT
VX5A1HU22M3	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

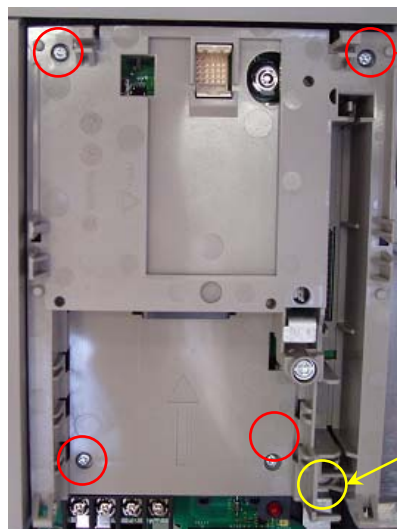
Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

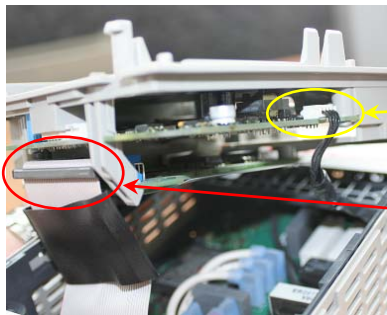


Remove 4 screws (S5 to S8).

Remove this screw (S38).

	Mark	Size	Torque
	S5-S8	M3x12	0,78Nm
	S38	M4x6	1,5Nm

Control Bloc: VX4A61100Y



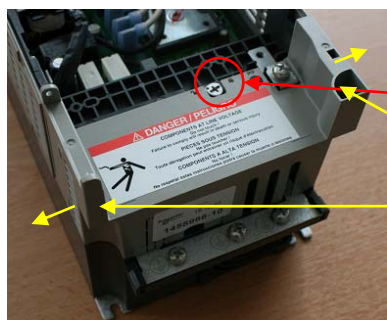
Disconnect the wire.
Application board (S103) → Power board (S103).

Disconnect the ribbon cable.

Plastic Parts: VY1A1203

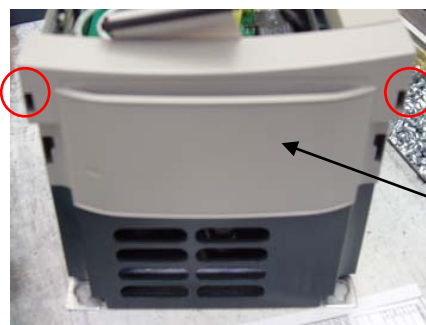


Remove the power trap.



Turn the screw in the unbolt position.

Push on the left and right to remove the wiring trap.



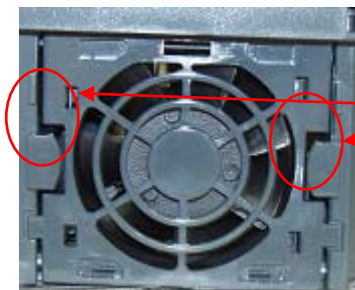
Push the 2 clips to remove the top cap.

Remove the IP4X top cover.

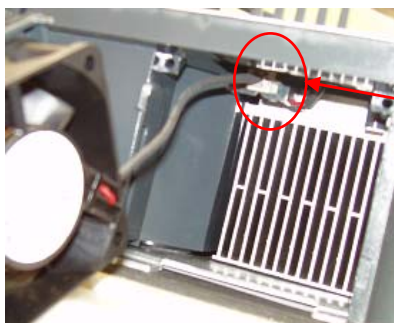
Plastic Parts: VY1A1203

Slide the side cap to the left or to the right to remove it.

Push the 2 clips to remove the front cap.

Fan KIT: VZ3V1209

Press on the 2 clips for remove the fan KIT.



Disconnect the wire.

Power Block Sub Assembly: VX5A1HU30M3

The power bloc sub assembly.

9.3.2 Product Assembling Drawing

Refer to following files: [Assembling_175409400A53_IED06.pdf](#)
[Assembling_175409500A53_IED03.pdf](#)
[Assembling_175409600A53_IED05.pdf](#)

9.3.3 Product Cabling Drawing

Refer to following file: [Cabling_175554400A53_IED04.pdf](#)

9.4 ATV61/71 Size 4 (size, refer to 1.2)

9.4.1 Dismantling and reassembling

**Size 4: ATV71HU55M3, ATV71HU55N4, ATV71HU75N4
ATV61HU55M3, ATV61HU55N4, ATV61HU75N4**

ATV71HU55M3	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU55M3	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU55N4	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU55N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HU75N4	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU75N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU55M3	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU55M3	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU55N4	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU55N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HU75N4	
Reference	Designation
VZ3V1204	Fan KIT
VY1A1204	Plastic Parts KIT
VX5A1HU75N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

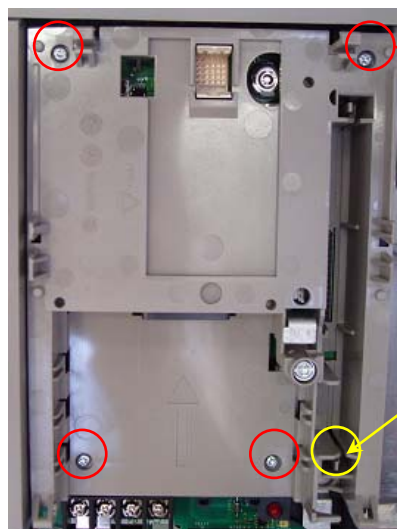
Control Bloc: VX4A61100Y



Press the two clips and pull forward to take out the display board.



For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

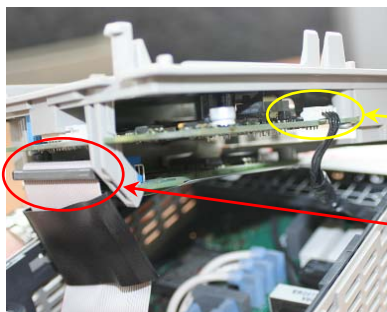


Remove 4 screws (S5 to S8).

Remove this screw (S38).

	Mark	Size	Torque
	S5-S8	M3x12	0,78Nm
	S38	M4x6	1,5Nm

Control Bloc: VX4A61100Y



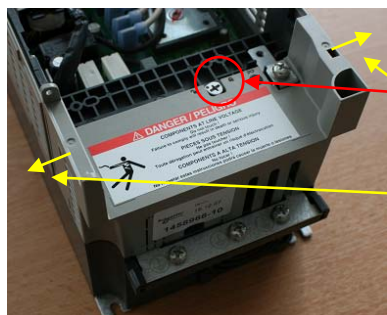
Disconnect the wire.
Application board (S103) → Power board (S103).

Disconnect the ribbon cable.

Plastic Parts: VY1A1204



Remove the power trap.



Turn the screw in the unbolt position.

Push on the left and right to remove the wiring trap.



Remove the IP4X top cover.

Push the 2 clips to remove the top cap.

Plastic Parts: VY1A1204

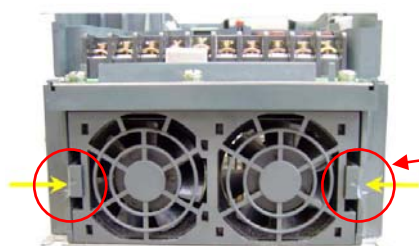


Slide the side cap to the left or to the right to remove it.

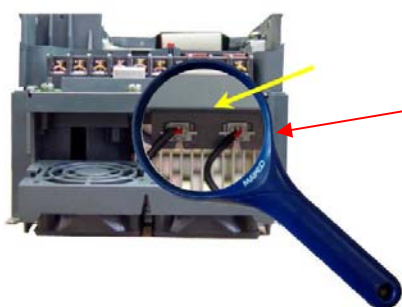


Push the 2 clips to remove the front cap.

Fan KIT: VZ3V1204



Press on the 2 clips for remove the fan KIT.



Disconnect 2 wires.

Power Block Sub Assembly: VX5A1HU55M3

The power bloc sub assembly.

9.4.2 Product Assembling Drawing

Refer to following files: [Assembling_175409600A53_IED05.pdf](#)
[Assembling_175577700A53_IED06.pdf](#)
[Assembling_175577800A53_IED03.pdf](#)

9.4.3 Product Cabling Drawing

Refer to following file: [Cabling_175554500A53_IED03.pdf](#)

9.5 *ATV61/71 Size 5A (size, refer to 1.2)*

9.5.1 Dismantling and reassembling

Size 5A: ATV71HU75M3, ATV71HD11N4, ATV61HU75M3, ATV61HD11N4

ATV71HU75M3	
Reference	Designation
VZ3V1210	Fan KIT
VY1A1215	Plastic Parts KIT
VX5A1HU75M3	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HD11N4	
Reference	Designation
VZ3V1210	Fan KIT
VY1A1215	Plastic Parts KIT
VX5A1HD11N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

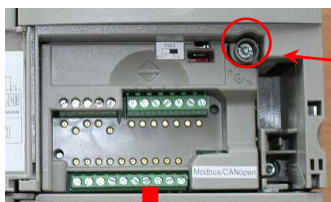
ATV61HU75M3	
Reference	Designation
VZ3V1210	Fan KIT
VY1A1215	Plastic Parts KIT
VX5A1HU75M3	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HD11N4	
Reference	Designation
VZ3V1210	Fan KIT
VY1A1215	Plastic Parts KIT
VX5A1HD11N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

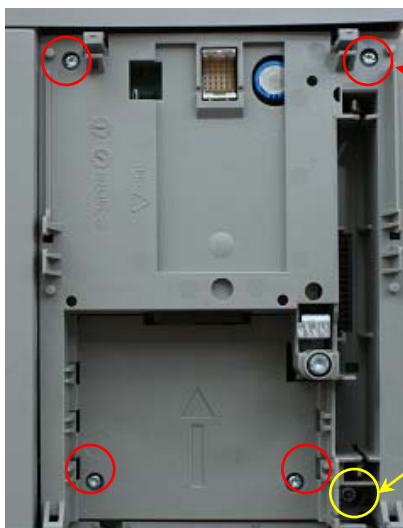
Control Bloc: VX4A71100Y



Press the two clips and pull forward to take out the display board.



For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

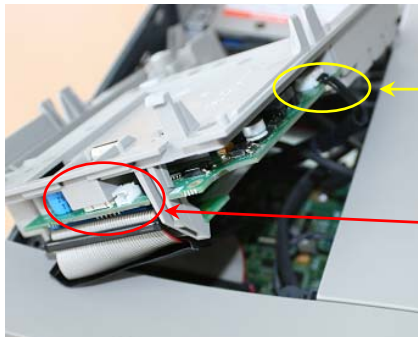


Remove 4 screws (S5 to S8).

Remove this screw (S38).

	Mark	Size	Torque
	S5-S8	M3x12	0,78Nm
	S38	M4x6	1,5Nm

Control Bloc: VX4A71100Y



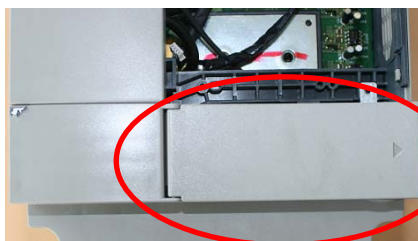
Disconnect the wire.
Application board (S103) → Power board (S103).

Disconnect the ribbon cable.

Plastic Parts: VY1A1215



Push the 2 clips to remove the top cap.



Disconnect the ribbon cable.



Turn the screw in the unbolt position.

Plastic Parts: VY1A1215



Push on the left and right to remove the wiring trap.

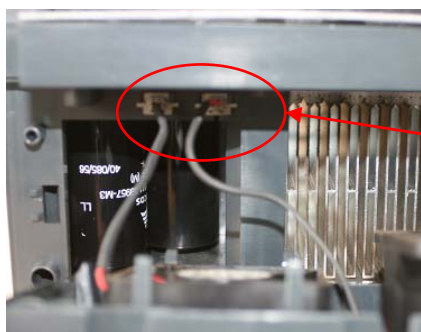


Slide the side cap to the left or to the right to remove it.

Fan KIT: VZ3V1210



Press on the 2 clips for remove the fan KIT.



Disconnect 2 wires.

Power Block Sub Assembly: VX5A1HU75M3

The power bloc sub assembly.

9.5.2 Product Assembling Drawing

Refer to following files: [Assembling_175409600A53_IED05.pdf](#)
[Assembling_175577700A53_IED06.pdf](#)
[Assembling_175577800A53_IED03.pdf](#)

9.5.3 Product Cabling Drawing

Refer to following file: [Cabling_175554500A53_IED03.pdf](#)

9.6 ATV61/71 Size 5B (size, refer to 1.2)

9.6.1 Dismantling and reassembling

Size 5B: ATV71HD11M3X, ATV71HD15M3X, ATV71HD15N4, ATV71HD18N4, ATV61HD11M3X, ATV61HD15M3X, ATV61HD15N4, ATV61HD18N4

ATV71HD11M3X	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD11M3X	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HD15M3X	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD15M3X	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HD15N4	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HD18N4	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HD11M3X	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD11M3X	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HD15M3X	
Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD15M3X	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

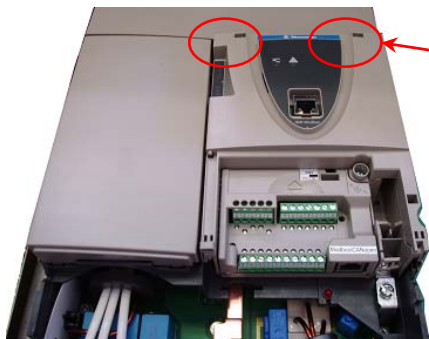
ATV61HD15N4

Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

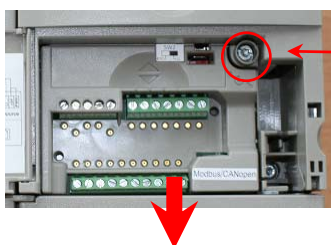
ATV61HD18N4

Reference	Designation
VZ3V1205	Fan KIT
VZ3N1205	Power Terminals
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

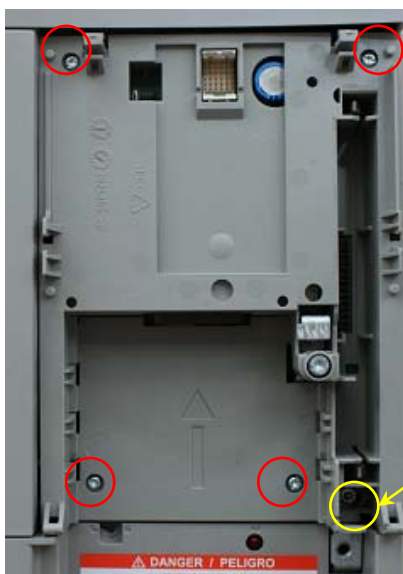
Control Bloc: VX4A61100Y



Press the two clips and pull forward to take out the display board.



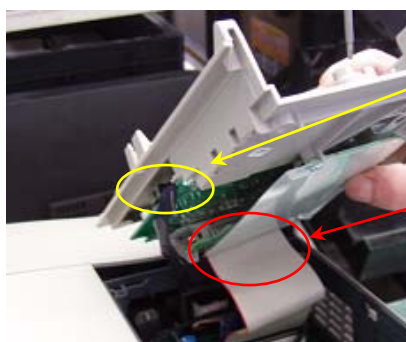
For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S5 to S8).

Remove this screw (S38).

Disconnect the wire.
Application board (S103) → Power board (S103).



Disconnect the ribbon cable.

	Mark	Size	Torque
	S5-S8	M3x12	0,78Nm
	S38	M4x6	1,5Nm

Plastic Parts: VY1A1205



Push the 2 clips to remove the top cap and IP4X top cover.



Open the power trap.

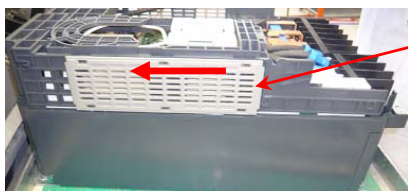


Turn the screw in the unbolt position.

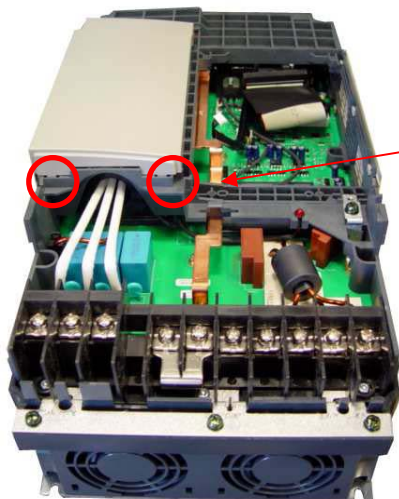
Push on the left and right to remove the wiring trap.



Slide the side cap to the left or to the right to remove it.

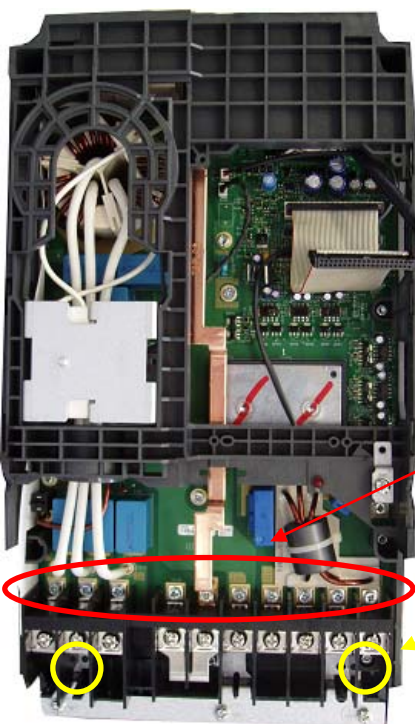


Plastic Parts: VY1A1205



Push the 2 clips to remove the front cap.

Power Terminals: VZ3N1205

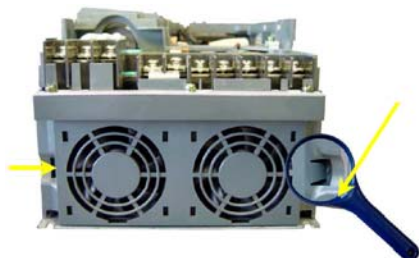


Remove 10 screws (S103 to S113).

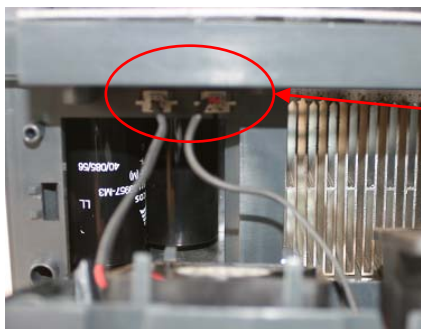
Remove the 2 screws (S29 & S30).

Mark	Size	Torque
S29 - S30	4x12	1Nm
S103 - S113	M4x10	1,5Nm

Fan KIT: VZ3V1205



← Press on the 2 clips for remove the fan KIT.



Disconnect 2 wires.

Power Block Sub-Assembly: VX5A1HD15N4



The power bloc sub assembly.

9.6.2 Product Assembling Drawing

Refer to following files: [Assembling_175409600A53_IED05.pdf](#)
[Assembling_175577800A53_IED03.pdf](#)
[Assembling_175577900A53_IED05.pdf](#)

9.6.3 Product Cabling Drawing

Refer to following file: [Cabling_175554600A53_04.pdf](#)

9.7 ATV61/71 Size 6 (size, refer to 1.2)

9.7.1 Dismantling and reassembling

Size 6: **ATV71HD18M3X, ATV71HD22M3X, ATV71HD22N4, ATV61HD18M3X, ATV61HD22M3X, ATV61HD22N4**

ATV71HD18M3X	
Reference	Designation
VZ3V1211	Fan KIT
VZ3N1301	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1103	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD18M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3N1206	Power Terminals
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V

ATV71HD22M3X	
Reference	Designation
VZ3V1211	Fan KIT
VZ3N1301	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1104	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD22M3X	Power Block Sub Assembly
VZ3N1206	Power Terminals
VX4A1103	Front Cover 4x7 Digits
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V
VX4A1104	Terminal Board

ATV71HD22N4	
Reference	Designation
VZ3V1211	Fan KIT
VZ3N1305	Mains Connexion KIT
VZ3N1304	Bars KIT
VZ3N1302	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1HD22N4	Power Block Sub Assembly
VX5A1101	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1107	Sub Assembly Filter Boards
VZ3N1206	Power Terminals
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD18M3X	
Reference	Designation
VZ3V1211	Fan KIT
VZ3N1301	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1103	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD18M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3N1206	Power Terminals
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V

ATV61HD22M3X

Reference	Designation
VZ3V1211	Fan KIT
VZ3N1301	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1104	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD22M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VZ3N1206	Power Terminals
VX4A1103	Front Cover 4x7 Digits
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V

ATV61HD22N4

Reference	Designation
VZ3V1211	Fan KIT
VZ3N1305	Mains Connexion KIT
VZ3N1304	Bars KIT
VZ3N1302	Wires KIT
VY1ADV1101	Screws KIT
VY1ADC1101	Lot of 2 capacitors (2700µF / 400V)
VY1A1206	Metal Parts KIT
VY1A1101	Current Sensor KIT
VX5A1HD22N4	Power Block Sub Assembly
VX5A1101	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1107	Sub Assembly Filter Boards
VX4A1104	Terminal Board
VZ3N1206	Power Terminals
VX4A1103	Front Cover 4x7 Digits

Metal Parts KIT: VY1A1206

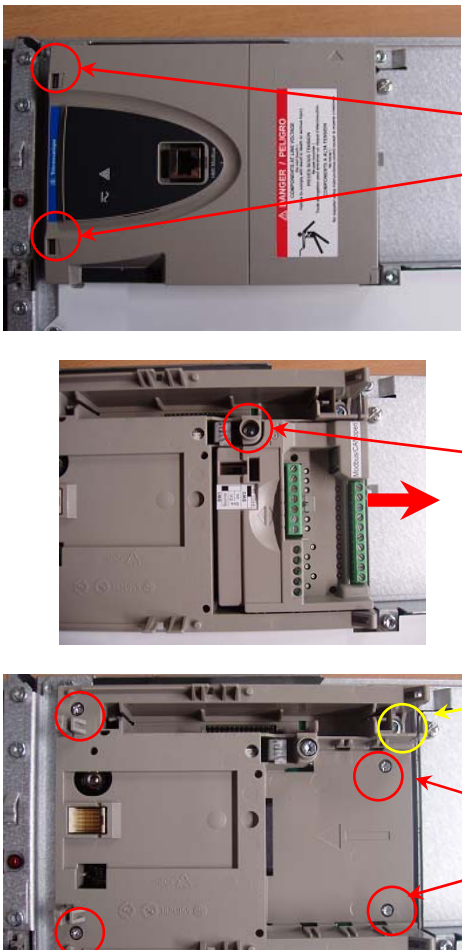


Remove 6 screws (S243 to S248).

Unscrew this screw and remove the control cap.

	Mark	Size	Torque
	S243-S248	M4x10	1,5Nm

Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

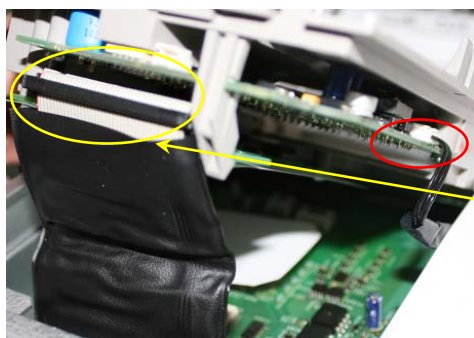
For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

Remove this screw (S153).

Remove 4 screws (S180 to S183).

	Mark	Size	Torque
	S180-S183	M3x8	0,78Nm
	S153	M4x8	1,5Nm

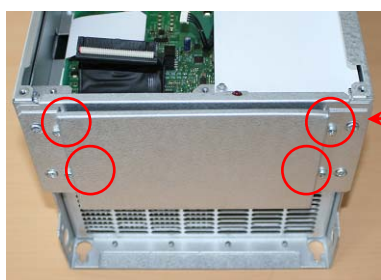
Control Bloc: VX4A61100Y



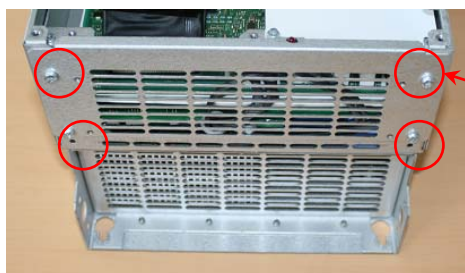
Disconnect the wire.
Application board (S103) → Power board (S103).

Disconnect the ribbon cable.

Metal Parts KIT: VY1A1206



Remove 4 screws (S29, S30, S33 and S34).



Remove 4 screws (S27, S28, S31 and S32).



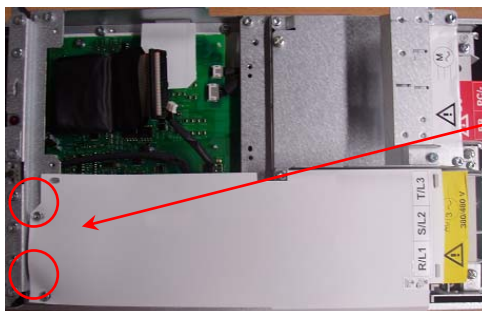
Remove 3 screws (S14, S15, and S18).



Remove 4 screws
(S11, S12, S13 and
S17).

	Mark	Size	Torque
	S29, S30 S33, S34	M4x10	2Nm
	S27, S28 S31, S32	M4x10	2Nm
	S14 S15 S18	M4x6	2Nm
	S11, S12 S13, S17	M4x6	2Nm

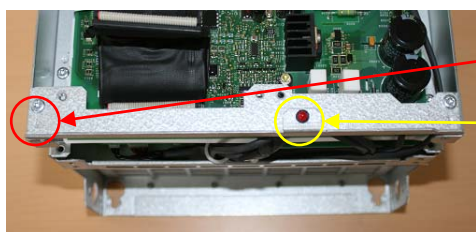
Screws KIT: VY1ADV1101



Remove 2 screws (S73 and S74).

	Mark	Size	Torque
	S73 S74	M4x10	2Nm

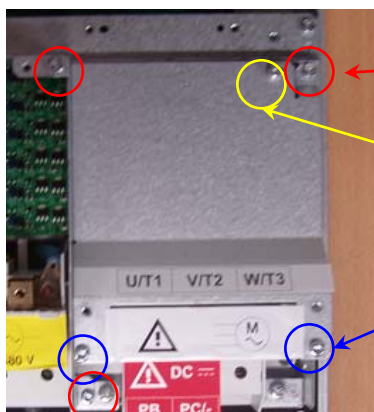
Metal Parts KIT: VY1A1206



Remove this screw (S75).

Remove this LED.

	Mark	Size	Torque
	S75	M4x10	2Nm



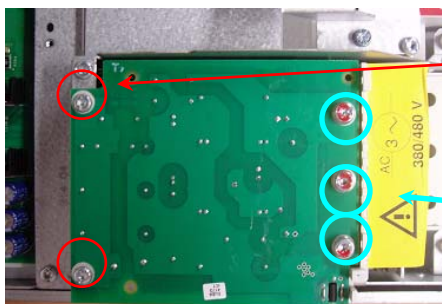
Remove 3 screws (S70 to S72).

Remove this screw

Remove 2 screws
(S7 and S8).

	Mark	Size	Torque
	S70- S72	M4x10	2Nm
	S22	M5x12	1,5Nm
	S7 S8	M4x12	2Nm

Sub Assembly Filter Boards: VX4A1107



Remove 2 screws (S68 and S69).

Remove 3 screws
(S124 and S126).

	Mark	Size	Torque
	S68 S69	M4x10	1,5Nm
	S124- S126	M5x12	2,5Nm

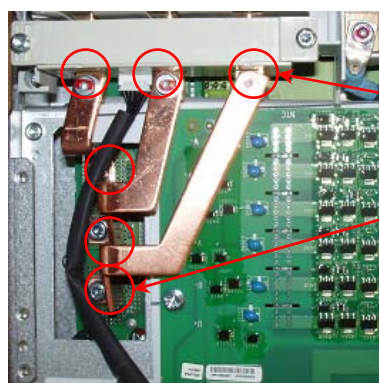
Metal Parts KIT: VY1A1206



Remove 2 screws (S66 and S67).

Mark	Size	Torque
S66 S67	M4x10	2Nm

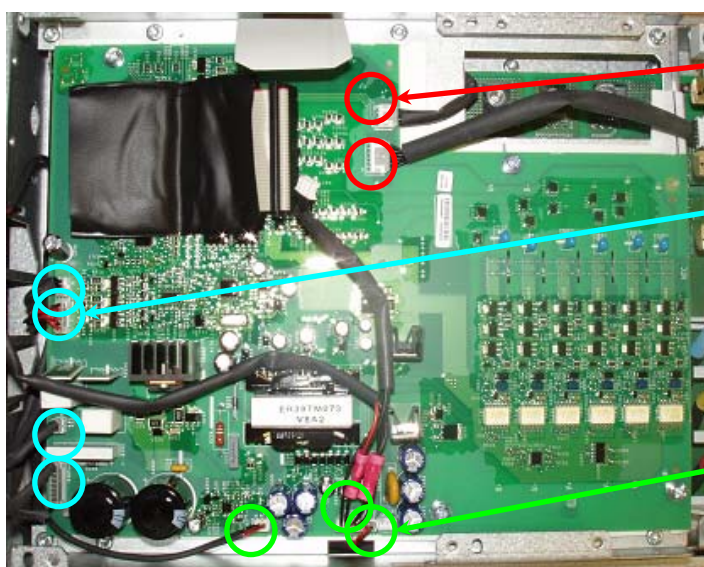
Connexion Bars KIT: VZ3N1304



Remove 6 screws (S118 to S123).

Mark	Size	Torque
S129-S134	M5x12	2,5Nm

Power Block Sub-Assembly: VX5A1HD22N4 (only 480V)



Disconnect 2 wires (from up to down).

- * Power board (S400) → Bus board (S400).
- * Power board (S500) → Current sensor.

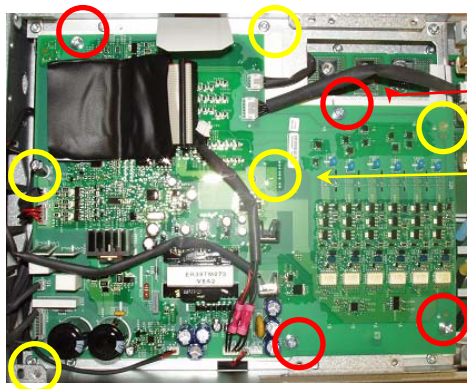
Disconnect 4 wires (from up to down).

- * Power board (S101) → Bus board (S101).
- * Power board (S104) → Bus board (S104).
- * Power board (S202) → Bus board (S202).
- * Power board (S201) → Bus board (S201).

Disconnect 3 wires (from left to right).

- * Power board (S200) → DEL.
- * Power board (S103) → Control Bloc.
- * Power board (S120) → Power fan.

Power Block Sub-Assembly: VX5A1HD22N4 (only 480V)

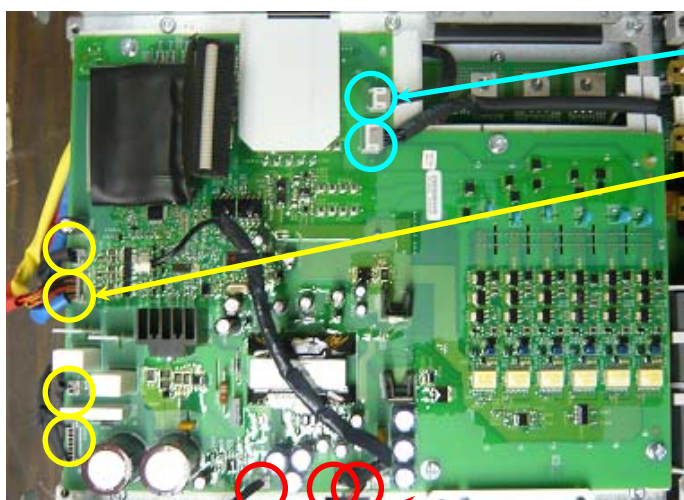


Remove 4 screws (S169 to S172).

Press 5 clips.

Mark	Size	Torque
S169-S172	M4x10	1,5Nm

Power Block Sub-Assembly: VX5A1HD22M3X (only 240V)



Disconnect 2 wires (from up to down).

- * Power board (S400) → Bus board (S400).
- * Power board (S500) → Current sensor.

Disconnect 4 wires (from up to down).

- * Power board (S101) → Bus board (S101)
- * Power board (S104) →
 1. Red girdle → THY/DIO(18):
red wire → pin 4 ; black wire → pin 5
 2. Yellow girdle → THY/DIO(19):
red wire → pin 4 ; black wire → pin 5
 3. Blue girdle → THY/DIO(20):
red wire → pin 4 ; black wire → pin 5

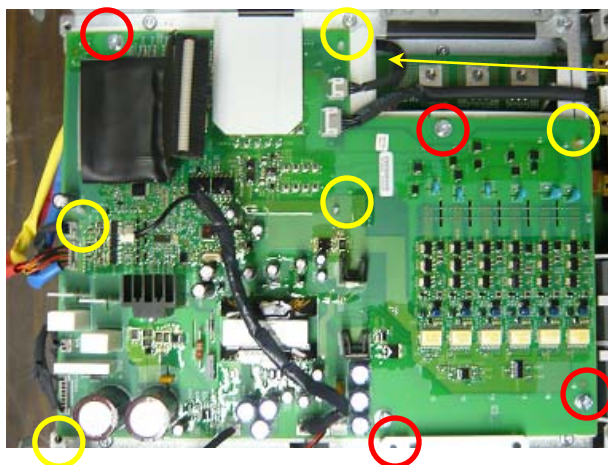
- * Power board (S202) → Bus board (S202)
- * Power board (S201) → Bus board (S201)

Disconnect 3 wires (from left to right).

- * Power board (S200) → DEL
- * Power board (S103) → Control Bloc
- * Power board (S120) → Power fan

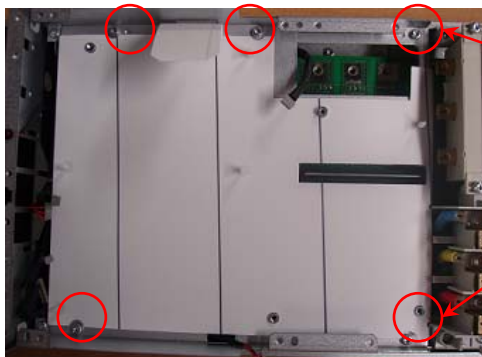
Press 5 clips.

Remove 4 screws (S169 to S172).



Mark	Size	Torque
S169-S172	M4x10	1,5Nm

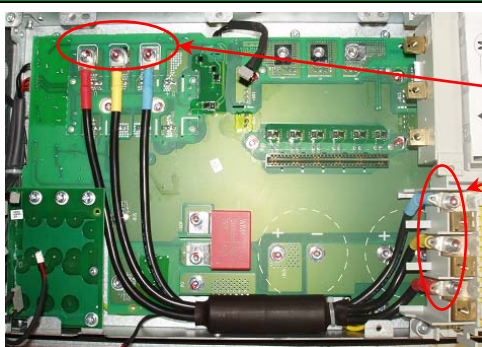
Power Block Sub-Assembly: VX5A1HD22N4



Remove 5 screws (S61 to S65).

Mark	Size	Torque
 S61-S65	M4x10	1,5Nm

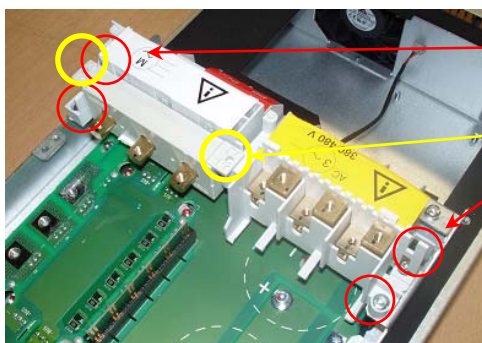
Mains Connexion KIT : VZ3N1305



Remove 6 screws (S112 to S117).

Mark	Size	Torque
 S112-S117	M5x12	1,5Nm

Current Sensor KIT : VY1A1101 (only 480V) Power Terminals : VZ3N1206 (only 480V)

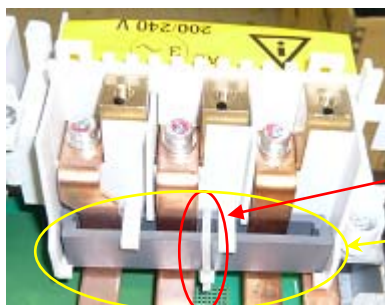


Remove 4 screws (S57 to S60).

Remove 2 screws (S5 and S6).

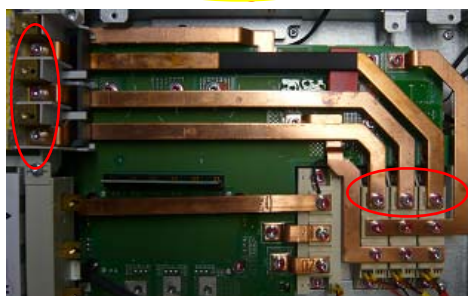
Mark	Size	Torque
 S57-S60	M4x10	1,5Nm
 S5-S6	M4x12	1,5Nm

Bars KIT: VZ3N1303 (only 240V)



Cut the colson tie.

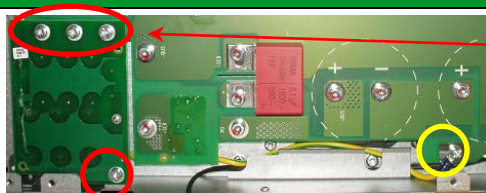
Remove the ferrite core.



Remove 6 screws (S112 to S117).

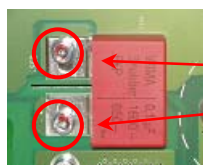
	Mark	Size	Torque
	S112-S117	M5x12	1,5Nm

DC BUS Board: VX5A1101 (only 480V)

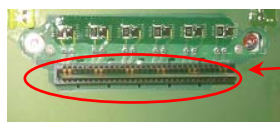


Remove 4 screws (S165 to S168).

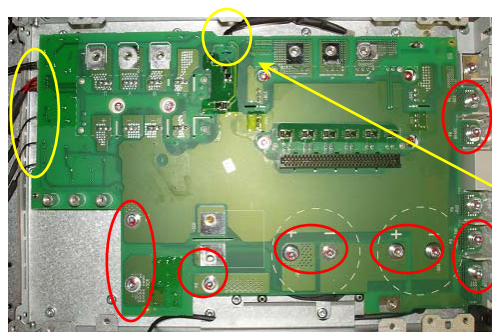
Remove this screw (S264) and remove the cable (filter board).



Remove 2 screws (S103 and S127).



Remove the interconnect PCB (no screw).

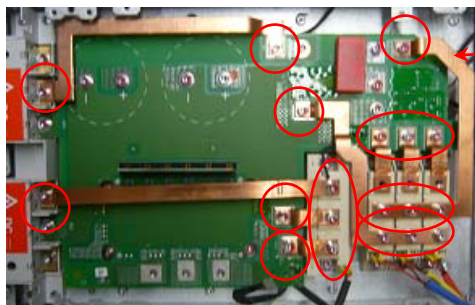


Remove 11 screws (S100 and S102 to S111).

Disconnect the wires (for the wiring, see the Power Block Sub-Assembly: **VX5A1HD22N4**

	Mark	Size	Torque
	S165-S168	M4x10	1,5Nm
	S264	M4x10	1,5Nm
	S103-S127	M5x12	2,5Nm
	S100-S111	M5x12	2,5Nm

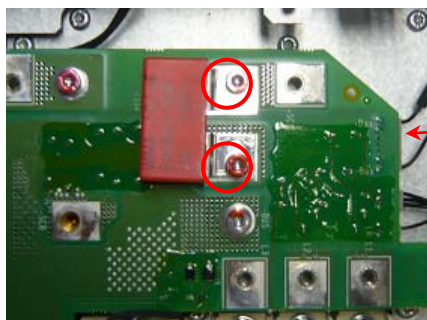
Bars KIT: VZ3N1303 (only 240V)



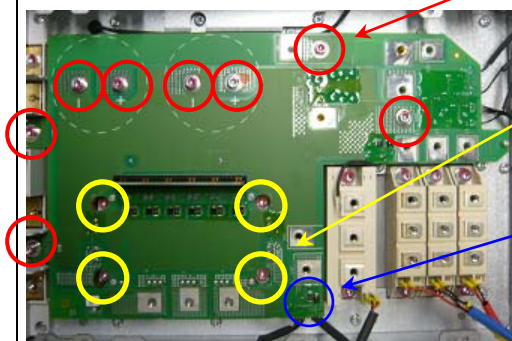
Remove 19 screws
(S110, S111 and
S127 to S143).

	Mark	Size	Torque
	S110, S111 S127- S143	M5x12	2,5Nm

DC BUS Board: VX5A1104 (only 240V)



Remove 2 screws (S100 and S101).



Remove 8 screws (S102 to S109).

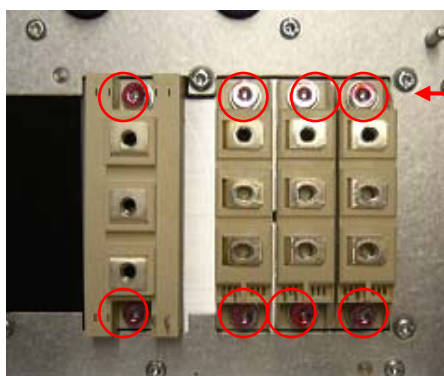
Remove 4 screws (S149 to S152).

Disconnect the wire:

*Bus board→Module IGBT 21
Red wire→pin 6
Black wire→pin 7*

	Mark	Size	Torque
	S100 S101	M 5x12	2,5Nm
	S102- S109	M 5x12	2,5Nm
	S149- S152	M 5x20	2,5Nm

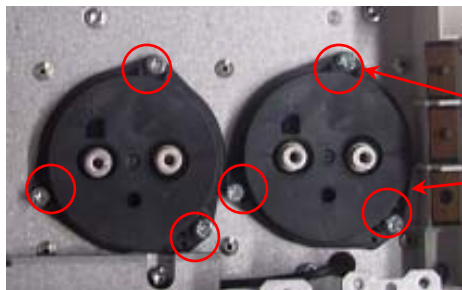
Rectifier (Thyristor / Diode): VZ3TD1092M1671 (only 240V)



Remove 6 screws (S219 to S226).

	Mark	Size	Torque
	S219- S226	M 6x20	0,78Nm

Lots of 2 capacitors: VY1ADC1101



Remove 6 screws (S159 to S164).

	Mark	Size	Torque
	S159-S164	M4x10	0,78Nm

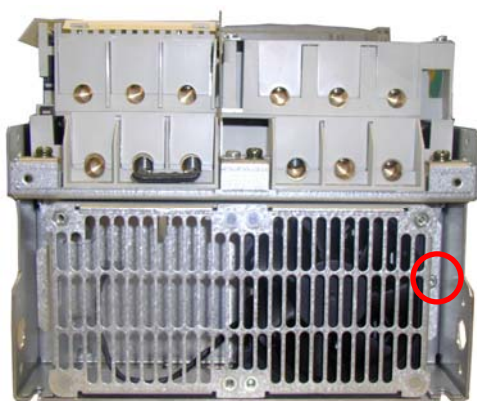
Power Terminals: VZ3N1206



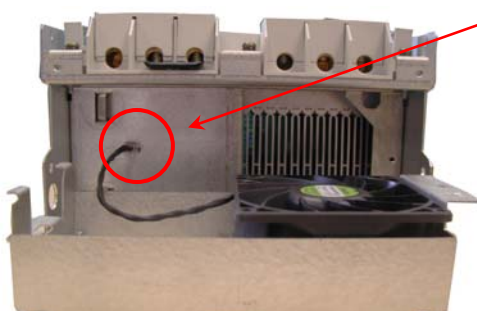
Remove 6 screws (S53 to S56).

	Mark	Size	Torque
	S53-S56	M4x10	2Nm

Fan Kit: VZ3V1211



Remove this screw (S35).



Disconnect this wire.

	Mark	Size	Torque
	S35	M4x10	1,5Nm
	S173 S174	M4x16	Nm

9.7.2 Product Assembling Drawing

Refer to following files: [Assembling_175578100A53_05.pdf](#)
[Assembling_175578500A53_06.pdf](#)

9.7.3 Product Cabling Drawing

Refer to following file: [Cabling_175578800A53_05.pdf](#)

9.8 *ATV61/71 Size 7A (size, refer to 1.2)*

9.8.1 Dismantling and reassembling

Size 7A: ATV71HD30N4, ATV71HD37N4, ATV61HD30N4, ATV61HD37N4

ATV71HD30N4	
Reference	Designation
VZ3V1206	Fan KIT
VZ3TD1057M1671	Rectifier (Thyristor / Diode) 55A / 1600V
VZ3N1307	Wires KIT
VZ3N1306	Mains Connexion KIT
VZ3N1207	Power Terminals
VZ3IM1145M1271	Module Braking IGBT (175A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1101	Screws KIT
VY1ADC1102	Lot of 2 capacitors (4800µF / 400V)
VY1A1207	Metal Parts KIT
VY1A1102	Current Sensor KIT
VX5A1102	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5IM2145M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A1108	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

ATV71HD37N4	
Reference	Designation
VZ3V1206	Fan KIT
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V
VZ3N1307	Wires KIT
VZ3N1306	Mains Connexion KIT
VZ3N1207	Power Terminals
VZ3IM2195M1271	Modules IGBT Inverter 220A / 1200V
VZ3IM1145M1271	Module Braking IGBT (175A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1101	Screws KIT
VY1ADC1103	Lot of 2 capacitors (5700UF / 400V)
VY1A1207	Metal Parts KIT
VY1A1102	Current Sensor KIT
VX5A1HD37N4	Power Block Sub Assembly
VX5A1102	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1108	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

ATV61HD30N4	
Reference	Designation
VZ3V1206	Fan KIT
VZ3TD1057M1671	Rectifier (Thyristor / Diode) 55A / 1600V
VZ3N1307	Wires KIT
VZ3N1306	Mains Connexion KIT
VZ3N1207	Power Terminals
VZ3IM1145M1271	Module Braking IGBT (175A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1101	Screws KIT
VY1ADC1102	Lot of 2 capacitors (4800µF / 400V)
VY1A1207	Metal Parts KIT
VY1A1102	Current Sensor KIT
VX5A1102	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5IM2145M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A1108	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

ATV61HD37N4	
Reference	Designation
VZ3V1206	Fan KIT
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V
VZ3N1307	Wires KIT
VZ3N1306	Mains Connexion KIT
VZ3N1207	Power Terminals
VZ3IM2195M1271	Modules IGBT Inverter 220A / 1200V
VZ3IM1145M1271	Module Braking IGBT (175A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1101	Screws KIT
VY1ADC1103	Lot of 2 capacitors (5700UF / 400V)
VY1A1207	Metal Parts KIT
VY1A1102	Current Sensor KIT
VX5A1HD37N4	Power Block Sub Assembly
VX5A1102	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1108	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

Metal Parts KIT: VY1A1207



Remove 6 screws (S243 to S248).

Unscrew this screw and remove the control cap.

	Mark	Size	Torque
	S243-S248	M4x10	Nm

Control Block: VX4A61100Y



For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

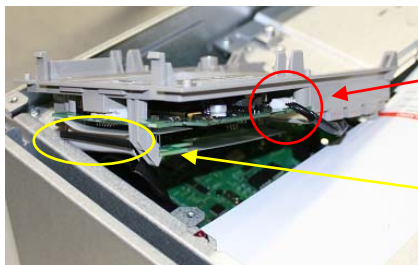
Press the two clips and pulls forward to take out the display board.

Remove 4 screws (S180 to S183).

Remove this screw (S153).

	Mark	Size	Torque
	S180-S183	M3x8	0,78Nm
	S153	M4x8	1,5Nm

Control Block: VX4A61100Y



Disconnect the wire.

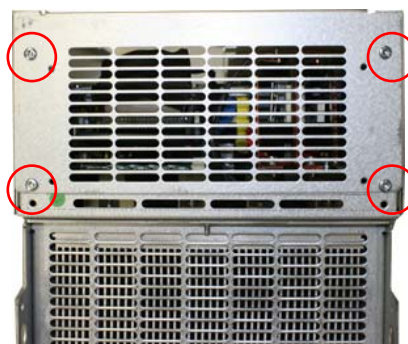
Application board (S103) → Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

Metal Parts KIT: VY1A1207



Remove 4 screws (S29, S30, S33 and S34).



Remove 4 screws (S27, S28, S31 and S32).



Remove 3 screws
(S14 to S16).

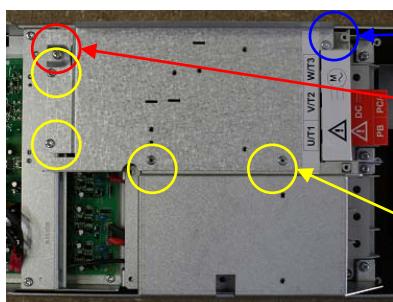
	Mark	Size	Torque
	S29,S30 S33,S34	M 4x6	2Nm
	S27,S28 S31,S32	M 4x10	2Nm
	S11- S16	M 4x6	2Nm

Metal Parts KIT: VY1A1207



Remove 3 screws
(S11 to S13).

	Mark	Size	Torque
	S29,S30 S33,S34	M 4x6	2Nm
	S27,S28 S31,S32	M 4x10	2Nm
	S11- S16	M 4x6	2Nm



Remove this screw (S7).

Remove this screw

Remove 4 screws
(S74 to S77).

	Mark	Size	Torque
	S7	M4x12	1,5Nm
	S22	M5x12	1,5Nm
	S74- S77	M4x10	1,5Nm

Screws KIT: VY1ADV1101



Remove 2 screws (S78 and S79).

	Mark	Size	Torque
	S78- S79	M4x10	1,5Nm

Mains Connexion KIT: VZ3N1306

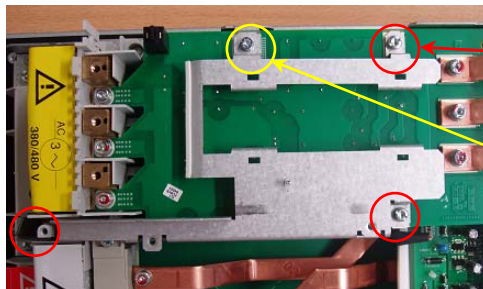


Remove 6 screws (S141 to S146).

Cut 2 colsons tie.

	Mark	Size	Torque
	S141- S146	M5x12	2,5Nm

Metal Parts KIT: VY1A1207

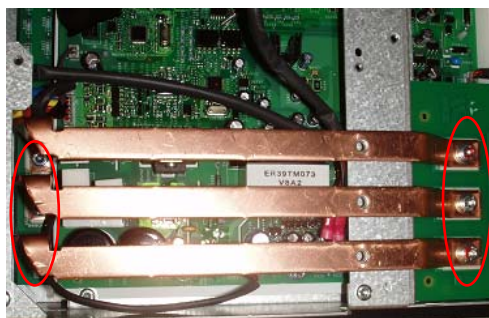


Remove 3 screws (S71 to S73).

Remove this screw (S166).

	Mark	Size	Torque
	S71-S73	M4x10	1,5Nm
	S166	M4x10	1,5Nm

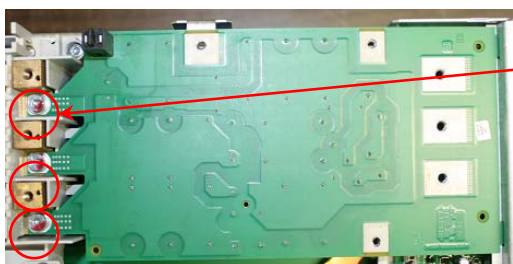
Mains Connexion KIT: VZ3N1306



Remove 6 screws (S132 to S137).

	Mark	Size	Torque
	S132-S137	M5x12	2,5Nm

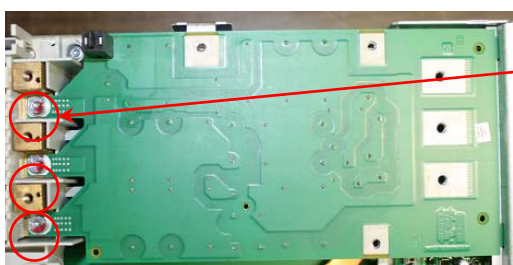
Filter Board: VX4A1108



Remove 3 screws (S138 to S140).

	Mark	Size	Torque
	S138-S140	M5x12	2,5Nm

Filter Board: VX4A1108



Remove 3 screws (S138 to S140).

	Mark	Size	Torque
	S138-S140	M5x12	2,5Nm

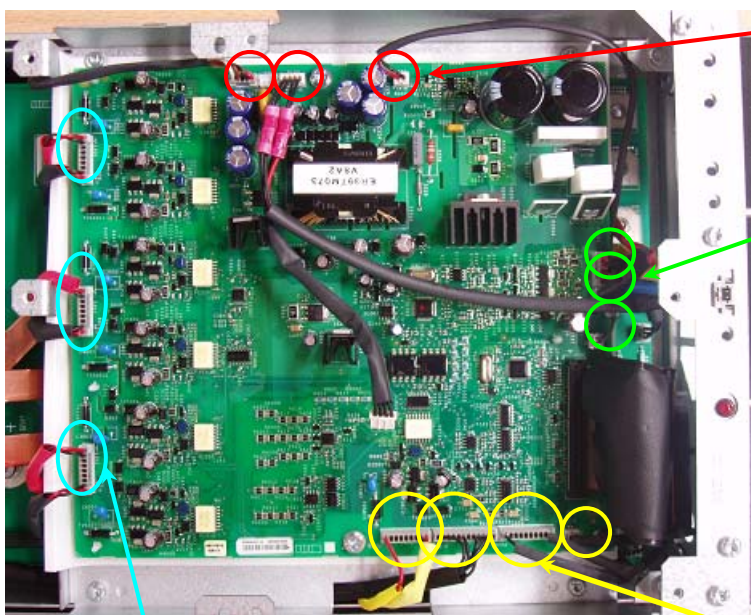
Metal Parts KIT: VY1A1207



Remove 2 screws (S69 and S70).

Mark	Size	Torque
S69-S70	M4x10	2Nm

Power Block Sub-Assembly: VX5A1HD37N4



Disconnect 3 wires (from left to right)

- * Power board (S203) → Power 2 fans
- * Power board (S103) → Power internal fan and control bloc (S103)
- * Power board (S205) → DEL

Disconnect 3 wires (from down to up)

- * Power board (S700) → heatsink, thermal sensor
 - * Power board (S101) → Bus board (S101)
 - * Power board (S102) → Modules THY/DIO:
- Blue girdle → IGBT (15):*
Black wire → pin 5; red wire → pin 4
Yellow girdle → IGBT (14):
Black wire → pin 5; red wire → pin 4
Red girdle → IGBT (13):
Black wire → pin 5; red wire → pin 4

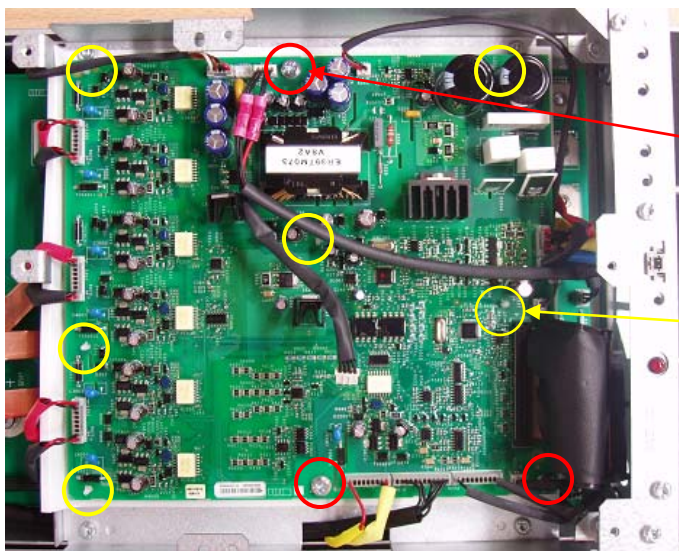
Disconnect 3 wires (from up to down)

- * Power board (S300) → Modules IGBT (12):
Red cable: red wire → pin 4; blue wire → pin 5
Black cable: red wire → pin 6; blue wire → pin 7
- * Power board (S301) → Modules IGBT (11):
Red cable: red wire → pin 4; blue wire → pin 5
Black cable: red wire → pin 6; blue wire → pin 7
- * Power board (S302) → Modules IGBT (10):
Red cable: red wire → pin 4; blue wire → pin 5
Black cable: red wire → pin 6; blue wire → pin 7

Disconnect 4 wires (from left to right)

- * Power board (S400) → Bus board (S100) and Module IGBT (9):
Red wire: pin 6 of IGBT;
Black wire: pin 7 of IGBT
- * Power board (S500) → Current sensor
- * Power board (S200) → Bus board (S200)
- * Power board (S202) → Bus board (S202)

Power Block Sub-Assembly: VX5A1HD37N4



Remove 3 screws
(S163 to S165).

Press 6 clips.

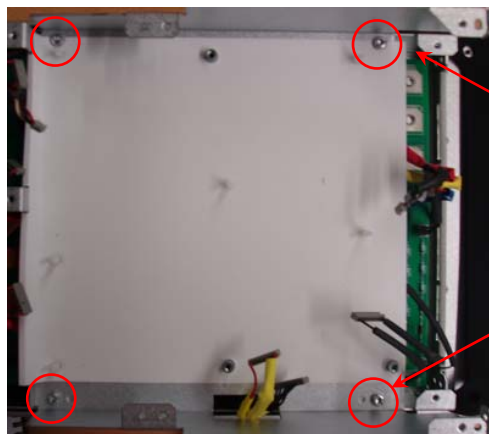
Mark	Size	Torque
S163- S165	M4x10	1,5Nm

Metal Parts KIT: VY1A1207



Remove 2 screws (S67 to S68).

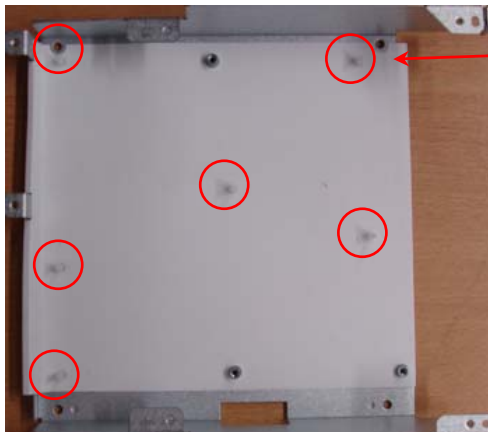
Disconnect the DEL.



Remove 4 screws (S63 to S66).

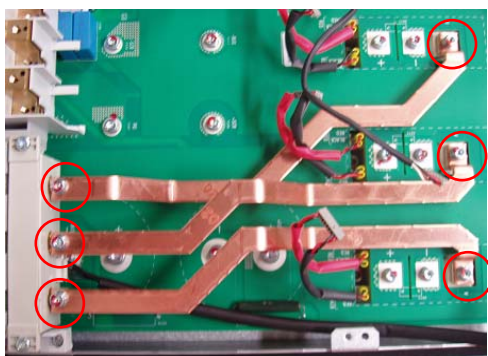
Mark	Size	Torque
S67- S68	M4x10	2Nm
S63- S66	M4x10	2Nm

Metal Parts KIT: VY1A1207



Remove 6 clips and remove the insulate sheet.

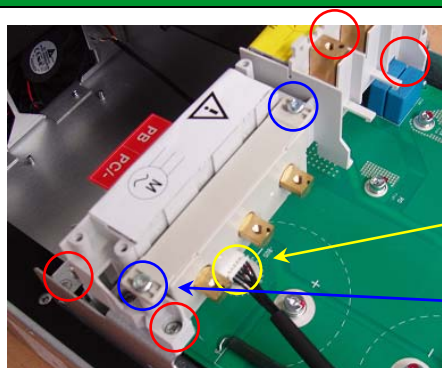
	Mark	Size	Torque
	S67-S68	M4x10	2Nm
	S63-S66	M4x10	2Nm



Remove 6 screws (S126 to S131).

	Mark	Size	Torque
	S126-S131	M5x12	2,5Nm

Current Sensor KIT: VY1A1102



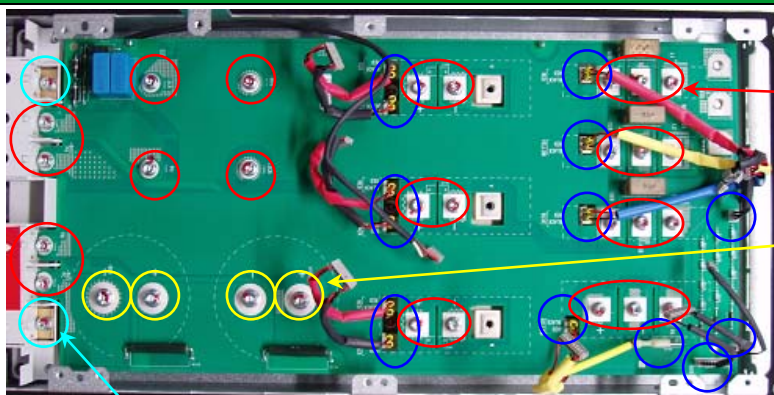
Remove 4 screws (S59 to S62).

Disconnect the wire.

Remove 2 screws (S5 and S6).

	Mark	Size	Torque
	S59-S62	M4x10	2Nm
	S5-S6	4x12	2Nm

DC BUS Board: VX5A1102



Remove 2 screws (S147 and S148).

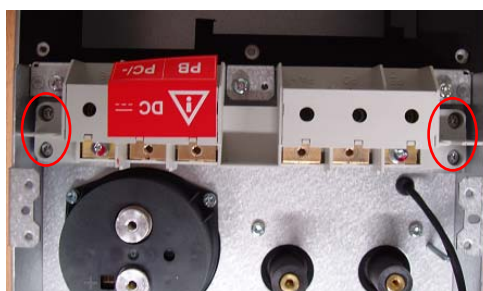
Disconnect 14 wires. For the wiring, see the Power Block Sub Assembly: **VX5A1HD37N4**

Remove 26 screws (S100 to S125).

Remove 4 screws (S184 to S187).

	Mark	Size	Torque
	S100-S125	M5x12	2,5Nm
	S184-S187	M6x12	3,5Nm
	S147-S148	M5x25	2,5Nm

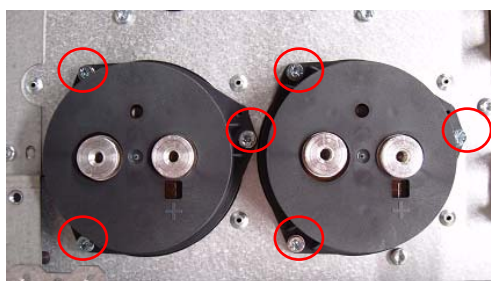
Power Terminals: VZ3N1207



Remove 4 screws (S55 to S58).

	Mark	Size	Torque
	S55-S58	M4x10	2Nm

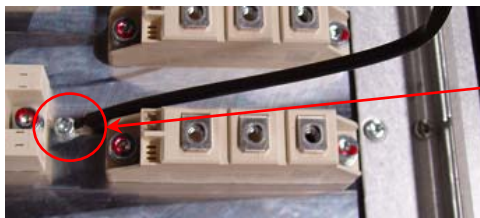
Lots of 2 capacitors: VY1ADC1103



Remove 6 screws (S157 to S162).

	Mark	Size	Torque
	S157-S162	M4x10	0,78Nm

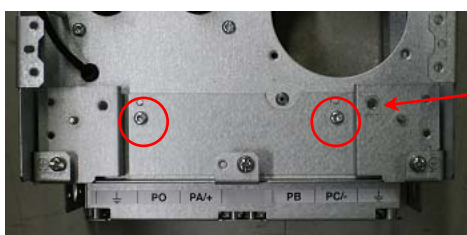
Thermal Sensor: VZ3G1101



Remove this screw (S54).

	Mark	Size	Torque
	S54	M4x10	2Nm

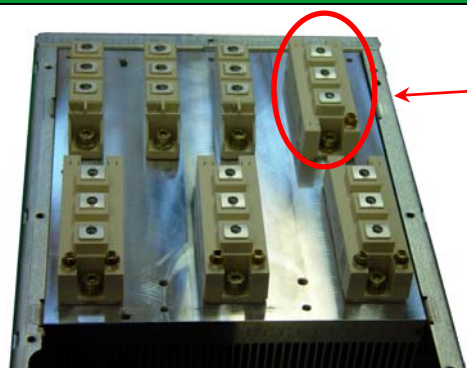
Metal Parts KIT: VY1A1207



Remove 2 screws (S52 and S53).

	Mark	Size	Torque
	S52 S53	M4x10	2Nm

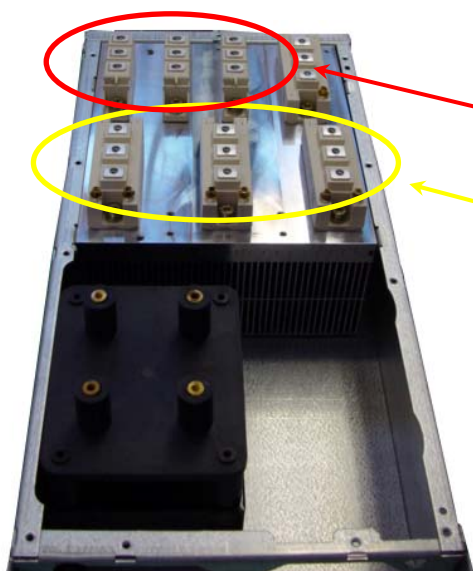
Module Braking IGBT: VZ3IM1145M1271



Remove 2 screws (S219 to S232).

	Mark	Size	Torque
	S219- S232	M6x20	4,4Nm

Modules IGBT Inverter: VZ3IM2195M1271
Rectifier (Thyristor / Diode): VZ3TD1072M1671



Remove 12 screws (S219 to S232).

Thyristors.

IGBT Modules.

Warning:

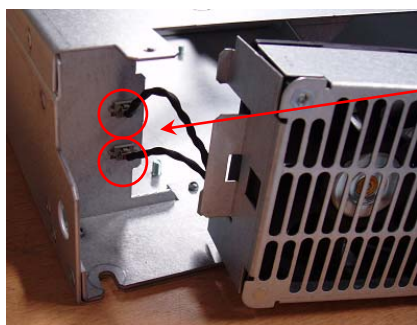
Do not forget to put the grease between the modules and the heatsink

Mark	Size	Torque
S219-S232	M6x20	4,4Nm

Fan KIT: VZ3V1206



Remove this screw (S35).



Disconnect 2 wires.

Mark	Size	Torque
S35	M4x10	1.5 Nm

9.8.2 Product Assembling Drawing

Refer to following files: [Assembling_175578200a53_05.pdf](#)
[Assembling_175578500A53_06.pdf](#)

9.8.3 Product Cabling Drawing

Refer to following file: [Cabling_175579200A53_IED04.pdf](#)

9.9 *ATV61/71 Size 7B (size, refer to 1.2)*

9.9.1 Dismantling and reassembling

**Size 7B: ATV71HD30M3X, ATV71HD37M3X, ATV71HD45M3X,
ATV61HD30M3X, ATV61HD37M3X, ATV61HD45M3X**

ATV71HD30M3X	
Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V
VZ3N1310	Wires KIT
VZ3N1309	Bus Bar KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2201M1271	Module IGBT Inverter 200A / 600V
VZ3IM1195M0671	Module Braking IGBT (250A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1104	Lot of 2 capacitors (4800µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD30M3X	Power Block Sub Assembly
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

ATV71HD37M3X	
Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1132M1671	Rectifier (Thyristor-Diode) 130A 1600V
VZ3N1310	Wires KIT
VZ3N1309	Bus Bar KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2301M1271	Module IGBT Inverter 300A / 600V
VZ3IM1300M0671	Module Braking IGBT (400A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1105	Lot of 2 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD37M3X	Power Block Sub Assembly
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1108	Filter Board
VX4A1104	Terminal Board

ATV71HD45M3X	
Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1162M1671	Rectifier (Thyristor-Diode) 160A 1600V
VZ3N1310	Wires KIT
VZ3N1309	Bus Bar KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2401M1271	Module IGBT Inverter 400A / 600V
VZ3IM1300M0671	Module Braking IGBT (400A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1106	Lot of 2 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD45M3X	Power Block Sub Assembly
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits

ATV61HD30M3X	
Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V
VZ3N1310	Wires KIT
VZ3N1309	Bus Bar KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2201M1271	Module IGBT Inverter 200A / 600V
VZ3IM1195M0671	Module Braking IGBT (250A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1104	Lot of 2 capacitors (4800µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD30M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits

ATV61HD37M3X

Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1132M1671	Rectifier (Thyristor-Diode) 130A 1600V
VZ3N1310	Wires KIT
VZ3N1309	Bus Bar KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2301M1271	Module IGBT Inverter 300A / 600V
VZ3IM1300M0671	Module Braking IGBT (400A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1105	Lot of 2 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD37M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits

ATV61HD45M3X

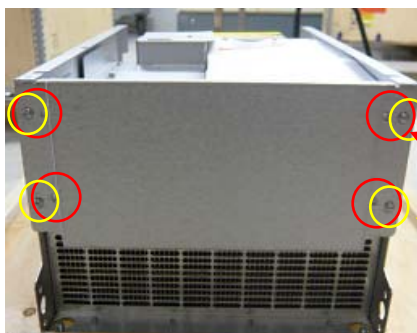
Reference	Designation
VZ3V1207	Fan KIT
VZ3TD1162M1671	Rectifier (Thyristor-Diode) 160A 1600V
VZ3N1310	Wires KIT
VZ3N1308	Power Interconnexion KIT
VZ3IM2401M1271	Module IGBT Inverter 400A / 600V
VZ3IM1300M0671	Module Braking IGBT (400A / 600V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1106	Lot of 2 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1103	Current Sensor KIT
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD45M3X	Power Block Sub Assembly
VX4A1104	Terminal Board
VY1A1208	Metal Parts KIT
VX4A1103	Front Cover 4x7 Digits
VY1A1208	Metal Parts KIT

Metal Parts KIT : VY1A1208



Unscrew this screw

Remove 6 screws (S243 to S248)



Remove 4 screws (S29, S30, S33 and S34).

Remove 4 screws (S27, S28, S31 and S32).



Remove 3 screws (S14 to S16).



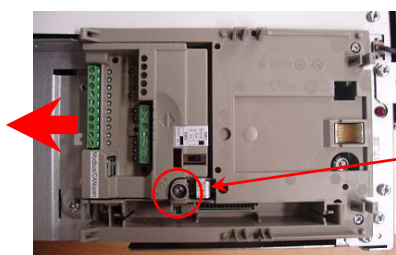
Remove 4 screws
(S11 to S13).

	Mark	Size	Torque
	S243-S248	M4x10	2Nm
	S29 S30 S33 S34	M4X6	1,5Nm
	S27 S28 S31 S32	M4x10	2Nm
	S14-S16	M4X6	1,5Nm
	S11-S13	M4X6	1,5Nm

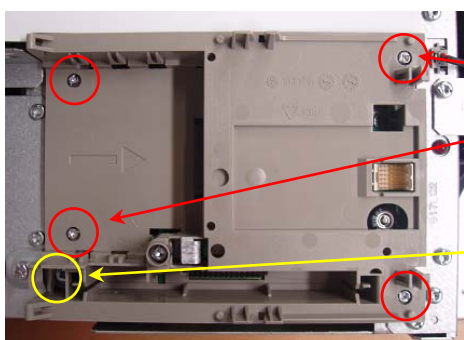
Control Bloc : VX4A61100Y



Press the two clips and pulls forward to take out the display board.

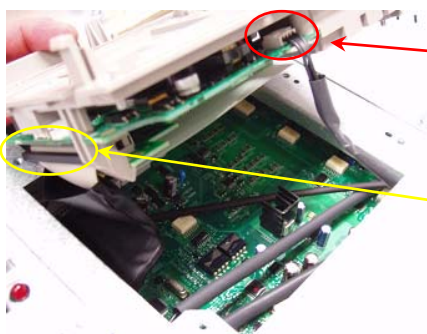


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S180 to S183).

Remove this screw (S153).

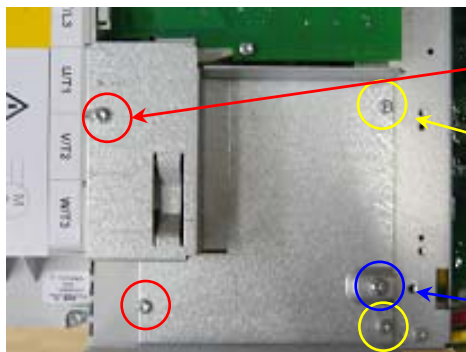


Disconnect the wire
Application board (S103)→E101→Power board (S103)

Disconnect the ribbon cable

	Mark	Size	Torque
	S180-S183	M3x8	0,78Nm
	S153	M4x8	1,5Nm

Metal Parts KIT : VY1A1208



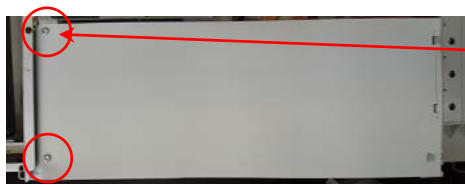
Remove 2 screws (S7 and S8).

Remove this screw
(S64 and S65).

Remove this screw (S19).

	Mark	Size	Torque
	S7 S8	4x12	1,5Nm
	S64 S65	M4x10	1,5Nm
	S19	M5x12	1,5Nm

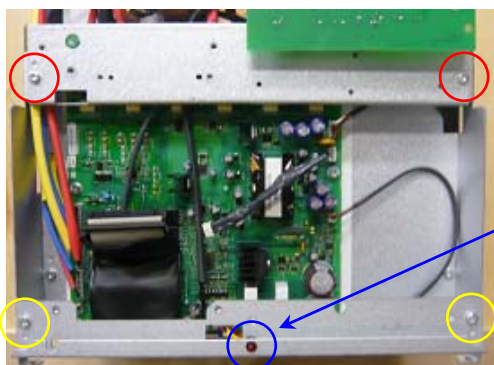
Screws KIT : VY1ADV1104



Remove 2 screws (S66 and S67).

	Mark	Size	Torque
	S66 S67	M4x10	1,5Nm

Metal Parts KIT : VY1A1208



Remove 2 screws (S61 and S63).

Disconnect the wire.

Remove 2 screws (S60 to S62).

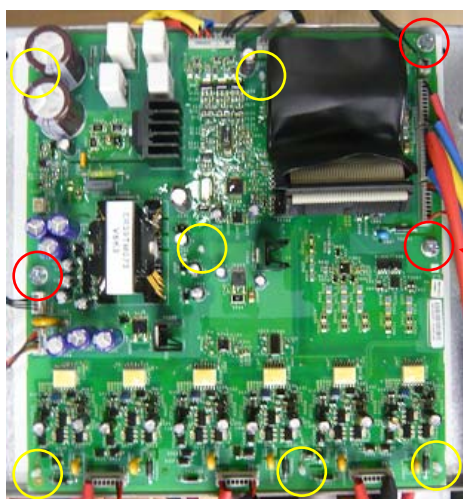
	Mark	Size	Torque
	S60 S62	M4x10	1,5Nm
	S61 S63	M4x10	1,5Nm

SUB ASSEMBY POWER : VX5A1HD30M3X



Disconnect 3 wires (from up to down)

- Power board (S205)→E102→DEL
- Power board (S103)→E101
→Power internal fan and control bloc (S103);
- Power board (S203)→E105→Power 2 fans



Disconnect 3 wires (from right to left)

- Power board (S700)→E118→heatsink, thermal sensor
- Power board (S101)→E112→Filter board (S101)
- Power board (S102)→E107→Modules THY/DIO :
 1. Blue cable→ THY/DIO (10) :
red wire→ pin 4 ; black wire→pin 5
 2. Yellow cable→ THY/DIO (9) :
red wire→ pin 4 ; black wire→pin 5
 3. Red cable→ THY/DIO (8) :
red wire→ pin 4 ; black wire→pin 5

Disconnect 4 wires (from up to down)

- Power board (S202)→E111→Filter board (S202)
- Power board (S200)→E109→Module IGBT (11) :
Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT
- Power board (S500)→E110→3 current sensors :
Blue wire : W (sensor 48) ; red : U (46) ; Yellow : V (47)
- Power board (S400)→E108→ Module IGBT (11)
Yellow cable : pin 1 of IGBT ;
Red wire : pin 6 of IGBT, Black wire : pin 7 of IGBT

Disconnect 3 wires (from left to right)

- Power board (S300)→E115→ Modules IGBT (14) :
Red cable : red wire→ pin G1 ; black wire→pin E1
Black cable : red wire→ pin G2 ; black wire→pin E2
- Power board (S301)→E116→ Modules IGBT (13) :
Red cable : red wire→ pin G1 ; black wire→pin E1
Black cable : red wire→ pin G2 ; black wire→pin E2
- Power board (S302)→E117→ Modules IGBT (12) :
Red cable : red wire→ pin G1 ; black wire→pin E1
Black cable : red wire→ pin G2 ; black wire→pin E2

Remove 3 screws (S166 to S168)

Press 6 clips

Mark	Size	Torque
S166- S168	M4x10	1,5Nm

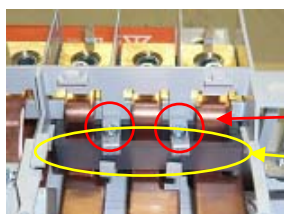
SUB ASSEMBY POWER : VX5A1HD30M3X



Remove 4 screws (S56 to S59).

	Mark	Size	Torque
	S104-S106	M4x10	2Nm

Interconnection power KIT: VZ3N1308



Remove 2 screws (S9 and S10).

Remove the ferrite type.

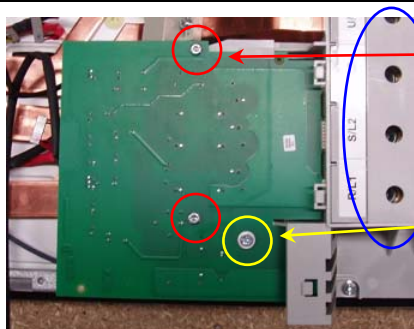


Remove 6 screws (S104 to S106).

Unscrew 3 screws.

	Mark	Size	Torque
	S9 S10	4x12	1,5Nm
	S104-S106	M5x12	2,5Nm

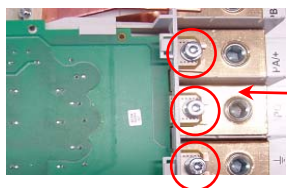
Filter Board : VX4A1109



Remove 3 screws (S5-S6).

Remove the mask of the power terminal.

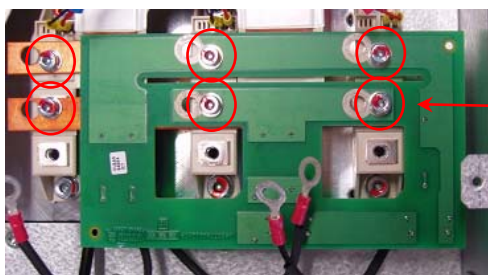
Remove this screw (S165).



Unscrew 3 screws.

	Mark	Size	Torque
	S5 S6	4x12	1Nm
	S165	M4x10	1,5Nm

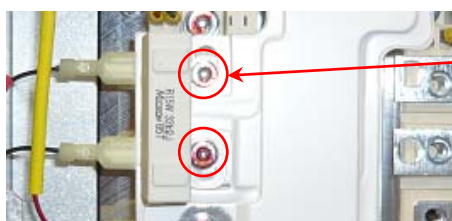
BUS Bar KIT: VZ3N1309



Remove 6 screws (S98 to S103)

Mark	Size	Torque
S98-S103	M5x12	2,5Nm

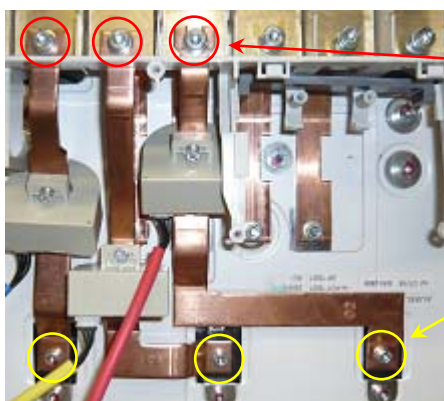
Lot of 2 capacitors : VY1ADC1104



Remove 2 screws (S201 and S204)

Mark	Size	Torque
S201 S204	M5x12	2,5Nm

Interconnection power KIT: VZ3N1308



Unscrew 3 screws.

From left to right : blue cable, yellow cable and red cable.

Remove 3 screws (S199, S202 and S207).

The U, V and W bars can be removed.

Mark	Size	Torque
S199 S202 S207	M5x12	2,5Nm

Interconnection power KIT: VZ3N1308



Remove 4 nuts (S90 and S93).

Unscrew 4 screws.

Mark	Size	Torque
S90-S93	M5X8	2,5Nm

Interconnection power KIT: VZ3N1308

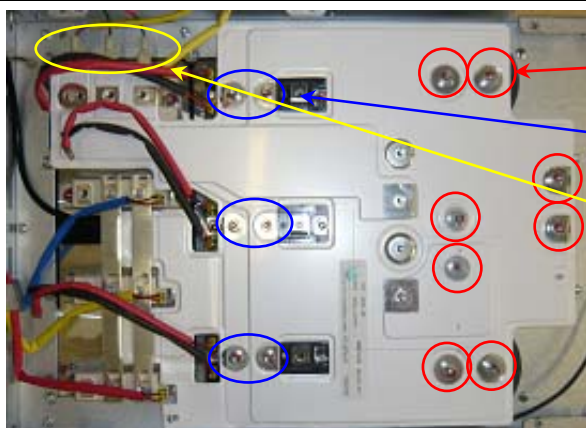


Remove 6 screws (S147 to S152).

Remove 2 screws (S25 to S26).

	Mark	Size	Torque
	S147-S152	M5x25	2,5Nm
	S25 S26	M8x20	1,5Nm

BUS Bar KIT : VZ3N1309



Remove 8 screws (S186 to S189, S190, S191, S194 and S195).

Remove 6 screws (S199, S200, S203, S204, S208 and S209)

Disconnect 2 wires

- Power board (S200)→E109
→Module IGBT (11) :

Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT

- Power board (S400)→E108
→ Module IGBT (11)

Yellow wire : pin 1 of IGBT ; Red wire : pin 6 of IGBT

Black wire : pin 7 of IGBT

Disconnect 3 wires

Power board (S102)→E107→Modules IGBT :

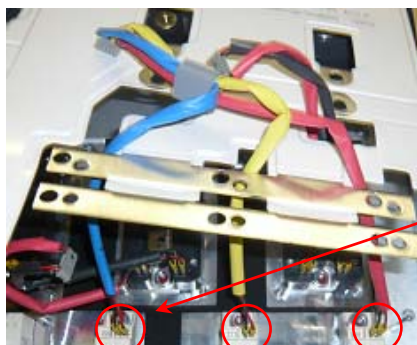
Blue cable→ IGBT (10) : red wire→ pin 5

blue wire→pin 4

Yellow cable→ IGBT (9) : red wire→ pin 5

blue wire→pin 4

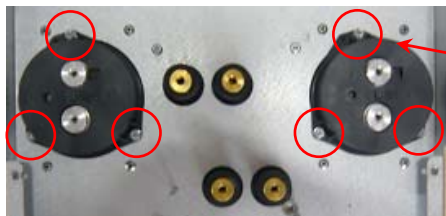
Red cable→ IGBT (8) : red wire→ pin 5 ; blue wire→pin 4



The bus bar can be removed.

	Mark	Size	Torque
	S186-S189 S190, S191 S194, S195	M6x12	2,5Nm
	S199, S200 S203, S204 S208, S209	M5x12	2,5Nm

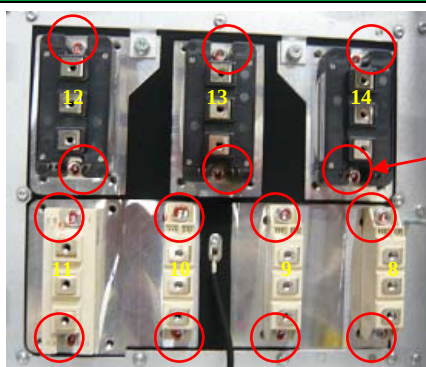
Lots of 2 capacitors : VY1ADC1104



Remove 6 screws (S159 to S164).

Mark	Size	Torque
S159- S164	M4x10	0,78Nm

Rectifier (Thyristor / Diode) : VZ3TD1092M1671 Module Braking IGBT : VZ3IM1195M0671 Module IGBT Inverter : VZ3IM2201M1271

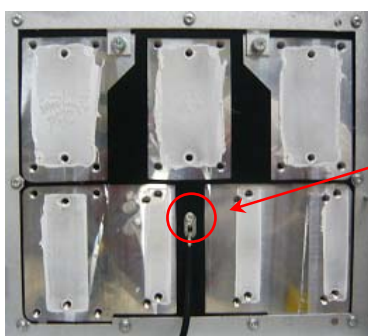


Remove 14 screws (S219 and S233).

The 7 modules can be removed
(**Warning:** do not forget
To put grease between the
modules and heatsink).

Mark	Size	Torque
S219- S233	M6x20	4,4Nm

Thermal Sensor : VZ3G1101



Remove this screw (S55).

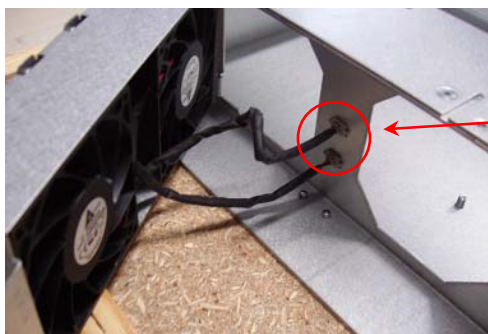
The heatsink can be removed.

Mark	Size	Torque
S55	M4x10	2Nm

Fan KIT : VZ3V1207



Remove this screw (S35).



Disconnect 2 wires.

	Mark	Size	Torque
	S35	M4x10	2Nm

9.9.2 Product Assembling Drawing

Refer to following files: [Assembling_175578400a53_06.pdf](#)
[Assembling_175578500A53_06.pdf](#)

9.9.3 Product Cabling Drawing

Refer to following file: [Cabling_175578900A53_04.pdf](#)

9.10 ATV61/71 Size 8 (size, refer to 1.2)

9.10.1 Dismantling and reassembling

Size 8: **ATV71HD45N4, ATV71HD55N4, ATV71HD75N4,**
ATV61HD45N4, ATV61HD55N4, ATV61HD75N4

ATV71HD45N4	
Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1107	Lot of 4 capacitors (3900µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2200M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A71100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VZ3N1311	Power Interconnexion KIT
VX4A1103	Front Cover 4x7 Digits

ATV71HD55N4	
Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1108	Lot of 4 capacitors (4800µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2300M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A71100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VZ3N1311	Power Interconnexion KIT
VX4A1103	Front Cover 4x7 Digits

ATV71HD75N4	
Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1132M1671	Rectifier (Thyristor-Diode) 130A 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1109	Lot of 4 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2400M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A71100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VZ3N1311	Power Interconnexion KIT
VX4A1103	Front Cover 4x7 Digits

ATV61HD45N4	
Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1072M1671	Rectifier (Thyristor / Diode) 80A / 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1107	Lot of 4 capacitors (3900µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2200M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A61100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VZ3N1311	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits

ATV61HD55N4

Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1092M1671	Rectifier (Thyristor / Diode) 92A / 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1108	Lot of 4 capacitors (4800µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2300M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A61100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3N1311	Interconnection KIT

ATV61HD75N4

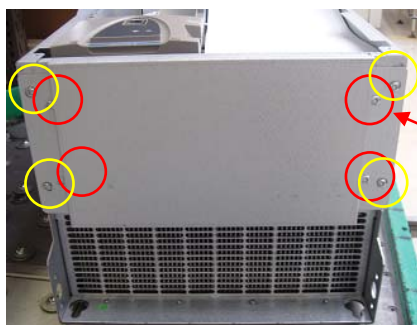
Reference	Designation
VZ3V1208	Fan KIT
VZ3TD1132M1671	Rectifier (Thyristor-Diode) 130A 1600V
VZ3N1313	Wires KIT
VZ3N1312	BUS Bar KIT
VZ3IM1300M1271	Module Braking IGBT (345A / 1200V)
VZ3G1101	Thermal Sensor
VY1ADV1104	Screws KIT
VY1ADC1109	Lot of 4 capacitors (5700µF / 400V + Balancing Resistor)
VY1A1209	Metal Parts KIT
VY1A1104	Current Sensor KIT
VX5IM2400M1271	Sub Assembly Power KIT (3 x IGBT + Power Board)
VX4A61100Y	Control Bloc P < 90Kw
VX4A1110	Filter Board
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3N1311	Interconnection KIT

Metal Parts KIT: VY1A1209



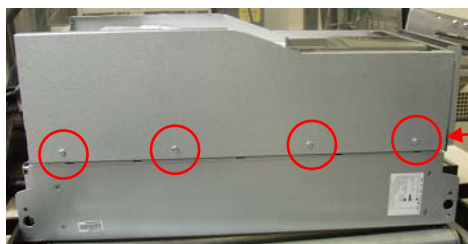
Unscrew this screw

Remove 6 screws (S243 to S248)

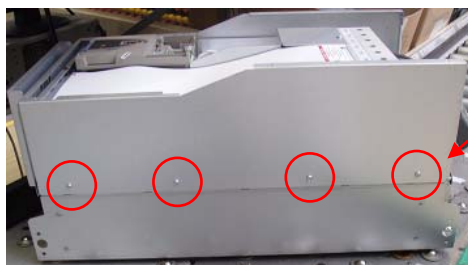


Remove 4 screws (S29, S30, S33 and S34).

Remove 4 screws (S27, S28, S31 and S32).



Remove 4 screws (S14 to S16 and S18).



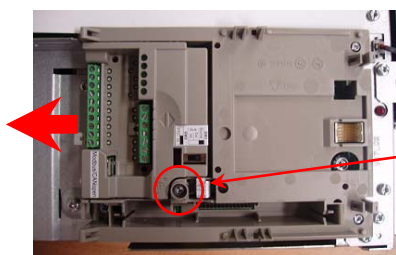
Remove 4 screws (S11 to S13 and S17).

	Mark	Size	Torque
	S243-S248	M4x10	2Nm
	S29 S30 S33 S34	M4X6	1,5Nm
	S27 S28 S31 S32	M4x10	2Nm
	S14-S16 S18	M4X6	1,5Nm
	S11-S13 S17	M4X6	

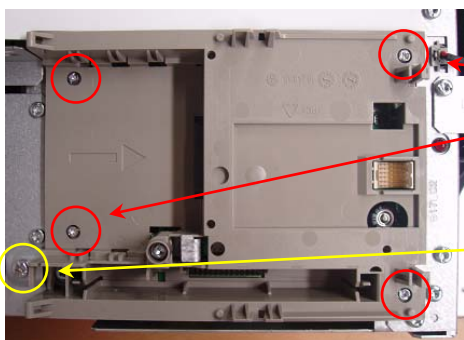
Control Bloc : VX4A61100Y



Press the two clips and pulls forward to take out the display board.

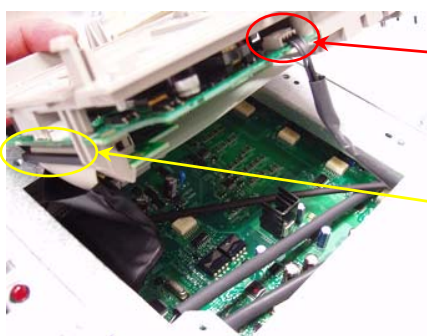


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S180 to S183).

Remove this screw (S153).

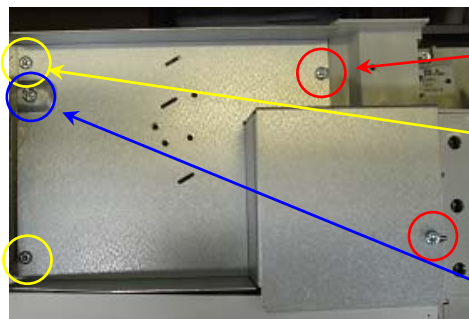


Disconnect the wire
Application board (S103)→E101→Power board (S103)

Disconnect the ribbon cable

	Mark	Size	Torque
	S180-S183	M3x8	0,78Nm
	S153	M4x8	1,5Nm

Metal Parts KIT : VY1A1209



Remove 2 screws (S7 and S8).

Remove this screw (S64 and S65).

Remove this screw (S19).

	Mark	Size	Torque
	S7 S8	4x12	1,5Nm
	S64 S65	M4x10	1,5Nm
	S19	M5x12	1,5Nm

Screws KIT : VY1ADV1104



Remove 2 screws (S66 and S67).

	Mark	Size	Torque
	S66 S67	M4x10	1,5Nm

Metal Parts KIT : VY1A1209



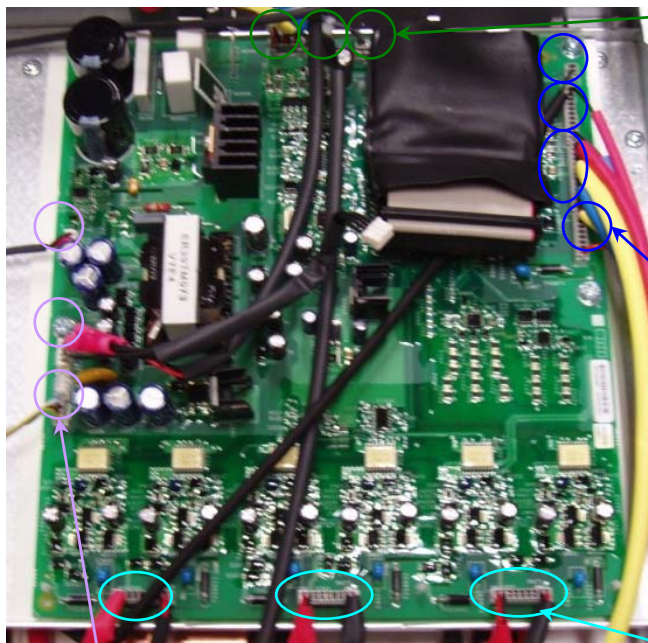
Remove 2 screws (S60 and S62).

Disconnect the wire.

Remove 2 screws (S61 to S63).

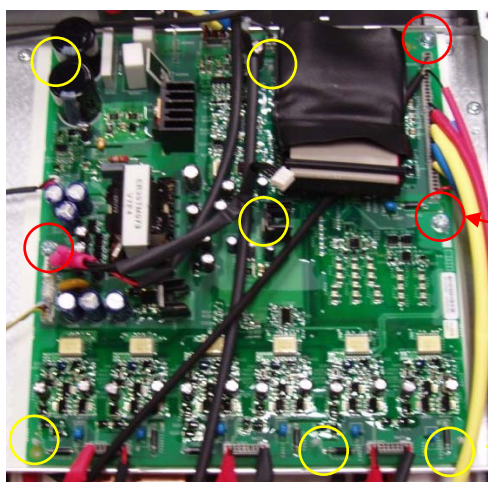
	Mark	Size	Torque
	S60 S62	M4x10	1,5Nm
	S61 S63	M4x10	1,5Nm

SUB ASSEMBY POWER : VX5IM2200M1271



Disconnect 3 wires (from up to down)

- Power board (S205)→E102→DEL
- Power board (S103)→E101
→Power internal fan and control bloc (S103):
- Power board (S203)→E105→Power 2 fans



Disconnect 3 wires (from right to left)

- Power board (S700)→E118→heatsink, thermal sensor
- Power board (S101)→E112→Bus board (S101)
- Power board (S102)→E107→Modules THY/DIO :
Blue cable→ THY/DIO (10) : red wire→ pin 5 ; blue wire→pin 4
Yellow cable→ THY/DIO (9) : red wire→ pin 5 ; blue wire→pin 4
Red cable→ THY/DIO (8) : red wire→ pin 5 ; blue wire→pin 4

Disconnect 4 wires (from up to down)

- Power board (S202)→E111→Filter board (S202)
- Power board (S200)→E109→Module IGBT (11) :
Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT
- Power board (S500)→E110→3 current sensors :
Blue wire : W (sensor 48) ; red : U (46) ; Yellow : V (47)
- Power board (S400)→E108→ Module IGBT (11)
Yellow wire : pin 1 of IGBT ; Red wire : pin 6 of IGBT
Black wire : pin 7 of IGBT

Disconnect 3 wires (from left to right)

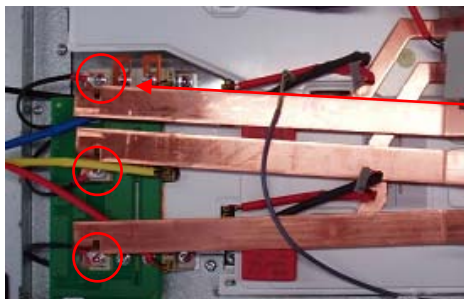
- Power board (S300)→E115→ Modules IGBT (14) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2
- Power board (S301)→E116→ Modules IGBT (13) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2
- Power board (S302)→E117→ Modules IGBT (12) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2

Remove 3 screws (S166 to S168)

Press 6 clips

Mark	Size	Torque
S166- S168	M4x10	1,5Nm

Interconnection power KIT: VZ3N1311



Remove 3 screws (S104 to S106) and remove 3 wires.

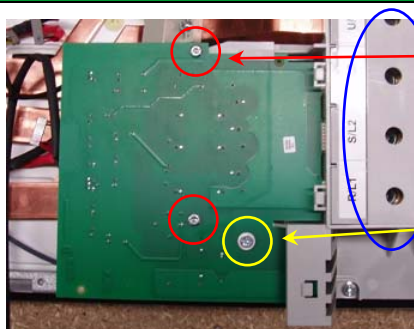
S134 to S136 :

ATV61 55Kw : 2,5Nm ; 75Kw and 90Kw : 3,5Nm

ATV71 45Kw and 55Kw : 2,5Kw ; 75Kw : 3,5Nm

	Mark	Size	Torque
	S104-S106	M5x12	2,5 or 3,5Nm

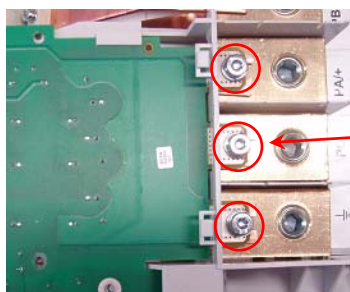
Filter Board : VX4A1110



Remove 3 screws (S5-S6).

Remove the mask of the power terminal.

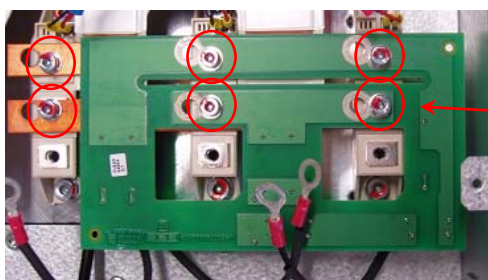
Remove this screw (S165).



Unscrew 3 screws.

	Mark	Size	Torque
	S5 S6	4x12	
	S165	M4x10	

BUS Bar KIT: VZ3N1312



Remove 6 screws (S98 to S103)

S107 to S112 :

ATV61 55Kw : 2,5Nm ; 75Kw and 90Kw : 3,5Nm

ATV71 45Kw and 55Kw : 2,5Kw ; 75Kw : 3,5Nm

	Mark	Size	Torque
	S98-S103	M5x12	2,5 or 3,5Nm

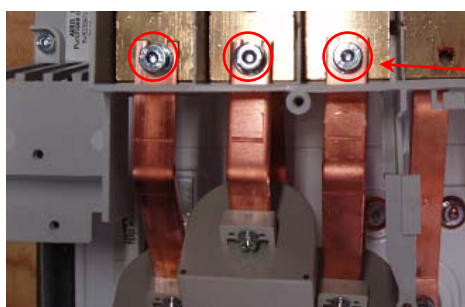
BUS Bar KIT: VZ3N1312



Remove 9 screws
(S198 to S200, S202, S203, S205 and S207 to S209)

Mark	Size	Torque
S198-S200 S202,S203 S205 S207-S209	M6x12	3,5Nm

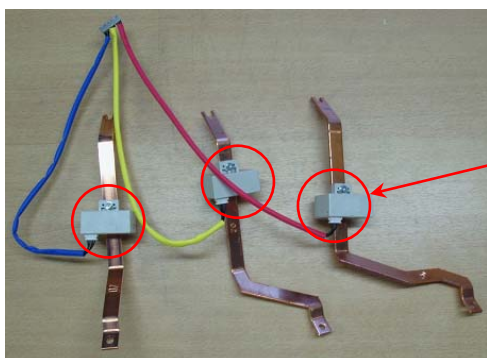
Interconnection power KIT: VZ3N1311



Unscrew 3 screws.
From left to right: blue cable, yellow cable and red cable.

The U, V and W bars can be removed.

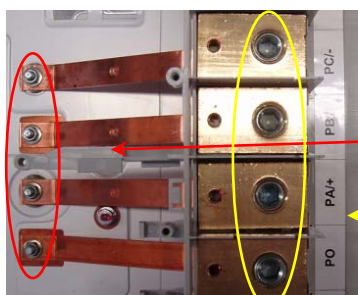
Current Sensor KIT : VY1A1104



Remove 3 screws (S100 and S102) and remove 3 wires.

3 current sensors→E110→ Power board (S500)

Interconnection power KIT: VZ3N1311



Remove 4 nuts (S90 and S93).

Unscrew 4 screws.

Mark	Size	Torque
S90-S93	M5X8	2,5Nm

Lots of 4 capacitors : VY1ADC1107



Remove 8 screws (S190 and S197).

	Mark	Size	Torque
	S190-S198	M6x12	3,5Nm

Interconnection power KIT: VZ3N1311

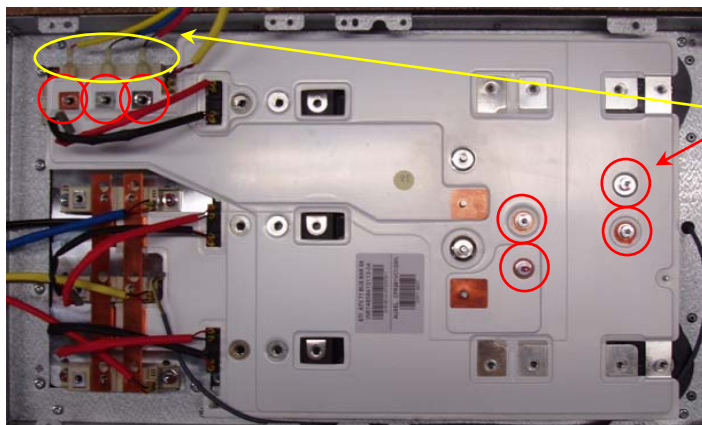


Remove 6 screws (S147 to S152).

Remove 2 screws (S25 to S26).

	Mark	Size	Torque
	S147-S152	M5x25	2,5Nm
	S25 S26	M8x20	1,5Nm

BUS Bar KIT : VZ3N1312



Remove 7 screws (S186 to S189, S201, S204 and S206)

Disconnect 2 wires

- Power board (S200)→E109
→Module IGBT (11) :

Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT

- Power board (S400)→E108
→ Module IGBT (11)

Yellow wire : pin 1 of IGBT ; Red wire : pin 6 of IGBT

Black wire : pin 7 of IGBT

Disconnect this wire

Power board (S102)→E107→Modules IGBT :

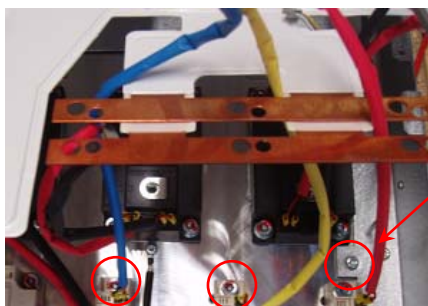
Blue cable→ IGBT (10) : red wire→ pin 5
blue wire→pin 4

Yellow cable→ IGBT (9) : red wire→ pin 5
blue wire→pin 4

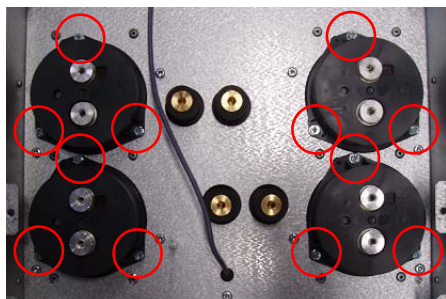
Red cable→ IGBT (8) : red wire→ pin 5 ; blue wire→pin 4

The bus bar can be removed.

	Mark	Size	Torque
	S186 - S189, S201, S204, S206	M6x12	3,5Nm



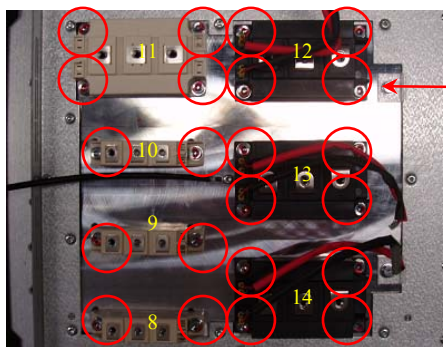
Lots of 4 capacitors : VY1ADC1107



Remove 12 screws
(S157 to S164 and S169 to S172).

Mark	Size	Torque
S157- S164 S169- S172	M4x10	0,78Nm

Rectifier (Thyristor / Diode) : VZ3TD1072M1671 Module Braking IGBT : VZ3IM1300M1271 Sub Assembly Power KIT : VX5IM2200M1271

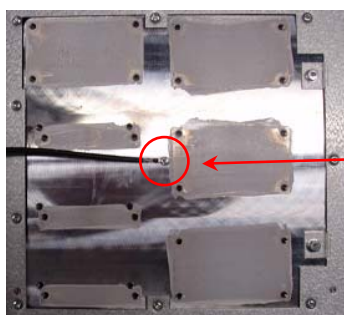


Remove 22 screws (S220 and S240).

The 7 modules can be removed
(**Warning**: do not forget to put the grease between the modules and heatsink).

Mark	Size	Torque
S220- S240	M6x20	4,4Nm

Thermal Sensor : VZ3G1101

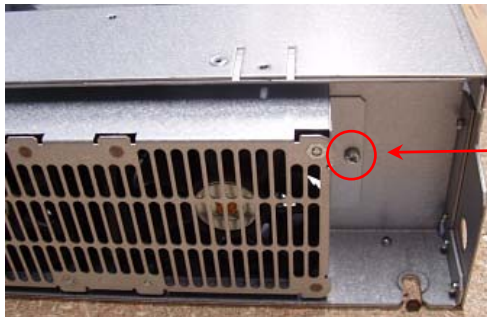


Remove this screw (S55).

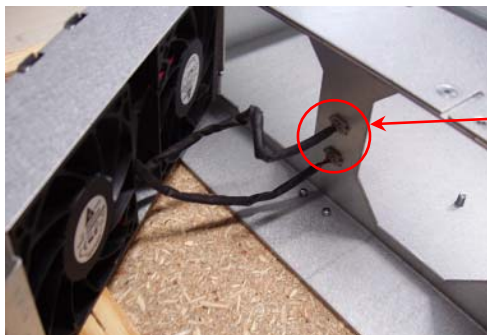
The heatsink can be removed.

Mark	Size	Torque
S55	M4x10	2Nm

Fan KIT : VZ3V1208



Remove this screw (S35).



Disconnect 2 wires.

	Mark	Size	Torque
	S35	M4x10	2Nm

9.10.2 Product Assembling Drawing

Refer to following files: [Assembling_175578400a53_06.pdf](#)
[Assembling_175578500A53_06.pdf](#)

9.10.3 Product Cabling Drawing

Refer to following file: [Cabling_175578900A53_04.pdf](#)

9.11 ATV61/71 Size A2 (size, refer to 1.2)

9.11.1 Dismantling and reassembling

Size A2: ATV71W075N4, ATV71WU15N4, ATV71WU22N4, ATV61W075N4, ATV61W075N4C, ATV61WU15N4, ATV61WU15N4C, ATV61WU22N4, ATV61WU22N4C, ATV61WU30N4, AT61WU30N4C

ATV71W075N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1H075N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WU15N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1H075N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WU22N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1HU22N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61W075N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1H075N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU15N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1HU15N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU22N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1HU15N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU30N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1203	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1202	Plastic Parts
VX5A1HU22N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1501



Remove 6 screws (S361 to S366).



Remove this screw (S353).

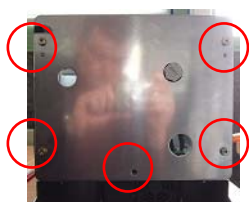


Disconnect the cable
RJ45 - RJ45.

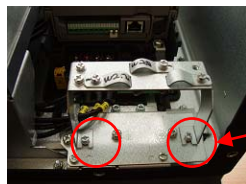
The front trap can be removed.

	Mark	Size	Torque
	S361- S366	M5x12	1,5Nm
	S353	M4x10	1,5Nm

Screws KIT: VY1ADV1200



Remove 5 screws (S378 to S382).



Remove 2 screws
(S347 and S348).

	Mark	Size	Torque
	S337 S338	M4x10	1,5Nm

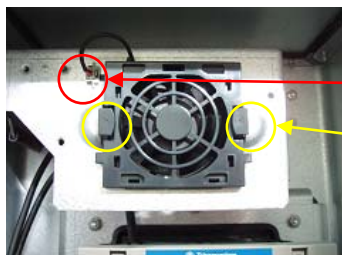
Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 screws (S337 and S338).

	Mark	Size	Torque
	S337 S338	M4x10	1,5Nm

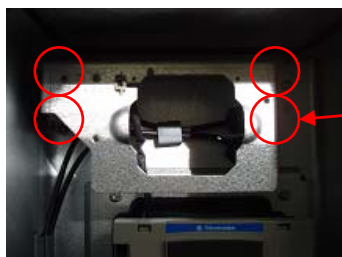
Internal Fan: VZ3V1217



Disconnect the wire.

Push on the two clips.

Screws KIT: VY1ADV1200



Remove 4 screws (S325 to S328).

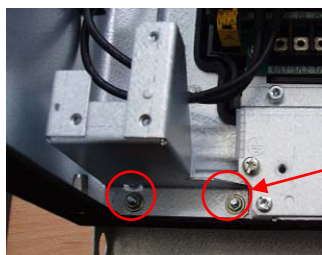


Remove 3 screws (S342 to S344).

Remove 2 screws (S329, S330).

	Mark	Size	Torque
	S325-S328	M4x10	2Nm
	S342-S344	M5x12	1,5Nm
	S329-S330	M4x10	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 nuts (S345 and S346).

	Mark	Size	Torque
	S345-S346	M6x8	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 6 screws (S332 to S337).

Disconnect the 3 wires
(from up to down):
L1, L3 and L3 ; Torque : 1,5Nm

	Mark	Size	Torque
	S332-S337	M4x10	2Nm

Screws KIT: VY1ADV1200



Remove 10 nuts (S309 to S318).

Remove 4 screws (S319, S321, S323 and S324).



Remove 4 nuts (S301 to S304).

The drive can be removed.

	Mark	Size	Torque
	S309-S318	M6x8	1,5Nm
	S319 S321 S323 S324	M4x10	2Nm
	S301-S304	M6x8	1,5Nm

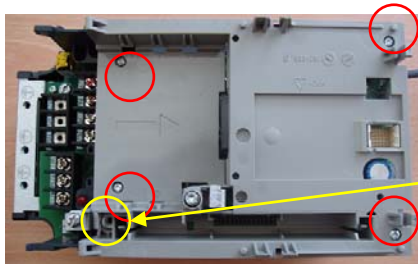
Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

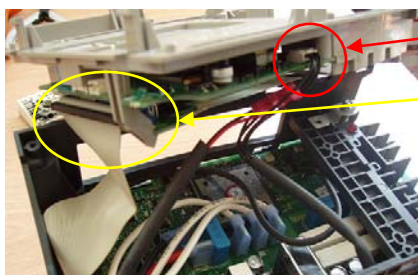


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S383 to S386).

Remove this screw (S387).



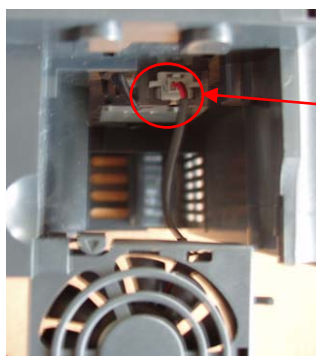
Disconnect the wire.

Disconnect the ribbon cable.

	Mark	Size	Torque
	S283-S386	M3x12	0,78Nm
	S387	M4x6	1,5Nm

Fan Kit : VZ3V1203

Push on the two clips.



Disconnect the wire.

Power Block Sub Assembly : VX5A1HU15N4

The Power Block Sub Assembly.

9.11.2 Product Assembling Drawing

Refer to following files: [Assembling_175409400A53_IED06.pdf](#)
[Assembling_175719900A53_IED05.pdf](#)

9.11.3 Product Cabling Drawing

Refer to following files: [Cabling_175554400A53_IED04.pdf](#)
[Cabling_176503300a53_02.pdf](#)

9.12 ATV61/71 Size A3 (size, refer to 1.2)

9.12.1 Dismantling and reassembling

Size A3: ATV71WU30N4, ATV71WU40N4, ATV61WU40N4, ATV61WU40N4C, ATV61WU55N4, ATV61WU55N4C

ATV71WU30N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1209	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1203	Plastic Parts KIT
VX5A1HU30N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WU40N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1209	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1203	Plastic Parts KIT
VX5A1HU40N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU40N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1209	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1203	Plastic Parts KIT
VX5A1HU30N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU55N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1209	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1000	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1501	Front Trap
VY1A1203	Plastic Parts KIT
VX5A1HU40N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1501



Remove 6 screws (S361 to S366).



Remove this screw (S353).

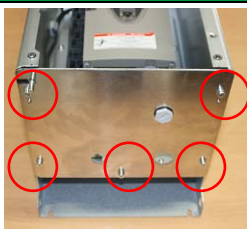


Disconnect the cable
RJ45 - RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S361-S366	M5x12	1,5Nm
	S353	M4x10	1,5Nm

Screws KIT: VY1ADV1200



Remove 5 screws (S378 to S382).

	Mark	Size	Torque
	S378-S382	M5x16	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 screws (S337 and S338).



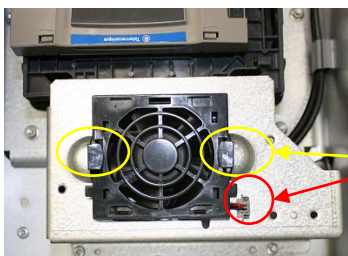
Remove 5 screws (S332 and S336).



Disconnect 3 wires (from left to right).
L1 to L3 with the torque: 1,5Nm

	Mark	Size	Torque
	S337-S338	M4x10	1,5Nm
	S332-S336	M4x10	2Nm

Internal Fan: VZ3V1217



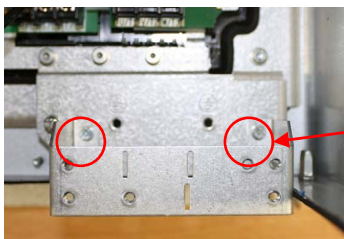
Disconnect the wire.

Push on the two clips.

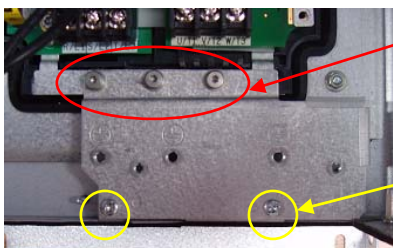
Screws KIT: VY1ADV1200



Remove 4 screws (S325 to S328).



Remove 3 screws (S347 to S348).

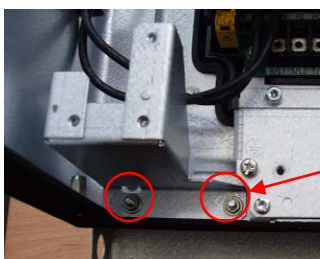


Remove 3 screws (S342 to S344).

Remove 2 screws (S329, S330).

	Mark	Size	Torque
	S325-S328	M4x10	2Nm
	S347-S348	M4x10	2Nm
	S342-S344	M5x12	2,6Nm
	S329-S330	M4x10	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 nuts (S345 and S346).

	Mark	Size	Torque
	S345-S346	M6x8	1,5Nm

Screws KIT: VY1ADV1200



Unscrew the 2 screws for disconnect the 2



Remove 10 nuts (309 to 318).

The support gasket and the molded gasket can be removed.



Remove 4 screws (S319, S321, S323 and S324).

Remove the 2 wires.

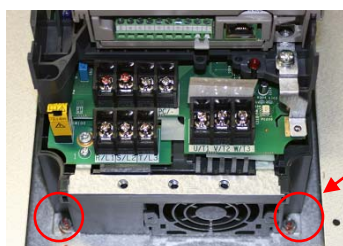
The enclosure can be removed.

	Mark	Size	Torque
	S309-S318	M6	1,5Nm
	S319 S321 S323 S324	M4x10	2Nm

Screws KIT: VY1ADV1200



Remove 4 nuts (S301 to S304).



	Mark	Size	Torque
	S301-S304	M6	1,5Nm

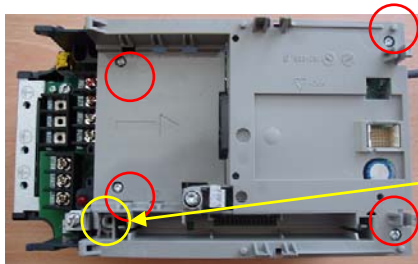
Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

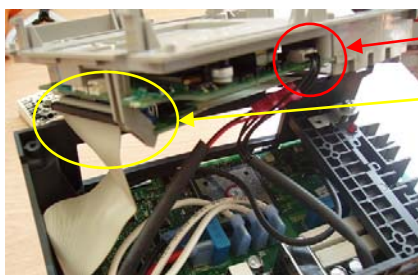


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S383 to S386).

Remove this screw (S347).

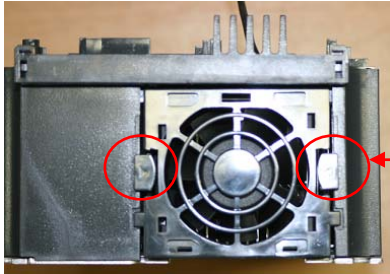


Disconnect the wire.

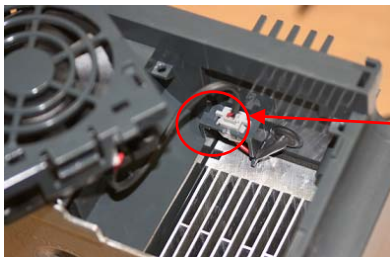
Disconnect the ribbon cable.

	Mark	Size	Torque
	S383-S386	M3x12	0,78Nm
	S47	M4x6	1,5Nm

Fan Kit: VZ3V1209



Push on the two clips.



Disconnect the wire.

Power Block Sub Assembly: VX5A1HU30N4



The Power Block Sub Assembly.

9.12.2 Product Assembling Drawing

Refer to following files: [Assembling_175409400A53_IED06.pdf](#)
[Assembling_175719900A53_IED05.pdf](#)

9.12.3 Product Cabling Drawing

Refer to following files: [Cabling_175554400A53_IED04.pdf](#)
[Cabling_176503300a53_02.pdf](#)

9.13 ATV61/71 Size B (size, refer to 1.2)

9.13.1 Dismantling and reassembling

Size B: **ATV71WU55N4, ATV71WU75N4, ATV61WU75N4,
ATV61WU75N4C, ATV61WD11N4, ATV61WD11N4C**

ATV71WU55N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1204	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1001	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1502	Front Trap
VY1A1204	Plastic Parts KIT
VX5A1HU55N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WU75N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1204	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1001	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1502	Front Trap
VY1A1204	Plastic Parts KIT
VX5A1HU55N4	Power Block Sub Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WU75N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1204	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1001	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1502	Front Trap
VY1A1204	Plastic Parts KIT
VX5A1HU55N4	Power Block Sub Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD11N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1204	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1001	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1200	Screws KIT
VY1A1502	Front Trap
VY1A1204	Plastic Parts KIT
VX5A1HU75N4	Power Block Sub-Assembly 7,5kW 400V
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1502



Remove 6 screws (S361 to S366).

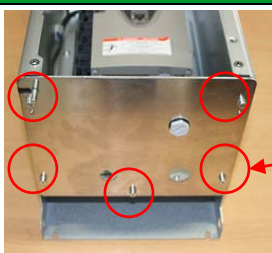
Remove this screw (S353).

Disconnect the cable
RJ45-RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S361-S366	M5x12	1,5Nm
	S353	M4x10	1,5Nm

Screws KIT: VY1ADV1200



Remove 5 screws (S378 to S382).

	Mark	Size	Torque
	S378-S382	M5x16	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



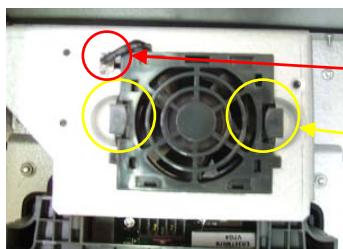
Remove 2 screws (S337 and S338).

Remove 5 screws (S332 and S336).

Disconnect 3 wires (from left to right).
L1 to L3 with the torque: 1,5Nm

	Mark	Size	Torque
	S337-S338	M4x10	1,5Nm
	S332-S336	M4x10	2Nm

Internal Fan: VZ3V1217



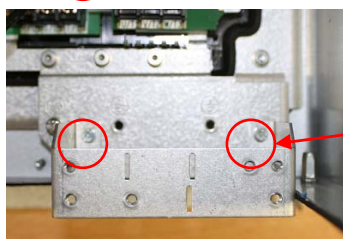
Disconnect the wire.

Push on the two clips.

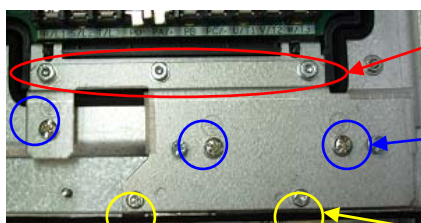
Screws KIT: VY1ADV1200



Remove 4 screws (S325 to S328).



Remove 3 screws (S347 to S348).



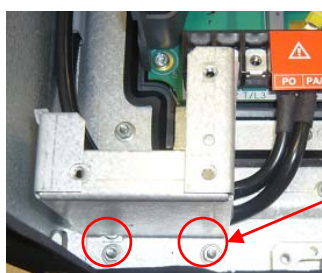
Remove 3 screws (S342 to S344).

Remove 3 screws (S339 to S341).

Remove 2 screws (S329, S330).

	Mark	Size	Torque
	S325-S328	M4x10	2Nm
	S347-S348	M4x10	1,5Nm
	S342-S344	M5x12	2,6Nm
	S329-S330	M4x10	2Nm
	S339-S341	M5x12	1,5Nm

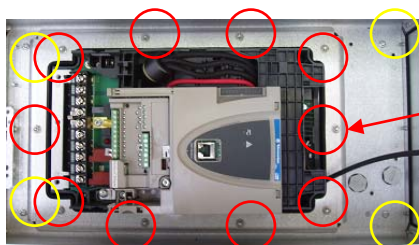
Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 nuts (S345 and S346).

	Mark	Size	Torque
	S345-S346	M6x8	1,5Nm

Screws KIT: VY1ADV1200



Remove 10 nuts (309 to 316).

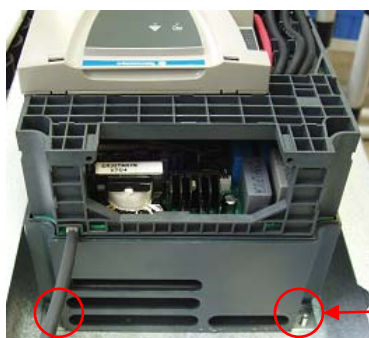
The support gasket and the molded gasket can be removed.

Remove 4 screws (S319, S321, S323 and S324).

The enclosure can be removed.

	Mark	Size	Torque
	S309-S318	M6	1,5Nm
	S319 S321 S323 S324	M4x10	2Nm

Screws KIT: VY1ADV1200



Remove 4 nuts (S301 to S304).



	Mark	Size	Torque
	S301-S304	M6	1,5Nm

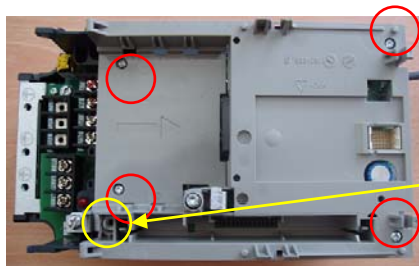
Control Bloc: VX4A71100Y



Press the two clips and pulls forward to take out the display board.

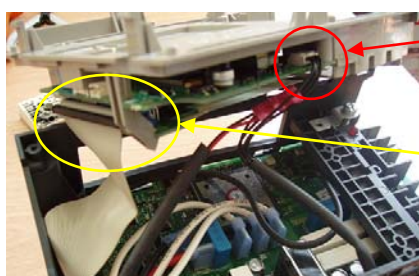


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S383 to S386).

Remove this screw (S347).



Disconnect the wire.

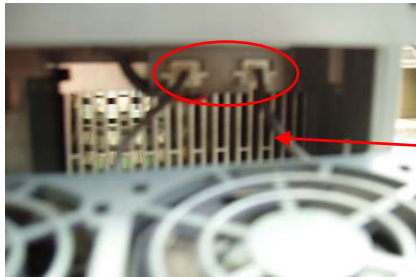
Application board (S103)→Power board (S103) and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S383-S386	M3x12	0,78Nm
	S47	M4x6	1,5Nm

Fan Kit: VZ3V1204

Push on the two clips.



Disconnect 2 wires.

Power Block Sub Assembly: VX5A1HU55N4

The Power Block Sub Assembly.

9.13.2 Product Assembling Drawing

Refer to following files: [Assembling_175577700A53_IED06.pdf](#)
[Assembling_175719900A53_IED05.pdf](#)

9.13.3 Product Cabling Drawing

Refer to following files: [Cabling_175554500A53_IED03.pdf](#)
[Cabling_176503300a53_02.pdf](#)

9.14 *ATV61/71 Size C (size, refer to 1.2)*

9.14.1 Dismantling and reassembling

Size C: ATV71WD11N4, ATV61WD15, ATV61WD15N4C

ATV71WD11N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1210	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1002	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1503	Front Trap
VY1A1215	Plastic Parts KIT
VX5A1HD11N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD15N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1210	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ2LB1002	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1503	Front Trap
VY1A1215	Plastic Parts KIT
VX5A1HD11N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1503



Remove 6 screws (S361 to S366).

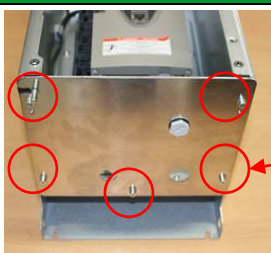
Remove this screw (S353).

Disconnect the cable
RJ45-RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S361-S366	M5x12	1,5Nm
	S353	M4x10	1,5Nm

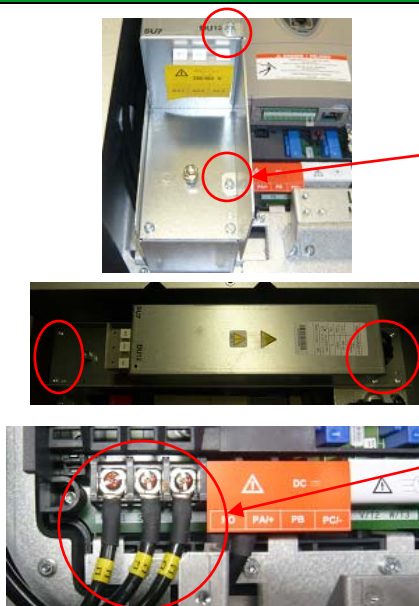
Screws KIT: VY1ADV1201



Remove 5 screws (S378 to S382).

	Mark	Size	Torque
	S378-S382	M5x16	1,5Nm

Screws KIT: VY1ADV1201 ONLY CLASS B



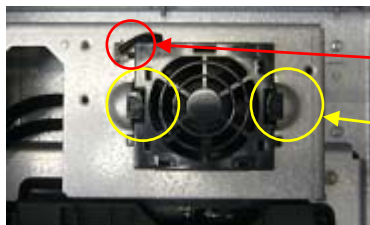
Remove 2 screws (S337 and S338).

Remove 5 screws (S332 and S336).

Disconnect 3 wires (from left to right).
L1, L3 and L3 ; Torque : 1,5Nm

	Mark	Size	Torque
	S337-S338	M4x10	1,5Nm
	S332-S336	M4x10	2Nm

Internal Fan: VZ3V1217



Disconnect the wire.

Push on the two clips.

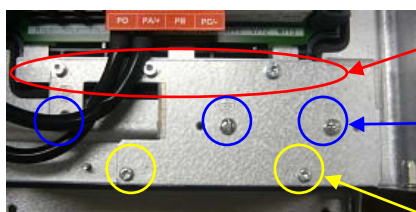
Screws KIT: VY1ADV1201



Remove 4 screws (S325 to S328).



Remove 3 screws (S347 to S348).



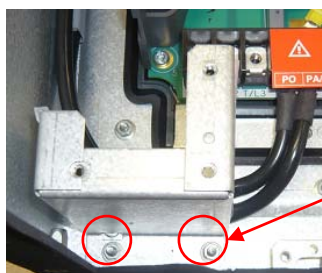
Remove 3 screws (S342 to S344).

Remove 3 screws
(S339 to S341).

Remove 2 screws (S329, S330).

	Mark	Size	Torque
	S325- S328	M4x10	2Nm
	S347- S348	M4x10	1,5Nm
	S342- S344	M5x12	2,6Nm
	S329 S330	M4x10	2Nm
	S339- S341	M5x12	1,5Nm

Screws KIT: VY1ADV1200 ONLY CLASS B



Remove 2 nuts (S345 and S346).

	Mark	Size	Torque
	S345 S346	M6x8	1,5Nm

Screws KIT: VY1ADV1201



Remove 10 nuts (309 to 318).

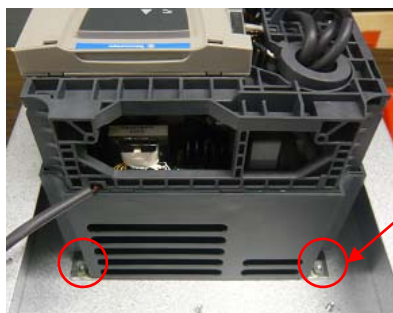
The support gasket and the molded gasket can be removed.

Remove 6 screws (S319 to S324).

The enclosure can be removed.

	Mark	Size	Torque
	S309-S318	M6x8	1,5Nm
	S319-S322	M4x10	2Nm

Screws KIT: VY1ADV1201



Remove 4 nuts (S301 to S304).

	Mark	Size	Torque
	S301-S304	M6	1,5Nm

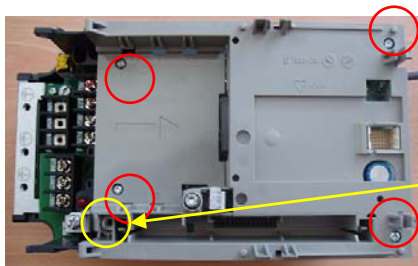
Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

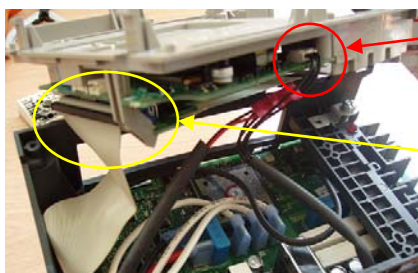


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S383 to S386).

Remove this screw (S387).



Disconnect the wire.
Application board (S103)→Power board (S103) and
Power internal fan

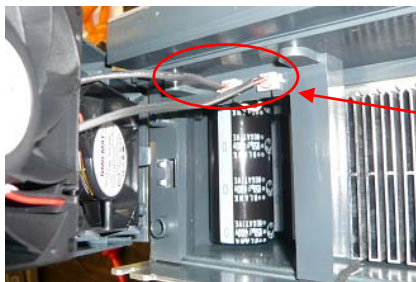
Disconnect the ribbon cable.

	Mark	Size	Torque
	S383-S386	M3x12	0,78Nm
	S387	M4x6	1,5Nm

Fan Kit: VZ3V1210



Push on the two clips.



Disconnect 2 wires.

Power Block Sub Assembly: VX5A1HD11N4



The Power Block Sub Assembly.

9.14.2 Product Assembling Drawing

Refer to following files: [Assembling_175577700A53_IED06.pdf](#)
[Assembling_175719900A53_IED05.pdf](#)

9.14.3 Product Cabling Drawing

Refer to following files: [Cabling_175554500A53_IED03.pdf](#)
[Cabling_176503300a53_02.pdf](#)

9.15 ATV61/71 Size D (size, refer to 1.2)

9.15.1 Dismantling and reassembling

Size D: **ATV71WD15N4, ATV71WD18N4, ATV61WD18N4, ATV61WD18N4C, ATV61WD22N4, ATV61WD22N4C**

ATV71WD15N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD18N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

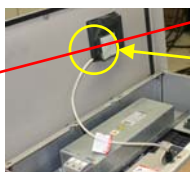
ATV61WD18N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD22N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A61100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1504



Remove 8 screws (S361 to S368).



Remove this screw (S353).

Disconnect the cable
RJ45-RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S361-S366	M5x12	1,5Nm
	S353	M4x10	1,5Nm

Screws KIT: VY1ADV1201 ONLY CLASS B



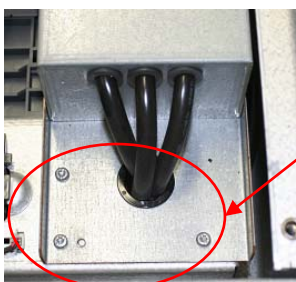
Remove 5 screws (S378 to S382).



Remove 2 screws (S337 and S338).

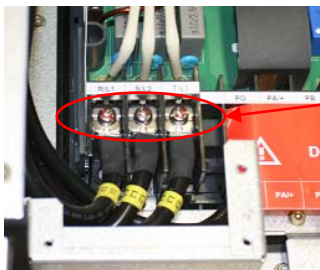


Remove 5 screws (S332 to S336).



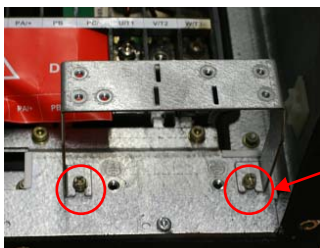
	Mark	Size	Torque
	S378-S382	M5x16	1,5Nm
	S337-S338	M4x10	1,5Nm
	S332-S336	M4x10	1,5Nm

Screws KIT: VY1ADV1201 ONLY CLASS B

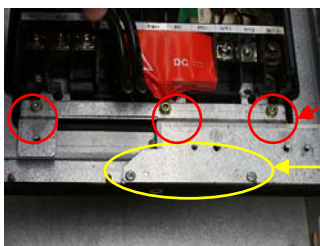


Unscrew the 3 screws, remove the 3 cables (Torque: 4,4Nm).

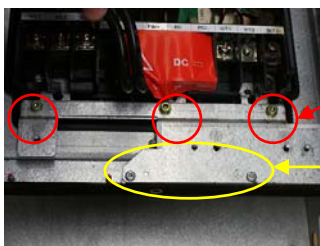
The filter can be removed.



Remove 2 screws (S347 and S348).



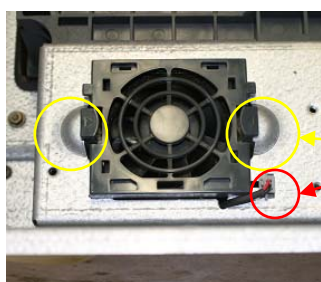
Remove 3 screws (S342 to S344).



Remove 2 screws (S329 and S330).

	Mark	Size	Torque
	S347 S348	M4x10	1,5Nm
	S342- S344	M5x12	2,6Nm
	S329 S330	M4x10	1,5Nm

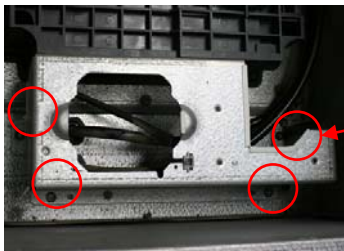
Internal Fan: VZ3V1217



Disconnect the wire.

Push on the two clips.

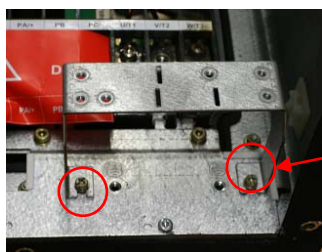
Screws KIT: VY1ADV1201



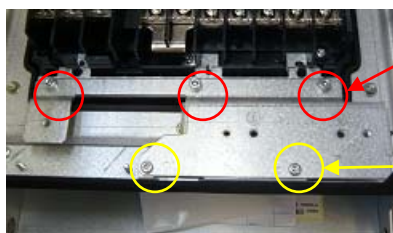
Remove 4 screws (S325 to S328).

	Mark	Size	Torque
	S325-S328	M4x10	2Nm

Screws KIT: VY1ADV1201 ONLY CLASS A



Remove 2 screws (S325 to S328).

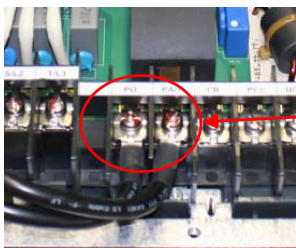


Remove 3 screws (S342 to S344).

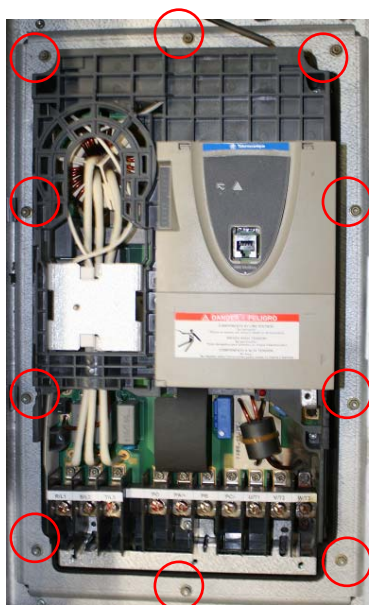
Remove 2 screws (S329 and S330).

	Mark	Size	Torque
	S325-S328	M4x10	1,5Nm
	S342-S344	M5x12	2,6Nm
	S329-S330	M4x10	1,5Nm

Screws KIT: VY1ADV1201



Unscrew 2 screws, remove 2 cables (Torque: 4,4Nm).



Remove 3 screws (S342 to S344).

	Mark	Size	Torque
	S309-S318	M6	1,5Nm

Screws KIT: VY1ADV1201



Remove 6 screws (S319 to S324).

The enclosure can be removed.



Remove 4 nuts (S301 to S304).



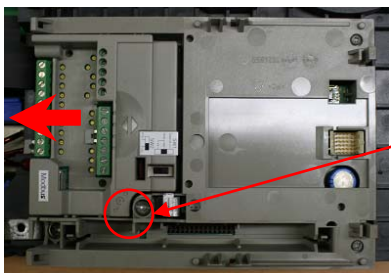
The drive can be removed.

	Mark	Size	Torque
	S319-S324	M4x10	2Nm
	S301-S304	M5	2,6Nm

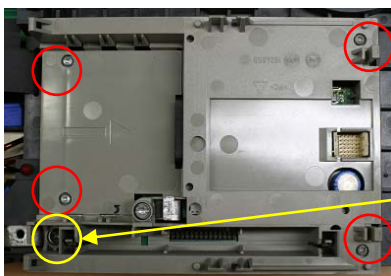
Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

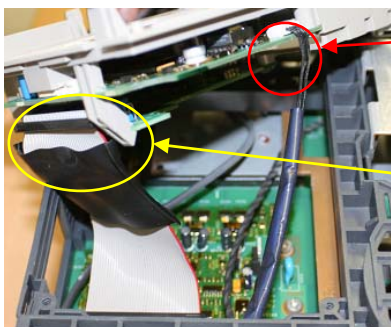


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S383 to S386).

Remove this screw (S387).



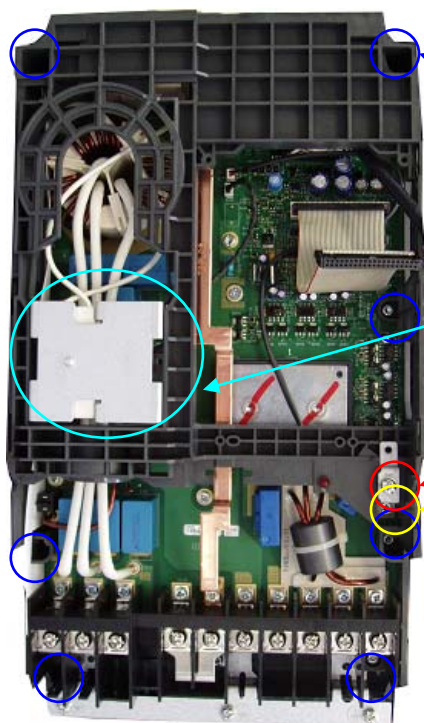
Disconnect the wire.

Application board (S103)→Power board (S103) and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S383-S386	M3x8	0,78Nm
	S387	M4x6	1,5Nm

Power Block Sub-Assembly: VX5A1HD15N4



Remove 7 screws (S31 to S35).

Remove the resistor (disconnect 2 wires : number 62).

Remove this screw (S64).

Remove this screw (S117).

Remove the ground bar.

The chassis can be removed.

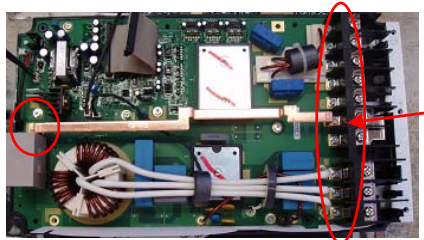
	Mark	Size	Torque
	S31-S35	4x12	1Nm
	S64	M5x12	1,5Nm
	S117	M4x16	1,5Nm

Screws KIT: VY1ADV1201



Remove the side insulate sheet.

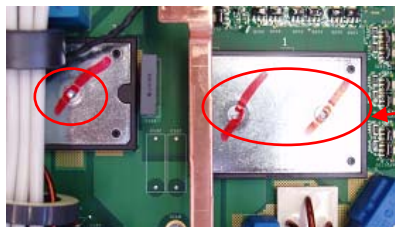
Power Terminals: VZ3N1205



Remove 11 screws (S101 to S111).

	Mark	Size	Torque
	S103-S113	M4x10	1,5Nm

Power Block Sub-Assembly: VX5A1HD15N4



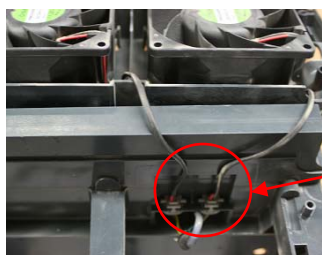
Remove 3 screws (S57 to S59).

	Mark	Size	Torque
	S57-S59	M4x30	2,2Nm

Fan Kit : VZ3V1205



Push on the two clips.



Disconnect the 2 wires.

The filter board can be removed.

9.15.2 Product Assembling Drawing

Refer to following files: [Assembling_175577900A53_05.pdf](#)
[Assembling_175719900A53_IED05.pdf](#)

9.15.3 Product Cabling Drawing

Refer to following files: [Cabling_175554600A53_04.pdf](#)
[Cabling_176503300a53_02.pdf](#)

9.16 ATV61/71 Size E (size, refer to 1.2)

9.16.1 Dismantling and reassembling

Size E: ATV61WD30N4, ATV61WD30N4, ATV71WD22N4

ATV61WD30N4 (C)	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD22N4	
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1505



Remove 8 screws (S187 to S194).

Remove this screw (S371).

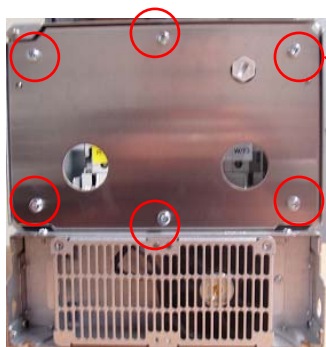


Disconnect the cable
RJ45-RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S187-S194	M6x16	2Nm
	S171	M4x10	1,5Nm

Metal Parts KIT: VY1A1206



Remove 6 screws (S203 to S208).

	Mark	Size	Torque
	S203-S208	M6x20	2Nm

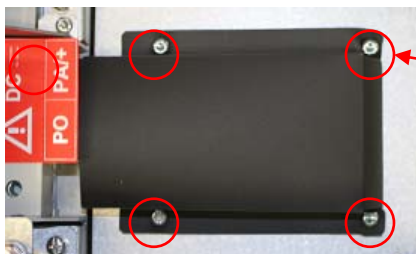
Metal Parts KIT: VY1A1206 ONLY CLASS B



Remove 5 screws (S104 to S107 and S170).

Mark	Size	Torque
S104- S107 S170	M4x10	2Nm

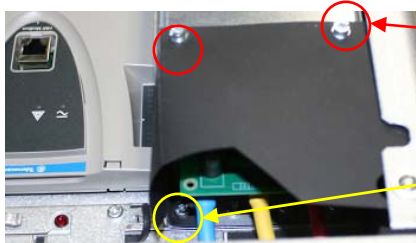
Insulate Sheet - Choke Shield: VY1A1511 ONLY CLASS B



Remove 4 screws (S19 to S22).

Mark	Size	Torque
S19- S22	4x12	2Nm

Insulate Sheet - Filter Cover: VY1A1508 ONLY CLASS B

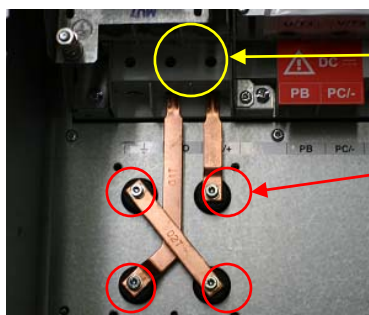


Remove 2 screws (S168 and S169).

Remove this screw (S84).

Mark	Size	Torque
S168 S169	M4x10	1,5Nm
S84	M4x10	2Nm

Screws KIT: VY1ADV1101 ONLY ATV61

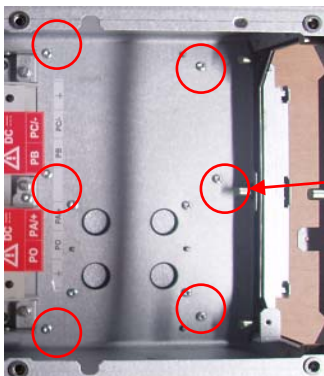


Unscrew 2 screws.

Remove 4 screws (S142 to S144 and S211).

Mark	Size	Torque
S142- S144, S211	M5x12	2,5Nm

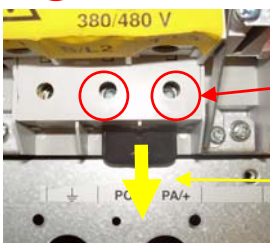
Metal Parts KIT: VY1A1206



Remove 6 screws (S90 to S91 and S96 to S99).



Remove 4 screws (S92 to S95).

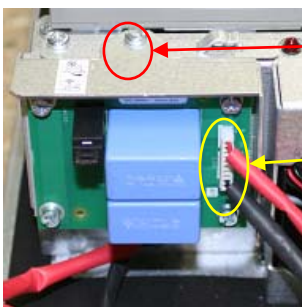


Unscrew these 2 screws.

And remove the shunt.

Mark	Size	Torque
S90-S99	M4x10	2Nm

Capacitor Board: VY1ADC1200

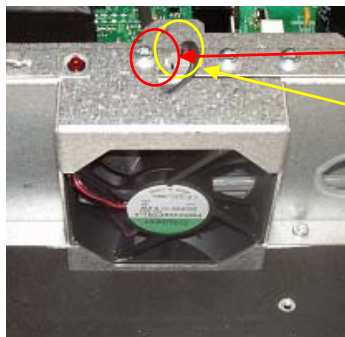


Remove this screw (S89).

Disconnect this wire.

Mark	Size	Torque
S89	M4x10	2Nm

Internal Fan: VZ3V1218

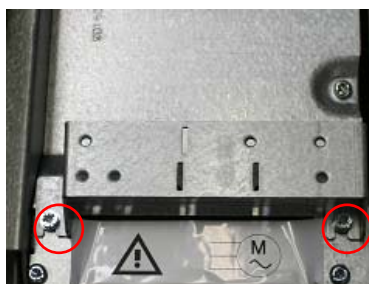


Disconnect the cable.

Remove this screw (S88).

	Mark	Size	Torque
	S88	M4x10	2Nm

Metal Parts KIT: VY1A1206



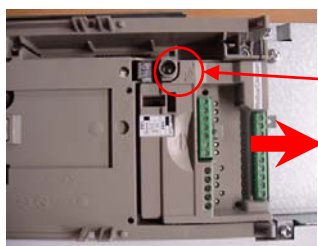
Remove 2 screws (S209 and S210).

	Mark	Size	Torque
	S209 S210	M4x10	1,5Nm

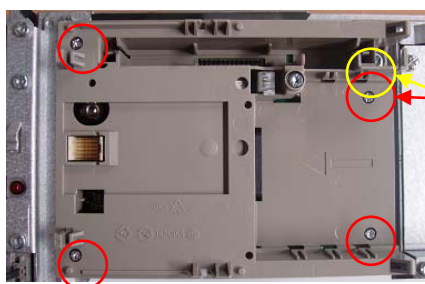
Control Bloc: VX4A71100Y



Press the two clips and pulls forward to take out the display board.

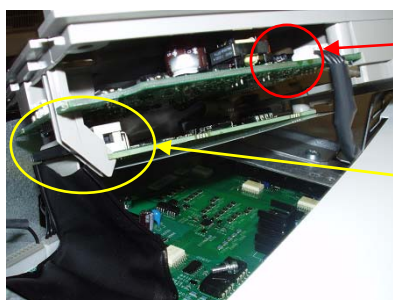


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S175 to S178).

Remove this screw (S163).

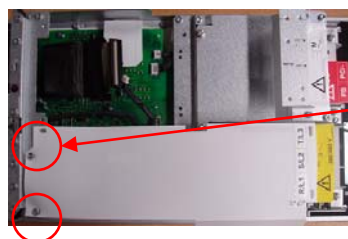


Disconnect the wire.
Application board (S103)→Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S175-S178	M3x8	0,78Nm
	S163	M4x8	1,5Nm

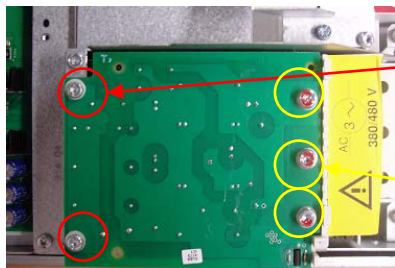
Screws KIT: VY1ADV1101 ONLY CLASS A



Remove 2 screws (S73 and S87).

	Mark	Size	Torque
	S73 S87	M4x10	2Nm

Sub Assembly Filter Boards: VX4A1107 ONLY CLASS A

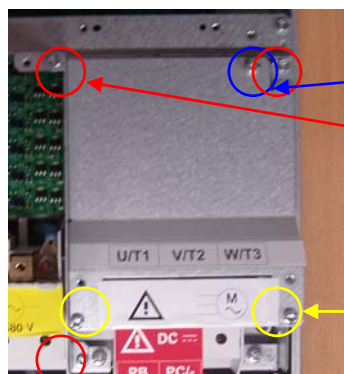


Remove 2 screws (S79 and S80).

Remove 3 screws
(S136 and S138).

	Mark	Size	Torque
	S79 S80	M4x10	1,5Nm
	S136- S138	M5x12	2,5Nm

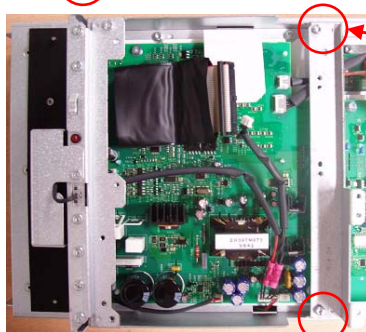
Metal Parts KIT: VY1A1206 ONLY CLASS A



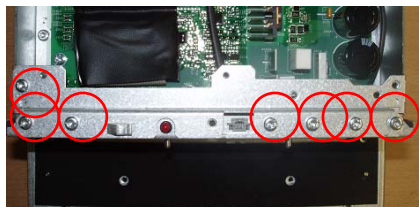
Remove this screw (S174).

Remove 3 screws (S74, S77 and S78).

Remove 2 screws (S17 and S18).



Remove 2 screws (S69 and S70).



Remove 7 screws
(S71, S72, S81 to S83,
S88 and S89).

	Mark	Size	Torque
	S174	M5x12	1,5Nm
	S74, S77, S78	M4x10	2Nm
	S17 S18	M4x12	2Nm
	S69 S70	M4x10	2Nm
	S71, S72, S81-S83, S88, S89	M4x10	2Nm

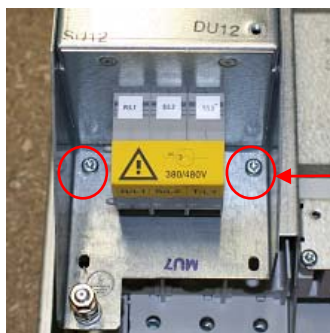
Screws KIT: VY1ADV1101 ONLY CLASS B



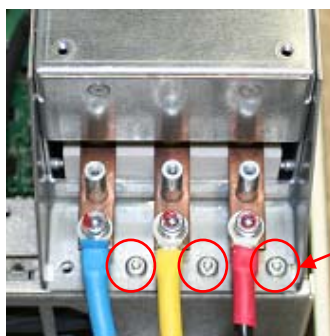
Remove 3 screws (S160 and S162).

Mark	Size	Torque
S160- S162	M4x10	1,5Nm

Screws KIT: VY1ADV1101 ONLY CLASS B



Remove 2 screws (S85 and S86).

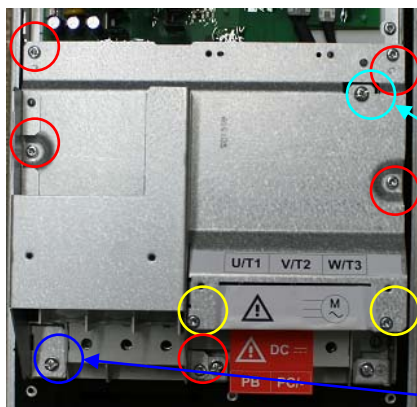


Remove 3 screws (S81 to S83).

The filter class B can be removed.

Mark	Size	Torque
S81-S83 S85,S86	M4x10	2Nm

Metal Parts KIT: VY1A1206 ONLY CLASS B



Remove 5 screws (S69, S70 and S74 to S76).

Remove this screw (S174).

Remove 2 screws (S17 and S18).

Remove this screw (S135).

	Mark	Size	Torque
	S69, S70 S74-S76	M4x10	2Nm
	S174	M5x12	1,5Nm
	S17 S18	4x12	2Nm
	S135	M5x12	2Nm

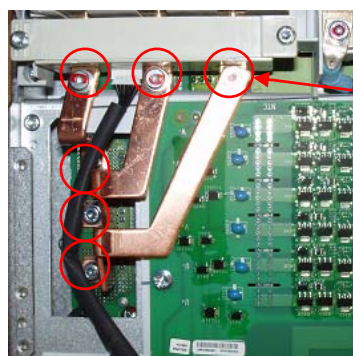
Metal Parts KIT: VY1A1206 ONLY CLASS B



Remove 3 screws (S71 and S73).

	Mark	Size	Torque
	S71- S73	M4x10	2Nm

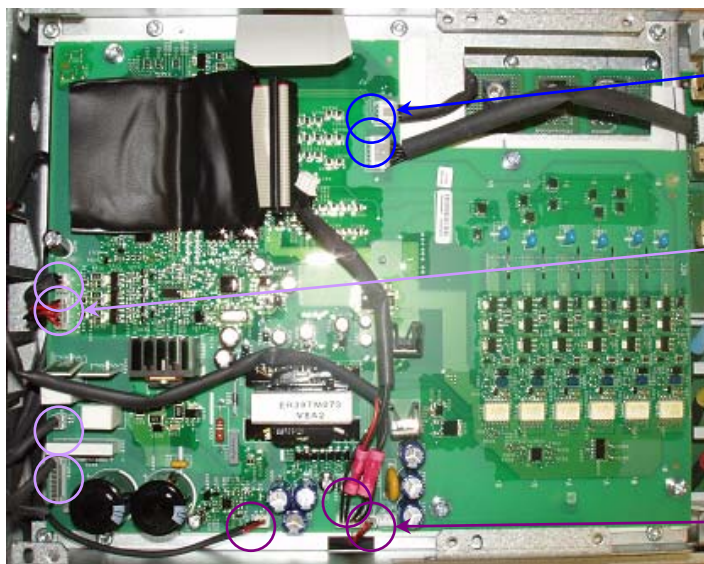
Connexion Bars KIT: VZ3N1304



Remove 6 screws (S129 to S134).

	Mark	Size	Torque
	S129- S134	M5x12	2,5Nm

Power Block Sub-Assembly: VX5A1HD22N4



Disconnect 2 wires (from up to down).

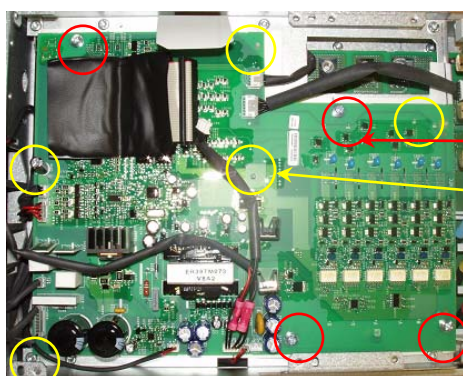
- Power board (S400)→E105
→Bus board (S400)
- Power board (S500)→E110
→Current sensor

Disconnect 4 wires (from up to down).

- Power board (S101)→E111
→Bus board (S101)
- Power board (S104)→E112
→Bus board (S104)
- Power board (S202)→E109
→Bus board (S202)
- Power board (S201)→E106
→Bus board (201)

Disconnect 3 wires (from left to right).

- Power board (S200)→E102
→Bus board (S101)
- Power board (S103)→E101
→Bus board (S104)
- Power board (S120)→E104
→Power fan



Remove 4 screws (S156 to S159).

Press 5 clips.

Mark	Size	Torque
S156- S159	M4x10	1,5Nm

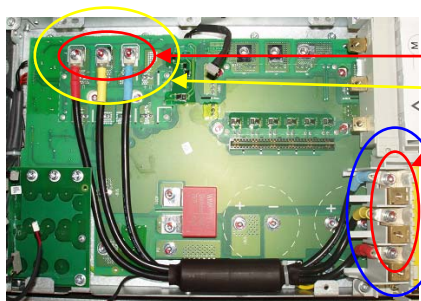
Power Block Sub-Assembly: VX5A1HD22N4



Remove 5 screws (S54 to S58).

Mark	Size	Torque
S54- S58	M4x10	2Nm

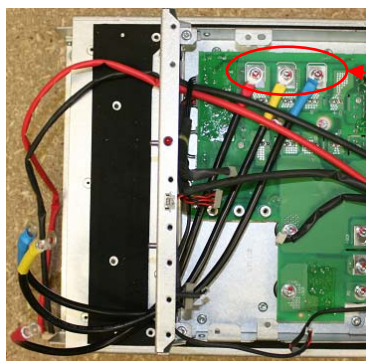
Mains Connexion KIT: VZ3N1305 ONLY CLASS A



Remove 6 screws (S123 to S128).
From left to right : red, yellow and blue.
From up to down : blue, yellow and red.

Mark	Size	Torque
S123-S128	M5x12	2,5Nm

Screws KIT : VY1ADV1101 ONLY CLASS B

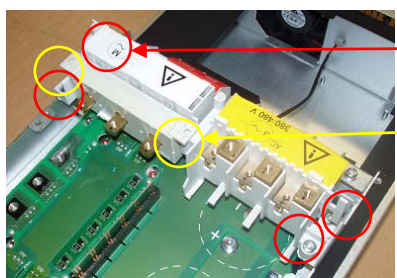


Remove 3 screws (S126 to S128).
From left to right : L1 red, L2 yellow and L3 blue.

Mark	Size	Torque
S126-S128	M5x12	2,5Nm

Current Sensor KIT: VY1A1101

Power Terminals: VZ3N1206



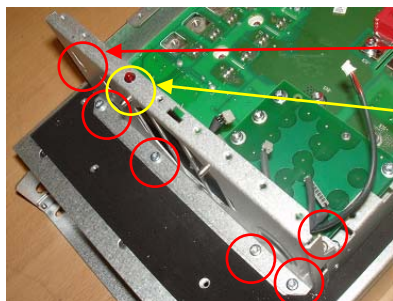
Remove 4 screws (S59 to S62).

Remove 2 screws (S15 and S16).

Remove the hall effect sensor

Mark	Size	Torque
S59 S62	M4x10	2Nm
S15 S16	M4x12	2Nm

Metal Parts KIT: VY1A1206

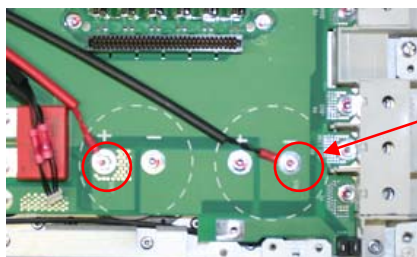


Remove 6 screws (S63 to S68).

Remove the DEL (E102).

	Mark	Size	Torque
	S63-S68	M4x10	2Nm

Screws KIT: VY1ADV1101



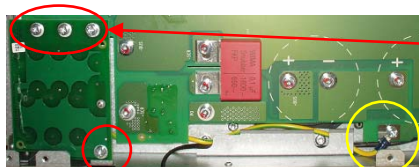
Remove 2 screws (S120 and S121).

Red wire : on + left capacitor.

Black wire : on – right capacitor.

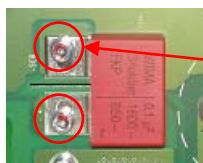
	Mark	Size	Torque
	S120-S121	M5x12	2,5Nm

DC BUS Board: VX5A1101

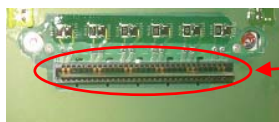


Remove 4 screws (S151 to S154).

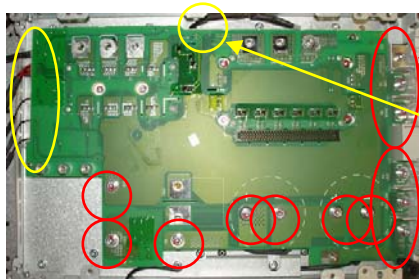
Remove this screw (S155) and remove the cable (filter board).



Remove 2 screws (S119 and S122).



Remove the interconnect PCB (no screw).

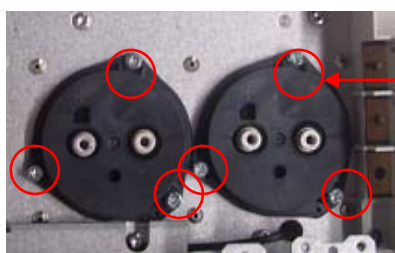


Remove 13 screws (S110 to S118 and S149 to S150).

Disconnect the wires (for the wiring, see the Power Block Sub-Assembly VX5A1HD22N4)

	Mark	Size	Torque
	S151-S154	M4x10	1,5Nm
	S155	M4x10	1,5Nm
	S119-S122	M5x12	2,5Nm
	S110-S118	M5x12	2,5Nm
	S149-S150	M5x20	2,5Nm

Lots of 2 capacitors: VY1ADC1101



Remove 6 screws (S44 to S49).

	Mark	Size	Torque
	S44-S49	M4x10	0,78Nm

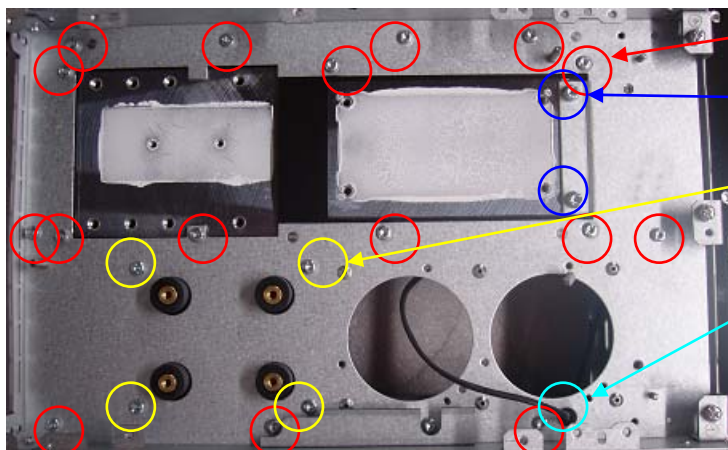
Power Terminals: VZ3N1206



Remove 6 screws (S50 to S53).

	Mark	Size	Torque
	S50-S53	M4x10	2Nm

Metal Parts KIT: VY1A1206



Remove 17 screws (S27 to S43).

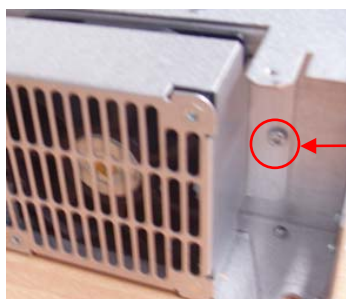
Remove 2 screws (S108 and S109).

Remove 4 screws (S11 to S14).

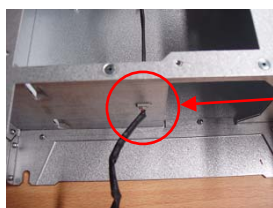
Remove the cable (E104).

	Mark	Size	Torque
	S27-S43	M4x10	2Nm
	S108-S109	M5x12	2,5Nm
	S11-S14	M4x12	2Nm

Fan Kit: VZ3V1211



Remove this screw (S26).



Disconnect this wire.

	Mark	Size	Torque
	S26	M4x10	1,5Nm

9.16.2 Product Assembling Drawing

Refer to following file: [Assembling_175719800A53_03.pdf](#)

9.16.3 Product Cabling Drawing

Refer to following files: [Cabling_175721500A53_IED03.pdf](#)

9.17 ATV61/71 Size F (size, refer to 1.2)

9.17.1 Dismantling and reassembling

**Size F: ATV61WD37N4, ATV61WD37N4C, ATV61WD45N4, ATV61WD45N4C
ATV71WD30N4, ATV71WD37N4**

ATV61WD37N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD45N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD30N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD37N4

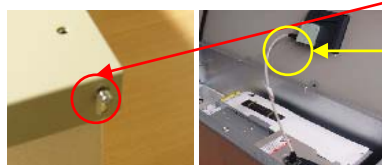
Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1506



Remove 8 screws (S219 to S226).

Remove this screw (S218).



Disconnect the cable
RJ45-RJ45.

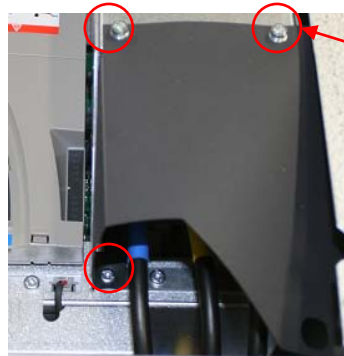
The front trap can be removed.

	Mark	Size	Torque
	S219-S226	M6x16	1,5Nm
	S218	M4x10	1,5Nm

Metal Parts KIT: VY1A1207 ONLY CLASS B



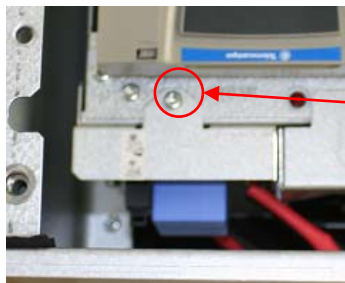
Remove 5 screws (S209 to S213).



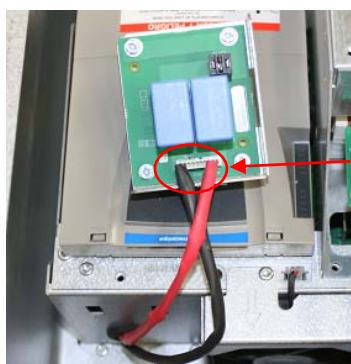
Remove 3 screws (S202 to S203).

	Mark	Size	Torque
	S209-S213	M4x10	2Nm
	S202-S203	M4x10	1,5Nm

Metal Parts KIT: VY1A1207 ONLY CLASS B



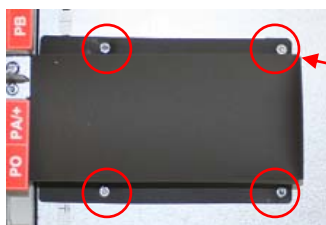
Remove this screw (S208).



Disconnect this wire.

	Mark	Size	Torque
	S208	M4x10	2Nm

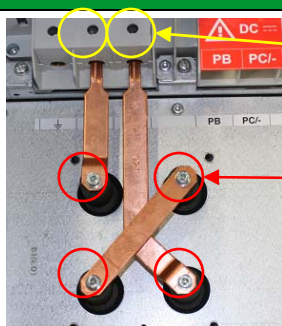
Insulate Sheet - Choke Shield: VY1A1512 ONLY CLASS B



Remove 4 screws (S198 to S201).

	Mark	Size	Torque
	S198-S201	M4x12	2Nm

Screws KIT : VY1ADV1101 ONLY CLASS B

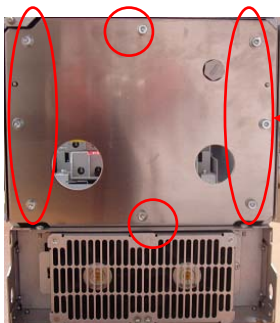


Unscrew 2 screws.

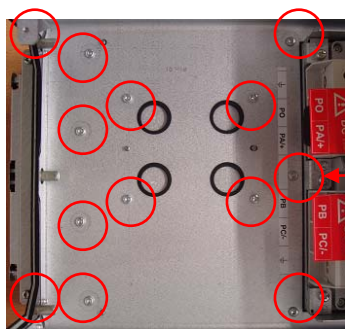
Remove 4 screws (S194 to S197).

	Mark	Size	Torque
	S194-S197	M5x12	2,5Nm

Metal Parts KIT: VY1A1207



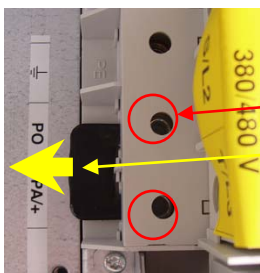
Remove 8 screws (S227 to S234).



Remove 13 screws
(S177 to S180 and S185 to S193).



Remove 4 screws (S181 to S184)



Unscrew these 2 screws.

And remove the shunt.

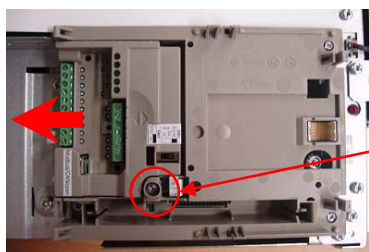
The enclosure can be removed.

	Mark	Size	Torque
	S227-S234	M6x20	2Nm
	S177-S193	M4x10	2Nm

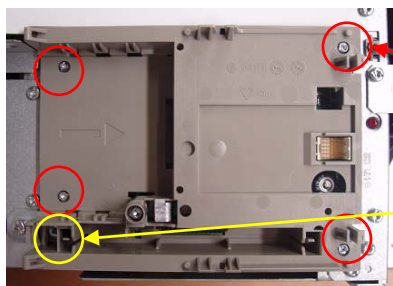
Control Bloc: VX4A71100Y



Press the two clips and pulls forward to take out the display board.

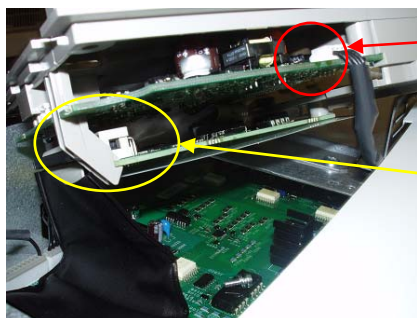


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S235 to S238).

Remove this screw (S239).

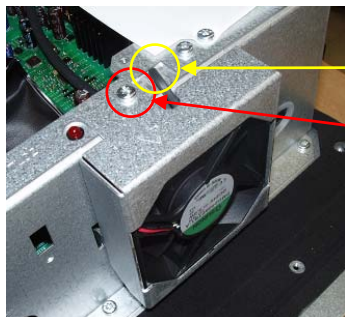


Disconnect the wire.
Application board (S103)→Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S235-S238	M3x8	0,78Nm
	S239	M4x8	1,5Nm

Internal Fan: VZ3V1218

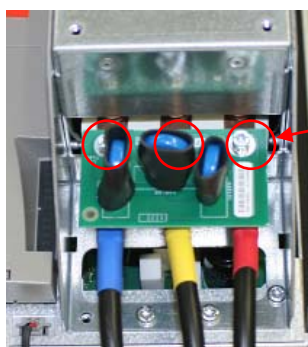


Disconnect the wire.

Remove this screw (S170).

	Mark	Size	Torque
	S170	M4x10	2Nm

Screws KIT: VY1ADV1101 ONLY CLASS B



Remove 3 screws (S174 to S176).

Remove 3 screws (S171 to S173).
L1 blue, L2 yellow and L3 red.

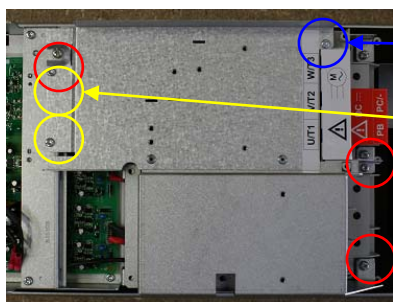
Remove 2 screws
(S164 and S165).

Remove 4 screws
(S127, S128, S167 and S168).

The filter class B can be removed.

	Mark	Size	Torque
	S174-S176	M4x10	1,5Nm
	S171-S173	M5x12	2,5Nm
	S164 S165	M4x10	1,5Nm
	S127, S128 S167, S168	M4x10	1,5Nm

Metal Parts KIT: VY1A1207 ONLY CLASS B



Remove this screw (S149).

Remove 2 screws
(S151 and S153).

Remove 3 screws
(S148, S144 and S157).

	Mark	Size	Torque
	S149	M4x12	1,5Nm
	S151 S153	M4x10	1,5Nm
	S148 S144 S157	M5x12	1,5Nm

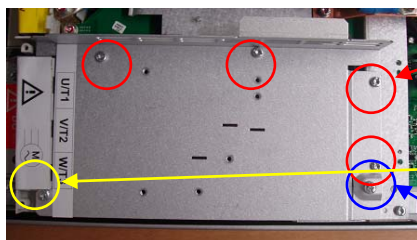
Screws KIT: VY1ADV1101 ONLY CLASSE A



Remove 2 screws (S154 and S155).

	Mark	Size	Torque
	S154-S155	M4x10	1,5Nm

Metal Parts KIT: VY1A1207 ONLY CLASSE A



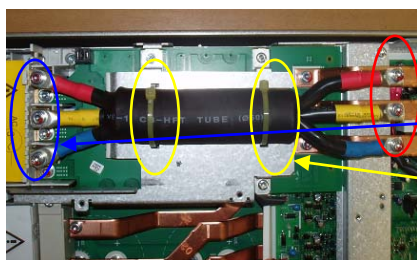
Remove 4 screws (S150 to S153).

Remove this screw (S149).

Remove this screw (S148).

	Mark	Size	Torque
	S150-S153	M4x10	1,5Nm
	S149	M4x12	1,5Nm
	S138	M5x12	1,5Nm

Mains Connexion KIT: VZ3N1306 ONLY CLASSE A



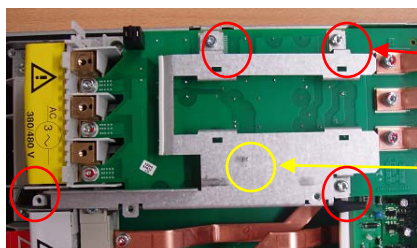
Remove 3 screws (S141 to S143).

Remove 3 screws (S132 to S134).

Cut 2 colsons tie.
The input harness can be removed.

	Mark	Size	Torque
	S141-S143	M5x12	2,5Nm
	S132-S134	M5x12	2,5Nm

Metal Parts KIT: VY1A1207 ONLY CLASSE A

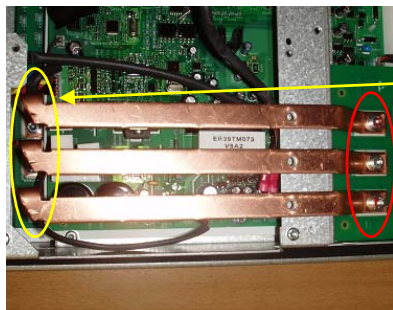


Remove 4 screws (S144 to S147).

Remove this clip.

	Mark	Size	Torque
	S144-S147	M4x10	1,5Nm

Mains Connexion KIT: VZ3N1306 ONLY CLASSE A



Remove 3 screws (S113 to S115).

Remove 3 screws (S138 to S140).

	Mark	Size	Torque
	S113-S115	M5x12	2,5Nm
	S138-S140	M5x12	2,5Nm

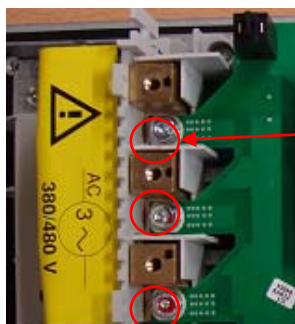
Metal Parts KIT: VY1A1207



Remove 2 screws (S123 and S124).

	Mark	Size	Torque
	S123-S124	M4x10	2Nm

Filter Board: VX4A1108

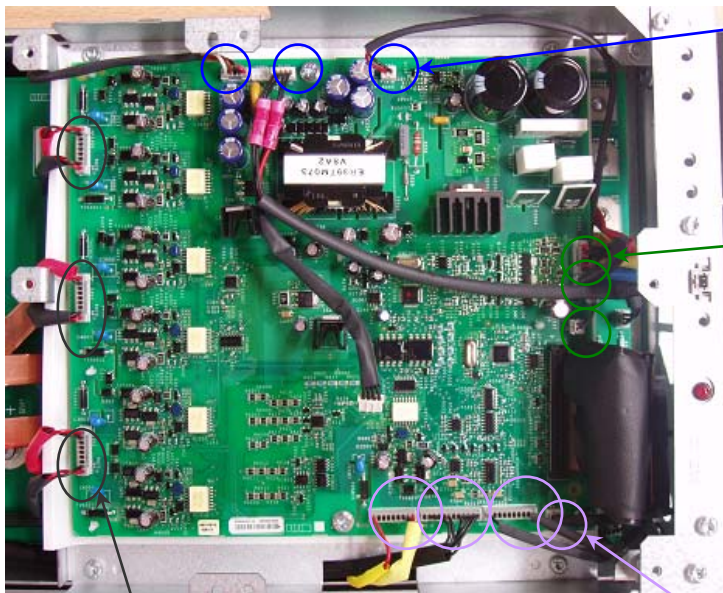


Remove 3 screws (S129 to S131).

The filter board can be removed.

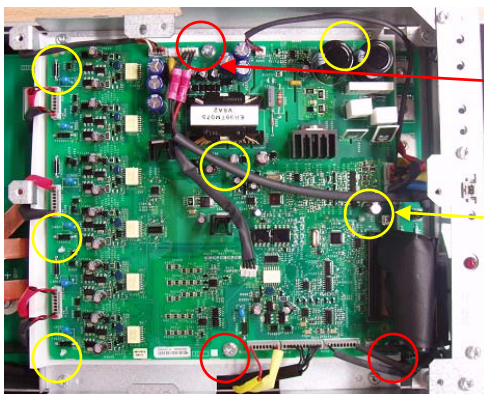
	Mark	Size	Torque
	S129-S131	M5x12	2,5Nm

Dismantling / reassembling Power Block Sub-Assembly: VX5A1HD37N4



Disconnect 3 wires (from up to down)

- Power board (S300)→E115→ Modules IGBT (12) :
Red cable : red wire→ pin 4 ; blue wire→pin 5
Black cable : red wire→ pin 6 ; blue wire→pin 7
- Power board (S301)→E116→ Modules IGBT (11) :
Red cable : red wire→ pin 4 ; blue wire→pin 5
Black cable : red wire→ pin 6 ; blue wire→pin 7
- Power board (S302)→E117→ Modules IGBT (10) :
Red cable : red wire→ pin 4 ; blue wire→pin 5
Black cable : red wire→ pin 6 ; blue wire→pin 7



Disconnect 3 wires (from left to right)

- Power board (S203)→E105
→Power 2 fans
- Power board (S103)→E101
→Power internal fan and control bloc (S103):
- Power board (S205)→E102→DEL

Disconnect 3 wires (from down to up)

- Power board (S700)→E118
→heatsink, thermal sensor
- Power board (S101)→E112
→Bus board (S101)
- Power board (S102)→E107→Modules
THY/DIO :

Blue cable→ IGBT (15) : Black wire→ pin 5
red wire→pin 4

Yellow cable→ IGBT (14) Black wire→ pin 5
red wire→pin 4

Red cable→ IGBT (13) : Black wire→ pin 5
red wire→pin 4

Disconnect 4 wires (from left to right)

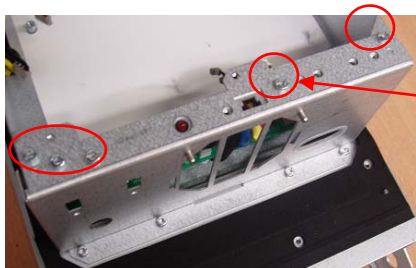
- Power board (S400)→E108→Bus
board (S100) and Module IGBT (9) :
Red wire : pin 6 of IGBT ; Black wire : pin 7 of
IGBT
- Power board (S500)→E110
→Current sensor
- Power board (S200)→E109
→ Bus board (S200)
- Power board (S202)→E111
→Bus board (S202)

Remove 3 screws (S120 to S122).

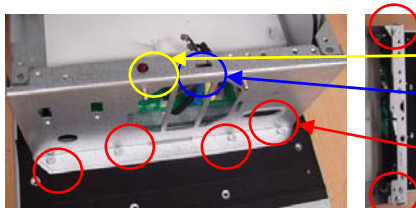
Press 6 clips.

Mark	Size	Torque
S120- S122	M4x10	1,5Nm

Metal Parts KIT: VY1A1207



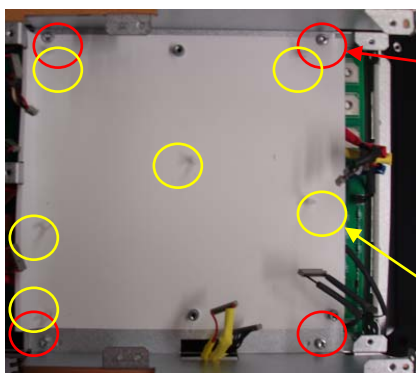
Remove 5 screws (S125 to S128).



Disconnect the DEL.

Disconnect the wire.

Remove 6 screws (S107 to S112).

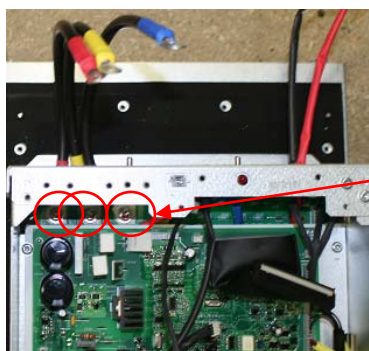


Remove 6 screws (S116 to S119).

Remove 6 clips and remove the insulate sheet.

	Mark	Size	Torque
	S125-S128	M4x10	2Nm
	S107-S112	M4x10	2Nm
	S116-S119	M4x10	2Nm

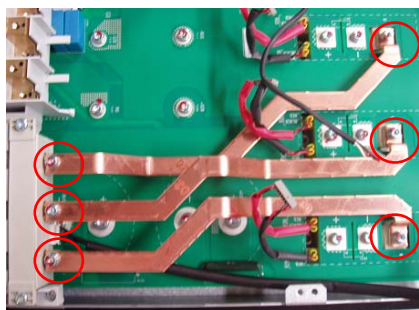
Metal Parts KIT: VY1A1207 ONLY CLASSE B



Remove 3 screws (S113 to S115).
L1 blue, L2 yellow and L3 red.

	Mark	Size	Torque
	S113-S115	M5x12	2,5Nm

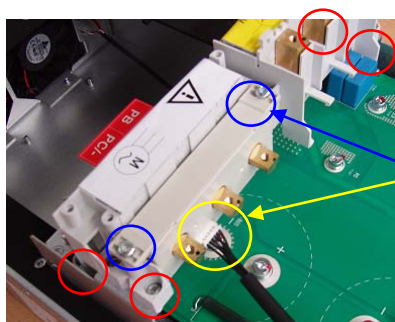
Metal Parts KIT: VY1A1207



Remove 6 screws (S101 to S106).

Mark	Size	Torque
S101-S106	M5x12	2,5Nm

Current Sensor KIT: VY1A1102



Remove 4 screws (S95 to S98).

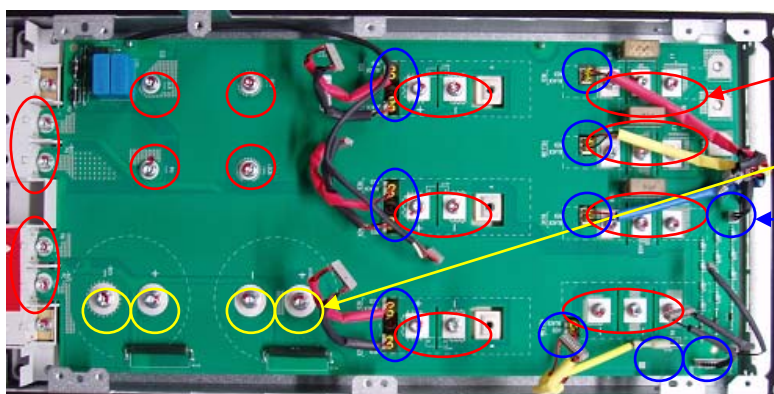
Disconnect the wire.

Remove 2 screws (S99 and S100).

The hall effect sensor can be removed.

Mark	Size	Torque
S95-S98	M4x10	2Nm
S99-S100	M4x10	2Nm

Current Sensor KIT: VY1A1102



Remove 26 screws (S69 to S90 and S64 to S67).

Remove 4 screws (S91 to S94).

Disconnect 14 wires.
(for the wiring, see the Power Block Sub-Assembly :
VX5A1HD37N4)

Mark	Size	Torque
S69 - S90 S64 - S67	M5x12	2,5Nm
S91-S94	M6x12	3,5Nm

Power Terminals: VZ3N1207

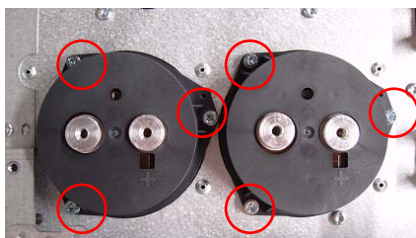


Remove 4 screws (S59 to S62).

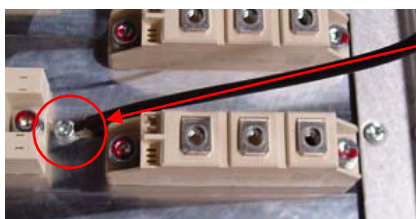
	Mark	Size	Torque
	S59-S62	M4x10	2Nm

Lots of 2 capacitors: VY1ADC1103

Thermal Sensor: VZ3G1101



Remove 6 screws (S53 to S57).

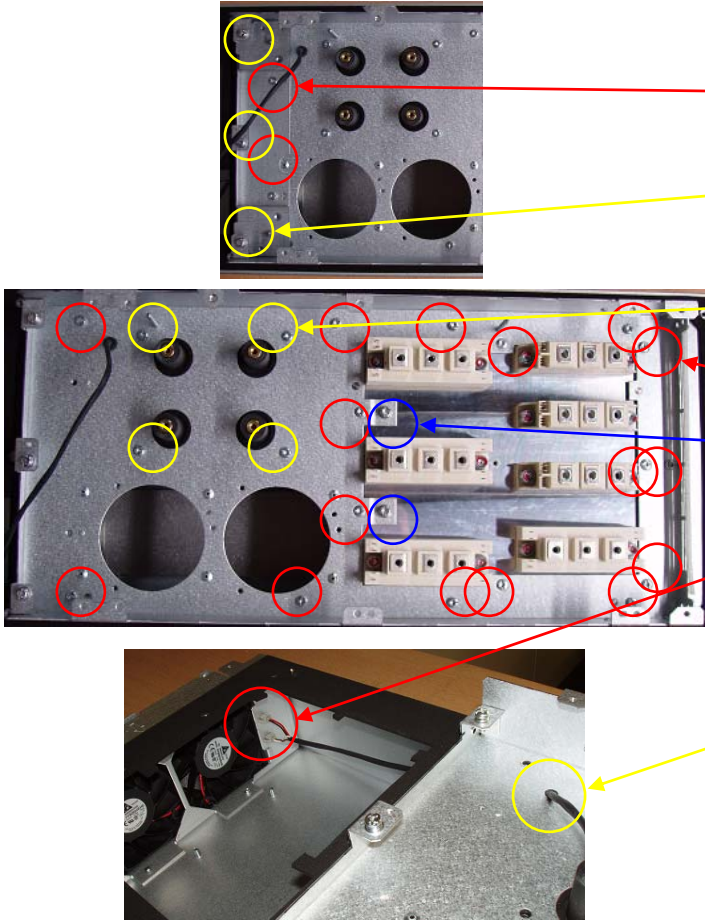


Remove this screw (S36).

The wire for the thermal sensor can be removed.

	Mark	Size	Torque
	S53-S57	M4x10	0,78Nm
	S36	M4x10	2Nm

Metal Parts KIT: VY1A1207



Remove 2 screws (S to S).

Remove 3 screws (S to S).

Remove 4 screws (S14 to S17).

Remove 22 screws (S18 to S33).

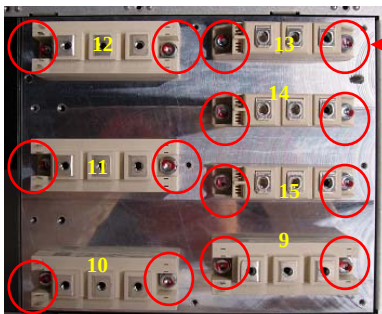
Remove 2 screws (S37 and S38).

Disconnect the wire.

Pass the wire in the hole.

	Mark	Size	Torque
	S-S		2Nm
	S-S		1,5Nm
	S14-S17	M4x12	2Nm
	S18-S33	M4x10	2Nm
	S37-S38	M5x12	2Nm

**Modules IGBT Inverter: VZ3IM2195M1271 Module Braking IGBT: VZ3IM1145M1271
Rectifier (Thyristor / Diode): VZ3TD1072M1671**



Remove 14 screws (S39 to S52).

The 7 modules can be removed (warning: grease between the modules and heatsink).

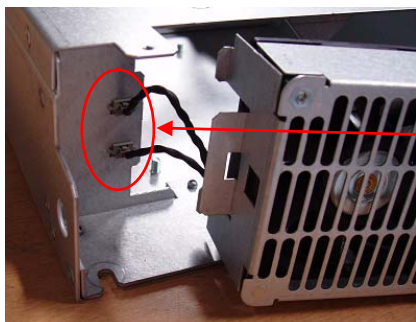
The heatsink can be removed.

	Mark	Size	Torque
	S39-S52	M6x20	1,5Nm

Fan KIT: VZ3V1206



Remove this screw (S05).



Disconnect 2 wires.

	Mark	Size	Torque
	S59-S62	M4x10	2Nm

9.17.2 Product Assembling Drawing

Refer to following file: [Assembling_175719700A53_03.pdf](#)

9.17.3 Product Cabling Drawing

Refer to following files: [Cabling_175721400A53_IED03.pdf](#)

9.18 ATV61/71 Size G (size, refer to 1.2)

9.18.1 Dismantling and reassembling

Size G: ATV61WD55N4, ATV61WD55N4C, ATV61WD75N4, ATV61WD75N4C, ATV61WD90N4, ATV61WD90N4C, ATV71WD45N4, ATV71WD55N4, ATV71WD75N4

ATV61WD55N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD75N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61WD90N4 (C)

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD45N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD55N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD15N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71WD75N4

Reference	Designation
VZ3V1217	Internal Fan
VZ3V1205	Fan KIT
VZ3N1400	Cable RJ45-RJ45
VZ3N1205	Power Terminals
VZ2LB1003	Labels KIT
VZ2GK1001	Roll of EPDM+ADH
VZ2GK1000	Roll of EPDM+ADH
VY1ADV1201	Screws KIT
VY1A1514	Insulate Sheet
VY1A1504	Front Trap
VY1A1205	Plastic Parts KIT
VX5A1HD18N4	Power Block Sub-Assembly
VX4A71100Y	Control Bloc P < 90Kw
VX4A24V250MA	Power Supply (24Volts / 250mA)
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Trap: VY1A1507



Remove 8 screws (S248 to S255)

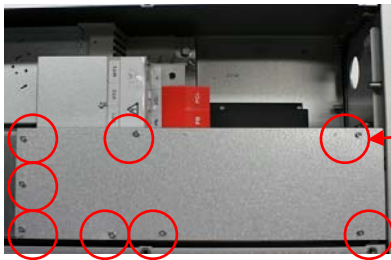
Remove this screw (S274).

Disconnect the cable
RJ45-RJ45.

The front trap can be removed.

	Mark	Size	Torque
	S248-S255	M6x16	2Nm
	S274	M4x10	1,5Nm

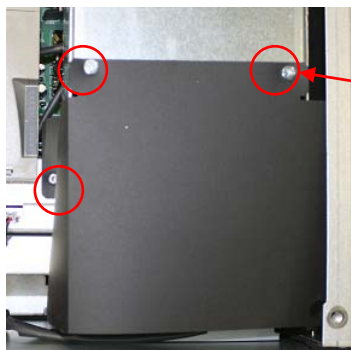
Metal Parts KIT: VY1A1209 ONLY CLASS B



Remove 8 screws (S235 to S242).

	Mark	Size	Torque
	S235-S242	M4x10	2Nm

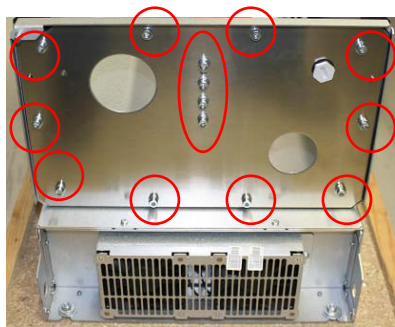
Insulate Sheet - Filter Cover: VY1A1510 ONLY CLASS B



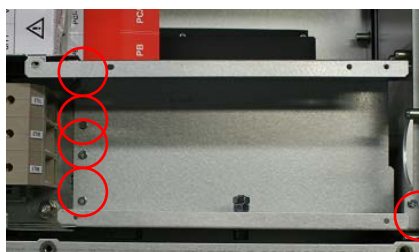
Remove 3 screws (S233 to S234).

	Mark	Size	Torque
	S233-S234	M4x10	1,5Nm

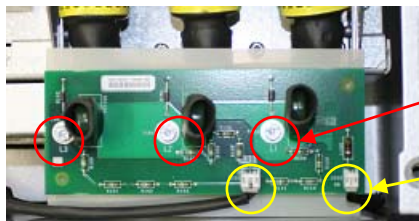
Metal Parts KIT: VY1A1209 ONLY CLASS B



Remove 14 screws (S256 to S267).



Remove 5 screws (S243 to S247).



Remove 3 screws
(S230 to S232).

Disconnect 2 wires.
(from left to right)
S101→E112
S202→E111

	Mark	Size	Torque
	S256-S267	M6x20	2Nm
	S243-S247	M4x10	2Nm
	S230-S232	M4x10	1,5Nm

Insulate Sheet - Choke Shield: VY1A1513 ONLY CLASS B

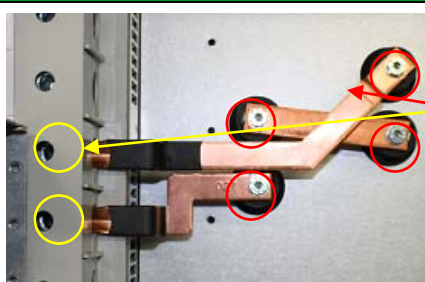


Remove 2 screws (S228 and S229).

And remove 2 screws (S226 and S227).

Mark	Size	Torque
S226- S229	M4x12	2Nm

Screws KIT : VY1ADV1104 ONLY CLASS B

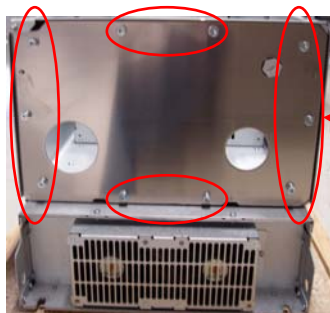


Unscrew 2 screws.

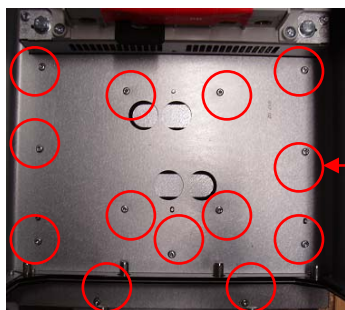
Remove 4 screws (S222 and S225).

Mark	Size	Torque
S222- S225	M6x12	3,5Nm

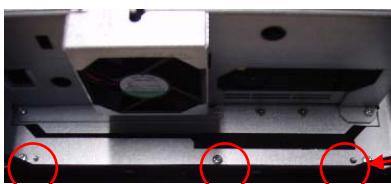
Metal Parts KIT: VY1A1209



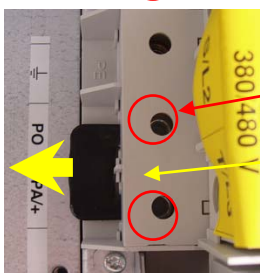
Remove 10 screws (S261 to S267).



Remove 13 screws
(S209 to S221).



Remove 3 screws (S206 to S208).



Unscrew these 2 screws.

And remove the shunt.

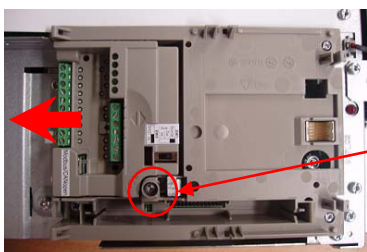
The enclosure can be removed.

	Mark	Size	Torque
	S261-S267	M6x20	2Nm
	S206-S217	M4x10	2Nm
	S218-S221	M4x10	2Nm

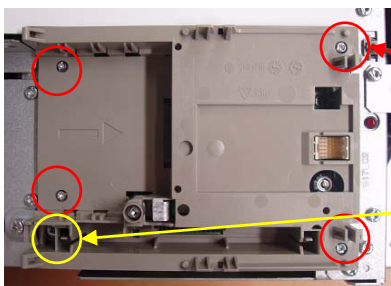
Control Bloc: VX4A71100Y



Press the two clips and pulls forward to take out the display board.

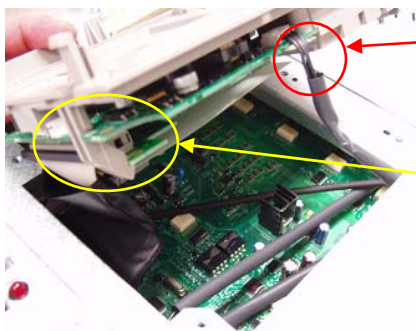


For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.



Remove 4 screws (S201 to S204).

Remove this screw (S205).

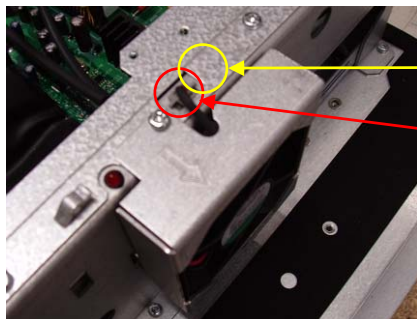


Disconnect the wire.
Application board (S103)→Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S201-S204	M3x8	0,78Nm
	S205	M4x8	1,5Nm

Internal Fan: VZ3V1218

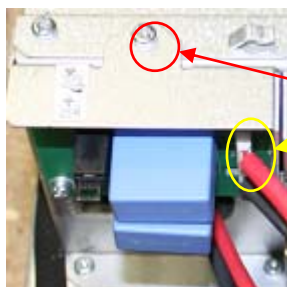


Disconnect the wire.

Remove this screw (S169).

	Mark	Size	Torque
	S169	M4x10	2Nm

Capacitor Board: VY1ADC1200 ONLY CLASS B



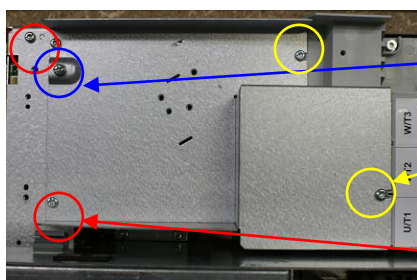
Disconnect the wire.

Capacitor PCBA (S100)→E120

Remove this screw (S195).

	Mark	Size	Torque
	S195	M4x10	2Nm

Screws KIT: VY1ADV1104 ONLY CLASS B



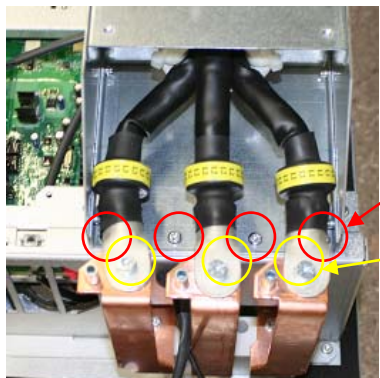
Remove this screw (S179)

Remove 2 screws
(S177 and S178).

Remove 2 screws
(S173 and S174).

	Mark	Size	Torque
	S179	M5x12	2Nm
	S177 S278	M4x12	2Nm
	S173 S174	M4x10	2Nm

Screws KIT: VY1ADV1104 ONLY CLASS B



Remove 4 screws (S166, S167, S185 and S186).

Remove 3 screws (S188 to S190).
From left to right : L3, L2 and L1



Remove 2 screws (S182 and S183).

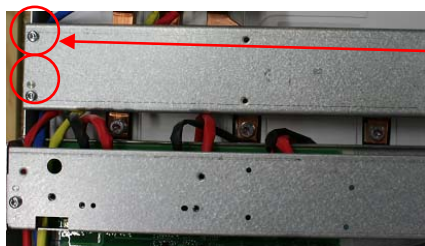


Remove 2 screws (S180 and S181).

The filter class B can be removed.

	Mark	Size	Torque
	S166, S167 S185, S186 S180 to S183	M4x10	3,5Nm
	S188- S190	M6x20	2Nm

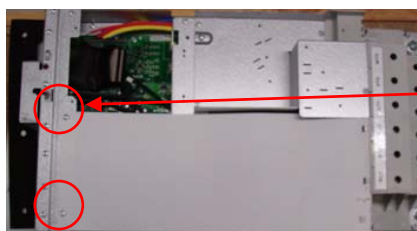
Screws KIT: VY1ADV1104 ONLY CLASS B



Remove 2 screws (S171 and S172).

	Mark	Size	Torque
	S171 S172	M4x10	2Nm

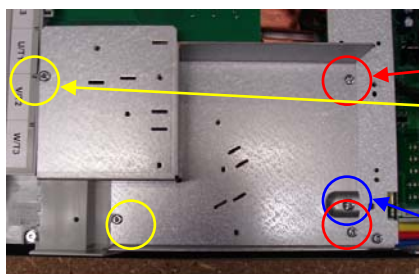
Screws KIT: VY1ADV1104 ONLY CLASS A



Remove 2 screws (S175 and S176).

	Mark	Size	Torque
	S175- S176	M4x10	1,5Nm

Screws KIT: VY1A1209 ONLY CLASS A



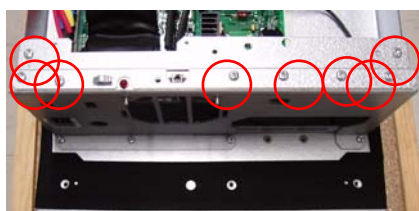
Remove 2 screws (S173 and S174).

Remove this screw (S177 and S178).

Remove this screw (S179).

	Mark	Size	Torque
	S173 S174	M4x10	2Nm
	S177 S178	M4x12	2Nm
	S179	M5x12	1,5Nm

Metal Parts KIT: VY1A1209



Remove 8 screws (S165 to S169).

	Mark	Size	Torque
	S165- S169	M4x10	2Nm

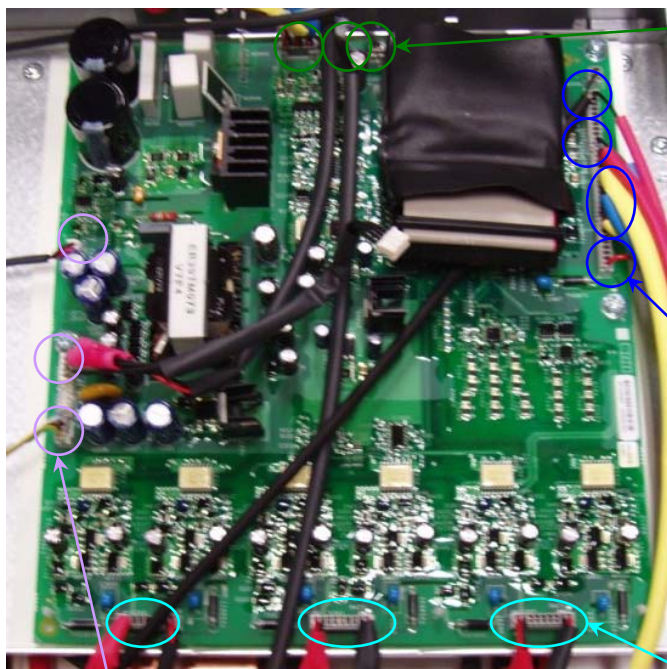
Metal Parts KIT: VY1A1209



Remove 8 screws (S165 to S169).

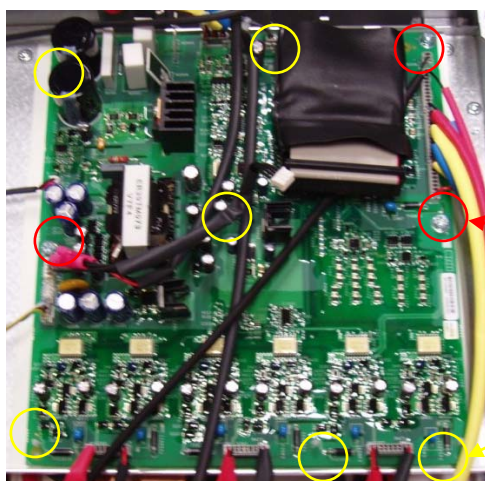
	Mark	Size	Torque
	S164 S170	M4x10	2Nm

SUB ASSEMBY POWER: AAV70185



Disconnect 3 wires (from up to down)

- Power board (S205)→E102→DEL
- Power board (S103)→E101
→Power internal fan and control bloc (S103):
- Power board (S203)→E105→Power 2 fans



Disconnect 3 wires (from right to left)

- Power board (S700)→E118→heatsink, thermal sensor
- Power board (S101)→E112→Bus board (S101)
- Power board (S102)→E107→Modules

THY/DIO :

Blue cable→ THY/DIO (10) : red wire→ pin 5 ; blue wire→pin 4

Yellow cable→ THY/DIO (9) : red wire→ pin 5 ; blue wire→pin 4

Red cable→ THY/DIO (8) : red wire→ pin 5 ; blue wire→pin 4

Disconnect 4 wires (from up to down)

- Power board (S202)→E111→Filter board (S202)
- Power board (S200)→E109→Module IGBT (11) :
Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT
- Power board (S500)→E110→3 current sensors :
Blue wire : W (sensor 48) ; red : U (46) ; Yellow : V (47)
- Power board (S400)→E108→ Module IGBT (11)
Yellow wire : pin 1 of IGBT ; Red wire : pin 6 of IGBT
Black wire : pin 7 of IGBT

Disconnect 3 wires (from left to right)

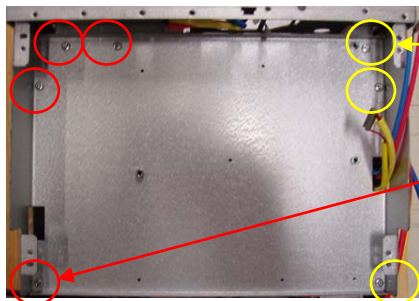
- Power board (S300)→E115→ Modules IGBT (14) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2
- Power board (S301)→E116→ Modules IGBT (13) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2
- Power board (S302)→E117→ Modules IGBT (12) :
Red cable : red wire→ pin G1 ; blue wire→pin E1
Black cable : red wire→ pin E2 ; blue wire→pin G2

Remove 3 screws (S156, S157 and S159).

Press 6 clips.

Mark	Size	Torque
S156 S157 S159	M4x10	1,5Nm

Metal Parts KIT: VY1A1209

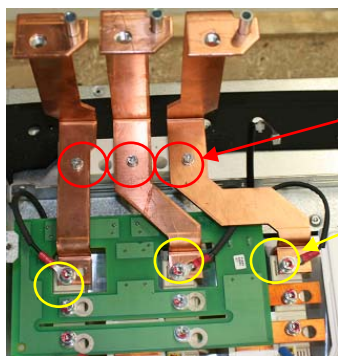


Remove 3 screws (S159 to S161).

Remove 4 screws (S152 to S155).

	Mark	Size	Torque
	S152-S155	M4x10	2Nm
	S159-S161	M4x10	2Nm

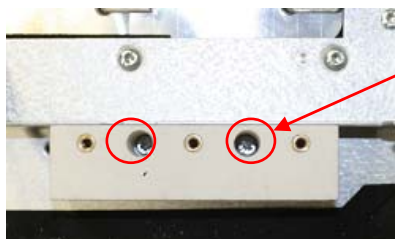
Screws KIT: VY1ADV1104 ONLY CLASS B



Remove 3 screws (S143 to S145).

Remove 3 screws (S134 to S136).
And disconnect 3 wires.

Remove 2 screws (S146 and S147).



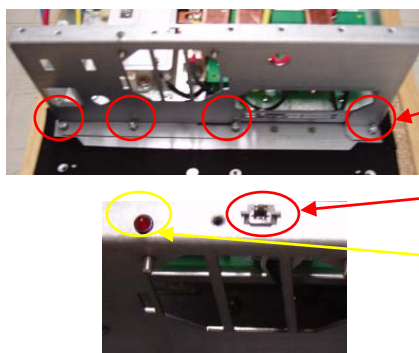
S146 and S147 :

ATV61 55Kw : 2,5Nm ; 75Kw and 90Kw : 3,5Nm

ATV71 45Kw and 55Kw : 2,5Kw ; 75Kw : 3,5Nm

	Mark	Size	Torque
	S143-S145 S146, S147	M4x20	1,5Nm
	S134-S136	M5x12	2,5 or 3,5Nm

Metal Parts KIT: VY1A1209



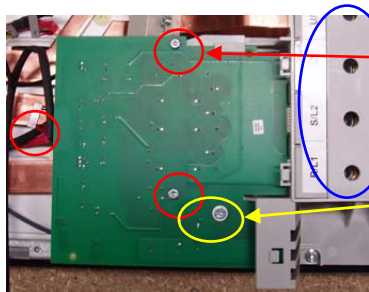
Remove 4 screws (S148 to S151).

Disconnect the wire

Disconnect the DEL.

	Mark	Size	Torque
	S148-S151	M4x10	1,5Nm

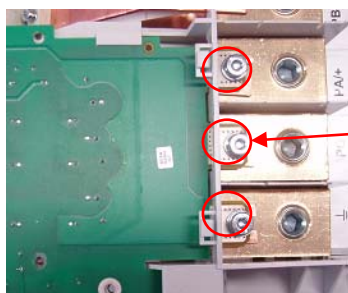
Filter Board: VX4A1110 ONLY CLASS A



Remove 3 screws (S131 - S133).

Remove this screws (S132).

Remove the mask of the power terminal.

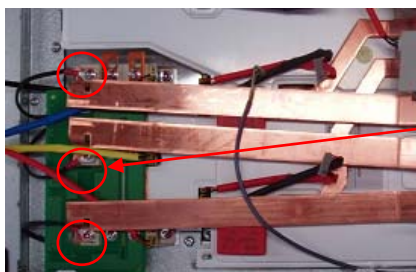


Unscrew 3 screws.

The filter board can be removed.

	Mark	Size	Torque
	S131-S133	M4x12	1,5Nm
	S132	M4x10	1,5Nm

Interconnection power KIT: VZ3N1311 ONLY CLASS A



Remove 3 screws (S134 and S136) and remove 3 wires.

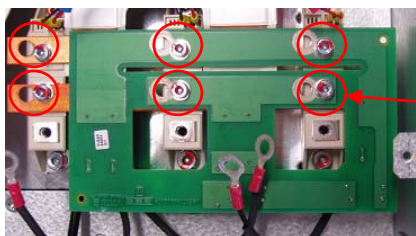
S134 to S136 :

ATV61 55Kw : 2,5Nm ; 75Kw and 90Kw : 3,5Nm

ATV71 45Kw and 55Kw : 2,5Kw ; 75Kw : 3,5Nm

	Mark	Size	Torque
	S134-S136	M5x12	2,5 or 3,5Nm

BUS Bar KIT: VZ3N1312



Remove 6 screws (S107 to S112).

S107 to S112 :

ATV61 55Kw : 2,5Nm ; 75Kw and 90Kw : 3,5Nm

ATV71 45Kw and 55Kw : 2,5Kw ; 75Kw : 3,5Nm

	Mark	Size	Torque
	S107-S112	M5x12	2,5 or 3,5Nm

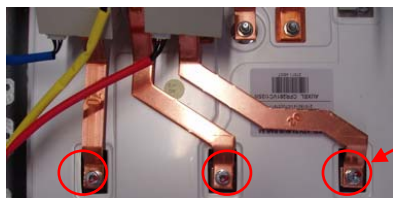
BUS Bar KIT: VZ3N1312



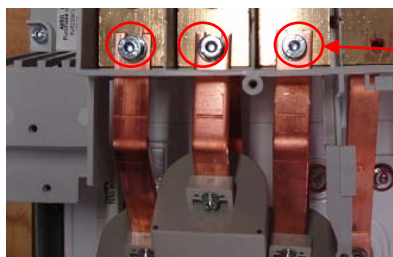
Remove 6 screws
(S113, S115, S118, S120, S123 and S124).

	Mark	Size	Torque
	S113, S115, S118, S120, S123 and S124	M6x12	3,5Nm

Interconnection power KIT: VZ3N1311



Remove 3 screws (S114, S117 and S122).
From left to right : blue cable, yellow cable and red cable.

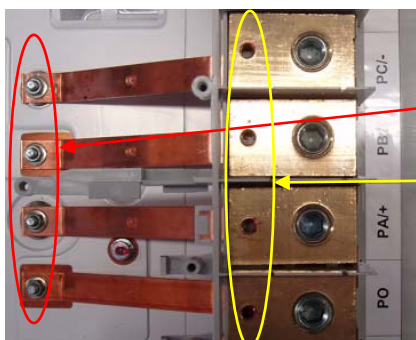


Unscrew 3 screws.

The U, V and W bars
can be removed.

	Mark	Size	Torque
	S114, S117, S122	M6x12	2,5Nm

Interconnection power KIT: VZ3N1311



Remove 4 nuts (S103 and S106).

Unscrew 4 screws.

The DC link bus
can be removed.

	Mark	Size	Torque
	S103- S106	M5	2,5Nm

Interconnection power KIT: VZ3N1311

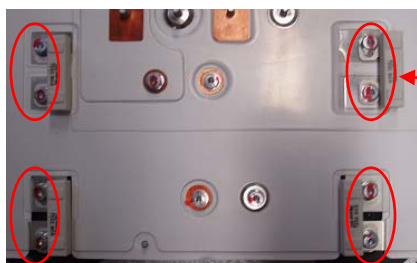


Remove 6 screws (S88 to S93).

Remove 2 screws (S278 to S279).

	Mark	Size	Torque
	S88-S93	M5x20	2,5Nm
	S278-S279	M8x20	1,5Nm

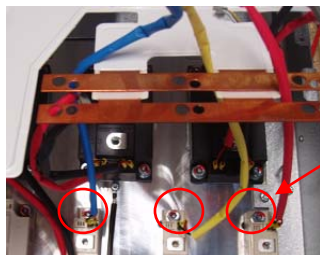
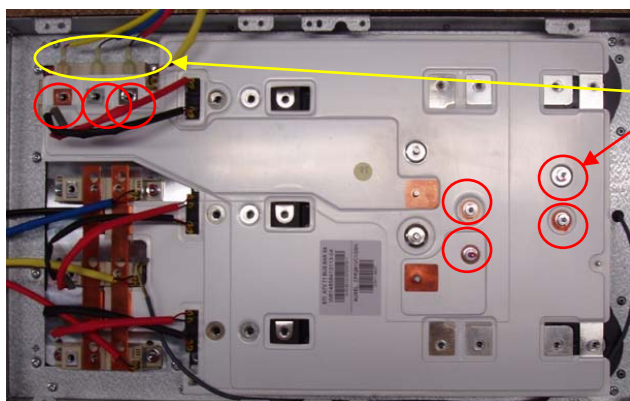
Lots of 4 capacitors: VY1ADC1107



Remove 8 screws (S80 and S87).

	Mark	Size	Torque
	S80-S87	M6x12	3,5Nm

BUS Bar KIT: VZ3N1312



Remove 7 screws (S76 to S79, S116, S119 and S121)

Disconnect wires

- Power board (S200)→E109 →Module IGBT (11) :

Blue wire : pin 2 of IGBT ; Red wire : pin 3 of IGBT

- Power board (S400)→E108 → Module IGBT (11)

Yellow wire : pin 1 of IGBT ; Red wire : pin 6 of IGBT

Black wire : pin 7 of IGBT

Disconnect this wire

Power board (S102)→E107→Modules IGBT :

Blue cable→ IGBT (10) : red wire→ pin 5
blue wire→pin 4

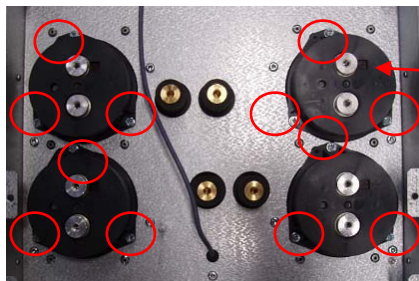
Yellow cable→ IGBT (9) : red wire→ pin 5
blue wire→pin 4

Red cable→IGBT (8) : red wire→ pin 5 ; blue wire→pin 4

The bus bar can be removed

	Mark	Size	Torque
	S76 - S79 S116, S119, S121	M6x12	3,5Nm

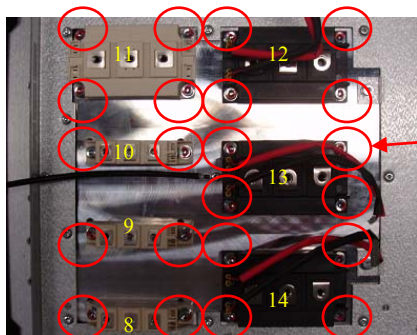
Lots of 4 capacitors: VY1ADC1107



Remove 12 screws (S65 and S75).

Mark	Size	Torque
S65-S75	M4x10	0,78Nm

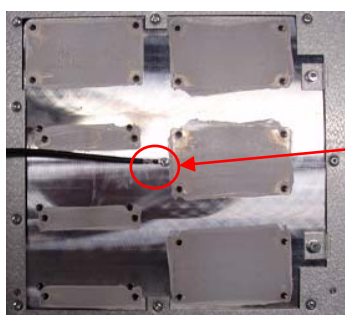
Rectifier (Thyristor / Diode): VZ3TD1072M1671 Module Braking IGBT: VZ3IM1300M1271



Remove 22 screws (S41 and S62).

Mark	Size	Torque
S41-S62	M6x20	4,4Nm

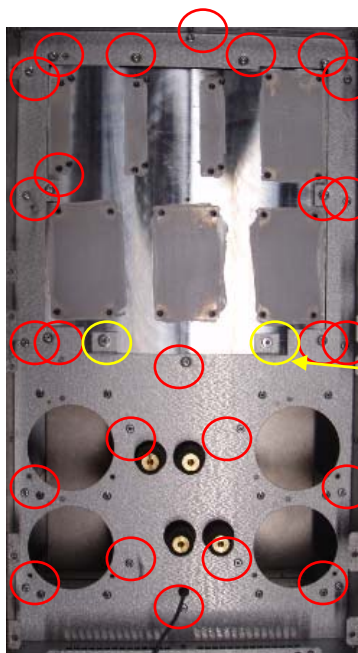
Thermal Sensor: VZ3G1101



Remove this screw (S63).

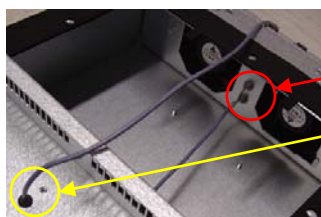
Mark	Size	Torque
S63	M4x10	2Nm

Metal Parts KIT: VY1A1209



Remove 25 screws (S16 to S40).

Remove 2 screws (S14 and S15).



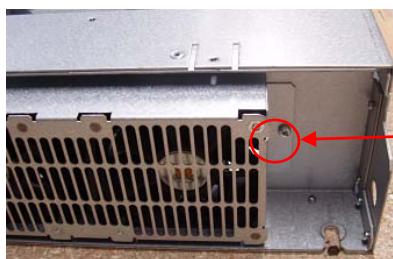
Disconnect 2 wires.

Remove the wire.

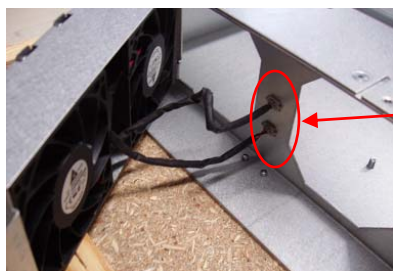
The chassis can be removed.

	Mark	Size	Torque
	S16-S40	M4x10	2Nm
	S14-S15	M6x12	3,5Nm

Fan KIT: VZ3V1208



Remove this screw (S13).



Disconnect 2 wires.

	Mark	Size	Torque
	S13	M4x10	1,5Nm

9.18.2 Product Assembling Drawing

Refer to following file: [Assembling_175719600A53_04.pdf](#)

9.18.3 Product Cabling Drawing

Refer to following files: [Cabling_175721300A53_IED04.pdf](#)

9.19 ATV61/71 Size 6Y (size, refer to 1.2)

9.19.1 Dismantling and reassembling

Size S6Y: ATV71HU22Y, ATV71HU30Y, ATV71HU40Y, ATV71HU55Y, ATV71HU75Y, ATV71HD11Y, ATV71HD15Y, ATV71HD18Y, ATV71HD22Y, ATV71HD30Y
ATV61HU30Y, ATV61HU40Y, ATV61HU55Y, ATV61HU75Y, ATV61HD11Y, ATV61HD15Y, ATV61HD18Y, ATV61HD22Y, ATV61HD30Y

ATV61HU30Y	
Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HU22Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HU40Y	
Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HU30Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HU55Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HU40Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HU75Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HU55Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD11Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HU75Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD15Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD11Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD18Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD15Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD22Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD18Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV61HD30Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A61100Y	Control Bloc P < 90Kw
VX5A1HD22Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3G1101	Thermal Sensor

ATV71HU22Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VX4A1123	Filter Board
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HU22Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HU30Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HU30Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HU40Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HU40Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HU55Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HU55Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HU75Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HU75Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HD11Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD11Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HD15Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD15Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HD18Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars kIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD18Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HD22Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD22Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HD30Y

Reference	Designation
VZ3V1219	Fan KIT
VZ3N1355	Power Input Torus
VZ3N1352	Bars KIT
VZ3N1350	Wires KIT
VX4A1123	Filter Board
VX4A1111	Filter Board
VZ3IM2145M1271Y	Modules KIT
VW3A4527	DC Choke
VY1ADV1115	Screws KIT
VY1ADC1201	Lot of 2 capacitors (1100uF / 690V)
VY1A1310	Column KIT and Plastic Support
VZ3N1208	Power Terminals
VY1A1110	Current Sensor KIT
VX5A1107	DC BUS Board
VX4A71100Y	Control Bloc P < 90Kw
VX5A1HD30Y	Power Board KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board

Screws KIT: VY1ADV1115



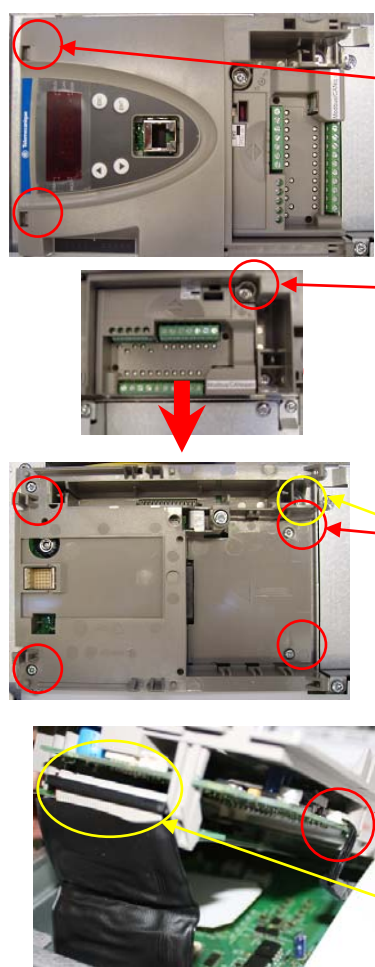
Remove 6 screws (S21).

Unscrew this screw and remove the control cap.

The front cover can be removed.

	Mark	Size	Torque
	S21	M4x10	1,5Nm

Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

Remove 4 screws (S13).

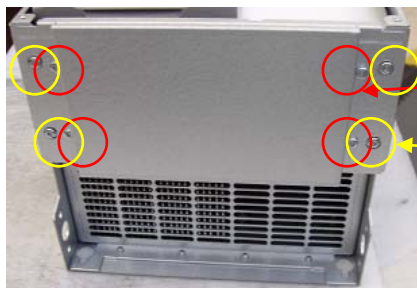
Remove this screw (S20).

Disconnect the wire.
Application board (S103)→Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

	Mark	Size	Torque
	S13	M3x8	0,78Nm
	S20	M4x8	1,5Nm

Screws KIT: VY1ADV1115



Remove 4 screws (S12).

Remove 4 screws (S03).

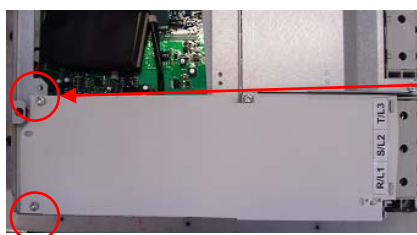


Remove 6 screws (S12).



	Mark	Size	Torque
	S12	M4x6	2Nm
	S03	M4x10	2Nm
	S12	M4x6	2Nm

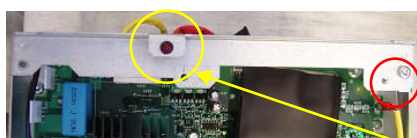
Screws KIT: VY1ADV1115



Remove 2 screws (S03).

	Mark	Size	Torque
	S03	M4x10	2Nm

Screws KIT: VY1ADV1115

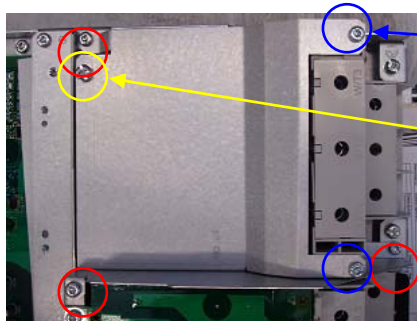


Remove this screw (S03).

Remove this DEL.

	Mark	Size	Torque
	S03	M4x10	2Nm

Screws KIT: VY1ADV1115



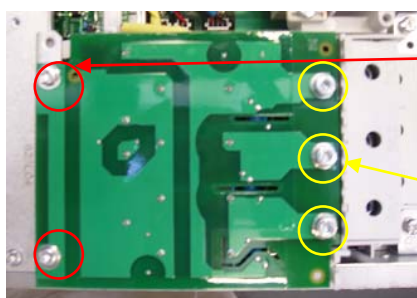
Remove 2 screws (S04).

Remove this screw (S07).

Remove 3 screws (S03).

	Mark	Size	Torque
	S04	4x12	2Nm
	S07	M5x12	1,5Nm
	S03	M4x10	2Nm

Filter Board 1: VX4A1123



Remove 2 screws (S03).

Remove 3 screws (S05).

	Mark	Size	Torque
	S03	M4x10	2Nm
	S05	M5x12	2,5Nm

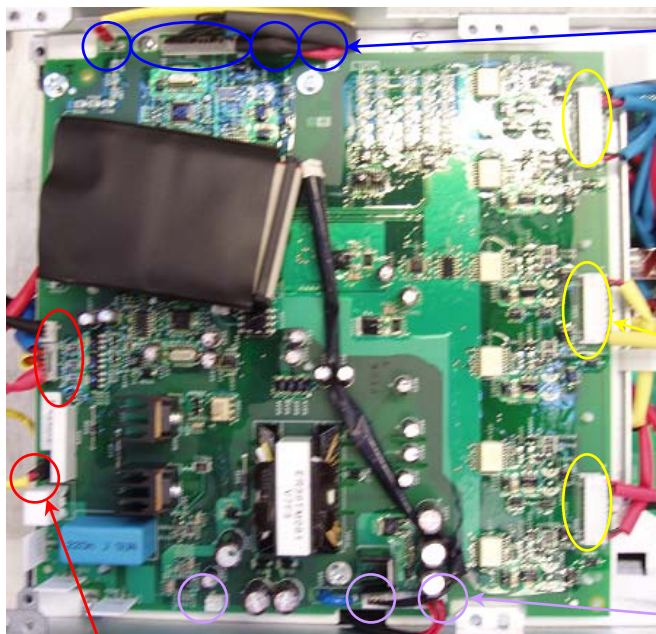
Screws KIT: VY1ADV1115



Remove 2 screws (S03).

	Mark	Size	Torque
	S03	M4x10	2Nm

Power Board KIT: VX5A1HD22Y



Disconnect 3 wires (from up to down)

- Power board (S101)→E112 (Black)→ Filter 2 board (L30R)
- Power board (S102)→E104→ Modules THY/DIO :
Red cable (module 19) : red wire→ pin 4 ; black wire→pin 5
Yellow cable (module 20) : red wire→ pin 4 ; black wire→pin 5
Blue cable (module 21) : red wire→ pin 4 ; black wire→pin 5
- Power board (S202)→E113 (Yellow)→ Filter 2 board (LR)

Disconnect 4 wires (from left to right)

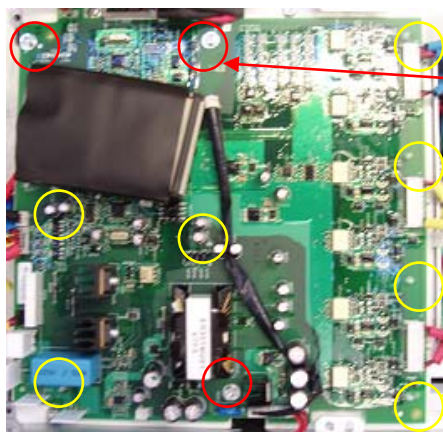
- Power board (S200)→E106 (Yellow)
→ Modules IGBT 22 pin 3
- Power board (S500)→E114 (Black)
→Hall effect sensor
- Power board (S800)→E105→Module IGBT 22 :
Blue cable→pin 2
Black cable : red wire→ pin 7 ; black wire→pin 6
- Power board (S801)→E107 (Red)
→Module IGBT 22 pin 1

Disconnect 3 wires (from up to down)

- Power board (S302)→E108 (Blue)
→Bus board (Blue)
- Power board (S301)→E109 (Yellow)
→Bus board (Yellow)
- Power board (S300)→E110 (Red)
→Bus board (Red)

Disconnect 3 wires (from left to right)

- Power board (S205)→E115→DEL
- Power board (S103)→E116 (Black)
→Application board (S103)
- Power board (S203)→E103 (Red)→Power fan

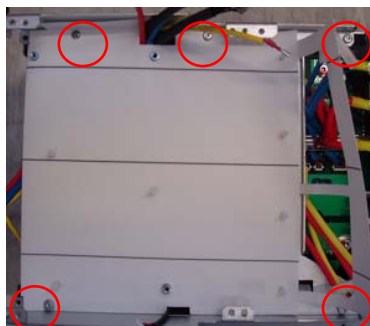


Remove 3 screws (S06).

Press 7 clips.

Mark	Size	Torque
S06	M4x10	1,5Nm

Screws KIT: VY1ADV1115

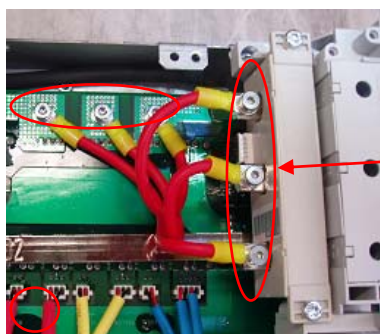


Remove 5 screws (S03).

The power insulate sheet and the PCBA chassis can be removed.

	Mark	Size	Torque
	S03	M4x10	1,5Nm

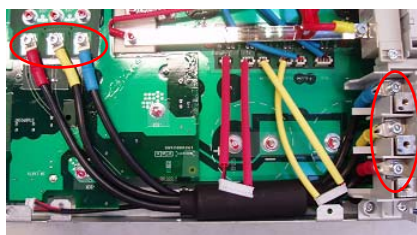
Current Sensor: VY1A1110



Remove 6 screws (S05).

	Mark	Size	Torque
	S05	M5x12	2,5Nm

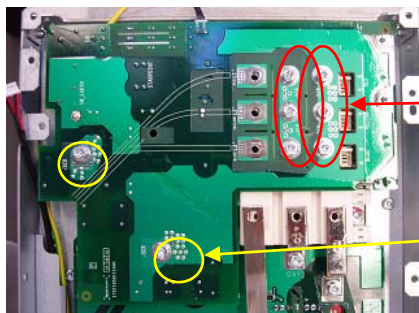
Power Input Torus: VZ3N1355



Remove 6 screws (S05).

	Mark	Size	Torque
	S05	M5x12	2,5Nm

Filter Board 2: VX4A1111



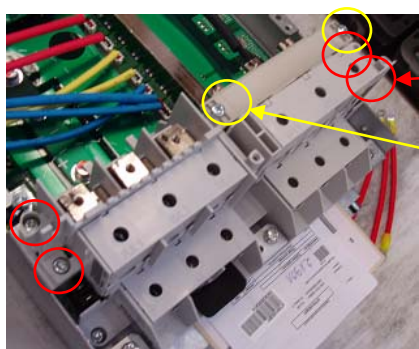
Remove 8 screws (S05).

Remove 2 screws (S10).

	Mark	Size	Torque
	S05	M5x12	2,5Nm
	S10	M5x10	1,5Nm

Current Sensor: VY1A1110

Power terminals: VZ3N1208



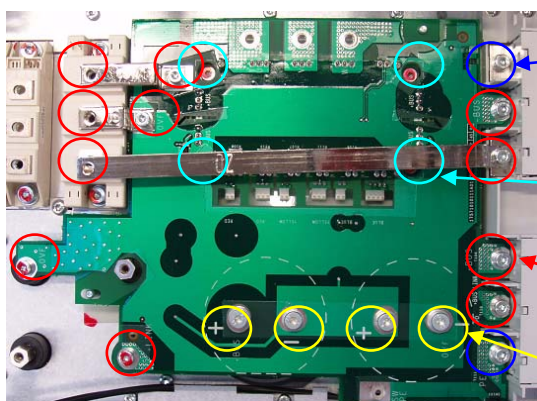
Remove 4 screws (S03).

Remove 2 screws (S04).

Remove the hall effect sensor and the top power terminal.

	Mark	Size	Torque
	S03	M4x10	2Nm
	S04	M4x12	1,5Nm

DC BUS Board: VX5A1107



Remove 2 screws (S22).

Remove 4 screws (S09).

Remove 11 screws (S05).

Remove 2 screws (S11).

	Mark	Size	Torque
	S22	M5x25	2,5Nm
	S09	M5x20	2,5Nm
	S05	M5x12	2,5Nm
	S11	M6x12	3,5Nm

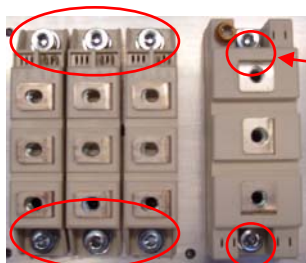
Power terminals: VZ3N1208



Remove 4 screws (S03).

	Mark	Size	Torque
	S03	M4x10	2Nm

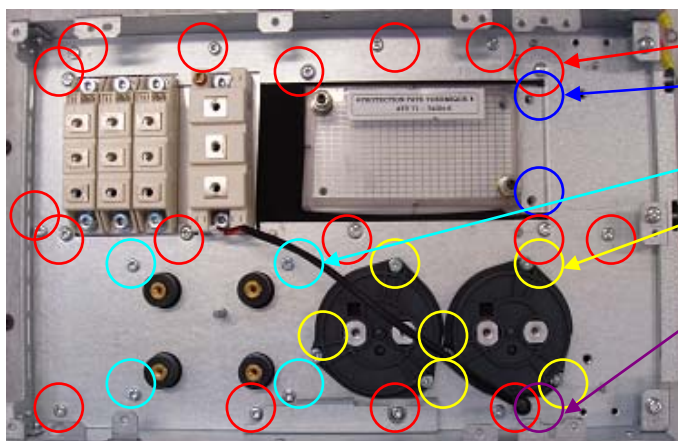
Module kit: VZ3IM2145M1271Y



Remove 8 screws (S08).

	Mark	Size	Torque
	S08	M6x20	4,4Nm

Screws KIT : VY1ADV1115



Remove 17 screws (S03).

Remove 2 screws (S05).

Remove 4 screws (S04).

Remove 6 screws (S06).

Remove the cable (E103).

The chassis
can
be removed.

	Mark	Size	Torque
	S03	M4x10	1,5Nm
	S05	M5x12	1,5Nm
	S04	4x12	2,5Nm
	S06	M4x10	2,5Nm

Fan KIT: VZ3V1219



Remove this screw (S03).



Disconnect 2 wires.

	Mark	Size	Torque
	S03	M4x10	0,78Nm

9.19.2 Product Assembling Drawing

Refer to following file: [Assembling_Cabling_aav3467800-5.1.pdf](#)

9.19.3 Product Cabling Drawing

Refer to following file:

9.20 ATV61/71 Size 8Y (size, refer to 1.2)

9.20.1 Dismantling and reassembling

Size S8Y: ATV71HD37Y, ATV71HD45Y, ATV71HD55Y, ATV71HD75Y, ATV71HD90Y, ATV61HD37Y, ATV61HD45Y, ATV61HD55Y, ATV61HD75Y, ATV61HD90Y

ATV71HD37Y	
Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD37Y	Power Board KIT
VZ3IM2200M3745Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT

ATV71HD45Y	
Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD45Y	Power Board KIT
VZ3IM2200M3745Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV71HD55Y	
Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD55Y	Power Board KIT
VZ3IM2200M55Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV71HD75Y	
Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD75Y	Power Board KIT
VZ3IM2200M7590Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV71HD90Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4529	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A71100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD90Y	Power Board KIT
VZ3IM2200M7590Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV61HD37Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD37Y	Power Board KIT
VZ3IM2200M3745Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VY1ADV1116	Screws KIT

ATV61HD45Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD37Y	Power Board KIT
VZ3IM2200M3745Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV61HD55Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4528	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD45Y	Power Board KIT
VZ3IM2200M55Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

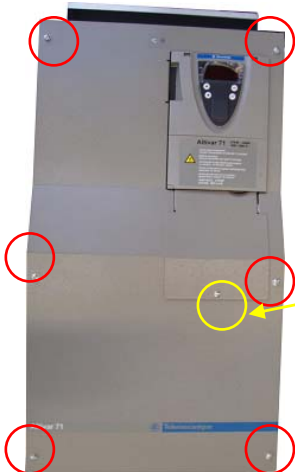
ATV61HD75Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD55Y	Power Board KIT
VZ3IM2200M7590Y	Power Modules KIT
VZ3IM2200M7590Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

ATV61HD90Y

Reference	Designation
VZ3N1354	Bus Bar KIT
VZ3N1351	Wires KIT
VZ3N1209	Power Terminal
VW3A4529	DC Choke
VY1ADV1116	Screws KIT
VY1ADC1202	Lot of 4 capacitors (1800uF / 690V + Balancing Resistor)
VY1A1111	Current Sensor KIT
VX4A1208	Snubber board
VX4A61100Y	Control Bloc P < 90Kw
VX4A1122	EMC Filter
VX5A1HD75Y	Power Board KIT
VZ3IM2200M7590Y	Power Modules KIT
VZ3N1353	Interconnection KIT
VX4A1103	Front Cover 4x7 Digits
VX4A1104	Terminal Board
VZ3V1208	Fan KIT
VZ3G1101	Thermal Sensor

Screws KIT: VY1ADV1116



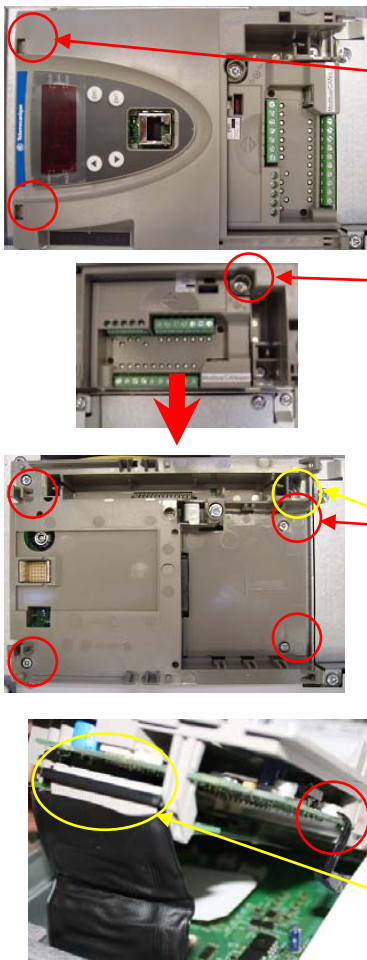
Remove 6 screws (S12).

Unscrew this screw and remove the control cap.

The front cover can be removed.

	Mark	Size	Torque
	S12	M4x10	1,5Nm

Control Bloc: VX4A61100Y



Press the two clips and pulls forward to take out the display board.

For remove the terminal board, turn the screw in the unbolt position, and pulls forward to take out the terminal board.

Remove 4 screws (S14).

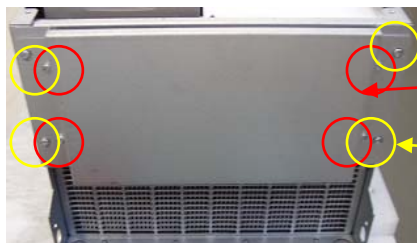
Remove this screw (S7).

Disconnect the wire.
Application board (S103)→Power board (S103)
and Power internal fan

Disconnect the ribbon cable.

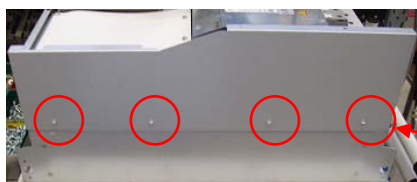
	Mark	Size	Torque
	S14	M3x8	0,78Nm
	S7	M4x8	1,5Nm

Screws KIT: VY1ADV1116

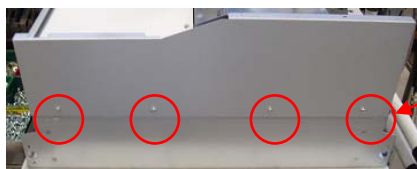


Remove 4 screws (S16).

Remove 4 screws (S03).



Remove 8 screws (S16).



	Mark	Size	Torque
	S16	M4x6	2Nm
	S03	M4x10	2Nm
	S16	M4x6	2Nm

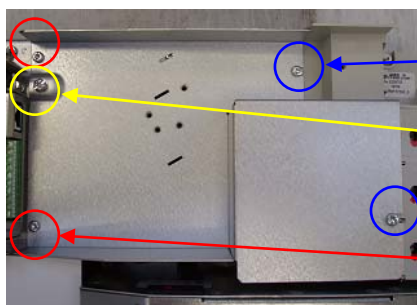
Screws KIT: VY1ADV1116



Remove 2 screws (S03).

	Mark	Size	Torque
	S03	M4x10	2Nm

Screws KIT: VY1ADV1116



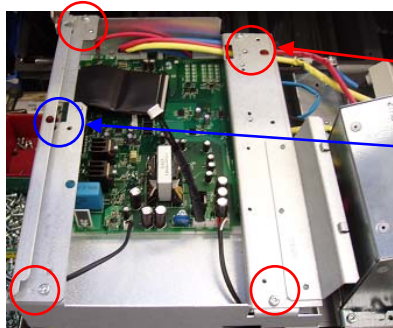
Remove 2 screws (S04).

Remove this screw (S13).

Remove 3 screws (S03).

	Mark	Size	Torque
	S04	4x12	1,5Nm
	S13	M5x12	1,5Nm
	S03	M4x10	1,5Nm

Screws KIT: VY1ADV1116

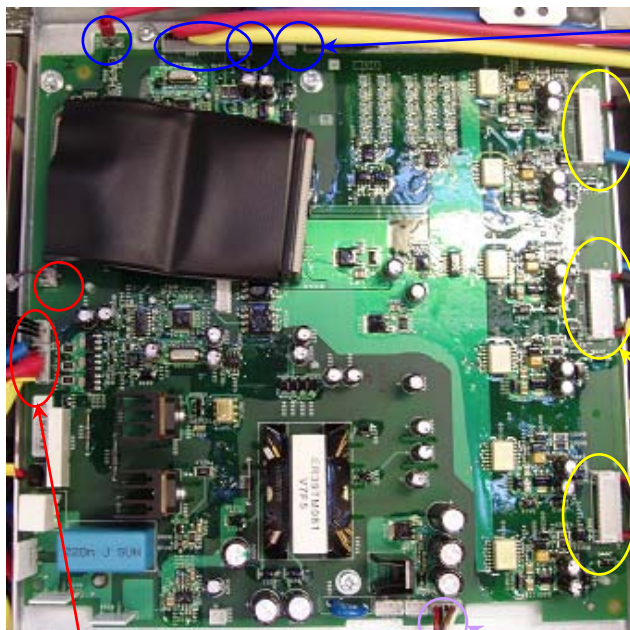


Remove 2 screws (S03).

Remove this DEL.

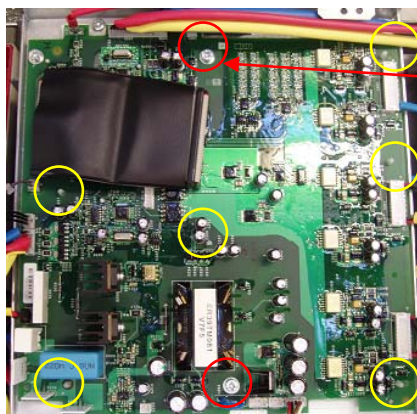
	Mark	Size	Torque
	S03	M4x10	1,5Nm

Power Board KIT: VX5A1HD37Y



Disconnect 3 wires (from up to down)

- Power board (S700)→E103→Thermal Sensor (Heatsink)
- Power board (S101)→E113→Application board (S103)
 - Power board (S102)→E105→ :
Blue cable→module 10 : black wire→ pin 5 ; red wire→pin 4
Yell. cable→module 9 : black wire→ pin 5 ; red wire→pin 4
Red cable→module 8 : black wire→ pin 5 ; red wire→pin 4



Disconnect 4 wires (from left to right)

- Power board (S200)→E107→ Module IGBT 11 :
Blue cable : pin 2
Blue cable : red wire→ pin 8 ; black wire→pin 9
 - Power board (S500)→E111
Blue cable : hall effect sensor W
Yellow cable : hall effect sensor V
Red cable : hall effect sensor U
- Black cable : red wire→ pin 8 ; black wire→pin 9
- Power board (S800)→E115 (Red)
→ Module IGBT 11 pin 3
- Power board (S801)→E106 (Yellow)
→ Module IGBT 11 pin 1

Disconnect 3 wires (from up to down)

- Power board (S302)→E108→ Module IGBT 12 :
Blue cable : red wire→ pin 10 ; black wire→pin 11
Black cable : red wire→ pin 8 ; black wire→pin 9
- Power board (S302)→E109→ Module IGBT 13 :
Yellow cable : red wire→ pin 10 ; black wire→pin 11
Black cable : red wire→ pin 8 ; black wire→pin 9
- Power board (S302)→E110→ Module IGBT 14 :
Red cable : red wire→ pin 10 ; black wire→pin 11
Black cable : red wire→ pin 8 ; black wire→pin 9

Disconnect this wire

- Power board (S203)→E104→Power fans

Remove 3 screws (S07).

Press 7 clips.

Mark	Size	Torque
S07	M4x10	1,5Nm